Communications

Detailed Solutions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Communications (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

1. **Random Processes** 1
2. **Digital Communications** 69
3. **Information Theory** 195
4. **Analog Communications** 221

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SOLUTIONS

1

Random Processes

Topics

1. Autocorrelation and Power Spectral Density
2. Properties of White Noise
3. Filtering of Random Signals Through LTI Systems

Unit Snapshot*

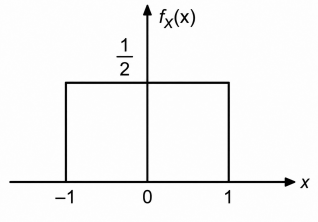
Stream: EC	**Questions:** 69	**MCQ:** 51	**MSQ:** 1	**NAT:** 17
1M: 22	**2M:** 47	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2021)–8M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 2M Ans: A ● ○ ○

Given:

$$X \sim U(-1, 1)$$



Since X and N are independent, $E[X] = E[N] = 0$ and

$$E[X^2] = \text{Var}(X) = \frac{1}{3}$$

for X uniformly distributed over $[-1, 1]$.

For $Y = X + N$,

$$E[XY] = E[X(X + N)] = E[X^2] + E[XN] = \frac{1}{3} + E[X]E[N] = \frac{1}{3}.$$

For $Z = X^2 + N$,

$$E[XZ] = E[X(X^2 + N)] = E[X^3] + E[XN].$$

Since X has a symmetric distribution about zero,

$$E[X^3] = 0, \qquad E[XN] = E[X]E[N] = 0.$$

Hence,

$$E[XZ] = 0.$$

Therefore, the required pair is

$$\boxed{\text{(A) } \frac{1}{3} \text{ and } 0}$$

Q1.25.1 MCQ 2025 2M Ans: A ● ○ ○

Given:

$$f(t) = \sum_{n=1}^{N} a_n \, p(t - nT)$$

where

$$E[a_n] = 0, \qquad \text{Var}(a_n) = 1$$

and a_n's are independent random variables.

Also,

$$p(t) = \begin{cases} 1, & 0 \le t \le \dfrac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$

(i) Mean of the process

$$E[f(t)] = E\left[\sum_{n=1}^{N} a_n p(t - nT)\right]$$

Using linearity of expectation,

$$E[f(t)] = \sum_{n=1}^{N} p(t - nT)\, E[a_n]$$

Since

$$E[a_n] = 0$$

therefore,

$$E[f(t)] = 0$$

Hence, mean is constant and independent of time.

(i) TRUE

(ii) Autocorrelation function

$$R_f(t, t+\tau) = E[f(t) f(t+\tau)]$$

Now,

$$f(t) = \sum_{n=1}^{N} a_n p(t - nT)$$

and

$$f(t+\tau) = \sum_{m=1}^{N} a_m p(t + \tau - mT)$$

Therefore,

$$R_f(t, t+\tau) = E\left[\left(\sum_{n=1}^{N} a_n p(t - nT)\right)\left(\sum_{m=1}^{N} a_m p(t + \tau - mT)\right)\right]$$

Expanding,

$$R_f(t, t+\tau) = \sum_{n=1}^{N}\sum_{m=1}^{N} p(t - nT) p(t + \tau - mT)\, E[a_n a_m]$$

Since a_n's are independent and zero mean,

$$E[a_n a_m] = 0, \qquad n \neq m$$

and

$$E[a_n^2] = 1$$

Hence only diagonal terms remain:

$$R_f(t, t+\tau) = \sum_{n=1}^{N} p(t-nT)\, p(t+\tau-nT)$$

The above expression explicitly depends on t.
Therefore, autocorrelation is not solely a function of τ.
Hence the process is not WSS.

(ii) FALSE

$\therefore$ (a) (i) is TRUE and (ii) is FALSE

Mistakes to be avoided

- **Autocorrelation Confusion:** Do not assume that

$$R_f(t, t+\tau)$$

automatically depends only on τ. A process is WSS only if the autocorrelation is independent of t.

- **Cross Terms:** While expanding

$$E[a_n a_m],$$

remember that for $n \neq m$,

$$E[a_n a_m] = E[a_n]E[a_m] = 0$$

because the variables are independent and zero mean.

Q1.24.1 MCQ | 2024 | 1M | Ans: A | ● ○ ○

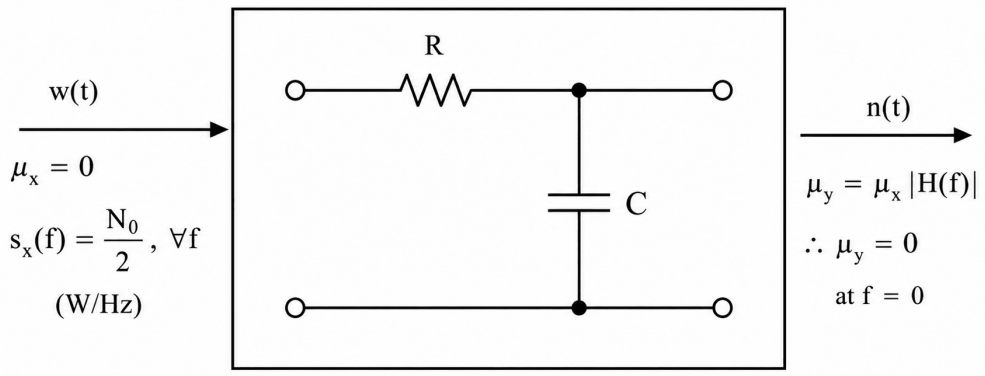


Fig. Solution Q1.24.1

$$H(f) = \frac{1}{1 + j\left(\frac{f}{f_c}\right)}$$

$$|H(f)|^2 = \frac{1}{1+\left(\frac{f}{f_c}\right)^2}, \qquad \forall f$$

where

$$f_c = \frac{1}{2\pi RC} \text{ Hz}$$

$$\text{Variance of } n(t_k) = \text{power of } n(t)$$

$$= \int_{-\infty}^{\infty} S_y(f)\, df = \int_{-\infty}^{\infty} S_x(f)\, |H(f)|^2\, df$$

$$= \int_{-\infty}^{\infty} \frac{N_0}{2} \cdot \frac{1}{1+\left(\frac{f}{f_c}\right)^2}\, df$$

$$= \frac{N_0}{2} \int_{-\infty}^{\infty} \frac{1}{1+\left(\frac{f}{f_c}\right)^2}\, df$$

Let

$$u = \frac{f}{f_c}$$

Then

$$df = f_c\, du$$

Hence,

$$= \frac{N_0}{2} f_c \int_{-\infty}^{\infty} \frac{1}{1+u^2}\, du$$

Using

$$\int_{-\infty}^{\infty} \frac{1}{1+u^2}\, du = \pi$$

$$= \frac{N_0}{2} \times f_c \times \pi$$

Substituting

$$f_c = \frac{1}{2\pi RC}$$

$$= \frac{N_0}{2} \times \frac{1}{2\pi RC} \times \pi$$

$$= \frac{N_0}{4RC} \text{ Watts}$$

$$\boxed{\text{(a)} \quad \sigma_n^2 = \frac{N_0}{4RC}}$$

Key Points

- **PSD Relation:**

$$S_y(f) = S_x(f)\,|H(f)|^2$$

- **White Noise PSD:**

$$S_x(f) = \frac{N_0}{2}, \qquad \forall f$$

- **Noise Power:** Output noise variance is obtained using

$$\sigma_n^2 = \int_{-\infty}^{\infty} S_y(f)\,df$$

Q1.24.2 NAT 2024 2M Ans: 0.5 ● ○ ○

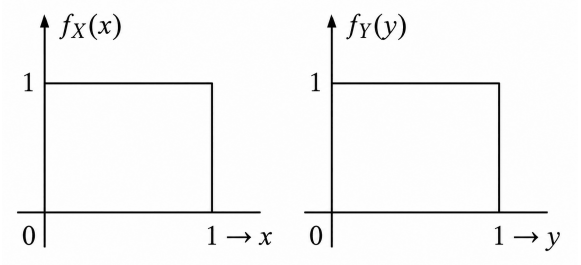


Fig. Solution Q1.24.2

$$f(X,Y) = f(X) \times f(Y)$$

Since X and Y are independent random variables,

$$P[X \geq Y] = P[Y < X] = \int_{x=0}^{1}\int_{y=0}^{x} f(X,Y)\,dy\,dx$$

$$= \int_{x=0}^{1}\int_{y=0}^{x} 1 \times 1\,dy\,dx$$

$$P[X \geq Y] = \int_{x=0}^{1} (Y)\Big|_{Y=0}^{Y=X} dx = \int_{x=0}^{1} X\,dx = \frac{1}{2}$$

$$\boxed{(0.5)}$$

Q1.23.1 MSQ 2023 1M Ans: A,B ● ○ ○

PSD functions are REAL and EVEN

$\therefore$ (a) and (b) are correct

(c) is ODD function hence wrong

(d) is neither EVEN nor ODD hence wrong

(a) $S_X(\omega) = \dfrac{2}{9+\omega^2}$ (b) $S_X(\omega) = e^{-\omega^2}\cos^2\omega$

Q1.23.2 MCQ 2023 2M Ans: B ● ○ ○

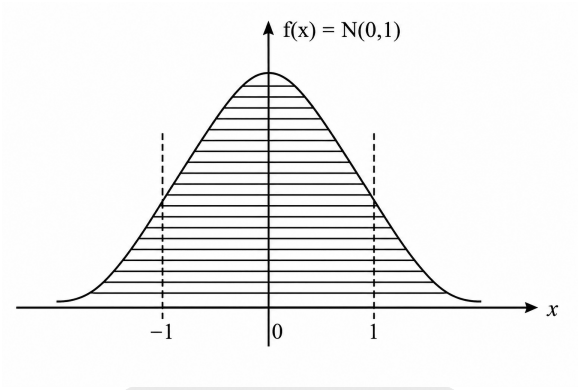


Fig. Solution Q1.23.2

$$f_X(x) = N(0,1)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad -\infty < X < \infty$$

$$P[X > 1] = 1 - P[X \le 1] = 1 - \left[1 - Q\left(\frac{1-0}{1}\right)\right]$$

$$= Q(1)$$

$$\therefore\ P[-1 < X < 1] = 1 - 2Q(1)$$

$$\therefore\ P[Y = 0] = P[-1 < X < 1] = 1 - 2Q(1)$$

$\Rightarrow$ Any values of $X \ge 2$ and $X \le -2$ are mapped to $Y = 1$ and $Y = -1$

$$P[X > 2] = 1 - P[X \le 2] = 1 - [1 - Q(2)] = Q(2)$$

$$\therefore\ P[X > 2] = P[Y = 1] = Q(2)$$

$$P[X \le -2] = 1 - Q(-2)$$

$$= 1 - [1 - Q(2)] = Q(2)$$

$$\therefore\ P[X \le -2] = P[Y = -1] = Q(2)$$

$\Rightarrow$ from $X = 1$ to 2, Y is CRV

$\therefore$ Y is MRV taking values $Y = 0,\ Y = 1,\ Y = -1$ as discrete and $Y = 0$ to $Y = 1$ as continuous

Hence pdf of Y is

$$\boxed{\text{(b) } a\delta(Y+1) + b\delta(Y) + c\delta(Y-1) + g(Y)}$$

Q1.23.3 NAT | 2023 | 2M | Ans: 0.25 | ● ○ ○

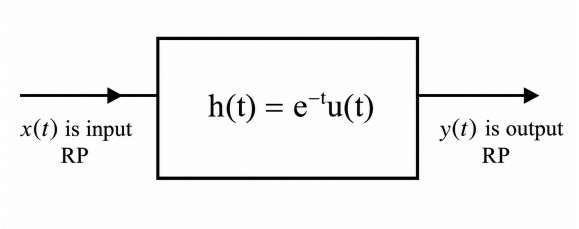


Fig. Solution Q1.23.3

For an LTI system, the output PSD is given by

$$S_Y(f) = S_X(f)\,|H(f)|^2$$

Given,

$$h(t) = e^{-t}u(t)$$

Taking Fourier Transform,

$$H(f) = \frac{1}{1 + j2\pi f}$$

Hence,

$$|H(f)|^2 = \frac{1}{1 + 4\pi^2 f^2}, \qquad -\infty < f < \infty$$

Since the input PSD is

$$S_X(f) = \frac{1}{2}$$

therefore,

$$S_Y(f) = \frac{1}{2} \cdot \frac{1}{1 + 4\pi^2 f^2}, \qquad -\infty < f < \infty$$

Output power is obtained by integrating the output PSD

$$P_0 = \int_{-\infty}^{\infty} S_Y(f)\,df$$

$$P_0 = \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{1 + (2\pi f)^2}\,df$$

Let

$$2\pi f = y$$

$$2\pi\, df = dy$$

$$df = \frac{dy}{2\pi}$$

Substituting,

$$P_0 = \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{1+y^2} \cdot \frac{dy}{2\pi}$$

$$= \frac{1}{4\pi}\int_{-\infty}^{\infty} \frac{1}{1+y^2}\, dy$$

Using

$$\int \frac{1}{1+y^2}\, dy = \tan^{-1}(y)$$

$$P_0 = \frac{1}{4\pi}\left[\tan^{-1}(y)\right]_{-\infty}^{\infty}$$

$$= \frac{1}{4\pi}\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right]$$

$$= \frac{1}{4\pi}(\pi)$$

$$\boxed{P_0 = \frac{1}{4} = 0.25}$$

Q1.22.1 NAT | 2022 | 2M | Ans: 1.17 | ● ● ●

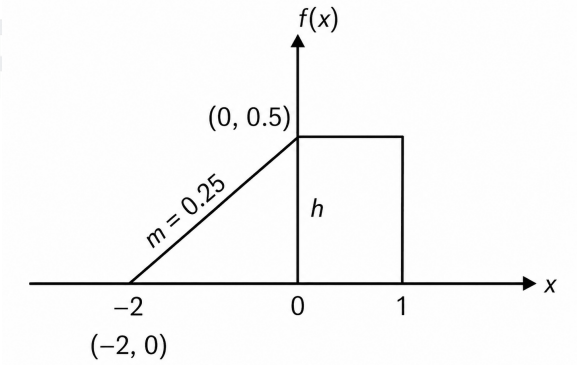


Fig. Solution Q1.22.1

Using the property that total area under a pdf must be equal to 1

$$\frac{1}{2} \times 2 \times h + 1 \times h = 1$$

$$2h = 1$$

$$h = 0.5$$

Hence,

$$f(x) = \frac{1}{4}(x+2), \qquad -2 \le x \le 0$$

$$f(x) = 0.5, \qquad 0 \le x \le 1$$

$$\text{For } 0 \le x \le 1, \quad x_q = \alpha \quad \text{and} \quad \theta_e = x - \alpha$$

$$\text{For } -2 \le x \le 0, \quad x_q = \beta \quad \text{and} \quad \theta_e = x - \beta$$

Quantization noise power is minimized using mean square error criterion

$$N_q = E[\theta_e^2] = \int_0^1 (x-\alpha)^2 \frac{1}{2}\, dx$$

$$= \frac{1}{2} \cdot \frac{1}{3}\left[(x-\alpha)^3\right]_0^1$$

$$= \frac{1}{6}\left[(1-\alpha)^3 + \alpha^3\right]$$

For minimum noise power,

$$\frac{dN_q}{d\alpha} = 0$$

$$\Rightarrow \frac{1}{6}\left[3(1-\alpha)^2(-1) + 3\alpha^2\right] = 0$$

$$3\alpha^2 = 3(1-\alpha)^2$$

$$\alpha^2 = 1 + \alpha^2 - 2\alpha$$

$$2\alpha = 1$$

$$\boxed{\alpha = 0.5}$$

Now,

$$N_q = E[\theta_e^2] = \int_{-2}^0 (x-\beta)^2 \frac{1}{4}(x+2)\, dx$$

$$= \frac{1}{4}\int_{-2}^0 (x^2 + \beta^2 - 2x\beta)(x+2)\, dx$$

Expanding,

$$= \frac{1}{4}\int_{-2}^0 \left(x^3 + \beta^2 x - 2x^2\beta + 2x^2 + 2\beta^2 - 4x\beta\right) dx$$

$$= \frac{1}{4}\left[\frac{1}{4}(-16) + \frac{\beta^2}{2}(-4) - \frac{2\beta}{3}(8) + \frac{2}{3}(8) + 2\beta^2(2) - 2\beta(-4)\right]$$

$$= -1 - \frac{\beta^2}{2} - \frac{4\beta}{3} + \frac{4}{3} + \beta^2 + 2\beta$$

For minimum noise power,

$$\frac{dN_q}{d\beta} = 0$$

$$\Rightarrow -\beta - \frac{4}{3} + 2\beta + 2 = 0$$

$$\beta = \frac{4}{3} - 2$$

$$\boxed{\beta = -\frac{2}{3}}$$

Finally,

$$\alpha - \beta = \frac{1}{2} - \left(-\frac{2}{3}\right)$$

$$= \frac{1}{2} + \frac{2}{3} = \frac{3+4}{6} = \frac{7}{6}$$

$$\boxed{\alpha - \beta = \frac{7}{6} \approx 1.17}$$

Q1.21.1 NAT | 2021 | 1M | Ans: 2 | ● ○ ○

For a wide-sense stationary (WSS) random process,

$$\boxed{\text{Average power} = R_X(0)}$$

From the given autocorrelation function graph,

$R_X(\tau)$ has a maximum value of 2 at $\tau = 0$

Therefore,

$$\boxed{R_X(0) = 2}$$

Hence,

$$\boxed{\text{Average power of } X(t) = 2}$$

Q1.20.1 NAT | 2020 | 1M | Ans: 6 | ● ● ○

Given,

$$\phi(t) = \begin{cases} 1, & 5 \le t \le 7 \\ 0, & \text{otherwise} \end{cases}$$

$$S_W(f) = 3 \text{ Watts/Hz}$$

$$R_W(\tau) = 3\delta(\tau) = 3\delta(t_1 - t_2)$$

Variance is defined as

$$\text{Var}[y] = E[y^2] - E[y]^2$$

The output of the correlator is

$$y = \int_{-\infty}^{\infty} W(t)\phi(t)\,dt$$

Since white noise has zero mean, $E[W(t)] = 0$.
Therefore,

$$E[y] = \int_{-\infty}^{\infty} E[W(t)]\phi(t)\,dt = 0$$

Now,

$$E[y^2] = E[y \cdot y]$$

$$= E\left[\int_{-\infty}^{\infty} W(t_1)\phi(t_1)\,dt_1 \int_{-\infty}^{\infty} W(t_2)\phi(t_2)\,dt_2\right]$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} E[W(t_1)W(t_2)]\phi(t_1)\phi(t_2)\,dt_1 dt_2$$

Using $E[W(t_1)W(t_2)] = R_W(t_1 - t_2)$,

$$E[y^2] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} R_W(t_1 - t_2)\phi(t_1)\phi(t_2)\,dt_1 dt_2$$

Substituting $R_W(t_1 - t_2) = 3\delta(t_1 - t_2)$,

$$E[y^2] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} 3\delta(t_1 - t_2)\phi(t_1)\phi(t_2)\,dt_1 dt_2$$

Using the sifting property of delta function, $t_1 = t_2 = t$.
Hence,

$$E[y^2] = 3\int_{-\infty}^{\infty} \phi^2(t)\,dt$$

$$= 3 \times \text{Energy}[\phi(t)]$$

Now, $\phi(t) = 1$ for $5 \le t \le 7$.
Therefore,

$$\text{Energy}[\phi(t)] = \int_5^7 1^2\,dt = 2$$

Thus,

$$E[y^2] = 3 \times 2 = 6$$

Also, $E[y]^2 = 0$.
Therefore,

$$\text{Var}[y] = E[y^2] - E[y]^2 = 6 - 0$$

$$\boxed{\text{Var}[y] = 6}$$

Equivalent PSD approach:

$$S_Y(f) = |H(f)|^2 S_W(f)$$

$$\sigma_Y^2 = \int_{-\infty}^{\infty} S_Y(f)\,df = \frac{N_0}{2}\int_{-\infty}^{\infty} |H(f)|^2\,df$$

Using Parseval energy theorem,

$$\int_{-\infty}^{\infty} |H(f)|^2\, df = \int_{-\infty}^{\infty} |\phi(t)|^2\, dt = E_\phi$$

Hence,

$$\sigma_Y^2 = \left(\frac{N_0}{2}\right) E_\phi = 3 \times 2 = 6$$

$$\boxed{\sigma_Y^2 = 6}$$

Key Points

- **Output Variance of White Noise:** For white noise passed through a correlator or LTI system,

$$\sigma_Y^2 = \frac{N_0}{2} E_\phi$$

where E_ϕ is the energy of the impulse response or basis function.

Mistakes to be avoided

- **Confusing Power and Energy:** Do not directly substitute signal duration in place of energy; first compute

$$E_\phi = \int |\phi(t)|^2 dt$$

- **Incorrect Delta Function Handling:** While using

$$\delta(t_1 - t_2)$$

ensure the sifting property is applied correctly to reduce the double integral.

- **Missing Zero Mean Property:** White Gaussian noise has zero mean, so

$$E[Y]^2 = 0$$

should not be ignored in variance calculations.

Q1.19.1 NAT | 2019 | 2M | Ans: 17.5 | ● ○ ○

Let,

$$X_1(t) = h(t) * X(t)$$

For an LTI system,

$$S_{X_1}(f) = |H(f)|^2 S_X(f)$$

Since $Z(t)$ and $X(t)$ are uncorrelated,

$$R_{XZ}(\tau) = R_{ZX}(\tau) = 0$$

Hence,

$$R_{YY}(\tau) = R_{X_1X_1}(\tau) + R_{ZZ}(\tau)$$

Taking Fourier Transform,

$$S_{YY}(f) = S_{X_1}(f) + S_{ZZ}(f)$$

Power in $y(t)$ is equal to the total area under $S_{YY}(f)$.

$$S_{X_1}(f) = |H(f)|^2 S_X(f) = (0.5)^2 \times 5 = 1.25 \text{ W/Hz}, \qquad -5 \le f \le 5$$

This flat band runs from $f = -5$ Hz to $f = +5$ Hz, so its width is $5 - (-5) = 10$ Hz. Area under $S_{X_1}(f)$ (width × height):

$$= (10)(1.25) = 12.5$$

Therefore,area under $S_Z(f)$:

$$= \frac{1}{2} \times 10 \times 1 = 5$$

Therefore,

$$P_Y = 12.5 + 5 = 17.5 \text{ W}$$

$$\boxed{P_Y = 17.5 \text{ W}}$$

Q1.19.2 NAT | 2019 | 1M | Ans: 11 | ● ○ ○

Given X, Y are random variables,

$$E(2X + Y) = 0$$

$$E(X + 2Y) = 33$$

Using linearity of expectation,

$$2E(X) + E(Y) = 0 \qquad \cdots (1)$$

$$E(X) + 2E(Y) = 33 \qquad \cdots (2)$$

By solving (1) and (2),

$$E(Y) = 22$$

$$E(X) = -11$$

Therefore,

$$E(X) + E(Y) = 22 - 11$$

$$\boxed{E(X) + E(Y) = 11}$$

Q1.19.3 NAT | 2019 | 1M | Ans: 0.3679 | ● ○ ○

Given that Z is an exponential random variable with cumulative distribution function

$$F_Z(x) = \begin{cases} 1 - e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

The probability density function is obtained by differentiating the CDF.

$$f_Z(x) = F_Z'(x)$$

Hence,

$$f_Z(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Using conditional probability,

$$P(Z > 2 \mid Z > 1) = \frac{P((Z > 2) \cap (Z > 1))}{P(Z > 1)}$$

Since $Z > 2 \subset Z > 1$,

$$P(Z > 2 \mid Z > 1) = \frac{P(Z > 2)}{P(Z > 1)}$$

$$= \frac{\int_2^\infty e^{-x}\, dx}{\int_1^\infty e^{-x}\, dx}$$

$$= \frac{[-e^{-x}]_2^\infty}{[-e^{-x}]_1^\infty}$$

$$= \frac{-(e^{-\infty} - e^{-2})}{-(e^{-\infty} - e^{-1})}$$

Since $e^{-\infty} = 0$,

$$= \frac{e^{-2}}{e^{-1}}$$

$$= e^{-1}$$

$$\boxed{P(Z > 2 \mid Z > 1) = 0.3679}$$

Key Points

- **Conditional Probability Formula:**
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
- **Subset Simplification:** If $A \subseteq B$, then
$$A \cap B = A$$
which simplifies the conditional probability expression.
- **Exponential Distribution Property:** Exponential random variables exhibit the memoryless property:
$$P(Z > s + t \mid Z > s) = P(Z > t)$$

Q1.18.1 MCQ | 2018 | 2M | Ans: B | ● ○ ○

Given,

$$S_N(f) = 0.5 \text{ W/Hz}$$

For an LTI system,

$$S_Y(f) = |H(f)|^2 S_N(f)$$

The output power is obtained by integrating the output PSD.

$$P_Y = \int_{-\infty}^{\infty} S_Y(f)\, df$$

Substituting,

$$P_Y = \int_{-\infty}^{\infty} |H(f)|^2 (0.5)\, df$$

$$= 0.5 \int_{-\infty}^{\infty} |H(f)|^2 \, df$$

Using Parseval's theorem,

$$\int_{-\infty}^{\infty} |h(t)|^2 dt = \int_{-\infty}^{\infty} |H(f)|^2 df$$

Hence,

$$P_Y = 0.5 \int_{-\infty}^{\infty} |h(t)|^2 dt$$

Given,

$$h(t) = 0.5e^{-t^2/2}$$

Therefore,

$$P_Y = 0.5 \int_{-\infty}^{\infty} \left(0.5e^{-t^2/2}\right)^2 dt$$

$$= \frac{1}{8} \int_{-\infty}^{\infty} e^{-t^2} \, dt$$

Using the Gaussian integral,

$$\int_{-\infty}^{\infty} e^{-at^2} \, dt = \sqrt{\frac{\pi}{a}}$$

For $a = 1$,

$$\int_{-\infty}^{\infty} e^{-t^2} \, dt = \sqrt{\pi}$$

Thus,

$$P_Y = \frac{1}{8}\sqrt{\pi}$$

$$P_Y \approx 0.22$$

$$\boxed{(B) P_Y = 0.22}$$

Key Points

- **Output PSD of LTI System:** For an LTI system excited by a random process,

$$S_Y(f) = |H(f)|^2 S_X(f)$$

- **Power from PSD:** Average output power is equal to the total area under the output PSD curve.
- **Parseval's Theorem:** Energy can be computed equivalently in time domain, frequency domain f, or angular frequency domain ω:

$$\int_{-\infty}^{\infty} |h(t)|^2 dt = \int_{-\infty}^{\infty} |H(f)|^2 df = \frac{1}{2\pi} \int_{-\infty}^{\infty} |H(\omega)|^2 d\omega$$

- **Gaussian Integral Formula:**

$$\int_{-\infty}^{\infty} e^{-at^2} dt = \sqrt{\frac{\pi}{a}}$$

which is frequently used with Gaussian-shaped signals.

Q1.18.2 NAT | 2018 | 1M | Ans: 0.25 | ● ○ ○

Given X_1, X_2, X_3, X_4 are independent, zero mean, unit variance random variables.
Since all random variables are identically distributed and independent, each variable has equal probability of being the smallest.

$$P(X_1 \text{ smallest}) = P(X_2 \text{ smallest}) = P(X_3 \text{ smallest}) = P(X_4 \text{ smallest})$$

Also,

$$P(X_1 \text{ smallest}) + P(X_2 \text{ smallest}) + P(X_3 \text{ smallest}) + P(X_4 \text{ smallest}) = 1$$

Hence,

$$4\,P(X_1 \text{ smallest}) = 1$$

$$P(X_1 \text{ smallest}) = \frac{1}{4} = 0.25$$

$$\boxed{P(X_1 \text{ smallest}) = 0.25}$$

Q1.17.1 MCQ | 2017 | 2M | Ans: A | ● ● ○

(I) $E[x(t)] = E[y(t)]$
For an LTI system, the output mean is scaled by the DC gain $H(0)$.

$$E[y(t)] = H(0)\,E[x(t)]$$

Given $H(0) = 1$,

$$E[y(t)] = E[x(t)]$$

Hence,

$$\boxed{E[x(t)] = E[y(t)]}$$

(II) $E[x^2(t)] \neq E[y^2(t)]$
Proof:
Average power of a WSS random process is obtained from the area under its PSD.

$$E[y^2(t)] = \int_{-\infty}^{\infty} S_y(f)\,df$$

For an LTI system,

$$S_y(f) = S_x(f)|H(f)|^2$$

Therefore,

$$E[y^2(t)] = \int_{-\infty}^{\infty} S_x(f)|H(f)|^2\,df$$

$$= \int_{-1/2}^{1/2} S_x(f)\,df$$

$$= 2 - 2e^{-0.5} \qquad \cdots (1)$$

Now,

$$E[x^2(t)] = \int_{-\infty}^{\infty} S_x(f)\,df$$

$$= 2 \qquad \cdots (2)$$

Since,

$$(1) \neq (2)$$

Hence,

$$\boxed{E[x^2(t)] \neq E[y^2(t)]}$$

(III) $E[y^2(t)] \neq 2$
Proof:
From part (II),

$$E[x^2(t)] = 2$$

but,

$$E[y^2(t)] = 2 - 2e^{-0.5} \neq 2$$

Hence,

$$\boxed{E[y^2(t)] \neq 2}$$

$$\boxed{\text{(A) only I is true}}$$

Q1.17.2 NAT | 2017 | 1M | Ans: 2 | ● ○ ○

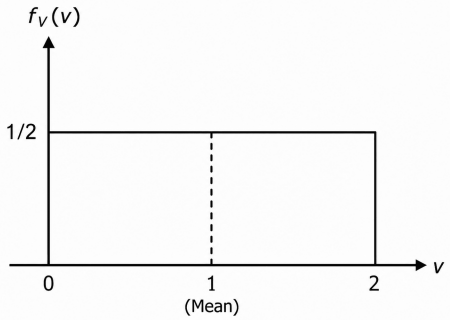


Fig. Solution Q1.17.2

Given,

$$X(t) = U + Vt$$

Also,

$$E[U] = 0, \qquad E[V] = 1$$

Using linearity of expectation,

$$E[X(t)] = E[U + Vt]$$

$$= E[U] + E[V]\,t$$

Substituting the given values,

$$E[X(t)] = 0 + 1 \times t$$

$$E[X(t)] = t$$

At $t = 2$,

$$\boxed{E[X(t)]_{t=2} = 2}$$

Q1.16.1 NAT | 2016 | 2M | Ans: -1 | ● ○ ○

Theory: For a WSS random process, the PSD is obtained from the Fourier transform of the autocorrelation sequence. A spectral null occurs when the PSD becomes zero.

The PSD of $X(t)$ is

$$S_X(f) = S(f) = \frac{|G(f)|^2}{T}\left(\sum_{m=-\infty}^{\infty} R_b(m)e^{-j2\pi fmT}\right)$$

The autocorrelation sequence is

$$R_b(\tau) = E[\beta_n \beta_{n-\tau}]$$

Given,

$$\beta_n = \alpha_n + k\alpha_{n-3}$$

Therefore,

$$R_b(\tau) = E\left[(\alpha_n + k\alpha_{n-3})(\alpha_{n-\tau} + k\alpha_{n-\tau-3})\right]$$

Expanding,

$$= E[\alpha_n\alpha_{n-\tau}] + kE[\alpha_n\alpha_{n-\tau-3}] + kE[\alpha_{n-3}\alpha_{n-\tau}] + k^2E[\alpha_{n-3}\alpha_{n-\tau-3}]$$

Using independence of α_n,

$$= R(\tau) + kR(\tau+3) + kR(\tau-3) + k^2R(\tau)$$

Hence,

$$R_b(\tau) = (1+k^2)R(\tau) + kR(\tau+3) + kR(\tau-3)$$

Therefore,

$$R_b(\tau) = \begin{cases} 1+k^2, & \tau = 0 \\ k, & \tau = \pm 3 \\ 0, & \text{otherwise} \end{cases}$$

The corresponding power spectral density is

$$S_b(f) = 1 + k^2 + 2k\cos(2\pi f \cdot 3T) \text{ (lag 3 symbols= 3T)}$$

Here the lag is counted in symbols: a lag of $m = \pm 3$ symbols is a time shift of $3T$ seconds, so the cosine contains $3T$ and not just 3.

For spectral null,

$$S_b(f) = 0$$

Given that null occurs at

$$f = \frac{1}{3T}$$

Substituting,

$$1 + k^2 + 2k \cos\left(2\pi\left(\frac{1}{3T}\right)3T\right) = 0$$

$$1 + k^2 + 2k \cos(2\pi) = 0$$

Since,

$$\cos(2\pi) = 1$$

we get

$$1 + k^2 + 2k = 0$$

$$(k+1)^2 = 0$$

Hence,

$$\boxed{k = -1}$$

Q1.16.2 NAT 2016 2M Ans: 100 ● ○ ○

Theory: For a random process, the mean-square value is obtained using expectation properties. Also, expectation of a constant is the constant itself.

$$E[x^2(t)] = E[(3V(t) - 8)^2]$$

Expanding,

$$= E[(3V(t))^2 + 64 - 2(3V(t))(8)]$$

$$= E[9V^2(t) + 64 - 48V(t)]$$

Using linearity of expectation,

$$= 9E[V^2(t)] + E[64] - 48E[V(t)]$$

Given,

$$E[V(t)] = 0$$

and

$$E[V^2(t)] = \text{MS} = R(0) = 4e^{-5(0)} = 4$$

Also,

$$E[\text{constant}] = \text{constant}$$

Therefore,

$$E[x^2(t)] = 9 \times 4 + 64$$

$$= 36 + 64$$

$$\boxed{E[x^2(t)] = 100}$$

Q1.16.3 MCQ 2016 2M Ans: B ● ○ ○

Theory: For a WSS random process, the autocorrelation of delayed signals is obtained by shifting the argument of the original autocorrelation function.

Given,

$$Y(t) = X(t) - X(t - T_0)$$

The autocorrelation function (ACF) of the output is

$$R_y(\tau) = E[y(t)y(t + \tau)]$$

Substituting $y(t)$,

$$R_y(\tau) = E\Big[(X(t) - X(t - T_0))(X(t + \tau) - X(t + \tau - T_0))\Big]$$

Expanding,

$$= E[X(t)X(t + \tau)] - E[X(t - T_0)X(t + \tau)]$$

$$-E[X(t)X(t + \tau - T_0)] + E[X(t - T_0)X(t + \tau - T_0)]$$

Using the WSS property,

$$E[X(t)X(t + \tau)] = R_x(\tau)$$

$$E[X(t - T_0)X(t + \tau)] = R_x(\tau + T_0)$$

$$E[X(t)X(t + \tau - T_0)] = R_x(\tau - T_0)$$

$$E[X(t - T_0)X(t + \tau - T_0)] = R_x(\tau)$$

Therefore,

$$R_y(\tau) = R_x(\tau) - R_x(\tau - T_0) - R_x(\tau + T_0) + R_x(\tau)$$

Hence,

$$\boxed{(B)R_y(\tau) = 2R_x(\tau) - R_x(\tau - T_0) - R_x(\tau + T_0)}$$

Q1.15.1 MCQ | 2015 | 2M | Ans: A | ● ○ ○

Theory: Passing a Gaussian random process through an LTI system preserves Gaussianity. Also, for zero-mean WSS noise, variance equals output power.

A Gaussian random process $X(t)$ is passed through an LTI system to produce $Y(t)$.
Sampling a Gaussian random process gives a Gaussian random variable. Hence, $Y(t_i)$ has Gaussian distribution with mean and variance same as the Gaussian random process $Y(t)$.
The output PSD is

$$S_Y(f) = |H(f)|^2 S_X(f)$$

where $S_X(f) = \frac{N_0}{2}$.
The variance is given by

$$\text{Variance} = \text{mean square value} - (\text{mean})^2$$

The mean of the output process is

$$m_Y(t) = H(0)m_X(t)$$

Since the input has zero mean,

$$m_X(t) = 0$$

Therefore,

$$m_Y(t) = 0$$

Hence, variance becomes equal to output power.
The output power is

$$P_Y = \int_{-\infty}^{\infty} S_Y(f)\, df$$
$$= \int_{-\infty}^{\infty} |H(f)|^2 \frac{N_0}{2}\, df$$
$$= \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2\, df$$

Using Parseval's theorem,

$$\int_{-\infty}^{\infty} |H(f)|^2\, df = \int_{-\infty}^{\infty} |h(t)|^2\, dt$$

Given,

$$\int_{-\infty}^{\infty} |h(t)|^2\, dt = 3A^2$$

Therefore,

$$P_Y = \frac{N_0}{2}(3A^2)$$

$$= \frac{3}{2}N_0A^2$$

Thus, the variance at $t = 4$ or equivalently at $\tau = 0$ is

$$\boxed{(A)\sigma_Y^2 = \frac{3}{2}N_0A^2}$$

Q1.15.2 MCQ 2015 2M Ans: B ●●○

For the discrete random variables X_n, the autocorrelation function is

$$R_X(k) = \sum_{i=1}^{m}[X_nX_{n+k}]_iP_i \tag{1}$$

where P_i is the probability of occurrence of the i^{th} symbol.
Since X_n is equally likely to be 1 or -1,

$$P(1) = P(-1) = \frac{1}{2}$$

For $k = 0$,

$$R_X(0) = \sum_{i=1}^{2}(X_nX_n)_iP_i$$

$$= (1)(1)\left(\frac{1}{2}\right) + (-1)(-1)\left(\frac{1}{2}\right)$$

$$= 1$$

Hence,

$$R_X(k) = 1, \qquad k = 0$$

For $k \neq 0$, there are four possible combinations of X_n and X_{n+k}.

X_n	X_{n+k}	P_i
1	-1	1/4
-1	1	1/4
-1	-1	1/4
1	1	1/4

$$R_X(k) = \sum_{i=1}^{4}(X_nX_{n+k})_iP_i$$

$$= (1)(-1)\left(\frac{1}{4}\right) + (-1)(1)\left(\frac{1}{4}\right) + (-1)(-1)\left(\frac{1}{4}\right) + (1)(1)\left(\frac{1}{4}\right)$$

$$= -\frac{1}{4} - \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

$$= 0$$

Therefore,

$$R_X(k) = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0 \end{cases} \tag{2}$$

Given,

$$Y_n = X_n + 0.5X_{n-1}$$

The autocorrelation of Y_n is

$$R_Y(k) = \sum_{i=1}^{m} (Y_n Y_{n+k})_i P_i$$

Substituting Y_n,

$$R_Y(k) = \sum_{i=1}^{m} \left[(X_n + 0.5X_{n-1})(X_{n+k} + 0.5X_{n-1+k}) \right]_i P_i$$

Expanding,

$$R_Y(k) = R_X(k) + 0.5R_X(k-1) + 0.5R_X(k+1) + 0.25R_X(k)$$

$$R_Y(k) = 1.25R_X(k) + 0.5R_X(k-1) + 0.5R_X(k+1)$$

For $k = 0$,

$$R_Y(0) = R_X(0) + 0.5R_X(-1) + 0.5R_X(1) + 0.25R_X(0)$$

Using (2),

$$R_Y(0) = 1 + 0 + 0 + 0.25$$

$$= 1.25$$

For $k = 1$,

$$R_Y(1) = R_X(1) + 0.5R_X(0) + 0.5R_X(2) + 0.25R_X(1)$$

$$= 0 + 0.5(1) + 0 + 0$$

$$= 0.5$$

Similarly,

$$R_Y(-1) = 0.5$$

For all other values of k,

$$R_Y(k) = 0$$

Therefore,

$$(B) R_Y(k) = \begin{cases} 1.25, & k = 0 \\ 0.5, & k = \pm 1 \\ 0, & \text{otherwise} \end{cases}$$

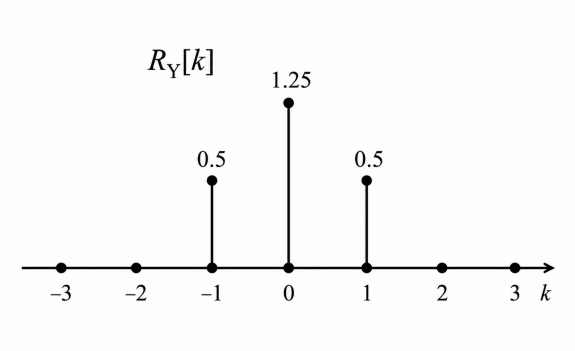


Fig. Solution Q1.15.2

Q1.15.3 MCQ 2015 2M Ans: D ● ● ○

Given,

$$P(X = 0) = p$$

$$P(Y = 0) = q$$

Since X and Y are independent random variables,

$$P(X + Y \geq 1) = 1 - P(X + Y < 1)$$

The only possibility for $X + Y < 1$ is

$$X = 0, \quad Y = 0$$

Therefore,

$$P(X + Y \geq 1) = 1 - P(X = 0,\ Y = 0)$$

Using independence,

$$P(X = 0,\ Y = 0) = P(X = 0)P(Y = 0) = pq$$

Hence,

$$P(X + Y \geq 1) = 1 - pq$$

$$\boxed{(D)P(X + Y \geq 1) = 1 - pq}$$

Q1.15.4 NAT 2015 2M Ans: 0.25 ● ● ○

Theory: The autocorrelation depends on the overlap between two shifted pulses. Greater overlap gives larger correlation.

Given,

$$y(t) = \sum_{n=-\infty}^{\infty} X_n\, p(t - nT - \phi)$$

The autocorrelation function is

$$R_{yy}(t_i, t_j) = E[y(t_i)y(t_j)]$$

$$= E[X_i X_j]\, p(t_i - nT - \phi)\, p(t_j - nT - \phi)$$

(a) If $|t_i - t_j| > T$
The pulses are either adjacent or distinct, hence the random variables are uncorrelated.

$$E[X_i X_j] = E[X_i]E[X_j]$$

Given,

$$E[X_n] = 0.5(1) + (-1)(0.5) = 0$$

Therefore,

$$E[X_i X_j] = 0 \tag{1}$$

(b) If $|t_i - t_j| < T$
The pulses overlap, hence correlation exists.

$$E[X_i X_j] = E[X_i^2]$$

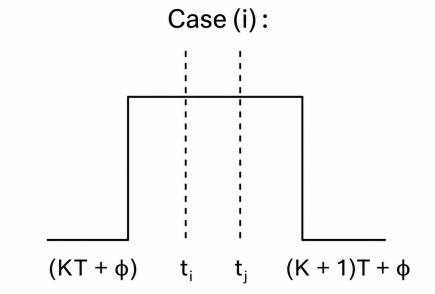
$$= (1)^2(0.5) + (-1)^2(0.5)$$

$$= 1 \tag{2}$$

The random variable ϕ is uniformly distributed over $(0, T)$.
Hence,

$$f_\phi(\phi) = \frac{1}{T}, \qquad 0 < \phi < T$$

If t_i and t_j lie on the same pulse, then



Case (i):

$$KT + \phi < t_i$$

$$(K + 1)T + \phi > t_j$$

Therefore,

$$\phi < t_i - KT$$

and

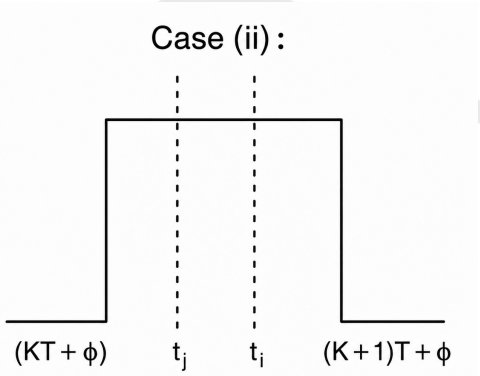
$$\phi > t_j - (K+1)T$$

Hence,

$$E[p(t_i - nT - \phi)\, p(t_j - nT - \phi)] = \int_{t_j-(K+1)T}^{t_i-KT} \left(\frac{1}{T}\right) d\phi$$

$$= 1 - \frac{t_j - t_i}{T}$$

Similarly for Case(ii),



$$E[p(t_i - nT - \phi)\, p(t_j - nT - \phi)] = 1 - \frac{t_i - t_j}{T}$$

Let,

$$\tau = t_j - t_i$$

Then,

$$R_{yy}(t_i, t_j) = R_{yy}(\tau)$$

$$= E[y(t)y(t+\tau)]$$

Using (2),

$$R_{yy}(\tau) = E[X_i^2]\left(1 - \frac{|\tau|}{T}\right), \qquad |\tau| \le T$$

$$= 1\left(1 - \frac{|\tau|}{T}\right)$$

Now,

$$\tau = \frac{3T}{4}$$

Therefore,

$$R_{yy}\left(\frac{3T}{4}\right) = 1\left(1 - \frac{3T/4}{T}\right)$$

$$= 1\left(1 - \frac{3}{4}\right)$$

$$= 0.25$$

$$\boxed{R_{yy}\left(\frac{3T}{4}\right) = 0.25}$$

Q1.14.1 MCQ | 2014 | 2M | Ans: D | ● ○ ○

Theory: For a random phase sinusoidal process, the autocorrelation depends only on the time difference $(t_1 - t_2)$, which makes the process WSS.

Given,

$$x(t) = \sqrt{2}\sin(2\pi t + \phi)$$

The autocorrelation function is

$$E[x(t_1)x(t_2)]$$

$$= E\left[\sqrt{2}\sin(2\pi t_1 + \phi) \cdot \sqrt{2}\sin(2\pi t_2 + \phi)\right]$$

$$= 2E\left[\sin(2\pi t_1 + \phi)\sin(2\pi t_2 + \phi)\right]$$

Using the identity,

$$2\sin A \sin B = \cos(A - B) - \cos(A + B)$$

we get

$$E[x(t_1)x(t_2)] = E\left[\cos\left(2\pi(t_1 - t_2)\right) - \cos\left(2\pi t_1 + 2\pi t_2 + 2\phi\right)\right]$$

$$= E\left[\cos\left(2\pi(t_1 - t_2)\right)\right] - E\left[\cos\left(2\pi t_1 + 2\pi t_2 + 2\phi\right)\right]$$

The expectation of the second term is zero because ϕ is uniformly distributed.
Hence,

$$E[x(t_1)x(t_2)] = \cos\left(2\pi(t_1 - t_2)\right)$$

Therefore,

$$\boxed{(D)\, R_x(t_1, t_2) = \cos\left(2\pi(t_1 - t_2)\right)}$$

Q1.14.2 NAT | 2014 | 2M | Ans: 1.414 | ● ● ○

Theory: When two independent AWGN channels are combined, the signal amplitude doubles while the noise variances add.

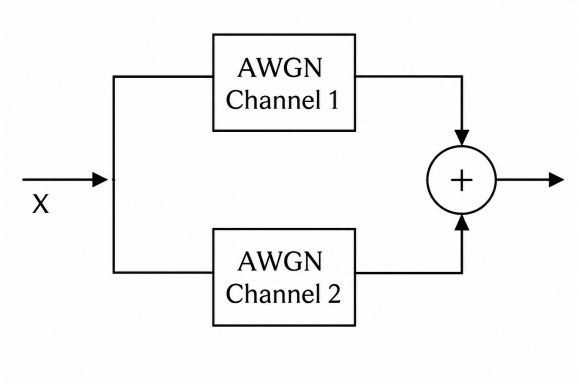


Fig. Solution Q1.14.2

Given BER for a single AWGN channel,

$$\text{BER} = Q\left(\sqrt{r}\right)$$

At the receiver input, the signal amplitude becomes doubled.
When two independent Gaussian random variables are added, the resultant random variable is also Gaussian and the pdf is obtained by convolution.

$$f_z(z) = \int_{-\infty}^{\infty} f_y(z-x) f_x(x)\, dx$$

For two Gaussian random variables,

$$f_z(z) = \frac{1}{\sqrt{2\pi(\sigma_x^2 + \sigma_y^2)}} \exp\left[-\frac{(z-(\mu_x+\mu_y))^2}{2(\sigma_x^2+\sigma_y^2)}\right]$$

Given,

$$\sigma_x^2 = \sigma_y^2 = \sigma^2$$

therefore,

$$f(z) = \frac{1}{\sqrt{2\pi(2\sigma^2)}} \exp\left[-\frac{(z-2\mu)^2}{4\sigma^2}\right]$$

Hence, the noise variance becomes doubled.

$$\text{Noise Power} \propto \sigma^2$$

So,

Noise power increases by a factor of 2

Since the signal amplitude doubles,

Signal power increases by a factor of 4

Therefore,

$$\text{SNR increases by a factor of } \frac{4}{2} = 2$$

Thus,

$$b = \sqrt{2} = 1.414$$

Hence the new BER becomes

$$\text{BER} = Q\left(\sqrt{2r}\right)$$

$$= Q\left(\sqrt{2}\sqrt{r}\right)$$

$$= Q\left(1.414\sqrt{r}\right)$$

Therefore,

$$\boxed{\text{BER} = Q\left(1.414\sqrt{r}\right)}$$

and

$$\boxed{b = 1.414}$$

Q1.14.3 NAT 2014 2M Ans: 4 ●●○

Theory: The autocorrelation function is obtained by taking the inverse Fourier transform of the PSD. For a rectangular PSD, the ACF becomes a sinc function.

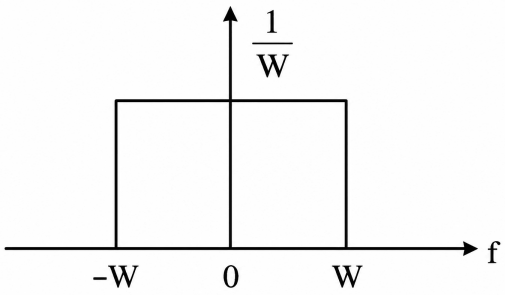


Fig. Solution Q1.14.3

$$R(\tau) = 2\,\text{sinc}(2W\tau)$$

Therefore,

$$E\left[\pi X(t)\, X\left(t - \frac{1}{4W}\right)\right] = \pi R\left(\frac{1}{4W}\right)$$

Substituting in the ACF expression,

$$= 2\pi\,\text{sinc}\left(2W \times \frac{1}{4W}\right)$$

$$= 2\pi \operatorname{sinc}\left(\frac{1}{2}\right)$$

Using,

$$\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

$$= 2\pi \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}}$$

$$= 2\pi \cdot \frac{1}{\pi/2}$$

$$= 4$$

Hence,

$$\boxed{E\left[\pi X(t)\, X\left(t - \frac{1}{4W}\right)\right] = 4}$$

Q1.14.4 MCQ 2014 2M Ans: C ● ○ ○

Theory: Time scaling changes the PSD according to the scaling property, while time shifting does not affect the PSD.

$$X(t) \overset{\text{PSD}}{\longleftrightarrow} S(f)$$

For time scaling,

$$X(2t) \overset{\text{PSD}}{\longleftrightarrow} \frac{1}{2} S\left(\frac{f}{2}\right)$$

Now,

$$X\left[2\left(t - \frac{1}{2}\right)\right] = X(2t - 1)$$

Applying the time-scaling property,

$$X\left[2\left(t - \frac{1}{2}\right)\right] \overset{\text{PSD}}{\longleftrightarrow} \frac{1}{2} S\left(\frac{f}{2}\right)$$

Since time shifting does not change the PSD,

$$X(2t - 1) \overset{\text{PSD}}{\longleftrightarrow} \frac{1}{2} S\left(\frac{f}{2}\right)$$

Hence,

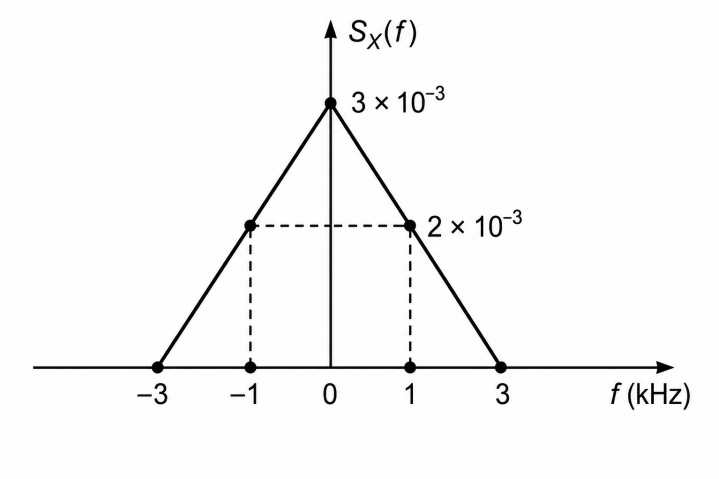
$$\boxed{(C) S_Y(f) = \frac{1}{2} S_X\left(\frac{f}{2}\right)}$$

Q1.14.5 NAT | 2014 | 2M | Ans: 2.5 | ● ● ○

Theory: Multiplication of a random process by a cosine shifts its PSD to $\pm f_c$. The output PSD of an LTI system is given by $|H(f)|^2 S_X(f)$.

Given,

$$S_X(f) = \begin{cases} 10^{-6}(3000 - |f|) \text{ Watts/Hz}, & |f| \le 3\,\text{kHz} \\ 0, & \text{otherwise} \end{cases}$$



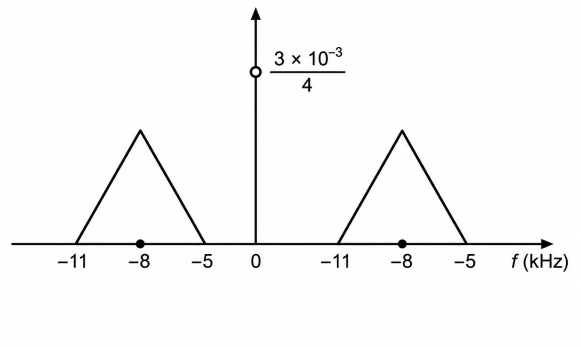
If

$$X(t) \overset{\text{PSD}}{\longleftrightarrow} S_X(f)$$

then,

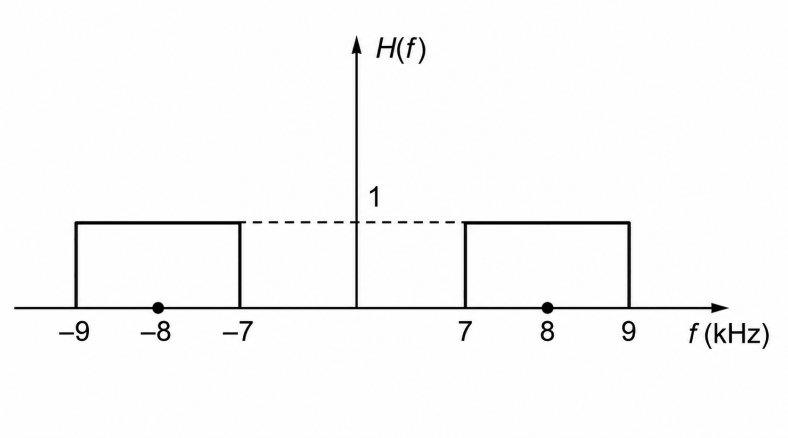
$$X(t)\cos(2\pi f_c t) \overset{\text{PSD}}{\longleftrightarrow} \frac{S_X(f - f_c) + S_X(f + f_c)}{4}$$

For

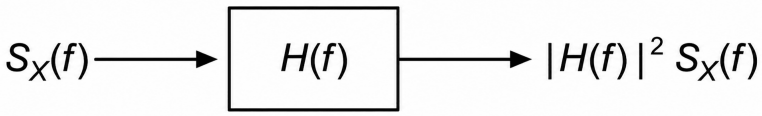


$$f_c = 8\,\text{kHz}$$

the PSD becomes shifted around ±8 kHz.



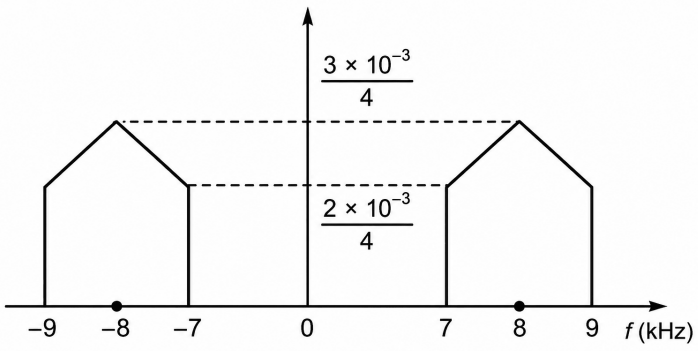
Now, the Band Pass Filter output PSD is



$$S_Y(f) = |H(f)|^2 S_X(f)$$

Since the filter passband lies between 7 kHz and 9 kHz, only the corresponding portions of the shifted PSD are passed.

Output power is equal to the total area under the output PSD.



$$P_o = 2\left[(2000)\left(\frac{2 \times 10^{-3}}{4}\right)\right] + 2\left[\frac{1}{2}(1000)\left(\frac{10^{-3}}{4}\right)\right]$$

$$= 2\left[1 + \frac{1}{4}\right]$$

$$= 2.5 \text{ Watts}$$

Hence,

$$\boxed{P_o = 2.5 \text{ Watts}}$$

Q1.14.6 NAT 2014 2M Ans: 0.1666 ● ○ ○

Theory: The sum or difference of independent random variables is obtained using convolution of their PDFs.

X_1, X_2, and X_3 are identical independent random variables uniformly distributed in $[0, 1]$.

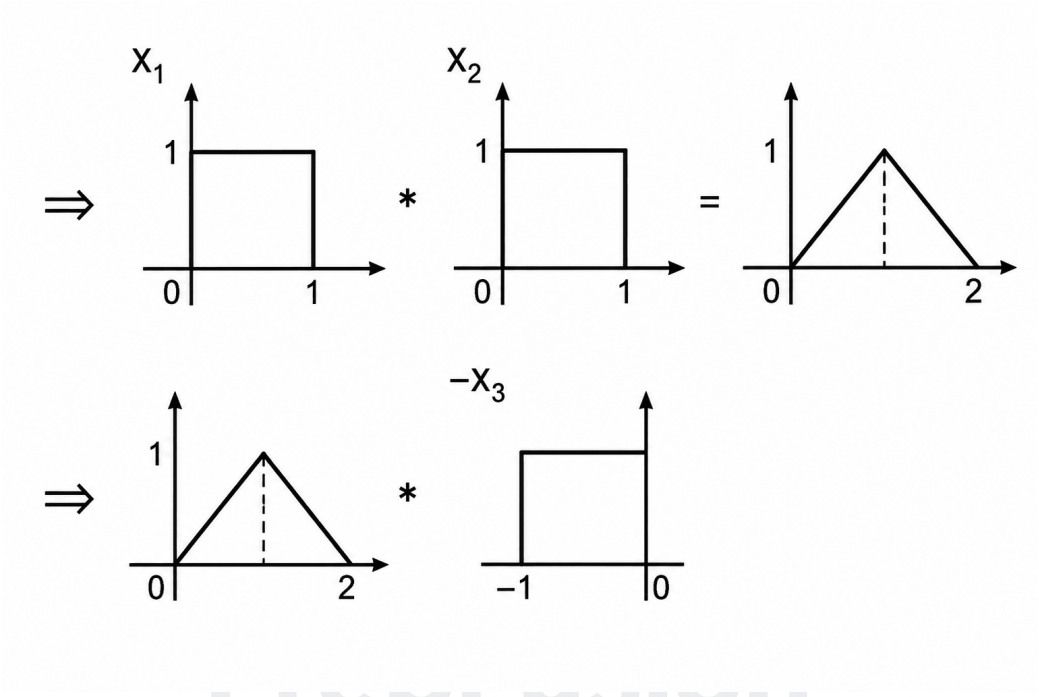


Fig. Solution Q1.14.6

$$P(X_1 + X_2 \le X_3)$$

can be written as

$$P(X_1 + X_2 - X_3 \le 0)$$

Let,

$$X = X_1 + X_2 - X_3$$

Then,

$$f_X(x) = f_{X_1}(x) * f_{X_2}(x) * f_{X_3}(-x)$$

The convolution of two uniform PDFs gives a triangular PDF.
Hence,

$$f_X(x) = \big(r(x) - 2r(x-1) + r(x-2)\big) * \big(u(x+1) - u(x)\big)$$

Therefore,

$$f_X(x) = \frac{(x+1)^2}{2}u(x+1) - \frac{3}{2}x^2u(x) + \frac{3}{2}(x-1)^2u(x-1) + \frac{1}{2}(x-2)^2u(x-2)$$

Now,

$$P(X \le 0) = \int_{-1}^{0} f_X(x)\,dx$$

Over the interval $-1 \le x \le 0$,

$$f_X(x) = \frac{(x+1)^2}{2}$$

Hence,

$$P(X \le 0) = \int_{-1}^{0} \frac{(x+1)^2}{2}\,dx$$

$$= \frac{1}{2}\left[\frac{(x+1)^3}{3}\right]_{-1}^{0}$$

$$= \frac{1}{2}\left(\frac{1}{3}\right)$$

$$= \frac{1}{6}$$

Therefore,

$$\boxed{P(X_1 + X_2 \le X_3) = \frac{1}{6}}$$

Q1.14.7 MCQ 2014 2M Ans: B ● ○ ○

Theory: The variance of a random variable is always non-negative, which directly gives a fundamental inequality relating $E[X]$ and $E[X^2]$.

The variance of X is

$$\sigma_X^2 = E[X^2] - (E[X])^2.$$

Since variance can never be negative,

$$\sigma_X^2 \ge 0.$$

Hence,

$$E[X^2] - (E[X])^2 \ge 0.$$

Therefore,

$$\boxed{(B)\,E[X^2] \ge (E[X])^2.}$$

Mistakes to be avoided

- Equality holds only when the random variable is deterministic (zero variance).
- Option $E[X^2] > (E[X])^2$ is not always true because equality is possible.
- Always remember the identity

$$\sigma_X^2 = E[X^2] - (E[X])^2.$$

Q1.13.1 MCQ | 2013 | 2M | Ans: D | ● ○ ○

Theory: The probability of error is obtained from the overlap area between the conditional PDFs beyond the detection threshold.

BER is given by

$$P_e = P(0)\,P(1/0) + P(1)\,P(0/1)$$

If the detection threshold is at

$$z = 1$$

then,

$$P(0) = P(1) = \frac{1}{2}$$

Since $f(z/0)$ has no area beyond $z = 1$,

$$P\left(\frac{Y=1}{X=0}\right) = \int_1^{\infty} f(z/0)\,dz = 0$$

Now,

$$P\left(\frac{Y=0}{X=1}\right) = \int_0^{1} f(z/1)\,dz$$

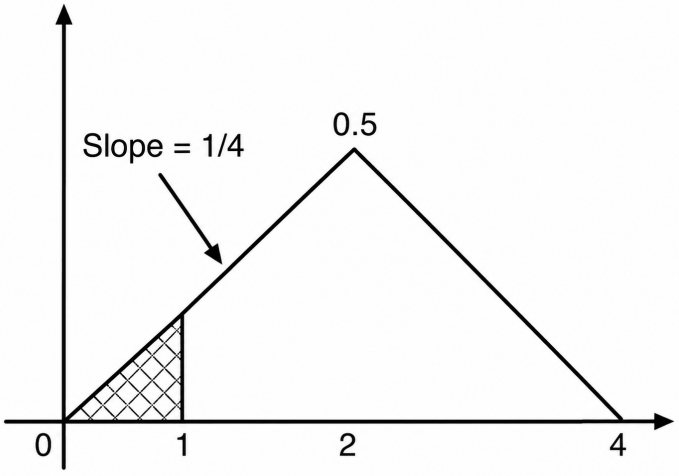


Fig. Solution Q1.13.1

From the figure 1.13.1, the shaded region is a triangle having

$$\text{base} = 1$$

and

$$\text{height} = \frac{1}{4}$$

Therefore,

$$P\left(\frac{Y=0}{X=1}\right)=\frac{1}{2}\times 1\times\frac{1}{4}=\frac{1}{8}$$

Hence,

$$P_e=\frac{1}{2}\times 0+\frac{1}{2}\times\frac{1}{8}$$

$$=\frac{1}{16}$$

Therefore,

$$\boxed{(D)P_e=\frac{1}{16}}$$

Q1.13.2 MCQ 2013 2M Ans: B

Theory: For equiprobable signals, the optimum threshold occurs at the intersection point of the two conditional PDFs.

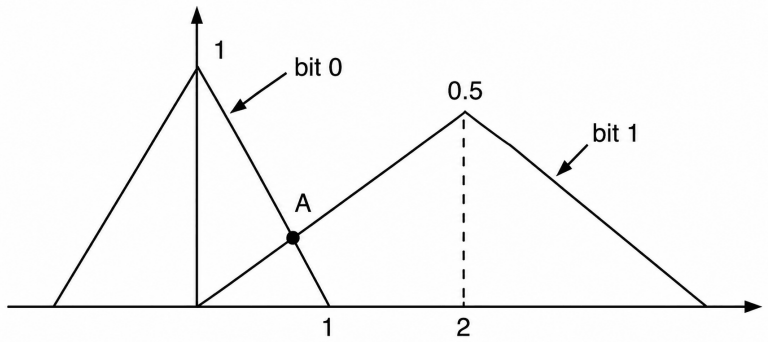


Fig. Solution Q1.13.2

The line equation for the pdf of received signal corresponding to bit 0 is

$$x+y=1$$

For bit 1,

$$y=\frac{0.5}{2}x$$

$$\Rightarrow y=0.25x$$

Substituting in

$$x+y=1$$

we get,

$$x+0.25x=1$$

$$1.25x=1$$

$$x = \frac{1}{1.25} = 0.8$$

Now,

$$y = 1 - 0.8 = 0.2$$

Hence, the intersection point is

$$A(0.8,\ 0.2)$$

Therefore, the optimum threshold is

$$0.8 = \frac{4}{5}$$

$$\boxed{(B)\text{Optimum Threshold} = \frac{4}{5} = 0.8}$$

Q1.13.3 MCQ 2013 2M Ans: D ● ○ ○

Theory: Scaling a random variable changes its CDF, and the sign of x determines the ordering between $F(x)$ and $G(x)$.
Given,

$$F(x) = P(U \le x),$$

and

$$G(x) = P(2V \le x).$$

Since V and U are identically distributed,

$$G(x) = P\left(V \le \frac{x}{2}\right) = F\left(\frac{x}{2}\right).$$

For $x > 0$,

$$\frac{x}{2} < x.$$

Since a CDF is a non-decreasing function,

$$F\left(\frac{x}{2}\right) \le F(x),$$

or,

$$G(x) \le F(x),$$

which gives

$$F(x) - G(x) \ge 0.$$

For $x < 0$,

$$\frac{x}{2} > x,$$

therefore,

$$F\left(\frac{x}{2}\right) \ge F(x),$$

or,

$$G(x) \ge F(x),$$

which gives

$$F(x) - G(x) \le 0.$$

Hence,

$$(F(x) - G(x))\, x \ge 0,$$

for all values of x.

$$\boxed{\text{(D) } (F(x) - G(x))\, x \ge 0}$$

Key Points

- $G(x) = F\left(\frac{x}{2}\right)$.
- A CDF is always a non-decreasing function.
- The sign of x determines the sign of $F(x) - G(x)$.

Mistakes to be avoided

- Do not assume $F(x) \geq G(x)$ for all x.
- Remember that dividing a negative number by 2 makes it larger (less negative).
- Use the monotonic property of the CDF instead of assuming symmetry from the zero-mean condition.

Q1.13.4 MCQ 2013 2M Ans: B ●○○

Theory: A linear combination of independent Gaussian random variables is also Gaussian.
Let

$$W = 3V - 2U.$$

Then,

$$P(3V \geq 2U) = P(3V - 2U \geq 0) = P(W \geq 0).$$

Given,

$$U \sim \mathcal{N}\left(0, \frac{1}{4}\right), \qquad V \sim \mathcal{N}\left(0, \frac{1}{9}\right).$$

Since U and V are independent,

$$W \sim \mathcal{N}\left(0,\ 3^2 \cdot \frac{1}{9} + (-2)^2 \cdot \frac{1}{4}\right) = \mathcal{N}(0, 2).$$

Thus, W is a zero-mean Gaussian random variable. Hence,

$$P(W \geq 0) = \frac{1}{2}.$$

Therefore,

$$\boxed{(B) P(3V \geq 2U) = \frac{1}{2}}$$

Key Points

- A linear combination of independent Gaussian random variables is Gaussian.
- A zero-mean Gaussian random variable is symmetric about zero.
- Therefore, $P(W \geq 0) = \frac{1}{2}$.

Mistakes to be avoided

- Do not add the standard deviations; add the variances after scaling.
- Remember that $\text{Var}(aX) = a^2\text{Var}(X)$.
- Use the symmetry property only after confirming that the Gaussian random variable has zero mean.

Q1.12.1 MCQ 2012 1M Ans: B ● ○ ○

Theory: The mean square value of a random process is equal to the total area under its power spectral density (PSD).

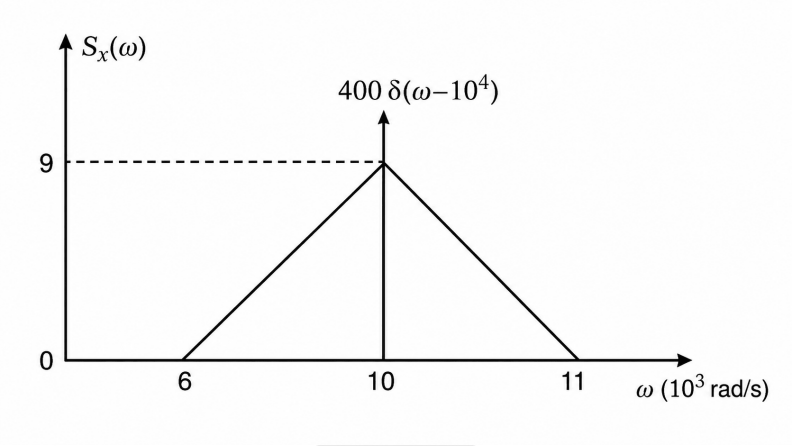


Fig. Solution Q1.12.1

$$E[X^2(t)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(\omega)\,d\omega$$

Since PSD is an even function,

$$E[X^2(t)] = \frac{1}{2\pi}\cdot 2\int_{0}^{\infty} S_X(\omega)\,d\omega$$

$$= \frac{1}{\pi}\left[400 + \frac{1}{2}(6)(2\times 10^3)\right]$$

$$= \frac{1}{\pi}[400 + 6000]$$

$$E[X^2(t)] = \frac{6400}{\pi}$$

Theory: The DC component of a PSD corresponds to the mean value of the process.

Since $S_X(\omega)$ has no impulse at $\omega = 0$,

$$E[X(t)] = 0$$

Therefore,

$$\boxed{(B)\ \frac{6400}{\pi},\ 0}$$

Q1.12.2 MCQ 2012 1M Ans: B ● ○ ○

Theory: For independent random variables, the joint probability becomes the product of individual probabilities.

$$P[\max(X,Y)] = P[X \le x,\ Y \le y]$$

$$= P[X \le x]\ P[Y \le y]$$

since X and Y are independent.

$$= \int_{-1}^{1/2} \frac{1}{2}\, dx \int_{-1}^{1/2} \frac{1}{2}\, dy$$

$$= \frac{9}{16}$$

$$\boxed{(B)\frac{9}{16}}$$

Q1.11.1 MCQ 2011 2M Ans: A ● ○ ○

Theory: Differentiation in time domain corresponds to multiplication by $j2\pi f$ in frequency domain.

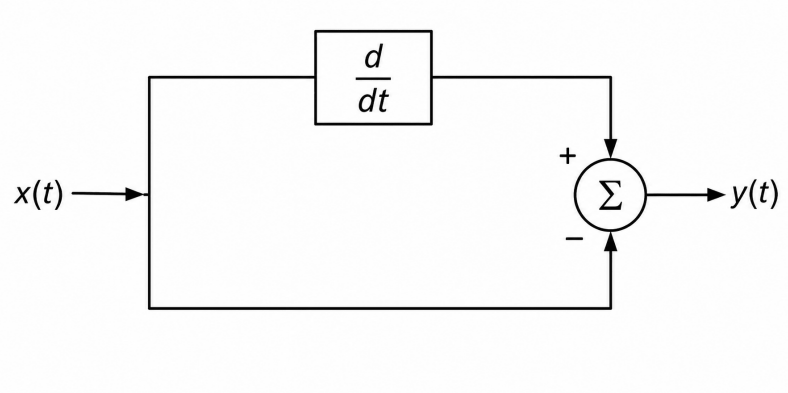


Fig. Solution Q1.11.1

$$y(t) = \frac{d}{dt}x(t) - x(t)$$

Applying Fourier Transform,

$$Y(f) = j2\pi f\, X(f) - X(f)$$

$$Y(f) = (j2\pi f - 1)\, X(f)$$

Using PSD relation,

$$S_Y(f) = |j2\pi f - 1|^2\, S_X(f)$$

$$S_Y(f) = \left(4\pi^2 f^2 + 1\right) S_X(f)$$

Given,

$$S_X(f) = \mathcal{F}\{R_X(\tau)\}$$

$$S_X(f) = e^{-\pi f^2}$$

Since Fourier Transform of a Gaussian is also Gaussian,

$$S_Y(f) = \left(4\pi^2 f^2 + 1\right) e^{-\pi f^2}$$

$$(A) S_Y(f) = \left(4\pi^2 f^2 + 1\right) e^{-\pi f^2}$$

Q1.10.1 MCQ 2010 2M Ans: D ● ● ○

Theory: A delay in time domain introduces a phase shift in frequency domain, while differentiation multiplies the spectrum by $j2\pi f$.

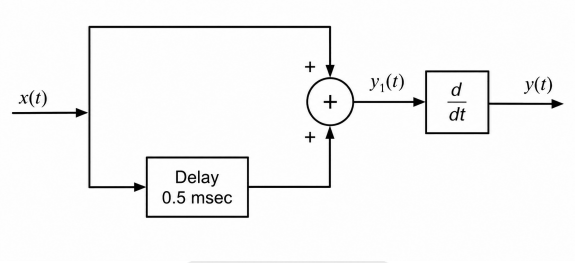


Fig. Solution Q1.10.1

$$y_1(t) = x(t) + x\left(t - 0.5 \times 10^{-3}\right)$$

Applying Fourier Transform,

$$Y_1(f) = X(f)\left[1 + e^{-j2\pi f(0.5\times10^{-3})}\right]$$

$$= X(f)\left[1 + e^{-j\pi f\times10^{-3}}\right]$$

Hence,

$$H_1(f) = 1 + e^{-j\pi f\times10^{-3}}$$

For the differentiator,

$$H_2(f) = j2\pi f$$

Therefore,

$$H(f) = H_1(f)H_2(f)$$

$$H(f) = (j2\pi f)\left[1 + e^{-j\pi f\times10^{-3}}\right]$$

Now,

$$|H(f)|^2 = 4\pi^2 f^2\left[2 + 2\cos(\pi f \times 10^{-3})\right]$$

$$= 8\pi^2 f^2\left[1 + \cos(\pi f \times 10^{-3})\right]$$

Using PSD relation,

$$S_Y(f) = |H(f)|^2 S_X(f)$$

$$S_Y(f) = 8\pi^2 f^2\left[1 + \cos(\pi f \times 10^{-3})\right] S_X(f)$$

For

$$f = (2n+1)f_0, \qquad f_0 = 1 \text{ kHz}$$

we get

$$\pi f \times 10^{-3} = (2n+1)\pi$$

Hence,

$$\cos((2n+1)\pi) = -1$$

Therefore,

$$1 + \cos(\pi f \times 10^{-3}) = 0$$

$$\boxed{\text{(D) } S_Y(f) = 0 \text{ for } f = (2n+1)f_0,\ f_0 = 1 \text{ kHz},\ n \text{ any integer}}$$

Q1.09.1 MCQ 2009 1M Ans: B ●○○

Theory: The power spectral density and autocorrelation function form a Fourier transform pair (Wiener–Khinchin theorem). Given that the power spectral density is a sine-squared function,

$$S_X(f) = \sin^2(\cdot).$$

Using the Fourier transform relation,

$$S_X(f) = \mathcal{F}\{R_X(\tau)\}.$$

The Fourier transform pair is

$$\text{tri}\left(\frac{\tau}{T}\right) \overset{\mathcal{F}}{\longleftrightarrow} T\ \text{sinc}^2(Tf).$$

Hence, if the PSD is of the form $\text{sinc}^2(f)$ (sine-squared/sinc-squared spectrum), the corresponding autocorrelation function is a triangular function.
Therefore, the autocorrelation has the shape shown in

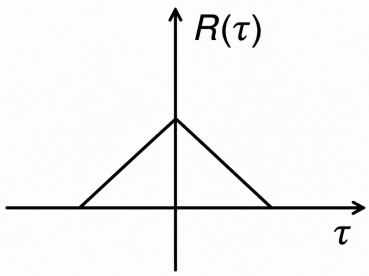


Fig. Solution Q1.09.1

(B)

Key Points

- PSD and autocorrelation are Fourier transform pairs.
- $\text{tri}(t) \leftrightarrow \text{sinc}^2(f)$.
- A sinc-squared PSD corresponds to a triangular autocorrelation.

Mistakes to be avoided

- Do not confuse $\sin^2(\cdot)$ with $\text{sinc}^2(\cdot)$; the standard Fourier transform pair is with sinc^2.
- Remember that the question asks for the shape of the autocorrelation, not the PSD.
- Use the Wiener–Khinchin theorem to relate PSD and autocorrelation.

Q1.09.2 MCQ 2009 1M Ans: B ● ○ ○

Theory: For a white noise input, the output PSD is obtained by multiplying the input PSD with the filter magnitude-squared response:

$$S_Y(f) = |H(f)|^2 S_X(f).$$

The output power is the area under the output PSD.

Given,

$$S_X(f) = 1 \times 10^{-10} \text{ W/Hz}.$$

Let

$$k = 1 \times 10^{-10} \text{ W/Hz}.$$

Hence,

$$S_Y(f) = k|H(f)|^2.$$

The output power is

$$P_Y = \int_{-\infty}^{\infty} S_Y(f)\, df = k \int_{-\infty}^{\infty} |H(f)|^2\, df.$$

Since $|H(f)|^2$ is a triangle with

$$\text{height} = 1$$

and

$$\text{base} = 20\,\text{kHz} = 2 \times 10^4 \text{ Hz},$$

its area is

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times (2 \times 10^4) \times 1$$

$$= 10^4.$$

Therefore,

$$P_Y = kA = (10^{-10})(10^4)$$

$$= 10^{-6} \text{ W}.$$

Hence,

$$\boxed{(B) P_Y = 1 \times 10^{-6} \text{ W}}$$

Q1.08.1 MCQ 2008 2M Ans: C ● ○ ○

Theory: The output PSD of an LTI system is obtained by multiplying the input PSD with $|H(f)|^2$.

$$H(f) = \frac{1}{1 + j2\pi fRC} = \frac{1}{1 + j\dfrac{f}{f_c}}$$

Therefore,

$$|H(f)|^2 = \frac{f_c^2}{f^2 + f_c^2}$$

Hence,

$$\text{Output PSD} = |H(f)|^2 \times \text{Input PSD}$$

$$= \frac{f_c^2}{f^2 + f_c^2} K$$

The output noise power is

$$= \int_{-\infty}^{\infty} (\text{Output PSD})\, df$$

$$= K \int_{-\infty}^{\infty} \frac{f_c^2}{f^2 + f_c^2}\, df$$

Using substitution

$$f = f_c \tan\theta$$

we get

$$= K\pi f_c$$

$$\boxed{K\pi f_c}$$

Q1.08.2 MCQ 2008 1M Ans: A ●○○

Theory: The cumulative distribution function (CDF) is obtained by integrating the probability density function (PDF). From the given figure,

$$f_X(x) = \begin{cases} 0, & x < -1, \\ x + 1, & -1 \le x \le 0, \\ 1 - x, & 0 \le x \le 1, \\ 0, & x > 1. \end{cases}$$

Therefore,

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt.$$

Evaluating over each interval,

$$F_X(x) = \begin{cases} 0, & x < -1, \\ \displaystyle\int_{-1}^{x} (t+1)\, dt = \frac{(x+1)^2}{2}, & -1 \le x \le 0, \\ \displaystyle\frac{1}{2} + \int_{0}^{x} (1-t)\, dt = \frac{1}{2} + x - \frac{x^2}{2}, & 0 \le x \le 1, \\ 1, & x \ge 1. \end{cases}$$

Hence, the CDF consists of an increasing parabola over $[-1, 0]$, followed by a decreasing parabola (concave downward) over $[0, 1]$, which matches **Option (A)**.

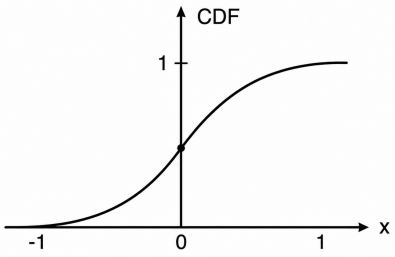


Fig. Solution Q1.08.2

(A)

Key Points

- The CDF is the integral of the PDF.
- The integral of an increasing linear function is an increasing parabola.
- The integral of a decreasing linear function is a concave-down parabola.

Mistakes to be avoided

- Do not confuse the PDF with the CDF.
- Ensure the CDF is continuous and non-decreasing.

- Verify that $F_X(-1) = 0$ and $F_X(1) = 1$.

Q1.07.1 MCQ 2007 1M Ans: B ● ○ ○

Theory: The power spectral density (PSD) of a real wide-sense stationary (WSS) random process is always a non-negative real function.
For a WSS random process,

$$S(f) = \mathcal{F}\{R(\tau)\},$$

where $R(\tau)$ is the autocorrelation function.
Since the PSD represents the distribution of average power over frequency,

$$S(f) \geq 0, \qquad \forall f.$$

Thus, the only statement that is always true is

$$\boxed{(B) S(f) \geq 0.}$$

Key Points

- PSD is the Fourier transform of the autocorrelation function.
- The PSD of a real WSS random process is always real and non-negative.
- For a real process, $S(-f) = S(f)$ (even symmetry).

Mistakes to be avoided

- Do not assume $S(0) \leq S(f)$; the PSD need not attain its minimum at $f = 0$.
- Do not confuse even symmetry with odd symmetry; for a real process, $S(-f) = S(f)$, not $-S(f)$.
- Remember that $\int_{-\infty}^{\infty} S(f)\, df = R(0)$, which equals the average power and is generally positive, not zero.

PrepFusion

Q1.07.2 MCQ 2007 1M Ans: B ● ○ ○

Theory: The autocorrelation function of a real wide-sense stationary (WSS) random process is an even function and attains its maximum magnitude at zero lag.
For a real WSS random process,

$$R(\tau) = R(-\tau),$$

hence the autocorrelation function is even.
Also,

$$|R(\tau)| \leq R(0),$$

and

$$R(0) = E[X^2(t)],$$

which is the mean-square value (average power) of the process.
Therefore, the statement

$$R(\tau) = -R(-\tau)$$

is incorrect since it implies that $R(\tau)$ is an odd function.
Hence ,

$$\boxed{(C) R(\tau) = -R(-\tau)}$$

Key Points

- The autocorrelation function of a real WSS process is an even function.
- $R(0)$ equals the mean-square value of the process.
- The maximum magnitude of the autocorrelation occurs at $\tau = 0$.

Mistakes to be avoided

- Do not confuse the autocorrelation function with an odd function; it is always even for a real WSS process.
- Remember that $R(0)$ represents the mean-square value, not the mean.
- Do not forget the inequality $|R(\tau)| \leq R(0)$.

Q1.06.1 MCQ 2006 2M Ans: A ●○○

Theory: For white Gaussian noise passed through an ideal LPF, the output autocorrelation is obtained from the inverse Fourier transform of the output PSD.

If $X(t_1)$ and $X(t_2)$ are two samples obtained from a random process $X(t)$ at t_1 and t_2 instants, then

1. $$E[X^k(t_1)X^j(t_2)] = E[X^k(t_1)]\,E[X^j(t_2)]$$

   $X(t_1)$ and $X(t_2)$ are statistically independent for all values of k and j.

2. $$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = 0$$

   Hence, $X(t_1)$ and $X(t_2)$ are orthogonal.

3. $$\text{Cov}(t_1, t_2) = R_{XX}(t_1, t_2) - E[X(t_1)]E[X(t_2)] = 0$$

   Therefore,

   $$E[X(t_1)X(t_2)] = E[X(t_1)]E[X(t_2)]$$

   Hence, $X(t_1)$ and $X(t_2)$ are uncorrelated.

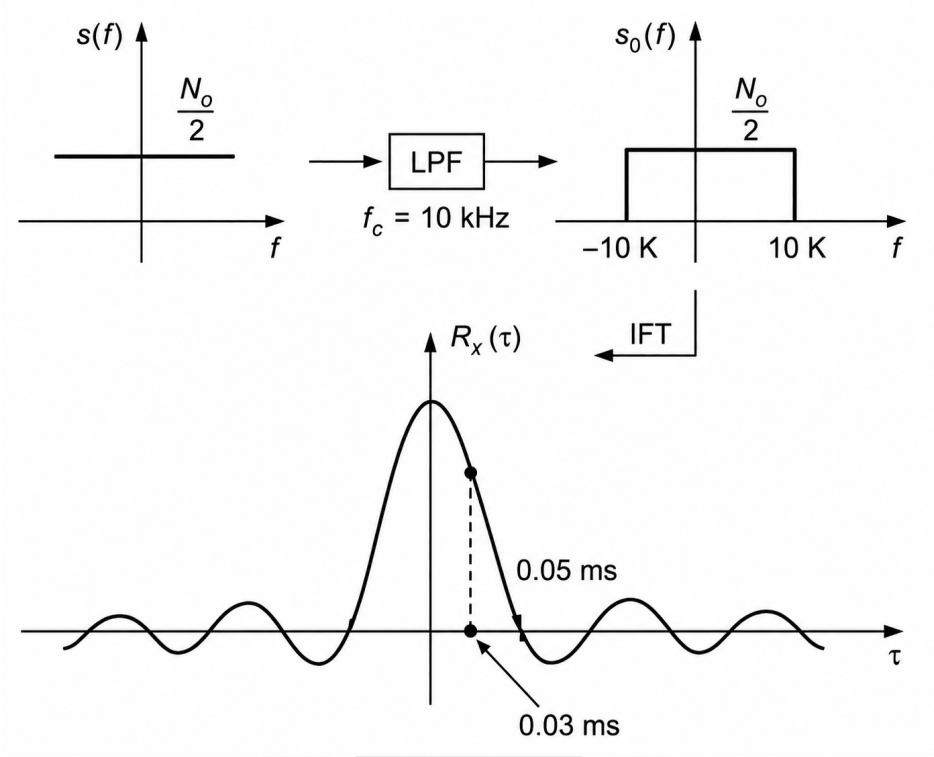


Fig. Solution Q1.06.1

Assume that the random process at the input of the LPF is $X(t)$, which is zero mean white Gaussian noise and the output of the LPF is $Y(t)$.
According to Wiener–Khinchin theorem,

$$R(\tau) \overset{\mathcal{F}}{\longleftrightarrow} S(f)$$

Hence, ACF and PSD form a Fourier transform pair.
Therefore, the ACF of $Y(t)$ is

$$R_Y(\tau) = N_0 B\,\text{Sinc}(2B\tau)$$

The mean value of the random process $Y(t)$ is

$$\mu_Y(t) = \mu_X(t)\,H(0)$$

where

$$\mu_Y(t) = E[Y(t)], \qquad \mu_X(t) = E[X(t)]$$

Since the input is zero mean,

$$\mu_Y(t) = 0 \times 1 = 0$$

Hence, $Y(t)$ is also a zero mean Gaussian random process.
Let $Y(t_s)$ and $Y(2t_s)$ be two consecutive samples obtained from $Y(t)$ at t_s and $2t_s$ instants.

$$E[Y(t_s)] = E[Y(2t_s)] = 0$$

Also,

$$R_{YY}(t_s) = R_{YY}(t, 2t_s) = E[Y(t_s)Y(2t_s)]$$

Using

$$R_{YY}(\tau) = E[Y(t)Y(t+\tau)]$$

we get

$$E[Y(t_s)Y(2t_s)] = R_{YY}(t_s) = N_0 B\,\text{Sinc}(2Bt_s)$$

Given

$$B = 10\text{ kHz}, \qquad t_s = 0.03\text{ ms}$$

Therefore,

$$R_{YY}(t_s) = N_0(10\times 10^3)\text{Sinc}\left(2\times 10\times 10^3\times 0.03\times 10^{-3}\right)$$

$$= 10^4 N_0\,\text{Sinc}(0.6)$$

Using

$$\text{Sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

Since

$$R_{YY}(t_s) \neq 0$$

the samples are not orthogonal.
Also,

$$\text{Cov}(t_s, 2t_s) = R_{YY}(t_s) - E[Y(t_s)]E[Y(2t_s)]$$

$$= R_{YY}(t_s) - 0 \neq 0$$

Hence, the samples are correlated and not orthogonal.

$$\boxed{(A)\text{ correlated .}}$$

Q1.06.2 MCQ 2006 2M Ans: A ●○○

Theory: For an LTI system, output PSD is obtained by multiplying the input PSD with the magnitude square of the transfer function.

$$S_Y(\omega) = |H(\omega)|^2 \times \text{Input PSD}$$

Given,

$$|H(\omega)|^2 = \frac{16}{16+\omega^2}$$

Therefore,

$$H(\omega) = \frac{4}{4+j\omega}$$

The above transfer function corresponds to an R–L LPF of the form

$$H(\omega) = \frac{R}{R + j\omega L}$$

(A) First order lowpass R–L filter

Q1.06.3 MCQ 2006 2M Ans: A ●○○

The transfer function corresponds to an R–L LPF of the form

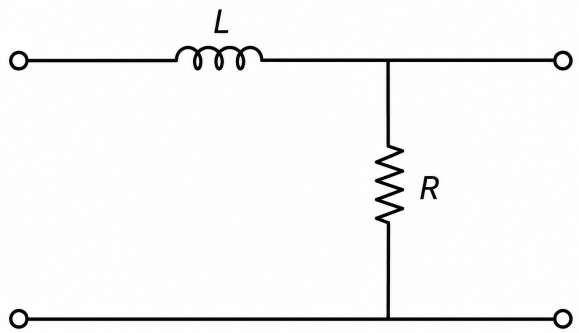


Fig. Solution Q1.06.3

$$H(\omega) = \frac{R}{R + j\omega L}$$

Comparing,

$$\frac{R}{R + j\omega L} = \frac{4}{4 + j\omega}$$

we get

$$R = 4\ \Omega, \qquad L = 1\ \text{H}$$

(A) First order R-L lowpass filter would have $R = 4\Omega,\ L = 1\text{H}$

Q1.06.4 MCQ 2006 2M Ans: B ●●○

Theory: When a continuous random variable is transformed into a finite set of values, the resulting random variable becomes discrete and its PDF is represented using Dirac delta functions.

Given,

$$f_X(x) = \frac{1}{10}\left[u(x+5) - u(x-5)\right],$$

so X is uniformly distributed over $[-5, 5]$.

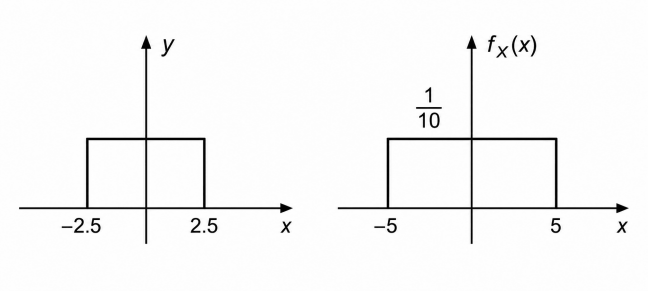


Fig. Solution Q1.06.4

From the given transformation,

$$Y = \begin{cases} 1, & -2.5 \le X \le 2.5, \\ 0, & |X| > 2.5. \end{cases}$$

Hence,

$$P(Y = 1) = P(-2.5 \le X \le 2.5) = \frac{5}{10} = \frac{1}{2},$$

and

$$P(Y = 0) = P(|X| > 2.5) = \frac{5}{10} = \frac{1}{2}.$$

Therefore, the PDF of Y is

$$\boxed{(B)\, f_Y(y) = \frac{1}{2}\delta(y) + \frac{1}{2}\delta(y-1)}$$

Key Points

- A continuous random variable can become discrete after a transformation.
- The probability masses are obtained by integrating the original PDF over the corresponding intervals.
- The PDF of a discrete random variable is expressed using Dirac delta functions.

Mistakes to be avoided

- Do not apply the Jacobian transformation formula; the mapping is many-to-one and produces a discrete random variable.
- Compute probabilities over the intervals before writing the delta functions.
- Ensure that the total probability is unity: $\frac{1}{2} + \frac{1}{2} = 1$.

Q1.06.5 MCQ 2006 1M Ans: B ● ○ ○

Theory: The total area under a probability density function (PDF) must be equal to one.

Given,

$$p(x) = Ke^{-\alpha|x|}, \qquad -\infty < x < \infty.$$

Using the normalization condition,

$$\int_{-\infty}^{\infty} p(x)\, dx = 1.$$

Substituting the given PDF,

$$K \int_{-\infty}^{\infty} e^{-\alpha|x|}\, dx = 1.$$

Since $e^{-\alpha|x|}$ is an even function,

$$2K\int_0^\infty e^{-\alpha x}\,dx = 1.$$

Now,

$$\int_0^\infty e^{-\alpha x}\,dx = \frac{1}{\alpha}.$$

Hence,

$$2K\left(\frac{1}{\alpha}\right) = 1,$$

which gives

$$K = \frac{\alpha}{2}.$$

Therefore,

$$\boxed{K = \frac{\alpha}{2}}$$

$$\boxed{(C)0.5\alpha}$$

Key Points

- The total area under a PDF is unity.
- Use symmetry since $e^{-\alpha|x|}$ is an even function.
- $\int_0^\infty e^{-\alpha x}\,dx = \frac{1}{\alpha}.$

Mistakes to be avoided

- Do not forget the factor of 2 due to even symmetry.
- Ensure $\alpha > 0$; otherwise, the integral does not converge.
- Apply the normalization condition before solving for K.

Q1.05.1 MCQ 2005 2M Ans: B ● ○ ○

Theory: For an LTI system, the output PSD is equal to the input PSD multiplied by the magnitude square of the transfer function.

$$PSD_{out} = |H(\omega)|^2\, PSD_{in} = 4N_0$$

Noise power is obtained by integrating the output PSD over the bandwidth B.

$$P_N = B \times S_{N_{out}}$$

$$P_N = B \times 4N_0$$

$$P_N = 4N_0B$$

$$\boxed{(B)P_N = 4N_0B}$$

Q1.05.2 MCQ | 2005 | 2M | Ans: C | ● ○ ○

Theory: The mean square value of a random variable is obtained using

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx$$

Given PDF is a straight line passing through the origin, hence

$$f_X(x) = mx + c$$

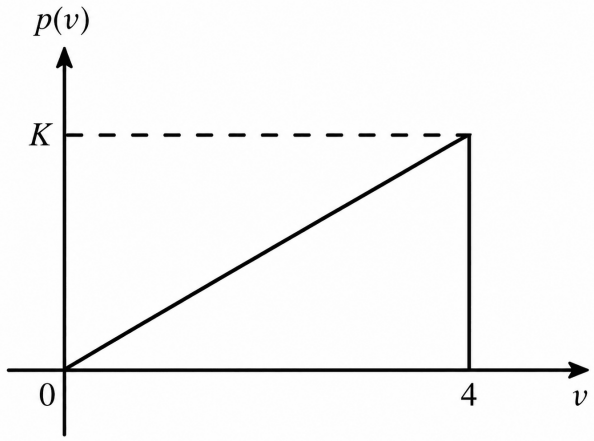


Fig. Solution Q1.05.2

Since the line passes through origin,

$$c = 0$$

Therefore,

$$f_X(x) = mx$$

Using total area property of PDF,

$$\frac{1}{2} \times 4 \times K = 1$$

$$2K = 1$$

$$K = \frac{1}{2}$$

Since at $x = 4$,

$$f_X(4) = 4m = \frac{1}{2}$$

$$m = \frac{1}{8}$$

Hence,

$$f_X(x) = \frac{x}{8}$$

Now,

$$E[X^2] = \int_0^4 x^2 \left(\frac{x}{8}\right) dx$$

$$= \frac{1}{8}\int_0^4 x^3 dx$$

$$= \frac{1}{8}\left[\frac{x^4}{4}\right]_0^4$$

$$= \frac{1}{32}(256)$$

$$= 8$$

$$\boxed{(C) E[X^2] = 8}$$

Q1.04.1 MCQ | 2004 | 2M | Ans: B | ● ● ○

Theory: The root-mean-square (RMS) quantization noise is the square root of the mean-square quantization error. The quantization error is

$$Q_e = x - x_q.$$

Hence, the mean-square quantization error is

$$E[Q_e^2] = E[(x - x_q)^2] = \int_0^1 (x - x_q)^2 f_X(x)\, dx.$$

Since X is uniformly distributed over $[0, 1]$,

$$f_X(x) = 1, \quad 0 \le x \le 1.$$

Therefore,

$$E[(x - x_q)^2] = \int_0^{0.3} (x - 0)^2\, dx + \int_{0.3}^1 (x - 0.7)^2\, dx.$$

Evaluating,

$$= \left[\frac{x^3}{3}\right]_0^{0.3} + \left[\frac{(x - 0.7)^3}{3}\right]_{0.3}^1$$

$$= \frac{(0.3)^3}{3} + \frac{(0.3)^3 - (-0.4)^3}{3}$$

$$= 0.009 + 0.030 = 0.039.$$

Hence, the RMS quantization noise is

$$\sqrt{0.039} = 0.198.$$

$$\boxed{\text{(B) RMS quantization noise} = 0.198}$$

Key Points

- Quantization error is $Q_e = x - x_q$.
- RMS error is the square root of the mean-square error.
- For a uniform random variable, $f_X(x) = 1$ over its support.

Mistakes to be avoided

- Do not compute the average absolute error instead of the mean-square error.
- Use the correct quantized value in each interval.
- Take the square root only after evaluating the mean-square error.

Q1.04.2 MCQ 2004 2M Ans: D ●○○

Theory: The probability mass at a point is equal to the jump in the cumulative distribution function (CDF) at that point. From the given CDF,

$$P(X = 1) = F_X(1) - F_X(1^-),$$

where $F_X(1^-)$ denotes the left-hand limit of the CDF at $x = 1$.
From the graph,

$$F_X(1^-) = 0.25, \qquad F_X(1) = 0.55.$$

Hence,

$$P(X = 1) = 0.55 - 0.25 = 0.30.$$

Therefore,

$$\boxed{(D) P(X = 1) = 0.30}$$

Key Points

- The probability mass at a point equals the jump in the CDF.
- $P(X = x_0) = F_X(x_0) - F_X(x_0^-)$.
- A continuous part of the CDF contributes no point mass.

Mistakes to be avoided

- Do not compute $P(X = 1)$ using the slope of the CDF.
- Always use the left-hand limit $F_X(1^-)$, not the value just before the next interval.
- Remember that only discontinuities in the CDF correspond to discrete probabilities.

Q1.03.1 MCQ 2003 2M Ans: C ●○○

Theory: The variance of the difference of two random variables is obtained using their variances and covariance.
Given,

$$W = Y - Z,$$

where

$$Y = X(2), \qquad Z = X(4).$$

Hence,

$$\mathrm{Var}(W) = \mathrm{Var}(Y) + \mathrm{Var}(Z) - 2\,\mathrm{Cov}(Y, Z).$$

Since

$$R_X(\tau) = 4\left(e^{-0.2|\tau|} + 1\right),$$

$$R_X(0) = 4(1 + 1) = 8.$$

Thus,

$$\text{Var}(Y) = \text{Var}(Z) = R_X(0) = 8.$$

The time difference is

$$|4 - 2| = 2,$$

so

$$\text{Cov}(Y, Z) = R_X(2) = 4\left(e^{-0.4} + 1\right).$$

Therefore,

$$\text{Var}(W) = 8 + 8 - 2 \times 4\left(e^{-0.4} + 1\right).$$

Using

$$e^{-0.4} \approx 0.6703,$$

$$\text{Var}(W) = 16 - 8(1.6703) \approx 2.64.$$

$$\boxed{(C)\ \text{Var}(W) = 2.64}$$

Key Points

- $\text{Var}(Y - Z) = \text{Var}(Y) + \text{Var}(Z) - 2\,\text{Cov}(Y, Z)$.
- For a WSS process, covariance depends only on the time difference.
- $R_X(0)$ gives the second moment of the process.

Mistakes to be avoided

- Do not use the mean while computing the variance formula.
- Use the time difference $|4 - 2| = 2$ when evaluating the autocorrelation.
- Do not confuse covariance with variance; covariance is $R_X(2)$, not $R_X(0)$.

Q1.03.2 MCQ 2003 2M Ans: B ● ○ ○

Theory: A Gaussian random variable is completely characterized by its mean and variance. Given (common data), the process has a constant mean

$$\mu = 2$$

and the auto-correlation function

$$R_X(\tau) = 4\left(e^{-0.2|\tau|} + 1\right).$$

The mean square value is

$$E[X^2] = R_X(0) = 4(1 + 1) = 8.$$

(As a check, $R_X(\tau) \to 4 = \mu^2$ as $|\tau| \to \infty$.) The variance is

$$\sigma^2 = R_X(0) - \mu^2 = 8 - 4 = 4,$$

so

$$\sigma = 2.$$

The required probability is

$$P(X \le 1) = F_X(1).$$

Using the Gaussian CDF,

$$P(X \le x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right),$$

we obtain

$$P(X \le 1) = 1 - Q\left(\frac{1-2}{2}\right) = 1 - Q(-0.5).$$

Since $Q(-\alpha) = 1 - Q(\alpha)$,

$$P(X \le 1) = 1 - [1 - Q(0.5)] = Q(0.5).$$

Therefore,

$$\boxed{(B)P(X \le 1) = Q(0.5)}$$

Key Points

- For a Gaussian random variable, $P(X \le x) = 1 - Q\left(\frac{x-\mu}{\sigma}\right)$.
- $R_X(0)$ is the mean square value; the variance is $R_X(0) - \mu^2$.
- $Q(-\alpha) = 1 - Q(\alpha)$.

Mistakes to be avoided

- Do not take $R_X(0)$ as the variance when the mean is not zero.
- Use the standard deviation σ, not the variance, in the Q-function argument.
- Do not ignore the mean: here $\mu = 2$, as given in the common data.

Q1.03.3 MCQ 2003 2M Ans: D ● ○ ○

Theory: The PDF of the sum of two independent random variables is obtained by the convolution of their individual PDFs.

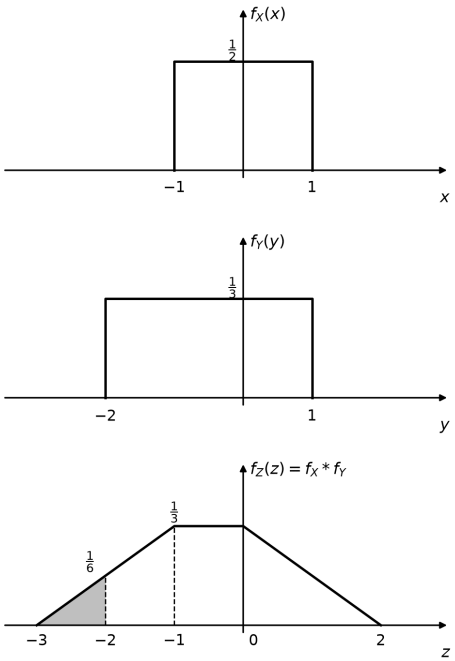


Fig. Solution Q1.03.3

Since

$$X \sim U(-1, 1), \qquad Y \sim U(-2, 1),$$

their PDFs are

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{3}, & -2 \le y \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Hence,

$$f_Z(z) = f_X(z) * f_Y(z).$$

The convolution gives a trapezoidal PDF over

$$-3 \le z \le 2,$$

rising linearly for $-3 \le z \le -1$, flat for $-1 \le z \le 0$ and falling linearly for $0 \le z \le 2$. Its maximum value is

$$\frac{1}{2} \times \frac{1}{3} \times 2 = \frac{1}{3}$$

(the two heights times the width 2 of the narrower pulse), and at $z = -2$

$$f_Z(-2) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}.$$

The required probability is

$$P(Z \le -2),$$

which corresponds to the left triangular portion of the PDF.
Therefore,

$$P(Z \le -2) = \frac{1}{2} \times (\text{base}) \times (\text{height}) = \frac{1}{2} \times 1 \times \frac{1}{6} = \frac{1}{12}.$$

Hence,

$$\boxed{(D) P(Z \le -2) = \frac{1}{12}}$$

Key Points

- The sum of independent random variables is obtained using convolution.
- The support of $Z = X + Y$ is the sum of the individual supports.
- The required probability equals the area under the PDF of Z.

Mistakes to be avoided

- Do not add the PDFs directly; use convolution.
- Use the correct support $[-3, 2]$ for Z.
- Compute probability as the area under the PDF, not the PDF value itself.

Q1.03.4 MCQ 2003 2M Ans: C ● ○ ○

Theory: Above threshold, an ideal frequency detector behaves like a differentiator, whose magnitude response is proportional to frequency.
The input noise PSD is white,

$$S_x(f) = \frac{N_0}{2}.$$

For an ideal frequency detector,

$$H(f) = j2\pi f.$$

Hence, the output noise PSD is

$$S_y(f) = S_x(f)\,|H(f)|^2\,.$$

Substituting,

$$S_y(f) = \frac{N_0}{2}\,(2\pi f)^2 = 2\pi^2 N_0 f^2.$$

Since

$$S_y(f) \propto f^2,$$

the output noise power spectral density has a **parabolic** shape.

(C) Parabolic

Key Points

- An ideal frequency detector behaves as a differentiator above threshold.
- Differentiation multiplies the spectrum by $(2\pi f)^2$.
- White noise at the input becomes parabolic noise at the output.

Mistakes to be avoided

- Do not assume white input noise remains white after FM detection.
- Remember that the differentiator transfer function is $H(f) = j2\pi f$.
- The output PSD is obtained by multiplying the input PSD with $|H(f)|^2$, not $H(f)$.

Q1.02.1 MCQ 2002 2M Ans: A ● ○ ○

Theory: Variance of the difference process is obtained using the autocorrelation relation of the random process.

$$\text{Var}[d(n)] = E[d^2(n)] - \{E[d(n)]\}^2$$

Since the process is zero mean,

$$E[d(n)] = E[x(n) - x(n-1)] = 0$$

Therefore,

$$\text{Var}[d(n)] = E[d^2(n)]$$

$$= E\{[x(n) - x(n-1)]^2\}$$

$$= E[x^2(n)] + E[x^2(n-1)] - 2E[x(n)x(n-1)]$$

Using

$$E[x^2(n)] = E[x^2(n-1)] = \sigma_x^2$$

and

$$E[x(n)x(n-1)] = R_{XX}(1)$$

we get

$$\sigma_d^2 = \sigma_x^2 + \sigma_x^2 - 2R_{XX}(1)$$

$$= 2\sigma_x^2 - 2R_{XX}(1)$$

Given,

$$2\sigma_x^2 - 2R_{XX}(1) = \frac{1}{10}\sigma_x^2$$

$$\Rightarrow 2 - \frac{2R_{XX}(1)}{\sigma_x^2} = \frac{1}{10}$$

$$\Rightarrow \frac{2R_{XX}(1)}{\sigma_x^2} = \frac{19}{10}$$

$$\Rightarrow \frac{R_{XX}(1)}{\sigma_x^2} = 0.95$$

$$\boxed{(A)\ \frac{R_{XX}(1)}{\sigma_x^2} = 0.95}$$

Q1.01.1 MCQ 2001 1M Ans: C ●○○

Theory: For a continuous random variable, the probability at any single point is always zero.

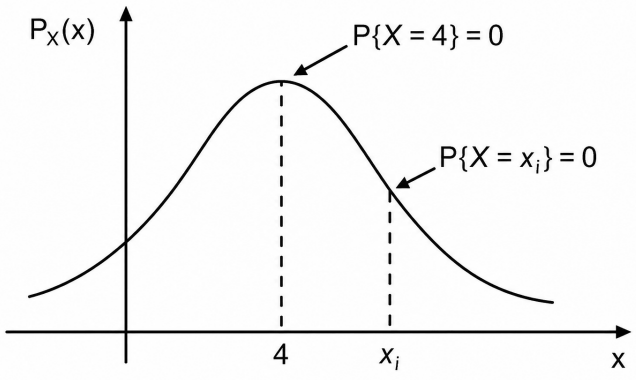


Fig. Solution Q1.01.1

Given,

$$P_X(x) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{(x-4)^2}{18}}$$

Since X is a continuous Gaussian random variable,

$$P(X = x) = 0$$

for every value of x.
Therefore,

$$P(X = 4) = 0$$

Similarly,

$$P(X = x_i) = 0$$

Hence, the probability of the event $\{X = 4\}$ is zero.

$$\boxed{(C) P(X = 4) = 0}$$

Q1.98.1 MCQ 1998 1M Ans: D ●○○

Theory: The autocorrelation of a rectangular pulse is obtained by convolving the signal with its time-reversed version, resulting in a triangular waveform.

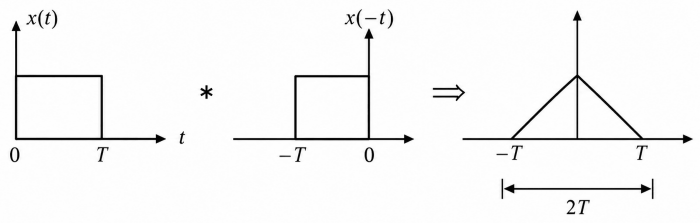


Fig. Solution Q1.98.1

For a rectangular pulse $x(t)$ of duration T,

$$R_{xx}(\tau) = x(t) * x(-t)$$

The convolution of two rectangular pulses produces a triangular pulse extending from $-T$ to T.
Hence, the ACF is triangular with total duration $2T$.

$$R_{xx}(\tau) = \left(1 - \frac{|\tau|}{T}\right), \qquad |\tau| \le T$$

and

$$R_{xx}(\tau) = 0, \qquad |\tau| > T$$

$$\boxed{(D) \text{The ACF of a rectangular pulse of duration } T \text{ is a triangular pulse of duration } 2T.}$$

Q1.98.2 MCQ 1998 1M Ans: C ● ○ ○

A very important property of the Fourier Transform is that a Gaussian function remains Gaussian in the frequency domain. If a Gaussian pulse exists in time domain, then its Fourier Transform is also Gaussian in frequency domain.

Gaussian Pulse in Time Domain

$$x(t) = e^{-at^2}, \qquad a > 0$$

Taking Fourier Transform,

$$X(f) = \sqrt{\frac{\pi}{a}}\, e^{-\frac{\pi^2 f^2}{a}}$$

which is again a Gaussian function in frequency.

$$(C)\text{Gaussian pulse in time domain} \xleftrightarrow{\mathcal{F}} \text{Gaussian pulse in frequency domain}$$

Key Points

- **Gaussian Fourier Transform Property:** A narrow pulse in time domain corresponds to a wide spectrum in frequency domain, and vice versa. Gaussian functions preserve their shape under Fourier Transform.

Q1.98.3 MCQ 1998 2M Ans: 2.51 ● ○ ○

Theory: The output noise power of an RC low-pass filter is obtained by integrating the input white noise PSD over the filter transfer function.

Given,

$$\frac{N_0}{2} = 2 \times 10^{-6}\ \text{V}^2/\text{Hz}, \qquad f_c = 4\ \text{kHz}.$$

For a first-order RC low-pass filter,

$$|H(f)|^2 = \frac{1}{1 + \left(\frac{f}{f_c}\right)^2}.$$

The mean squared output noise is

$$\sigma_n^2 = \int_{-\infty}^{\infty} \frac{N_0}{2} |H(f)|^2\, df.$$

Using

$$\int_{-\infty}^{\infty} \frac{df}{1 + \left(\frac{f}{f_c}\right)^2} = \pi f_c,$$

we get

$$\sigma_n^2 = \frac{N_0}{2}\, \pi f_c.$$

Substituting the given values,

$$\sigma_n^2 = 2 \times 10^{-6} \times \pi \times 4 \times 10^3 = 8\pi \times 10^{-3} \approx 2.513 \times 10^{-2}\ \text{V}^2.$$

Hence,

$$\boxed{\sigma_n^2 = 2.51 \times 10^{-2}\ \text{V}^2}$$

Therefore, the required answer is

$$\boxed{2.51}$$

Key Points

- Output noise power equals the integral of input PSD multiplied by the filter power response.
- For a first-order RC low-pass filter, $\sigma_n^2 = \frac{N_0}{2}\pi f_c$.
- Always use the two-sided PSD over the entire frequency axis.

Mistakes to be avoided

- Do not integrate only over positive frequencies when a two-sided PSD is given.
- Do not confuse $\frac{N_0}{2}$ with N_0.
- Use the power response $|H(f)|^2$, not the magnitude response $|H(f)|$.

Q1.97.1 MCQ 1997 1M Ans: A ●○○

A Gaussian random variable with zero mean and unit variance is called a standardized Gaussian random variable. Its probability density function is given by

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

where

$$\mu = 0, \qquad \sigma^2 = 1$$

$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is the probability density of a standardized Gaussian random variable.

Key Points

- **Gaussian PDF Property:** The Gaussian probability density function is symmetric about the mean and has the characteristic bell-shaped curve.

Q1.97.2 MCQ 1997 1M Ans: D ●○○

The autocorrelation function and power spectral density form a Fourier-transform pair:

$$R_x(\tau) \overset{\mathcal{F}}{\longleftrightarrow} S_x(f)$$

Hence,

$$R_x(\tau) = \mathcal{F}^{-1}\left\{\left[\frac{\sin(f)}{f}\right]^2\right\}$$

The Fourier transform pair

$$\left[\frac{\sin(f)}{f}\right]^2 \quad \overset{\mathcal{F}^{-1}}{\longleftrightarrow} \quad \text{triangular pulse}$$

shows that the autocorrelation function is a triangular pulse.

(D) A triangular pulse

Key Points

- By the Wiener–Khinchin relation, the autocorrelation function is the inverse Fourier transform of the power spectral density.
- The inverse Fourier transform of a squared sinc function is a triangular pulse.

Mistakes to be avoided

Do not confuse the transform pairs: a sinc function corresponds to a rectangular pulse, whereas a squared sinc function corresponds to a triangular pulse.

Mistakes to be avoided

- Do not confuse the sinc function in frequency with the sinc function in time.
- Do not assume the autocorrelation has the same shape as the PSD.
- Always identify the known Fourier transform pair before taking the inverse transform.

Q1.95.1 MCQ 1995 1M Ans: D ● ○ ○

A narrowband Gaussian noise signal can be represented as

$$n(t) = a(t)\cos(2\pi f_c t) - b(t)\sin(2\pi f_c t)$$

where $a(t)$ and $b(t)$ are independent zero mean Gaussian random variables having equal variance.
The envelope of the noise signal is

$$R = \sqrt{a^2(t) + b^2(t)}$$

Since the envelope is the magnitude of two independent Gaussian components, its probability density function follows the Rayleigh distribution.

Rayleigh Probability Density Function

$$f_R(r) = \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}, \qquad r \geq 0$$

(D) For a narrow band noise with Gaussian quadrature and inphase components, the probability density function of its envelope will be Rayleigh.

Key Points

- **Rayleigh Envelope Distribution:** The in-phase and quadrature components are Gaussian random variables. When their vector magnitude is taken,

$$R = \sqrt{a^2 + b^2}$$

the resulting envelope follows a Rayleigh distribution.

Q1.92.1 MCQ 1992 2M Ans: B ● ○ ○

Theory: Mean and variance of a random variable are obtained using the first and second moments of the probability density function.

$$E(X) = \int_{-\infty}^{\infty} x\, p(x)\, dx$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2\, p(x)\, dx$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Given,

$$p(x) = \frac{1}{4}, \qquad -1 \le x \le 3$$

$$E(X) = \int_{-1}^{3} x\, p(x)\, dx$$

$$= \frac{1}{4}\int_{-1}^{3} x\, dx$$

$$= \frac{1}{4}\left[\frac{x^2}{2}\right]_{-1}^{3}$$

$$= \frac{1}{4}\left(\frac{9}{2} - \frac{1}{2}\right)$$

$$= 1$$

$$E(X^2) = \int_{-1}^{3} x^2\, p(x)\, dx$$

$$= \frac{1}{4}\int_{-1}^{3} x^2\, dx$$

$$= \frac{1}{4}\left[\frac{x^3}{3}\right]_{-1}^{3}$$

$$= \frac{1}{4}\left(9 + \frac{1}{3}\right)$$

$$= \frac{7}{3}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{7}{3} - 1$$

$$= \frac{4}{3}$$

$$\boxed{(B) E(X) = 1, Var(X) = \frac{4}{3}}$$

Q1.89.1 MCQ | 1989 | 2M | Ans: B

Theory: In half-wave rectification, all negative values are clipped to zero while positive values remain unchanged. Hence the output PDF contains a delta function at the origin along with the positive half of the Gaussian PDF.
Half wave rectification is represented as

$$Y = X \qquad \text{for } x \geq 0$$

$$Y = 0 \qquad \text{for } x < 0$$

Therefore, the probability density function of Y is

$$f(y) = \frac{1}{2}\delta(y) + \frac{1}{\sqrt{2\pi N}} e^{-y^2/2N}$$

The mean square value is

$$E(Y^2) = \int_0^\infty y^2 f(y)\, dy$$

$$= \int_0^\infty y^2 \left[\frac{1}{2}\delta(y) + \frac{1}{\sqrt{2\pi N}} e^{-y^2/2N}\right] dy$$

Since,

$$y^2\delta(y) = 0$$

we get

$$E(Y^2) = \int_0^\infty \frac{y^2}{\sqrt{2\pi N}} e^{-y^2/2N}\, dy$$

Let

$$\frac{y}{\sqrt{N}} = t$$

Then,

$$y = \sqrt{N}\, t, \qquad dy = \sqrt{N}\, dt$$

Substituting,

$$E(Y^2) = \frac{1}{\sqrt{2\pi N}} \int_0^\infty N t^2 e^{-t^2/2} \sqrt{N}\, dt$$

$$= \frac{N}{\sqrt{2\pi}} \int_0^\infty t^2 e^{-t^2/2}\, dt$$

Using the standard Gaussian integral,

$$\int_0^\infty t^2 e^{-t^2/2}\, dt = \sqrt{\frac{\pi}{2}}$$

Therefore,

$$E(Y^2) = \frac{N}{\sqrt{2\pi}} \sqrt{\frac{\pi}{2}} = \frac{N}{2}$$

$$\boxed{(B) E(Y^2) = \frac{N}{2}}$$

Q1.87.1 MCQ | 1987 | 2M | Ans: D | ● ○ ○

Theory: Variance scales by the square of the multiplying constant. Also, variance is unaffected by sign inversion.

$$\text{Var}(aX) = a^2\sigma_x^2$$

Hence,

$$\text{Var}(-kX) = k^2\sigma_x^2$$

$$\text{Var}(-kX) = E\left[(-kX)^2\right] - [E(-kX)]^2$$

$$= k^2E(X^2) - [-kE(X)]^2$$

$$= k^2E(X^2) - k^2[E(X)]^2$$

$$= k^2\left(E(X^2) - [E(X)]^2\right)$$

$$= k^2\sigma_x^2$$

$$\boxed{(D)\text{Var}(-kX) = k^2\sigma_x^2}$$

PrepFusion

SOLUTIONS

11

Digital Communications

Topics

1. PCM and DPCM
2. Digital Modulation - ASK, PSK, FSK, QAM
3. Bandwidth and Inter-Symbol Interference
4. MAP and ML Detection
5. Matched Filter Receiver
6. SNR and BER Analysis
7. Error Correction - Hamming Codes and CRC

Unit Snapshot*

Stream: EC	**Questions:** 137	**MCQ:** 99	**MSQ:** 0	**NAT:** 38
1M: 41	**2M:** 96	**Descriptive:** 0	**Avg Weightage** ~5M	**Range:** 1M(2023)–8M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 NAT | 2026 | 1M | Ans: 0.95 to 1.05 | ● ○ ○

Given:

$$\text{Modulation scheme} = 16\text{-QAM}$$

$$R_b = 4 \times 10^6 \text{ bits/s}$$

Step 1: Find the Symbol Rate R_s

Theory:
In an M-ary digital modulation scheme, each symbol carries

$$\log_2 M$$

bits. Hence, the symbol rate is

$$R_s = \frac{R_b}{\log_2 M}.$$

For 16-QAM,

$$\log_2(16) = 4.$$

Therefore,

$$R_s = \frac{R_b}{\log_2 M} = \frac{4 \times 10^6}{4} = 1 \times 10^6$$

symbols/s.

Step 2: Find the Minimum Transmission Bandwidth

Theory:
For ideal Nyquist signaling, the minimum transmission bandwidth is equal to the symbol rate.

$$B_{\min} = R_s.$$

Substituting the value of R_s,

$$B_{\min} = 1 \times 10^6 \text{ Hz}$$

$$\boxed{B_{\min} = 1 \text{ MHz}}$$

Mistakes to be avoided

- **Bit Rate vs Symbol Rate:** Do not assume
$$R_s = R_b.$$
For M-ary modulation,
$$R_s = \frac{R_b}{\log_2 M}.$$
- **Bits per Symbol:** For 16-QAM,
$$\log_2(16) = 4,$$
not 16.
- **Units:** The symbol rate is measured in baud (symbols/s), while the bit rate is measured in bits/s.
- **Bandwidth Conversion:** Remember that
$$10^6 \text{ Hz} = 1 \text{ MHz}.$$

Q2.26.2 MCQ 2026 1M Ans: C ● ○ ○

Given:
A 4-level signal

$$s(t) \in \{-1, 0, +1, +2\}$$

with symbol rate

$$R_s = 10^4 \text{ symbols/s}.$$

Step 1: Determine the Equivalent Nyquist Bandwidth

Theory:
For a digital signal transmitted at a symbol rate R_s, the minimum Nyquist bandwidth required for zero intersymbol interference (ISI) is

$$B_{\text{Nyquist}} = \frac{R_s}{2}.$$

The number of signal levels does not affect the Nyquist bandwidth; it depends only on the symbol rate.

Substituting the given value,

$$B_{\text{Nyquist}} = \frac{10^4}{2} = 5 \times 10^3 \text{ Hz}.$$

$$\boxed{B_{\text{Nyquist}} = 5 \text{ kHz}}$$

Mistakes to be avoided

- **Confusing Bit Rate with Symbol Rate:** The given rate is
$$10^4 \text{ symbols/s},$$
not bits/s.
- **Using the Number of Levels:** Do not multiply the bandwidth by 4 just because the signal has four levels.
- **Forgetting the Factor of 2:** Nyquist bandwidth is
$$\frac{R_s}{2},$$
not R_s.

Q2.26.3 MCQ 2026 2M Ans: B ● ● ○

Step 1: Convert $\frac{E_b}{N_0}$ from dB to linear scale

Theory:
For BER calculations, the ratio $\frac{E_b}{N_0}$ must first be converted from dB to its linear value.

$$\frac{E_b}{N_0} = 10^{\frac{8.4}{10}} = 10^{0.84} \approx 6.92$$

Therefore,

$$\sqrt{\frac{E_b}{N_0}} = \sqrt{6.92} \approx 2.63$$

Step 2: Use the given approximation

Theory:
For large values of u, the complementary error function can be approximated as

$$\operatorname{erfc}(u) \approx \frac{1}{u\sqrt{\pi}} e^{-u^2}$$

Substituting $u = 2.63$,

$$\operatorname{erfc}(2.63) \approx \frac{1}{2.63\sqrt{\pi}} e^{-(2.63)^2}$$

Now,

$$(2.63)^2 \approx 6.92$$

and

$$e^{-6.92} \approx 9.9 \times 10^{-4}$$

Also,

$$2.63\sqrt{\pi} \approx 2.63 \times 1.772 \approx 4.66$$

Hence,

$$\operatorname{erfc}(2.63) \approx \frac{9.9 \times 10^{-4}}{4.66} \approx 2.1 \times 10^{-4}$$

Step 3: Compute the BER

Theory:
For QPSK,

$$P_b = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

Therefore,

$$P_b = \frac{1}{2}\left(2.1 \times 10^{-4}\right)$$

$$P_b \approx 1.05 \times 10^{-4}$$

Thus, the nearest BER is

$$\boxed{P_b \approx 10^{-4}}$$

Mistakes to be avoided

- **QPSK BER Formula:** Remember the correct BER expression:

$$P_b = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

  Missing the factor $\frac{1}{2}$ gives an answer twice the correct value.

Q2.26.4 NAT 2026 2M Ans: 0.10 to 0.20 ● ○ ○

Given:

$$p = 0.10$$

where p is the average bit error probability at the input of the decoder.

Theory:
A Hamming $(7, 4)$ code has minimum distance

$$d_{\min} = 3$$

and hence can correct

$$t = \left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor = 1$$

error per codeword. Therefore, decoding fails whenever two or more errors occur in a 7-bit codeword.

Probability of Decoder Failure

$$P(\text{failure}) = 1 - P(0 \text{ errors}) - P(1 \text{ error})$$

For zero errors,

$$P(0 \text{ errors}) = (1 - p)^7 = (0.9)^7 = 0.4783$$

For exactly one error,

$$P(1 \text{ error}) = {}^7C_1\, p(1 - p)^6 = 7p(1 - p)^6$$

$$= 7(0.1)(0.9)^6$$

$$= 7(0.1)(0.5314) = 0.3720$$

Hence,

$$P(\text{failure}) = 1 - 0.4783 - 0.3720$$

$$= 0.1497$$

$$\boxed{P(\text{failure}) \approx 0.15}$$

Q2.25.1 NAT 2025 1M Ans: 3 ● ○ ○

Given:
The generator matrix of a $(6, 3)$ binary linear block code is

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} = [I_3 \mid P]\,.$$

The minimum Hamming distance $d_{\min}$ of the code is required.

Theory:

For a linear block code, the minimum Hamming distance equals the minimum Hamming weight among all non-zero codewords:

$$d_{\min} = \min\{w(c) \mid c \neq 0\}.$$

Hence, we generate all possible codewords and find the smallest non-zero Hamming weight.

Step 1: Obtain the Parity Equations
From the parity part of G,

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$

the parity bits are

$$p_1 = d_1 \oplus d_3,$$

$$p_2 = d_2 \oplus d_3,$$

$$p_3 = d_1 \oplus d_2.$$

Therefore, each codeword is

$$c = [d_1\ d_2\ d_3\ p_1\ p_2\ p_3].$$

Step 2: Generate All Codewords

$(d_1 d_2 d_3)$	Codeword	Hamming Weight
000	000000	0
001	001110	3
010	010011	3
011	011101	4
100	100101	3
101	101011	4
110	110110	4
111	111000	3

Step 3: Find the Minimum Non-Zero Hamming Weight
Ignoring the all-zero codeword,

$$\{3, 3, 4, 3, 4, 4, 3\}.$$

Hence,

$$\min w(c) = 3.$$

Since the code is linear,

$$d_{\min} = \min w(c).$$

$$\boxed{d_{\min} = 3}$$

Key Points

- A generator matrix of a systematic linear block code is written as

$$G = [I_k \mid P],$$

where I_k is the $k \times k$ identity matrix and P is the parity matrix.

- The identity matrix portion directly carries the message bits, while the parity matrix generates the parity bits.
- For a systematic codeword,

$$c = [d_1\ d_2\ \cdots\ d_k\ p_1\ p_2\ \cdots\ p_{n-k}].$$

- For a (n, k) linear block code, the total number of possible codewords is

$$2^k.$$

- For a linear block code,

$$d_{\min} = \text{minimum non-zero Hamming weight.}$$

- The all-zero codeword is not considered while finding $d_{\min}$.
- Parity bits are generated using modulo-2 addition ($\oplus$):

$$1 \oplus 1 = 0, \qquad 1 \oplus 0 = 1.$$

Mistakes to be avoided

- **Using Ordinary Addition Instead of XOR:** Parity bits must be computed using modulo-2 addition:

$$1 \oplus 1 = 0.$$

Q2.25.2 MCQ | 2025 | 2M | Ans: D | ● ● ○

Given:

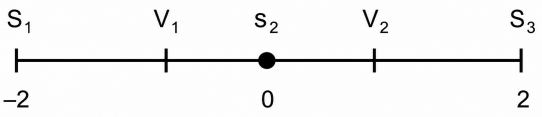


Fig. Solution Q2.25.2

The transmitted symbol S takes one of the three values

$$S \in \{-2, 0, 2\}$$

with equal probability,

$$P(S = -2) = P(S = 0) = P(S = 2) = \frac{1}{3}.$$

The received signal is

$$Y = S + N,$$

where N is zero-mean Gaussian noise.

Step 1: Determine the Decision Boundaries

Theory:
For Maximum Likelihood (ML) detection with equally likely symbols, the decision thresholds are placed midway between adjacent signal levels.
Threshold between -2 and 0:

$$V_1 = \frac{-2+0}{2} = -1$$

Threshold between 0 and 2:

$$V_2 = \frac{0+2}{2} = 1$$

Hence, the decision regions are

$$Y < -1 \Rightarrow -2,$$

$$-1 \le Y \le 1 \Rightarrow 0,$$

$$Y > 1 \Rightarrow 2.$$

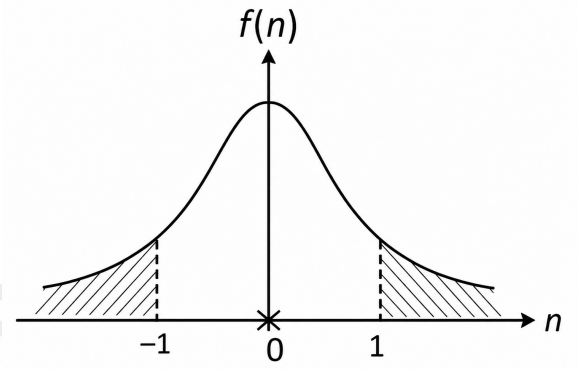


Fig. Solution Q2.25.2

Step 2: Error Probability when $S = -2$
When -2 is transmitted,

$$Y = -2 + N.$$

An error occurs if the received value crosses the threshold -1:

$$-2 + N > -1.$$

Therefore,

$$N > 1.$$

Hence,

$$P_e \mid S = -2 = P(N > 1).$$

Step 3: Error Probability when $S = 0$

When 0 is transmitted,

$$Y = N.$$

An error occurs if the received value crosses either threshold:

$$Y < -1$$

or

$$Y > 1.$$

Thus,

$$P_e \mid S = 0 = P(N < -1) + P(N > 1).$$

Since the Gaussian distribution is symmetric,

$$P(N < -1) = P(N > 1).$$

Therefore,

$$P_e \mid S = 0 = 2P(N > 1).$$

Step 4: Error Probability when $S = 2$
When 2 is transmitted,

$$Y = 2 + N.$$

An error occurs if

$$2 + N < 1.$$

Hence,

$$N < -1.$$

Using symmetry,

$$P_e \mid S = 2 = P(N < -1) = P(N > 1).$$

Step 5: Average Probability of Error
Since all symbols are equally likely,

$$P_e = \frac{1}{3}\left[P(N > 1) + 2P(N > 1) + P(N > 1)\right].$$

Simplifying,

$$P_e = \frac{4}{3}P(N > 1).$$

Comparing with

$$P_e = \alpha\, P(N > 1),$$

we obtain

$$\alpha = \frac{4}{3}.$$

$$\boxed{\alpha = \frac{4}{3}}$$

Key Points

- For equally likely symbols, ML decision thresholds are located at the midpoints between adjacent signal levels.
- The outer symbols (-2 and 2) have only one neighboring decision boundary, whereas the middle symbol (0) can be mistaken on either side.
- Due to symmetry of Gaussian noise,
$$P(N < -a) = P(N > a).$$
- The average error probability is obtained by weighting the conditional error probabilities by the symbol probabilities.

Mistakes to be avoided

- **Ignoring Two-Sided Error for the Middle Symbol:** For $S = 0$, errors occur on both sides:
$$P_e|S = 0 = P(N < -1) + P(N > 1).$$
- **Forgetting Equal Symbol Probabilities:** Since all symbols are equiprobable,
$$P(S_i) = \frac{1}{3}.$$

Q2.24.1 MCQ | 2024 | 1M | Ans: MTA | ● ○ ○

1: System model and zero ISI condition

$$z(t) = \sum_{n=-\infty}^{\infty} b[n]\, x(t - nT)$$

Sampling at $t = mT$:

$$z(mT) = \sum_{n=-\infty}^{\infty} b[n]\, x((m-n)T)$$

For zero ISI:

$$x(0) = 1, \quad x(kT) = 0 \ \ (k \neq 0)$$

This ensures that at each sampling instant only desired symbol contributes and other shifted pulses vanish.

2: Sampling representation and frequency-domain idea (from slide)

$$p(t) \cdot \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$p(t) \longleftrightarrow P(f), \quad \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow R_s \sum_{n=-\infty}^{\infty} \delta(f - nR_s), \quad R_s = \frac{1}{T_s}$$

Sampling in time produces periodic replicas of the spectrum in frequency, spaced by $R_s = 1/T_s$.

3: Convolution result (spectral repetition)

$$P(f) * \left\{ R_s \sum_{n=-\infty}^{\infty} \delta(f - nR_s) \right\} \Rightarrow \sum_{n=-\infty}^{\infty} P(f - nR_s) = \frac{A}{R_s} e^{-j2\pi f T_s}$$

$$\Rightarrow \sum_{n=-\infty}^{\infty} P(f - nR_s) = \text{constant (with phase)}$$

For zero ISI, the shifted spectra must combine to a constant so that distortion is avoided at sampling instants.

4: Apply to given system

$$P(f) \rightarrow X(f), \quad R_s = \frac{1}{T}$$

$$\sum_{n=-\infty}^{\infty} X\left(f - \frac{n}{T}\right) = T$$

Changing index $n \rightarrow -m$:

$$\boxed{\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = T}$$

Key Points

- **Zero ISI Condition:** The pulse must satisfy $x(0) = 1$ and $x(kT) = 0$ for $k \neq 0$ so that only the desired symbol is detected at sampling instants.
- **Sampling Effect:** Multiplication by $\sum \delta(t - nT)$ leads to spectral replicas spaced by $\frac{1}{T}$ in frequency domain.
- **Nyquist Criterion:** Shifted spectra must sum to a constant: $\sum X\left(f \pm \frac{m}{T}\right) = T$ to avoid ISI.
- **From Convolution Result:**

$$\boxed{\sum_{n=-\infty}^{\infty} P(f - nR_s) = \frac{A}{R_s} e^{-j2\pi f T_s}}$$

showing constant magnitude (with phase), which leads to the Nyquist condition.

Q2.24.2 MCQ 2024 2M Ans: B ●○○

Given:
The transmitted symbols are

$$S_1 = -4, \qquad S_2 = 0, \qquad S_3 = 4$$

with equal probabilities

$$P(S_1) = P(S_2) = P(S_3) = \frac{1}{3}.$$

The received signal is

$$r = s + w$$

where the noise is Gaussian with

$$\mu_w = 0, \qquad \sigma_w^2 = 4.$$

Hence,

$$\sigma_w = 2.$$

Theory:

For equally likely symbols in AWGN, the optimum ML decision thresholds are the midpoints between adjacent signal levels.

Step 1: Determine the optimum decision thresholds

Threshold between S_1 and S_2:

$$V_1 = \frac{-4 + 0}{2} = -2.$$

Threshold between S_2 and S_3:

$$V_2 = \frac{0 + 4}{2} = 2.$$

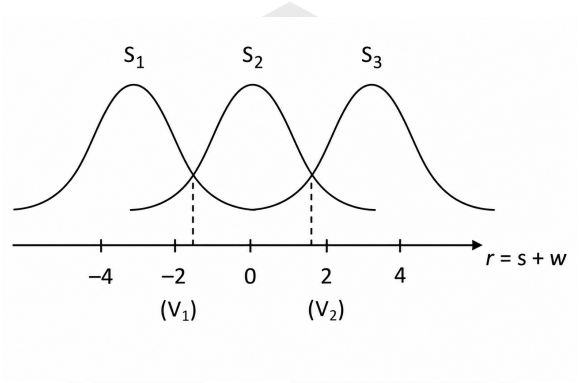


Fig. Solution Q2.24.2

Step 2: Error probability when $S_1 = -4$ is transmitted

The received signal is

$$r = -4 + w.$$

An error occurs if the received sample crosses the threshold $V_1 = -2$:

$$-4 + w > -2.$$

Therefore,

$$w > 2.$$

Thus,

$$P_{e|-4} = P(w > 2).$$

Using the Q-function,

$$P_{e|-4} = Q\left(\frac{2 - 0}{2}\right) = Q(1).$$

Step 3: Error probability when $S_2 = 0$ is transmitted

The received signal is

$$r = w.$$

An error occurs if

$$r < -2$$

or

$$r > 2.$$

Hence,

$$P_{e|0} = P(w < -2) + P(w > 2).$$

By symmetry of the Gaussian distribution,

$$P_{e|0} = Q(1) + Q(1) = 2Q(1).$$

Step 4: Error probability when $S_3 = 4$ is transmitted
The received signal is

$$r = 4 + w.$$

An error occurs if

$$4 + w < 2.$$

Therefore,

$$w < -2.$$

Thus,

$$P_{e|4} = P(w < -2) = Q(1).$$

Step 5: Average probability of error
Since all symbols are equally likely,

$$P_e = P(-4)P_{e|-4} + P(0)P_{e|0} + P(4)P_{e|4}.$$

Substituting the values,

$$P_e = \frac{1}{3}Q(1) + \frac{1}{3}\big(2Q(1)\big) + \frac{1}{3}Q(1).$$

$$P_e = \frac{4}{3}Q(1).$$

$$\boxed{(B)\, P_e = \frac{4}{3}Q(1)}$$

Key Points

- **ML Thresholds for Equal Priors:** For equally likely symbols in AWGN, the optimum decision threshold is the midpoint between adjacent signal levels.
- **Gaussian Tail Probability:** For noise $w \sim N(0, \sigma^2)$,

$$P(w > a) = Q\left(\frac{a}{\sigma}\right).$$

- **Symmetry Property:** A zero-mean Gaussian random variable satisfies

$$P(w > a) = P(w < -a).$$

- **Average Error Probability:** The overall symbol error probability is obtained by averaging the conditional error probabilities weighted by their prior probabilities.

Mistakes to be avoided

- **Wrong Threshold Locations:** Do not place thresholds at -1 and 1. The thresholds are the midpoints between adjacent symbols:

$$V_1 = \frac{-4+0}{2} = -2, \qquad V_2 = \frac{0+4}{2} = 2.$$

- **Using Variance Instead of Standard Deviation:** The Q-function argument uses the standard deviation:

$$\sigma = \sqrt{\sigma^2} = \sqrt{4} = 2.$$

- **Ignoring Two-Sided Errors for $S = 0$:** When $S = 0$, an error occurs on both sides:

$$r < -2 \quad \text{or} \quad r > 2.$$

Hence,

$$P_{e|0} = 2Q(1).$$

Q2.24.3 MCQ 2024 2M Ans: A

Given:
Information bit sequence

$$d = 111010101$$

Generator polynomial

$$C(x) = x^4 + x + 1$$

Corresponding binary generator sequence:

$$C = 10011$$

Theory:
In CRC encoding, if the generator polynomial has degree r, then r zeros are appended to the message. The appended sequence is divided (modulo-2) by the generator sequence, and the remainder forms the CRC bits.

Step 1: Append Zeros
The degree of the generator polynomial is

$$r = 4.$$

Hence, append 4 zeros to the message:

$$d' = 1110101010000.$$

Step 2: Modulo-2 Division
Dividend:

$$1110101010000$$

Divisor:

$$10011$$

Perform XOR division:

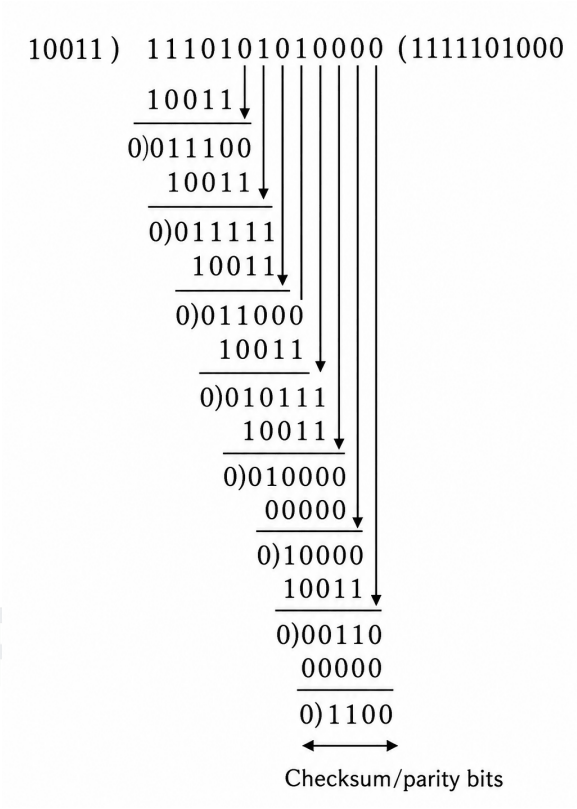


Fig. Solution Q2.24.3

CRC Remainder = 1100

Encoded Sequence = 1110101011100

Step 3: Form the CRC Codeword
The encoded sequence is obtained by appending the remainder to the original message:

$$\text{Codeword} = 111010101 \,||\, 1100$$

(A) Encoded Sequence = 1110101011100

Key Points

- If the generator polynomial degree is r, append r zeros before performing CRC division.
- The generator polynomial
$$C(x) = x^4 + x + 1$$
corresponds to the binary sequence
$$10011.$$
- The CRC remainder always contains exactly r bits, where r is the degree of the generator polynomial.
- The transmitted codeword is
$$\text{Message Bits} \,||\, \text{CRC Bits}.$$

Mistakes to be avoided

- **Appending Wrong Number of Zeros:** Append zeros equal to the degree of the generator polynomial, not the number of bits in the generator sequence.
- **Using Ordinary Binary Division:** CRC uses modulo-2 division where subtraction is replaced by XOR.
- **Appending the Remainder to the Wrong Sequence:** The remainder is appended to the original message, not to the zero-appended message.

Q2.24.4 NAT 2024 2M Ans: 45.00 to 45.30 ● ○ ○

Given:

$$x(t) = A\cos(2\pi f_0 t + \theta)$$

$$R_x(\tau) = \frac{2}{3}\cos(2\pi f_0 \tau)$$

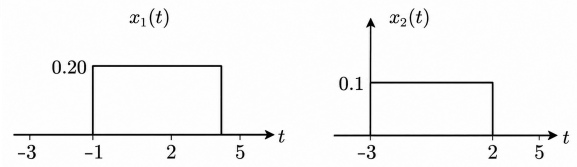


Fig. Solution Q2.24.4

An 8-bit mid-rise quantizer is used.

Theory:
For a uniform quantizer, the quantization error is assumed uniformly distributed over one quantization interval. Therefore,

$$N_q = \frac{\Delta^2}{12}.$$

Step 1: Signal Power
Signal power is obtained from the autocorrelation function at zero lag:

$$S = R_x(0).$$

Substituting $\tau = 0$,

$$S = \frac{2}{3}\cos(0) = \frac{2}{3}.$$

Hence,

$$S = \frac{2}{3}\ \text{W}.$$

Step 2: Quantization Noise Power

For an 8-bit quantizer,

$$\Delta = \frac{2-(-2)}{2^8} = \frac{4}{256} = \frac{1}{64}.$$

Therefore,

$$N_q = \frac{\Delta^2}{12} = \frac{\left(\frac{1}{64}\right)^2}{12} = \frac{1}{64\times 64\times 12}.$$

Step 3: Signal-to-Quantization Noise Ratio

$$\frac{S}{N_q} = \frac{\frac{2}{3}}{\frac{1}{64\times 64\times 12}}$$

$$= \frac{2\times 64\times 64\times 12}{3}$$

$$= 32768.$$

Step 4: Convert to dB

$$\left(\frac{S}{N_q}\right)_{\text{dB}} = 10\log_{10}(32768)$$

$$= 45.15\ \text{dB}.$$

$$\boxed{\left(\frac{S}{N_q}\right)_{\text{dB}} = 45.15\ \text{dB}}$$

Mistakes to be avoided

- **Using $R_x(\tau)$ Directly:** Signal power must be computed at

  $$\tau = 0.$$

  Hence,

  $$S = R_x(0).$$

- **Wrong Number of Quantization Levels:** For an 8-bit quantizer,

  $$L = 2^8 = 256,$$

not 8.

- **Incorrect Quantization Step Size:** Always use the full signal range:

$$\Delta = \frac{2-(-2)}{256}.$$

- **Using $\Delta/12$ Instead of $\Delta^2/12$:**

$$N_q = \frac{\Delta^2}{12}$$

is the quantization noise power.

- **Using 20 log Instead of 10 log:** Since SQNR is a power ratio,

$$\text{SQNR}_{dB} = 10\log_{10}\left(\frac{S}{N_q}\right).$$

Q2.23.1 NAT | 2023 | 1M | Ans: 10 | ● ○ ○

For an ideal n-bit ADC with a full-scale sinusoidal input,

$$\text{SNR}_{\text{dB}} = 6.02n + 1.76$$

Given,

$$61.96 = 6.02n + 1.76$$

Therefore,

$$6.02n = 61.96 - 1.76$$

$$6.02n = 60.20$$

$$n = \frac{60.20}{6.02}$$

Hence, the resolution of the ADC is

$$\boxed{10 \text{ bits}}$$

Q2.22.1 NAT | 2022 | 1M | Ans: 2.99 to 3.01 | ● ○ ○

Given:
The symbol rate is related to the bit rate by

$$R_b = R_s \log_2 M$$

where

$$R_b = \text{Bit rate}, \qquad R_s = \text{Symbol rate}, \qquad M = \text{Modulation order}.$$

Theory: Each symbol in an M-ary modulation scheme carries $\log_2 M$ bits.

Given symbol rate,

$$R_s = 1 \text{ Msymbol/s.}$$

For QPSK:

$$M = 4 = 2^2$$

$$R_{b1} = R_s \log_2 4 = 1 \times 2 = 2 \text{ Mbps.}$$

For 16-QAM:

$$M = 16 = 2^4$$

$$R_{b2} = R_s \log_2 16 = 1 \times 4 = 4 \text{ Mbps.}$$

Since the transmitted symbol stream alternates equally between QPSK and 16-QAM symbols, the average bit rate is

$$R_b = \frac{R_{b1} + R_{b2}}{2}$$

$$= \frac{2+4}{2}$$

$$= 3 \text{ Mbps.}$$

$$\boxed{R_b = 3 \text{ Mbps}}$$

Key Points

- Symbol rate and bit rate are related by
$$R_b = R_s \log_2 M.$$
- Each QPSK symbol carries
$$\log_2 4 = 2$$
bits.
- Each 16-QAM symbol carries
$$\log_2 16 = 4$$
bits.
- When two modulation schemes occur equally often, the overall bit rate is the arithmetic mean of their individual bit rates.

Q2.22.2 MCQ 2022 1M Ans: A ● ○ ○

Given: The frequency response $H(f)$ of a linear time-invariant system has magnitude:

$$|H(f)| = \begin{cases} 1, & -\alpha \le f \le \alpha \\ 0, & \text{otherwise} \end{cases}$$

Theory: A pure delay system with time delay T_d has the transfer function:

$$H(f) = e^{-j2\pi f T_d}$$

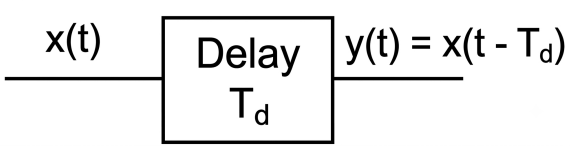


Fig. Solution Q2.22.2

This requires both a constant magnitude $|H(f)| = 1$ and a linear phase response $\angle H(f) = -2\pi f T_d$.

Knowing only the magnitude response $|H(f)|$ is **not sufficient** to classify a system as a pure delay system, because any arbitrary phase response $\angle H(f)$ can exist while keeping $|H(f)| = 1$.

For a Wide-Sense Stationary (WSS) process passed through a linear time-invariant system, the output power spectral density $S_Y(f)$ is related to the input power spectral density $S_X(f)$ by:

$$S_Y(f) = |H(f)|^2 S_X(f)$$

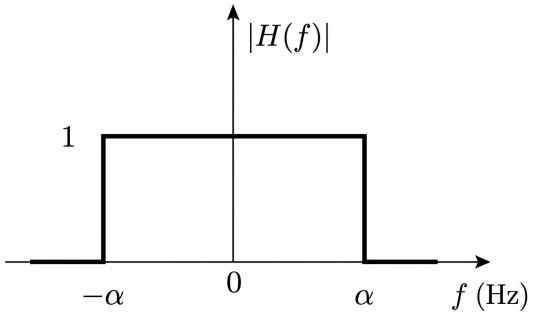


Fig. Solution Q2.22.2

Analysis of Statement I:
Without information about the phase response $\angle H(f)$, we cannot guarantee that the system is a pure delay system. Thus, Statement I is not necessarily true.

Statement I is **incorrect**.

Analysis of Statement II:
For inputs bandlimited to $-\alpha \le f \le \alpha$, we have $|H(f)| = 1$. Substituting this into the power spectral density formula:

$$S_Y(f) = |H(f)|^2 S_X(f) = (1)^2 S_X(f) = S_X(f) \quad \text{for } -\alpha \le f \le \alpha$$

Statement II is **correct**.

(C) Statement I is incorrect, and Statement II is correct.

Key Points

- A pure delay system $y(t) = x(t - T_d)$ requires both constant magnitude $|H(f)| = 1$ and linear phase response $\angle H(f) = -2\pi f T_d$.
- Magnitude spectrum alone does not define system behavior uniquely, as phase response can introduce phase distortion without altering $|H(f)|$.
- The relation between input and output Power Spectral Densities (PSD) depends solely on the squared magnitude response:
$$S_Y(f) = |H(f)|^2 S_X(f)$$
- Phase response has no effect on the output Power Spectral Density $S_Y(f)$.

Q2.22.3 NAT 2022 2M Ans: 0.84 to 0.86 ●○○

Given:
A $(7, 4)$ Hamming code is used over a Binary Symmetric Channel (BSC).

$$\epsilon = P(1|0) = P(0|1) = 0.1$$

The $(7, 4)$ Hamming code can correct **one bit error**. Therefore, decoding is correct if the received codeword contains **at most one error**.

Theory:
For a BSC, the probability of exactly r bit errors in an n-bit codeword follows the binomial distribution

$$P(r \text{ errors}) = {}^nC_r \epsilon^r (1 - \epsilon)^{n-r}.$$

Since the Hamming code corrects all patterns with 0 or 1 error,

$$P_{\text{correct}} = P(0 \text{ errors}) + P(1 \text{ error}).$$

Probability of zero errors

$$P(0 \text{ errors}) = {}^7C_0 (0.1)^0 (0.9)^7$$

$$= (0.9)^7$$

$$= 0.4782969.$$

Probability of exactly one error

$$P(1 \text{ error}) = {}^7C_1 \epsilon^1 (1 - \epsilon)^6$$

$$= {}^7C_1 (0.1)^1 (0.9)^6$$

$$= 7(0.1)(0.9)^6$$

$$= 0.3720087.$$

Probability of correct decoding

$$P_{\text{correct}} = 0.4782969 + 0.3720087$$

$$= 0.8503056.$$

$$P_{\text{correct}} \approx 0.8503.$$

$$\boxed{P_{\text{correct}} = (0.9)^7 + 7(0.1)(0.9)^6 = 0.8503}$$

Key Points

- For a linear block code, the notation (n, k) means:

$$k = \text{number of message (information) bits}$$

$$n = \text{number of bits in the encoded codeword}$$

$$r = n - k = \text{number of parity (redundancy) bits}$$

- Thus, the code rate is

$$R = \frac{k}{n}.$$

- For a $(7, 4)$ Hamming code,

$$n = 7, \qquad k = 4, \qquad r = n - k = 7 - 4 = 3.$$

- Hence, every 4-bit message is encoded into a 7-bit codeword by adding 3 parity bits.
- In general, for a Hamming code,

$$n = 2^r - 1,$$

$$k = 2^r - r - 1,$$

$$r = n - k.$$

- Example:

$$r = 3$$

$$n = 2^3 - 1 = 7,$$

$$k = 2^3 - 3 - 1 = 4.$$

Therefore, the $(7, 4)$ Hamming code contains 4 information bits and 3 parity bits.

Q2.22.4 NAT | 2022 | 2M | Ans: 0.23 | ● ○ ○

Given:

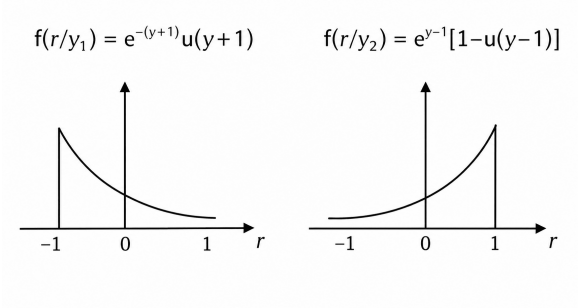


Fig. Solution Q2.22.4

Assume both hypotheses are equally likely.
For a Maximum Likelihood (ML) detector, the decision rule is based on comparing the likelihood functions:

$$\text{Decide } x_A \text{ if } f_{Y|x_A}(y) > f_{Y|x_B}(y)$$

$$\text{Decide } x_B \text{ if } f_{Y|x_B}(y) > f_{Y|x_A}(y)$$

Given the conditional probability density functions:

$$f_{Y|x_A}(y) = e^{-(y+1)}u(y+1)$$

$$f_{Y|x_B}(y) = e^{(y-1)}(1-u(y-1))$$

To find the decision boundaries, we equate the two PDFs where they both exist and overlap (i.e., for $-1 < y < 1$):

$$e^{-(y+1)} = e^{(y-1)}$$

$$-(y+1) = y-1$$

$$-y-1 = y-1 \implies 2y = 0 \implies y = 0$$

Now, we determine the decision regions by analyzing the intervals based on the given functions:

- For $y < -1$: $f_{Y|x_A}(y) = 0$ and $f_{Y|x_B}(y) > 0 \implies$ Decide x_B.
- For $-1 < y < 0$: $f_{Y|x_A}(y) > f_{Y|x_B}(y) \implies$ Decide x_A.
- For $0 < y < 1$: $f_{Y|x_B}(y) > f_{Y|x_A}(y) \implies$ Decide x_B.
- For $y > 1$: $f_{Y|x_B}(y) = 0$ and $f_{Y|x_A}(y) > 0 \implies$ Decide x_A.

Thus, the decision regions for each symbol are:

$$R_A \text{ (Region for } x_A) = (-1, 0) \cup (1, \infty)$$

$$R_B \text{ (Region for } x_B) = (-\infty, -1) \cup (0, 1)$$

Assuming equiprobable symbols for the overall probability of symbol error ($P(x_A) = P(x_B) = 0.5$), the total probability of error P_e is:

$$P_e = P(x_A)P(\text{error}|x_A) + P(x_B)P(\text{error}|x_B)$$

Calculating $P(\text{error}|x_A)$ (which occurs when y falls in R_B):

$$P(\text{error}|x_A) = \int_{R_B} f_{Y|x_A}(y)\,dy = \int_{-\infty}^{-1} 0\,dy + \int_{0}^{1} e^{-(y+1)}\,dy$$

$$P(\text{error}|x_A) = \left[-e^{-(y+1)}\right]_0^1 = -e^{-2} - (-e^{-1}) = e^{-1} - e^{-2}$$

Calculating $P(\text{error}|x_B)$ (which occurs when y falls in R_A):

$$P(\text{error}|x_B) = \int_{R_A} f_{Y|x_B}(y)\,dy = \int_{-1}^{0} e^{(y-1)}\,dy + \int_{1}^{\infty} 0\,dy$$

$$P(\text{error}|x_B) = \left[e^{(y-1)}\right]_{-1}^{0} = e^{-1} - e^{-2}$$

Therefore, the total probability of error is:

$$P_e = 0.5(e^{-1} - e^{-2}) + 0.5(e^{-1} - e^{-2}) = e^{-1} - e^{-2}$$

$$P_e = \frac{1}{e} - \frac{1}{e^2} \approx 0.36788 - 0.13534 = 0.23254$$

Rounding off to two decimal places, we get 0.23.

0.23

Key Points

- The Maximum Likelihood (ML) rule minimizes the probability of error by deciding on the symbol that maximizes the likelihood function $f_{Y|X}(y)$.
- The probability of error for a specific transmitted symbol is the area under its conditional PDF curve in the regions assigned to other symbols.
- $P_e = \sum P(x_i) \int_{y \notin R_i} f_{Y|x_i}(y)\,dy$.

Mistakes to be avoided

A common mistake is assuming standard single-threshold decision regions (e.g., $y > 0 \implies x_A$ and $y < 0 \implies x_B$). Always evaluate the PDFs across the entire range from $-\infty$ to ∞ because the presence of step functions $u(\cdot)$ can create disjoint decision regions, as seen in this problem.

Q2.21.1 NAT | 2021 | 1M | Ans: 9.24 to 9.28 | ● ○ ○

Given

$$f_m = 4 \text{ kHz}$$

$$f_s = 1.25 \text{ (Nyquist Rate)} = 1.25\,(2f_m) = 10 \text{ kHz}$$

$$n = 8 \text{ bits/sample}$$

$$\frac{S}{N} = 26 \text{ dB}$$

Converting SNR from dB to linear scale,

$$\frac{S}{N} = 10^{26/10} = 10^{2.6}$$

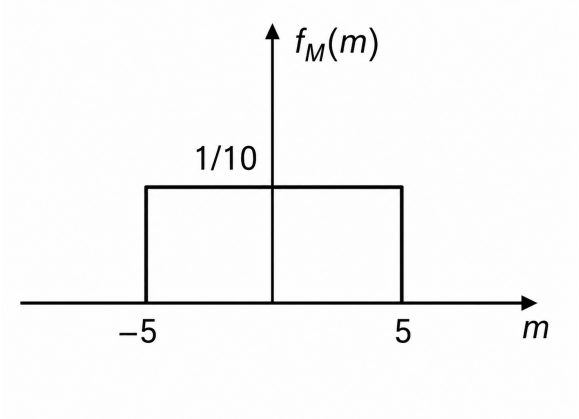


Fig. Solution Q2.21.1

Theory: For arbitrarily small probability of transmission error, the channel capacity must be at least equal to the source bit rate.

$$C \geq R_b$$

Using Shannon's capacity theorem,

$$C = B\log_2\left(1 + \frac{S}{N}\right)$$

and

$$R_b = nf_s$$

Therefore,

$$B\log_2\left(1 + \frac{S}{N}\right) \geq nf_s$$

Substituting the given values,

$$B\log_2\left(1 + 10^{2.6}\right) \geq 8 \times 10^4$$

$$B \geq \frac{8 \times 10^4}{\log_2(1 + 10^{2.6})}$$

$$B \geq 9.25 \times 10^3 \text{ Hz}$$

$$\boxed{B_{\min} = 9.25 \text{ kHz}}$$

Q2.21.2 NAT | 2021 | 1M | Ans: 2.80 to 3.20 | ● ○ ○

Given

$$f_m = 4 \text{ kHz}$$

$$A_m = 4 \text{ V}$$

$$f_s = 32 \text{ kHz}$$

Theory: To avoid slope-overload distortion in a delta modulator, the maximum slope of the message signal must not exceed the tracking slope of the modulator.

$$\left|\frac{dm(t)}{dt}\right|_{\max} \leq \frac{\Delta}{T_s}$$

For a sinusoidal message signal,

$$\left|\frac{dm(t)}{dt}\right|_{\max} = 2\pi f_m A_m$$

Therefore,

$$2\pi f_m A_m \leq \frac{\Delta}{T_s}$$

Since

$$T_s = \frac{1}{f_s},$$

we get

$$\Delta \geq A_m\, 2\pi f_m\, T_s = A_m\, 2\pi f_m \left(\frac{1}{f_s}\right)$$

$$\Delta_{\min} = \frac{A_m\, 2\pi f_m}{f_s}$$

Substituting the given values,

$$\Delta_{\min} = \frac{4 \times 2\pi \times 4 \times 10^3}{32 \times 10^3}$$

$$= \pi$$

$$\boxed{\Delta_{\min} = 3.14 \text{ V}}$$

Q2.21.3 MCQ | 2021 | 2M | Ans: C | ● ○ ○

Given codewords Given codewords:

$$C_1 = 0101100$$

$$C_2 = 0011110$$

$$C_3 = 1000110$$

in given above codewords, first $n - k = 3$ bits are parity bits and last $K = 4$ bits are message bits.
Given is $(7, 4)$ systematic linear Hamming code. For a linear code, sum of two codewords belonging to the code is also a codeword belonging to the code.

$$C_1 \oplus C_2 = 0110010 = C_4$$

$$C_2 \oplus C_3 = 1011000 = C_5$$

$$C_1 \oplus C_3 = 1101010 = C_6$$

$$C_3 \oplus C_4 = 1110100 = C_7$$

Based on codewords C_6 and C_7, options (a) and (d) will be incorrect.
Given codewords in the form of

$$\underset{\text{Parity bits}}{\underbrace{\begin{matrix} P_1 & P_2 & P_3 \\ \downarrow & \downarrow & \downarrow \end{matrix}}} \; \Big| \; \underset{\text{Message bits}}{\underbrace{\begin{matrix} d_1 & d_2 & d_3 & d_4 \\ \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}}}$$

From observation

$$P_1 = d_1 \oplus d_2 \oplus d_4$$

$$P_2 = d_2 \oplus d_3 \oplus d_4$$

$$P_3 = d_1 \oplus d_2 \oplus d_3$$

Checking the given options

P_0	P_1	P_2	d_0	d_1	d_2	d_3	
1	1	0	1	0	1	0	(Option a)
1	1	1	0	1	0	0	(Option b)
0	0	0	1	0	1	1	(Option c)
0	1	1	1	0	0	1	(Option d)

Therefore,

(C) 0001011 is a valid codeword.

Q2.21.4 NAT 2021 2M Ans: 30.00 to 34.00 ● ○ ○

Given:

$$V_{p-p} = 2 \text{ V}$$

$$V_{\text{rms}} = 0.1 \text{ V}$$

$$f_m = 5 \text{ kHz}$$

$$C = 50 \text{ kbps}$$

We need to find the maximum signal-to-quantization-noise ratio,

$$\left(\frac{S}{N_q}\right)_{\max}.$$

Theory: To maximize SQNR, the number of quantization bits n must be as large as possible while satisfying the channel bit-rate constraint.

Step 1: Signal Power
The signal power is

$$S = V_{\text{rms}}^2$$

$$S = (0.1)^2$$

$$S = 0.01.$$

Step 2: Maximum number of quantization bits
For PCM,

$$R_b = nf_s$$

where n is the number of bits/sample and f_s is the sampling frequency.
Using Nyquist sampling,

$$f_s = 2f_m$$

$$f_s = 2(5\text{ kHz})$$

$$f_s = 10\text{ kHz}.$$

Since the channel supports at most 50 kbps,

$$50 \times 10^3 \geq n(10 \times 10^3)$$

$$n \leq 5.$$

Hence,

$$n_{\max} = 5.$$

Step 3: Quantization step size
For a uniform quantizer,

$$\Delta = \frac{V_{p-p}}{2^n}$$

$$\Delta_{\min} = \frac{2}{2^5}$$

$$= \frac{1}{16}.$$

Step 4: Quantization noise power

For uniformly distributed quantization error,

$$N_q = \frac{\Delta^2}{12}.$$

Therefore,

$$(N_q)_{\min} = \frac{\left(\frac{1}{16}\right)^2}{12}$$

$$= \frac{1}{3072}$$

$$= 3.255 \times 10^{-4}.$$

Step 5: Maximum SQNR

$$\left(\frac{S}{N_q}\right)_{\max} = \frac{0.01}{3.255 \times 10^{-4}}$$

$$= 30.72.$$

$$\boxed{\left(\frac{S}{N_q}\right)_{\max} = 30.72}$$

Key Points

- Nyquist sampling requires
$$f_s \geq 2f_m.$$
- PCM bit rate is
$$R_b = nf_s.$$
- Increasing n decreases the quantization step size Δ, thereby reducing quantization noise.
- For a uniform quantizer,
$$N_q = \frac{\Delta^2}{12}.$$

Q2.21.5 NAT | 2021 | 2M | Ans: 0.03 to 0.05 | ● ○ ○

Given:

$$H_1 : \ X = +2 \text{ V}$$

$$H_0 : \ X = -2 \text{ V}$$

$$\sigma_n^2 = 0.4 \text{ V}^2$$

$$E[N] = 0$$

$$P(1) = 0.4, \qquad P(0) = 0.6$$

Theory: For binary detection in AWGN with unequal prior probabilities, the optimum threshold is obtained using the MAP (Maximum A Posteriori) criterion.

For MAP detection,

$$V_{\text{Th}}\frac{(a_1 - a_2)}{\sigma^2} - \frac{a_1^2 - a_2^2}{2\sigma^2} = \ln\left(\frac{P(0)}{P(1)}\right)$$

where

$$a_1 = E[2 + N] = E[2] + E[N] = 2$$

and

$$a_2 = E[-2 + N] = E[-2] + E[N] = -2.$$

Also,

$$\sigma^2 = \text{Var}[Y] = \text{Var}[X + N]$$

Since X is deterministic under each hypothesis,

$$\text{Var}[X] = 0$$

therefore,

$$\sigma^2 = 0 + \text{Var}[N] = 0.4.$$

Substituting the values,

$$V_{\text{Th}}\left(\frac{2 - (-2)}{0.4}\right) - \frac{2^2 - (-2)^2}{2(0.4)} = \ln\left(\frac{0.6}{0.4}\right)$$

$$V_{\text{Th}}\left(\frac{4}{0.4}\right) - \frac{4 - 4}{0.8} = \ln(1.5)$$

$$10V_{\text{Th}} = \ln(1.5)$$

$$V_{\text{Th}} = \frac{0.4}{4}\ln(1.5)$$

$$V_{\text{Th}} = 0.0405 \text{ V}.$$

Key Points

- MAP detection minimizes the probability of decision error when prior probabilities are unequal.
- If $P(0) = P(1)$, the MAP threshold reduces to the midpoint between the two signal levels.
- The threshold shifts toward the less probable symbol so that the overall probability of error is minimized.
- For binary signals s_0 and s_1 in AWGN, the optimum MAP threshold is

$$V_{\text{Th}} = \frac{s_0 + s_1}{2} + \frac{\sigma^2}{s_1 - s_0} \ln\left(\frac{P(0)}{P(1)}\right).$$

$$\boxed{V_{\text{Th}} \approx 0.0405 \text{ V}}$$

Q2.20.1 MCQ 2020 1M Ans: D ● ○ ○

Theory: Assuming independent bit errors, the probability that all N bits are received correctly is the product of the individual bit-correct probabilities.

Given that the probability of error in decoding a single bit is

$$P_e = \alpha,$$

the probability of correct decoding of a single bit is

$$P_c = 1 - \alpha.$$

For an N-bit block, the probability that all N bits are received correctly is

$$P(\text{no error in block}) = (1-\alpha)(1-\alpha)\cdots(1-\alpha)$$

$$= (1-\alpha)^N.$$

Therefore, the probability that the received block contains at least one error is

$$P(\text{erroneous block}) = 1 - P(\text{no error in block})$$

$$= 1 - (1-\alpha)^N.$$

Hence,

$$\boxed{(D) 1 - (1-\alpha)^N}$$

Q2.20.2 NAT 2020 2M Ans: 3 ● ○ ○

Given

$$S_1 = -3, \quad S_2 = -1, \quad S_3 = +1, \quad S_4 = +2$$

$$Y = S + W$$

$$W \sim \mathcal{N}(0, 1)$$

$$f_W(w) = \frac{1}{\sqrt{2\pi}} e^{-w^2/2}$$

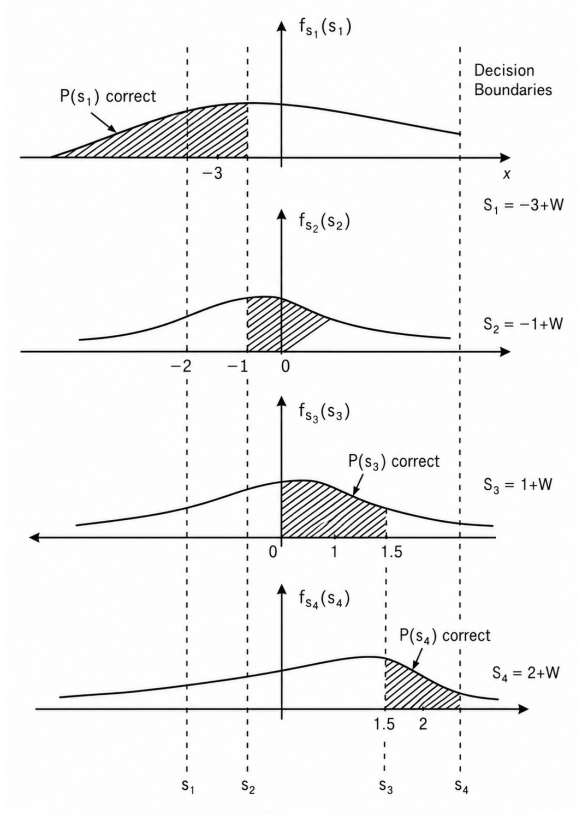


Fig. Solution Q2.20.2

Theory: Since all symbols are equally likely, the optimum decision boundaries are located at the midpoints between adjacent symbols.

$$\text{Boundary between } S_1 \text{ and } S_2 = \frac{-3 + (-1)}{2} = -2$$

$$\text{Boundary between } S_2 \text{ and } S_3 = \frac{-1 + 1}{2} = 0$$

$$\text{Boundary between } S_3 \text{ and } S_4 = \frac{1 + 2}{2} = 1.5$$

Hence,

$$S_1: \quad (-\infty, -2)$$

$$S_2: \quad (-2, 0)$$

$$S_3: \quad (0, 1.5)$$

$$S_4: \quad (1.5, \infty)$$

Probability of correct detection regions

For S_1, any received value below -2 is decoded correctly.

$$P(\text{correct} \mid S_1) = P(Y < -2 \mid S_1)$$

For S_2, the correct region is between -2 and 0.

$$P(\text{correct} \mid S_2) = P(-2 < Y < 0 \mid S_2)$$

For S_3, the correct region is between 0 and 1.5.

$$P(\text{correct} \mid S_3) = P(0 < Y < 1.5 \mid S_3)$$

For S_4, the correct region is greater than 1.5.

$$P(\text{correct} \mid S_4) = P(Y > 1.5 \mid S_4)$$

Observe that the decision interval for S_3 is the narrowest among all symbols.

$$\text{Width of region for } S_3 = 1.5 - 0 = 1.5$$

Therefore,

$$P(\text{correct} \mid S_3)$$

is the smallest, which implies that

$$P(\text{error} \mid S_3)$$

is the largest.

S_3 has the highest conditional symbol error probability

Q2.19.1 MCQ | 2019 | 1M | Ans: B | ● ○ ○

Theory: For a linear block code, the modulo-2 sum (XOR) of two valid messages maps to the modulo-2 sum of their corresponding codewords.

Given,

$$0001 \longrightarrow 0000111$$

and

$$0011 \longrightarrow 1100110.$$

Since the encoder is linear,

$$0010 = 0011 \oplus 0001.$$

Therefore, the corresponding codeword is

$$c(0010) = c(0011) \oplus c(0001).$$

Substituting the given codewords,

$$c(0010) = 1100110 \oplus 0000111.$$

Performing bitwise XOR,

$$\begin{array}{cccccccc} 1 & 1 & 0 & 0 & 1 & 1 & 0 \\ \oplus & & & & & & \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}$$

Hence,

$$c(0010) = 1100001.$$

Therefore,

$$\boxed{(B)1100001}$$

Q2.19.2 MCQ 2019 2M Ans: D ● ○ ○

Theory: For optimum binary detection in AWGN, the bit-error probability depends only on the Euclidean distance between the two signal points. Therefore, to obtain the same P_b, the minimum distance $d_{\min}$ must remain unchanged.

Case 1:
Given,

$$p(t) = \phi_1(t), \qquad q(t) = -\phi_1(t).$$

The corresponding signal points are located at

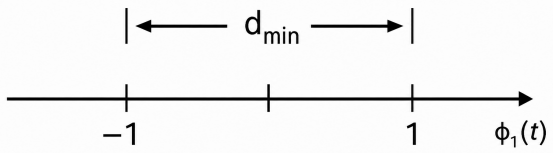


Fig. Solution Q2.19.2

$$(+1, 0) \quad \text{and} \quad (-1, 0).$$

Hence, the Euclidean distance between them is

$$d_{\min} = \sqrt{(1-(-1))^2} = 2.$$

Case 2:
Now,

$$p(t) = \phi_1(t), \qquad q(t) = \sqrt{E}\,\phi_2(t).$$

The signal points are

$$(1, 0) \quad \text{and} \quad (0, \sqrt{E}).$$

Therefore,

$$d_{\min} = \sqrt{(1-0)^2 + \left(0-\sqrt{E}\right)^2} = \sqrt{1+E}.$$

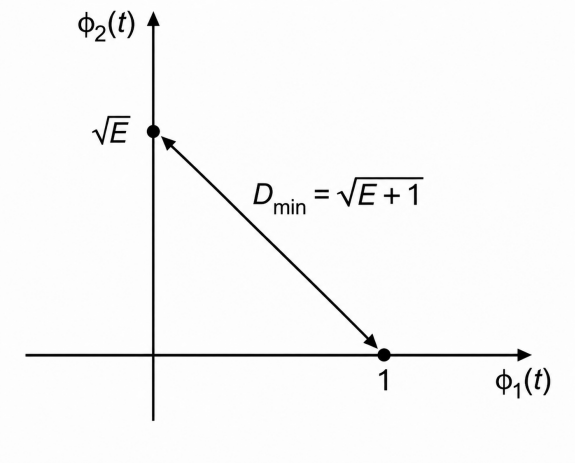


Fig. Solution Q2.19.2

For the same bit-error probability,

d_{min} must be the same in both cases.

Hence,

$$\sqrt{1+E} = 2.$$

Squaring both sides,

$$1 + E = 4.$$

Therefore,

$$\boxed{(D)3}$$

Q2.19.3 NAT 2019 2M Ans: 11.80 to 11.82 ● ○ ○

Theory: The coded bit rate is equal to the sampling rate multiplied by the number of bits used to represent each quantized sample.

The message signal occupies the frequency range

$$5 \text{ kHz} \le f \le 15 \text{ kHz}.$$

The carrier frequency is

$$f_c = 600 \text{ kHz}.$$

Hence, the AM signal occupies the frequency band

$$f_c - 15 \text{ kHz} \text{ to } f_c + 15 \text{ kHz}$$

$$= 585 \text{ kHz} \text{ to } 615 \text{ kHz}.$$

Therefore, the highest frequency component present is

$$f_{max} = 615 \text{ kHz}.$$

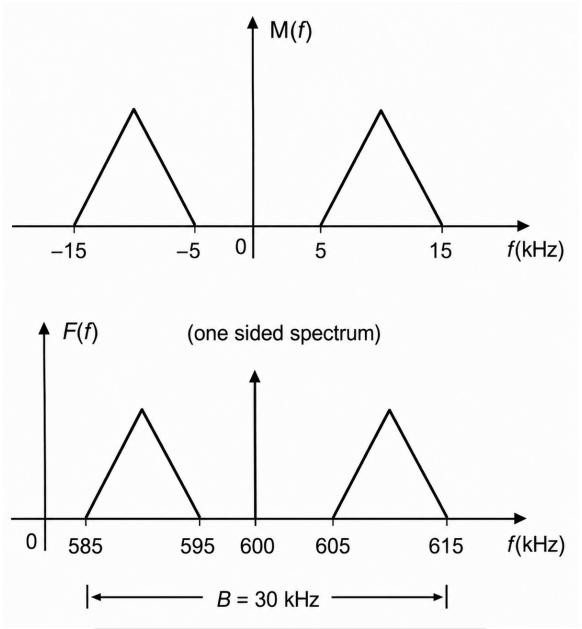


Fig. Solution Q2.19.3

The Nyquist sampling rate is

$$f_N = 2f_{\max}$$

$$= 2(615 \text{ kHz})$$

$$= 1230 \text{ kHz} = 1.23 \text{ MHz}.$$

Since sampling is performed at 1.2 times the Nyquist rate,

$$f_s = 1.2 f_N$$

$$= 1.2 \times 1.23$$

$$= 1.476 \text{ MHz}.$$

A 256-level quantizer requires

$$K = \log_2(256) = 8$$

bits per sample.
Hence, the coded bit rate is

$$R_b = K f_s$$

$$= 8 \times 1.476$$

Therefore,

$$\boxed{R_b = 11.81 \text{ Mbps}}$$

Q2.19.4 NAT | 2019 | 2M | Ans: 0.1 | ● ○ ○

Theory: To minimize the probability of error, the receiver should use the MAP (Maximum A Posteriori) decision rule.

Given,

$$P(X = +1) = 0.8, \qquad P(X = -1) = 0.2.$$

The noise N is uniformly distributed over

$$[-2, 2].$$

Hence,

$$f_N(n) = \frac{1}{4}, \qquad -2 \le n \le 2.$$

Since

$$Y = X + N,$$

the conditional densities are

$$f_Y(y|X = +1) = \frac{1}{4}, \qquad -1 \le y \le 3,$$

and

$$f_Y(y|X = -1) = \frac{1}{4}, \qquad -3 \le y \le 1.$$

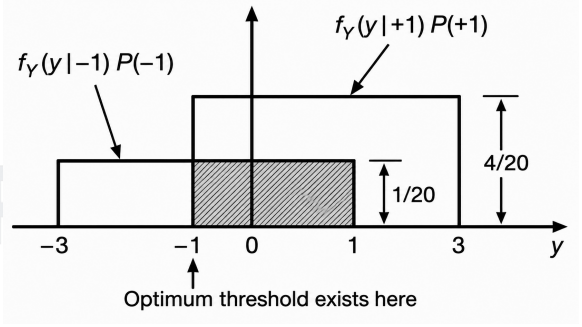


Fig. Solution Q2.19.4

The overlap region of the two densities is

$$-1 \le y \le 1.$$

In the overlap region,

$$f_Y(y| + 1)P(+1) = \frac{1}{4}(0.8) = 0.2,$$

$$f_Y(y| - 1)P(-1) = \frac{1}{4}(0.2) = 0.05.$$

Since

$$0.2 > 0.05,$$

the MAP rule always favors

$$X = +1$$

throughout the overlap region.
Therefore, the optimum threshold is placed at the left edge of the overlap region:

$$\boxed{\theta = -1}.$$

Now, an error occurs only when

$$X = -1$$

is transmitted but the receiver decides

$$X = +1.$$

That is,

$$P_e = P(X = -1)\, P(Y > -1 \mid X = -1).$$

For $X = -1$,

$$Y \sim \text{Uniform}[-3, 1].$$

Hence,

$$P(Y > -1 \mid X = -1) = \frac{1 - (-1)}{1 - (-3)} = \frac{2}{4} = \frac{1}{2}.$$

Therefore,

$$P_e = 0.2 \times \frac{1}{2} = 0.1.$$

Thus, the minimum probability of error is

$$\boxed{P_{e,\min} = 0.1}$$

Q2.18.1 NAT 2018 1M Ans: 16 ●○○

Theory: Plotkin bound gives an upper bound on the minimum distance of a binary code in terms of the code length and the number of codewords.

For a binary (n, k) code, the Plotkin bound is

$$2d_{\min} - 2 - \log_2 d_{\min} \le n - k.$$

Given,

$$d_{\min} = 2, \qquad n = 5.$$

Substituting,

$$2(2) - 2 - \log_2(2) \le 5 - k.$$

$$4 - 2 - 1 \le 5 - k.$$

$$1 \le 5 - k.$$

$$k \le 4.$$

Since the number of codewords in a binary linear code is

$$M = 2^k,$$

the maximum number of codewords is obtained for

$$k_{\max} = 4.$$

Hence,

$$M_{\max} = 2^4.$$

Therefore,

$$\boxed{16}$$

Q2.18.2 NAT 2018 1M Ans: 0.12 to 0.14 ●○○

Theory: The probability of error is the weighted sum of the conditional error probabilities. Since both symbols are equally likely, the total error probability is the average of the two conditional error probabilities.

Let

$$s_0 = -1, \qquad s_1 = +1.$$

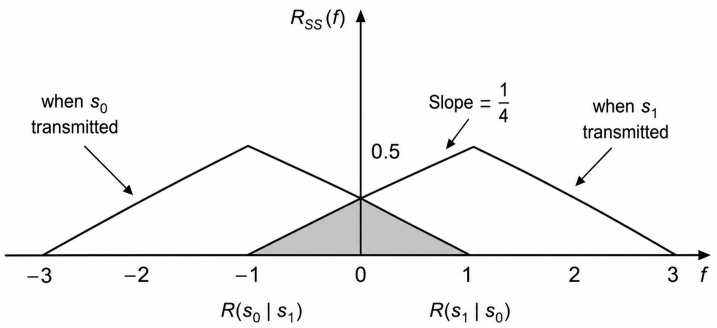


Fig. Solution Q2.18.2

Since the detection threshold is at zero,

$$r < 0 \Rightarrow s_0, \qquad r > 0 \Rightarrow s_1.$$

The probability of error is

$$P_e = P(s_1)P(r_0|s_1) + P(s_0)P(r_1|s_0).$$

From symmetry,

$$P(r_0|s_1) = P(r_1|s_0).$$

The error region is a triangle with

$$\text{base} = 1, \qquad \text{height} = \frac{1}{4}.$$

Hence,

$$P(r_0|s_1) = P(r_1|s_0) = \frac{1}{2} \times 1 \times \frac{1}{4} = \frac{1}{8}.$$

Since

$$P(s_0) = P(s_1) = \frac{1}{2},$$

we get

$$P_e = \frac{1}{2}\left(\frac{1}{8}\right) + \frac{1}{2}\left(\frac{1}{8}\right).$$

$$P_e = \frac{1}{8}.$$

$$\boxed{P_e = 0.125}$$

Q2.18.3 NAT 2018 2M Ans: -0.5 ● ○ ○

Theory: In a MAP detector, we choose the symbol having the larger posterior probability. Since the noise is uniform, the decision depends only on the weighted likelihoods.

Given,

$$P(X = -0.5) = \frac{1}{4}, \qquad P(X = 0.5) = \frac{3}{4}.$$

The noise Z is uniformly distributed over $(-1, 1)$, hence

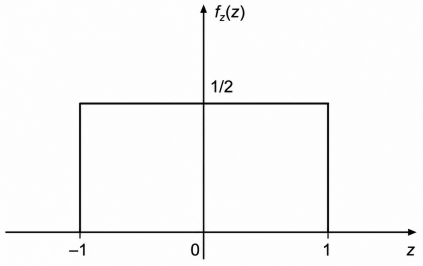


Fig. Solution Q2.18.3

$$f_Z(z) = \frac{1}{2}, \qquad -1 < z < 1.$$

Since

$$Y = X + Z,$$

the conditional PDFs are

$$f_Y(y|X = -0.5) = \frac{1}{2}, \qquad -1.5 < y < 0.5,$$

and

$$f_Y(y|X = 0.5) = \frac{1}{2}, \qquad -0.5 < y < 1.5.$$

Multiplying by the prior probabilities,

$$f_Y(y|X = -0.5)\, P(X = -0.5) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8},$$

$$f_Y(y|X = 0.5)\, P(X = 0.5) = \frac{1}{2} \times \frac{3}{4} = \frac{3}{8}.$$

Thus,

$$\frac{3}{8} > \frac{1}{8}.$$

Therefore, in the overlap region

$$-0.5 < y < 0.5,$$

the detector always prefers $X = 0.5$.

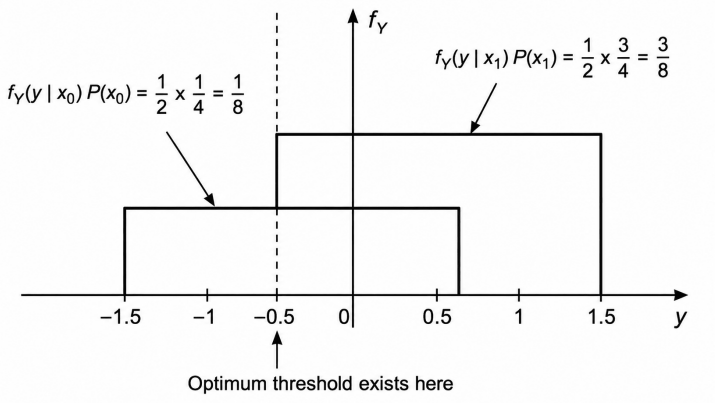


Fig. Solution Q2.18.3

For

$$y < -0.5,$$

only the PDF corresponding to $X = -0.5$ exists, so the decision is $X = -0.5$.Hence the optimum decision boundary occurs at
Therefore,

$$\boxed{\alpha = -0.5}$$

Q2.17.1 NAT 2017 1M Ans: 7 ● ○ ○

Theory: For a sinusoidal input and a uniform quantizer, the SQNR increases by approximately 6 dB for every additional bit.

For a PCM system with a sinusoidal input,

$$\text{SQNR}_{\text{dB}} = 6n + 1.8,$$

where n is the number of bits per sample.

Given,

$$\text{SQNR}_{\text{dB}} = 40 \text{ dB}.$$

Substituting,

$$6n + 1.8 \geq 40.$$

$$6n \geq 38.2.$$

$$n \geq \frac{38.2}{6}.$$

$$n \geq 6.3667.$$

Since the number of bits must be an integer,

$$\boxed{n_{\min} = 7 \text{ bits/sample}}$$

Q2.17.2 MCQ 2017 1M Ans: D ● ○ ○

In DPCM, difference of message signal sample with its prediction is quantized.

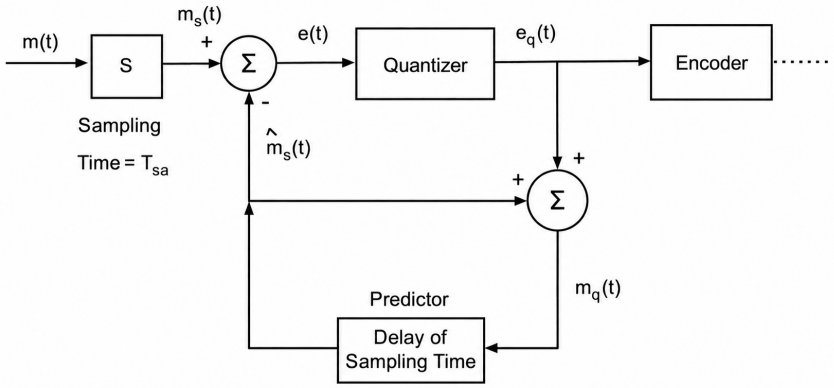


Fig. Solution Q2.17.2

where,

$m_s(t)$ = sampled message signal

$\hat{m}_s(t)$ = predicted value of the message signal

$e(t)$ = prediction error

$e_q(t)$ = quantizer output

$$m_q(t) = \text{output of the adder}$$

In DPCM, the quantizer operates on the prediction error,

$$e(t) = Q[m_s(t) - \hat{m}(t)]\,.$$

Thus, the transmitted information is the quantized difference between the actual sample and its predicted value.
Hence, the correct answer is

(D) The difference of message signal sample with its predictions is quantized

Key Points

- DPCM quantizes the prediction error instead of the actual sample.
- Prediction exploits correlation between successive samples.
- Quantizing a smaller prediction error reduces the required bit rate.

Mistakes to be avoided

- Do not confuse DPCM with PCM, where the message sample itself is quantized.
- Do not assume the sum of the sample and prediction is quantized.
- The prediction error is computed using the predicted sample, not a random signal.

Q2.17.3 MCQ 2017 1M Ans: B ●○○

Theory: For a pulse satisfying the Nyquist first criterion, the shifted spectra must add up to a constant value, ensuring zero intersymbol interference (ISI).

Given,

$$f_s = \frac{1}{T_s} = 2 \text{ k symbols/sec.}$$

If

$$P(f) \overset{\mathcal{F}^{-1}}{\longleftrightarrow} p(t),$$

then the condition for zero ISI is

$$\frac{1}{T_s}\sum_{n=-\infty}^{\infty} P\left(f - \frac{n}{T_s}\right) = p(0).$$

Multiplying both sides by T_s,

$$\sum_{n=-\infty}^{\infty} P\left(f - \frac{n}{T_s}\right) = p(0)T_s.$$

Since

$$p(0) = \int_{-\infty}^{\infty} P(f)\,df,$$

$p(0)$ is equal to the area under $P(f)$.

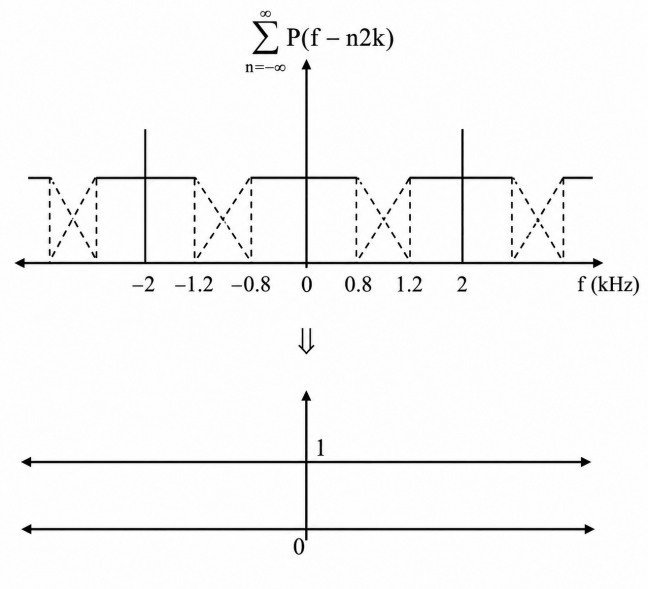


Fig. Solution Q2.17.3

The area under the given spectrum is

$$\text{Area} = 2\left(\frac{1}{2} \times 1 \times 0.4k\right) + 2(0.8k) = 2k.$$

Therefore,

$$p(0) = 2k.$$

Since

$$T_s = \frac{1}{2k},$$

we get

$$p(0)T_s = 2k\left(\frac{1}{2k}\right) = 1.$$

Hence,

$$\sum_{n=-\infty}^{\infty} P(f - n\,2k) = 1.$$

Thus, the Nyquist criterion is satisfied.

The above condition is satisfied only by **Option (b)**.

Remarks

- Option (a) would be correct if the pulse duration were from −1 to +1.
- Option (c) would be correct if the transition regions were from 0.8 to 1.2 and from −0.8 to −1.2.
- Option (d) would be correct if the triangular duration extended from −2 to +2.

(B)

Q2.17.4 MCQ | 2017 | 2M | Ans: B | ● ○ ○

Theory: For coherent binary FSK, the two waveforms are orthogonal (uncorrelated) when the frequency separation satisfies a specific relationship with the bit duration.

Given,

$$u_0(t) = 5\cos(20000\pi t), \qquad 0 \le t \le T,$$

$$u_1(t) = 5\cos(22000\pi t), \qquad 0 \le t \le T.$$

Comparing with

$$A\cos(2\pi f t),$$

the corresponding frequencies are

$$f_0 = \frac{20000\pi}{2\pi} = 10000 \text{ Hz},$$

$$f_1 = \frac{22000\pi}{2\pi} = 11000 \text{ Hz}.$$

Hence,

$$f_1 - f_0 = 11000 - 10000 = 1000 \text{ Hz}.$$

For binary FSK waveforms to be uncorrelated,

$$f_1 - f_0 = n\frac{R_b}{2}, \qquad n = 1, 2, 3, \ldots$$

Substituting the frequency separation,

$$1000 = n\frac{R_b}{2}.$$

Therefore,

$$R_b = \frac{2000}{n} \quad \text{bits/s}.$$

To obtain the smallest bit duration T,

$$T = \frac{1}{R_b},$$

we need the maximum possible bit rate.
This occurs for the minimum value of n,

$$n = 1.$$

Thus,

$$R_b^{(\max)} = 2000 \text{ bits/s}.$$

Hence,

$$T_{\min} = \frac{1}{2000} = 0.0005 \text{ s}.$$

Therefore,

$$\boxed{(B)T_{\min} = 0.5 \text{ ms}}$$

Q2.16.1 NAT 2016 1M Ans: 34.5 to 35.5 ● ○ ○

Theory: For binary PAM with a raised-cosine spectrum, the required transmission bandwidth depends on the bit rate and the roll-off factor.

Given,

$$R_b = 56 \text{ kbps},$$

$$\alpha = 0.25.$$

For a raised-cosine spectrum,

$$B = \frac{R_b}{2}(1 + \alpha).$$

Substituting the given values,

$$B = \frac{56}{2}(1 + 0.25).$$

$$B = 28(1.25).$$

Therefore,

$$\boxed{B = 35 \text{ kHz}}$$

Q2.16.2 NAT 2016 1M Ans: 1.74 to 1.76 ● ○ ○

Theory: For lossless (error-free) source coding, the minimum achievable average codeword length is equal to the source entropy.

The source probabilities are

$$P(a_1) = \frac{1}{2}, \qquad P(a_2) = \frac{1}{4}, \qquad P(a_3) = \frac{1}{8}, \qquad P(a_4) = \frac{1}{8}.$$

Hence,

$$L_{\min} = H.$$

The source entropy is

$$H = -\sum_{i=1}^{4} P(a_i) \log_2 P(a_i).$$

Substituting the given probabilities,

$$H = -\left[\frac{1}{2}\log_2\left(\frac{1}{2}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{8}\log_2\left(\frac{1}{8}\right) + \frac{1}{8}\log_2\left(\frac{1}{8}\right)\right].$$

Using

$$\log_2\left(\frac{1}{2}\right) = -1, \qquad \log_2\left(\frac{1}{4}\right) = -2, \qquad \log_2\left(\frac{1}{8}\right) = -3,$$

we get

$$H = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3).$$

$$H = \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{3}{8}.$$

$$H = 1.75.$$

Therefore,

$$\boxed{L_{\min} = H = 1.75 \text{ bits/symbol}}$$

Q2.16.3 NAT 2016 1M Ans: 15.9 to 16.1 ● ○ ○

Theory: For an M-ary PAM system, each symbol carries $\log_2 M$ bits. The minimum baseband bandwidth is

$$BW_{\min} = \frac{R_b}{2\log_2 M}.$$

Given,

$$f_s = 8 \text{ kHz},$$

$$n = 8 \text{ bits/sample},$$

$$M = 4.$$

The PCM bit rate is

$$R_b = n f_s$$

$$R_b = 8 \times 8 \times 10^3$$

$$R_b = 64 \times 10^3 \text{ bps} = 64 \text{ kbps}.$$

For 4-PAM,

$$\log_2 M = \log_2 4 = 2.$$

Hence,

$$BW_{\min} = \frac{R_b}{2\log_2 M} = \frac{64}{2 \times 2} = 16 \text{ kHz}.$$

Therefore,

$$\boxed{BW_{\min} = 16 \text{ kHz}}$$

Q2.16.4 MCQ 2016 1M Ans: C ●○○

Theory: The output of a matched filter is the convolution of the input pulse and the matched filter impulse response. For a rectangular pulse, this convolution produces a triangular waveform.

Given pulse,

$$P(t) = \begin{cases} \dfrac{1}{\sqrt{T_s}}, & 0 \le t \le T_s, \\ 0, & \text{otherwise.} \end{cases}$$

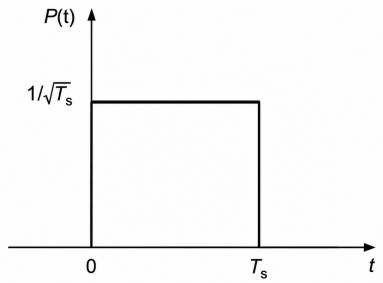


Fig. Solution Q2.16.4

The impulse response of the matched filter is

$$h(t) = P(T_s - t).$$

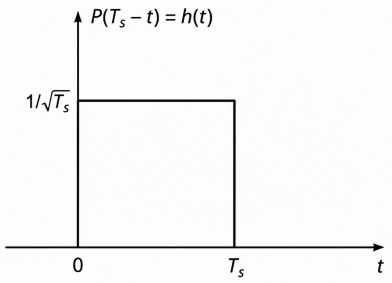


Fig. Solution Q2.16.4

Since $P(t)$ is a rectangular pulse, $h(t)$ is also a rectangular pulse of the same width T_s.
Hence, the matched filter output is

$$y(t) = P(t) * h(t).$$

Convolution of two rectangular pulses of duration T_s results in a triangular waveform:

$$y(t) = \begin{cases} \dfrac{t}{T_s}, & 0 \le t \le T_s, \\ \dfrac{2T_s - t}{T_s}, & T_s \le t \le 2T_s, \\ 0, & \text{otherwise.} \end{cases}$$

The peak occurs at

$$t = T_s,$$

with value

$$y(T_s) = 1.$$

Thus, the output is a symmetric triangular waveform extending from

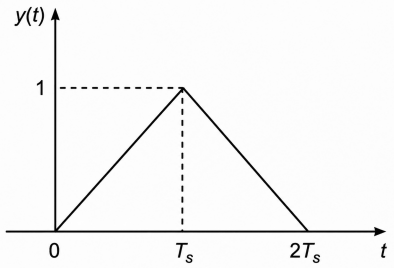


Fig. Solution Q2.16.4

$$t = 0 \quad \text{to} \quad t = 2T_s,$$

with its maximum at $t = T_s$.

(C)

Q2.16.5 NAT 2016 2M Ans: 30.0 to 33.0 ● ○ ○

Theory: For arbitrarily small probability of error, the information rate must not exceed the Shannon capacity of the AWGN channel.

Given,

$$B = 4 \text{ kHz},$$

$$\frac{\eta}{2} = 2.5 \times 10^{-5} \text{ W/Hz}$$

Hence,

$$N_0 = 2\left(2.5 \times 10^{-5}\right) = 5 \times 10^{-5} \text{ W/Hz}.$$

Also,

$$R_b = 52 \text{ kbps}.$$

From Shannon's capacity formula,

$$C = B \log_2 \left(1 + \frac{S}{N}\right).$$

Since

$$S = E_b R_b,$$

and

$$N = N_0 B,$$

we get

$$C = B \log_2 \left(1 + \frac{E_b R_b}{N_0 B}\right).$$

For reliable transmission,

$$C = R_b.$$

Therefore,

$$R_b = B \log_2 \left(1 + \frac{E_b R_b}{N_0 B}\right).$$

Dividing by B,

$$\frac{R_b}{B} = \log_2 \left(1 + \frac{E_b R_b}{N_0 B}\right).$$

Exponentiating both sides,

$$2^{R_b/B} = 1 + \frac{E_b R_b}{N_0 B}.$$

Hence,

$$E_b = \left(2^{R_b/B} - 1\right) \frac{N_0 B}{R_b}.$$

Substituting the given values,

$$E_b = \left(2^{52/4} - 1\right) \frac{(5 \times 10^{-5})(4)}{52}.$$

$$E_b = \left(2^{13} - 1\right) \frac{20 \times 10^{-5}}{52}.$$

$$E_b = 8191 \times \frac{20 \times 10^{-5}}{52}.$$

$$E_b = 0.031504 \text{ J/bit}.$$

Converting to mJ/bit,

$$\boxed{E_b \approx 31.5 \text{ mJ/bit}}$$

Q2.16.6 NAT 2016 2M Ans: 0.70 to 0.75 ● ○ ○

Theory: Since the channel is memoryless, bit errors occur independently. Therefore, the number of bit errors in n transmitted bits follows a binomial distribution.

Given,

$$p = 10^{-5}$$

(bit error probability)

$$n = 10^5$$

(number of transmitted bits)
If exactly r bits are in error, then

$$P(r) = nC_r\, p^r (1-p)^{n-r}.$$

We need the probability that *not more than one bit* is in error, i.e.,

$$P(r \le 1) = P(0) + P(1).$$

Hence,

$$P(r \le 1) = {}^nC_0\, p^0 (1-p)^n + {}^nC_1\, p (1-p)^{n-1}.$$

Substituting $n = 10^5$ and $p = 10^{-5}$,

$$P(r \le 1) = (1-10^{-5})^{10^5} + 10^5 (10^{-5})(1-10^{-5})^{10^5-1}.$$

$$P(r \le 1) = (1-10^{-5})^{10^5} + (1-10^{-5})^{99999}.$$

$$\boxed{P(\text{not more than one bit error}) \approx 0.7357}$$

Q2.16.7 NAT 2016 2M Ans: 0.24 to 0.26 ● ○ ○

Theory: For an M-QAM system with a raised-cosine spectrum, the required transmission bandwidth is

$$BW = \frac{R_b}{\log_2 M}(1+\alpha),$$

where α is the roll-off factor.

Given channel spectrum:

$$500 \text{ Hz to } 2000 \text{ Hz}$$

Hence, the available bandwidth is

$$BW = 2000 - 500 = 1500 \text{ Hz}.$$

Also,

$$R_b = 4800 \text{ bits/s},$$

$$M = 16.$$

Since

$$\log_2 16 = 4,$$

the bandwidth equation becomes

$$1500 = \frac{4800}{4}(1+\alpha).$$

$$1500 = 1200(1 + \alpha).$$

Therefore,

$$1 + \alpha = \frac{1500}{1200} = 1.25.$$

$$\alpha = 1.25 - 1 = 0.25.$$

Hence,

$$\boxed{\alpha = 0.25}$$

Q2.16.8 NAT 2016 2M Ans: 0.025 to 0.030 ● ○ ○

Theory: With a 3-repetition code and majority decoding, an error occurs only when at least two of the three transmitted bits are received incorrectly.

Given crossover probability,

$$p = 0.1.$$

For a transmitted bit, the decoder makes an incorrect decision if:

- Exactly two bits are received in error, or
- All three bits are received in error.

Therefore,

$$P_e = {}^3C_2 p^2(1-p) + {}^3C_3 p^3.$$

Substituting the values,

$$P_e = 3p^2(1-p) + p^3.$$

$$P_e = 3p^2 - 3p^3 + p^3.$$

$$P_e = 3p^2 - 2p^3.$$

Putting $p = 0.1$,

$$P_e = 3(0.1)^2 - 2(0.1)^3.$$

$$P_e = 3(0.01) - 2(0.001).$$

$$P_e = 0.03 - 0.002.$$

$$\boxed{P_e = 0.028}$$

Q2.16.9 MCQ 2016 2M Ans: A ● ○ ○

Theory: The output SNR is maximized when the receiver uses a matched filter. A matched filter is a time-reversed and delayed version of the transmitted signal.

When the signal-to-noise ratio is maximum,

$$h(t) = s(T - t).$$

Since time shifting and time reversal do not change signal energy,

$$E_h = E_s.$$

For an AWGN channel with two-sided PSD

$$\frac{N_0}{2},$$

the maximum output SNR of a matched filter is

$$(SNR)_{\max} = \frac{2E_s}{N_0}.$$

Therefore,

$$E_s = E_h$$

and

$$(SNR)_{\max} = \frac{2E_s}{N_0}.$$

Hence,

$$\boxed{(A)E_s = E_h, \qquad (SNR)_{\max} = \frac{2E_s}{N_0}}$$

Q2.15.1 NAT 2015 2M Ans: 0.39 to 0.41 ● ○ ○

Theory: Since the received symbol is known, the probability that a particular symbol was transmitted is obtained using Bayes' theorem.

Given,

$$P(X = 1) = 0.8,$$

$$P(X = 0) = 0.2,$$

and the crossover probability of the BSC is

$$p = \frac{1}{7}.$$

Hence,

$$P(Y = 0|X = 1) = \frac{1}{7},$$

$$P(Y = 0|X = 0) = \frac{6}{7}.$$

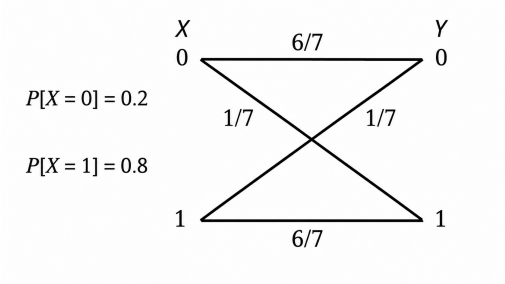


Fig. Solution Q2.15.1

Using Bayes' theorem,

$$P(X = 1|Y = 0) = \frac{P(Y = 0|X = 1)\,P(X = 1)}{P(Y = 0)}.$$

Now,

$$P(Y = 0) = P(Y = 0|X = 1)P(X = 1) + P(Y = 0|X = 0)P(X = 0).$$

Substituting the values,

$$P(X = 1|Y = 0) = \frac{\left(\frac{1}{7}\right)(0.8)}{\left(\frac{1}{7}\right)(0.8) + \left(\frac{6}{7}\right)(0.2)}.$$

$$= \frac{0.8}{0.8 + 1.2}.$$

$$= \frac{0.8}{2}.$$

Therefore,

$$\boxed{P(X = 1|Y = 0) = 0.4}$$

Q2.15.2 MCQ 2015 2M Ans: D ● ● ○

Theory: For minimum probability of error, the MAP detector chooses the symbol having the larger weighted likelihood, i.e., likelihood multiplied by its prior probability.

Given,

$$P(s_0) = \frac{1}{3}, \qquad P(s_1) = \frac{2}{3}.$$

The conditional PDFs are

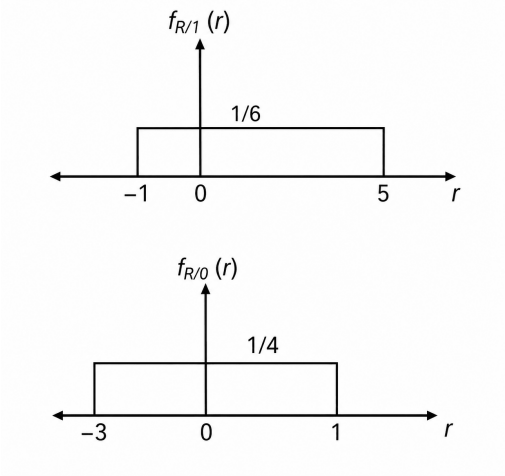


Fig. Solution Q2.15.2

$$f_R(r|s_0) = \frac{1}{4}, \qquad -3 \le r \le 1,$$

$$f_R(r|s_1) = \frac{1}{6}, \qquad -1 \le r \le 5.$$

Including the prior probabilities,

$$f_R(r|s_0)P(s_0) = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12},$$

$$f_R(r|s_1)P(s_1) = \frac{1}{6} \times \frac{2}{3} = \frac{1}{9}.$$

Since

$$\frac{1}{9} > \frac{1}{12},$$

the detector should decide in favor of s_1 throughout the overlap region $-1 \le r \le 1$.
Hence the optimum threshold is

$$r_{\text{th}} = -1.$$

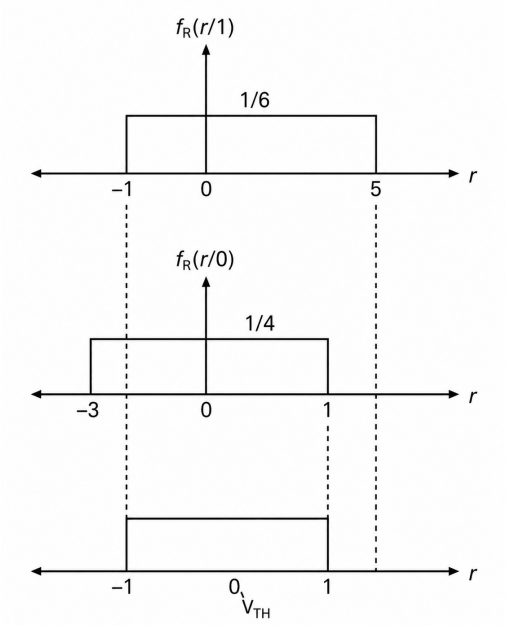


Fig. Solution Q2.15.2

For $r_{th} = -1$,

$$P(r_0|s_1)P(s_1) = 0,$$

because the entire support of $f_R(r|s_1)$ lies to the right of -1.
The only contribution to error comes from deciding s_1 when s_0 was transmitted:

$$P(r_1|s_0)P(s_0) = \frac{1}{12}\left[1 - (-1)\right].$$

$$= \frac{1}{12}(2) = \frac{1}{6}.$$

Therefore,

$$P_{e(\min)} = P(r_0|s_1)P(s_1) + P(r_1|s_0)P(s_0)$$

$$= 0 + \frac{1}{6} = \frac{1}{6}.$$

$$\boxed{(D)\, P_{e(\min)} = \frac{1}{6}}$$

Q2.15.3 NAT 2015 2M Ans: 400 ● ○ ○

Theory: In TDMA, multiple users share the same carrier bandwidth by transmitting in different time slots. Hence one GSM carrier of 200 kHz can support 8 users.

Given,

Bandwidth per GSM carrier = 200 kHz

Number of users supported per carrier = 8

$$\text{Number of active users} = 12$$

Since one carrier can accommodate only 8 users, 12 users require

$$\left\lceil \frac{12}{8} \right\rceil = 2$$

GSM carriers.
Therefore, the total bandwidth required is

$$B_{\text{total}} = 2 \times 200 = 400 \text{ kHz}$$

Hence,

$$\boxed{B_{\text{total}} = 400 \text{ kHz}}$$

Q2.15.4 MCQ 2015 2M Ans: A

Theory: For antipodal signaling $\{g(t), -g(t)\}$ over an AWGN channel with a matched filter receiver, the probability of error decreases as the signal energy increases. Hence, we choose the pulse with the maximum energy.

For a signal $g(t)$,

$$E = \int_0^1 g^2(t)\, dt$$

Option (A): Rectangular pulse

$$g(t) = 1, \qquad 0 \le t \le 1$$

$$E_A = \int_0^1 1^2\, dt = 1$$

Option (B): Triangular pulse

$$g(t) = \begin{cases} 2t, & 0 \le t \le \frac{1}{2} \\ 2(1-t), & \frac{1}{2} \le t \le 1 \end{cases}$$

$$E_B = 2\int_0^{1/2} (2t)^2 dt = 8\int_0^{1/2} t^2 dt$$

$$= 8\left[\frac{t^3}{3}\right]_0^{1/2} = \frac{1}{3}$$

Option (C): Decreasing ramp

$$g(t) = 1 - t, \qquad 0 \le t \le 1$$

$$E_C = \int_0^1 (1-t)^2 dt = \frac{1}{3}$$

Option (D): Increasing ramp

$$g(t) = t, \qquad 0 \le t \le 1$$

$$E_D = \int_0^1 t^2 dt = \frac{1}{3}$$

Thus,

$$E_A = 1, \qquad E_B = E_C = E_D = \frac{1}{3}$$

Since

$$E_A > E_B, E_C, E_D,$$

the rectangular pulse provides the maximum SNR and hence the minimum probability of error.

(A)

Q2.15.5 NAT 2015 1M Ans: 32 ●○○

Theory: For a full-scale sinusoidal input applied to a uniform quantizer, the signal-to-quantization noise ratio (SQNR) is approximately

$$\text{SQNR}_{\text{dB}} = 1.8 + 6n$$

where n is the number of bits per sample.

Given,

$$\text{SQNR} = 31.8 \text{ dB}$$

Substituting into the SQNR formula,

$$31.8 = 1.8 + 6n$$

$$6n = 30$$

$$n = 5$$

The number of quantization levels is

$$L = 2^n$$

$$L = 2^5$$

$$L = 32$$

Q2.15.6 MCQ 2015 1M Ans: A ●○○

Theory: To avoid slope overload distortion in a delta modulator, the maximum slope of the input signal must not exceed the maximum tracking slope of the modulator.

For a sinusoidal signal,

$$m(t) = A_m \sin(2\pi f_m t)$$

the maximum slope is

$$\left|\frac{dm(t)}{dt}\right|_{\max} = 2\pi f_m A_m.$$

The maximum tracking slope of the delta modulator is

$$\Delta f_s.$$

Hence, for slope-overload-free operation,

$$2\pi f_m A_m \leq \Delta f_s.$$

At the limiting condition,

$$2\pi f_m A_m = \Delta f_s.$$

Substituting the given values,

$$2\pi(2000)\, A_m = (0.1)(20000).$$

$$4000\pi\, A_m = 2000.$$

$$A_m = \frac{2000}{4000\pi}.$$

$$A_m = \frac{1}{2\pi}\ \text{V}.$$

Hence, the maximum permissible amplitude is

$$\boxed{A_m = \frac{1}{2\pi}\ \text{V}}$$

Q2.15.7 MCQ 2015 1M Ans: D ●○○

Theory: GSM uses Gaussian Minimum Shift Keying (GMSK), a constant-envelope digital modulation scheme that provides good spectral efficiency and allows the use of efficient non-linear power amplifiers. GSM employs

GMSK (Gaussian Minimum Shift Keying)

because it offers

- Constant envelope,
- High power efficiency,
- Reduced out-of-band radiation due to Gaussian filtering.

Hence, the correct answer is

(D) Gaussian Minimum Shift Keying (GMSK)

Key Points

- GSM uses GMSK modulation.
- GMSK is obtained by passing the data through a Gaussian filter before MSK modulation.
- Constant-envelope modulation enables efficient non-linear RF power amplifiers.

Mistakes to be avoided

- Do not confuse GSM with later cellular standards that use QPSK or QAM.
- MSK and GMSK are not identical; GMSK includes Gaussian pulse shaping.
- Walsh-Hadamard codes are spreading codes, not modulation schemes.

Q2.14.1 NAT 2014 2M Ans: 0.34 to 0.36 ● ● ○

Theory: For a 1-bit quantizer to produce equiprobable outputs, the threshold must divide the input distribution into two regions having equal probability.

Let the threshold be V_{TH}.
For equal-probability outputs,

$$P(X < V_{TH}) = P(X \geq V_{TH}) = \frac{1}{2}.$$

Given,

$$f_X(x) = \begin{cases} 2e^{-2x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Therefore,

$$\int_0^{V_{TH}} 2e^{-2x}\,dx = \int_{V_{TH}}^{\infty} 2e^{-2x}\,dx.$$

Since each side equals $\frac{1}{2}$, we can write

$$\int_{V_{TH}}^{\infty} 2e^{-2x}\,dx = \frac{1}{2}.$$

Evaluating the integral,

$$\left[-e^{-2x}\right]_{V_{TH}}^{\infty} = \frac{1}{2}.$$

$$e^{-2V_{TH}} = \frac{1}{2}.$$

Taking natural logarithm on both sides,

$$-2V_{TH} = \ln\left(\frac{1}{2}\right).$$

$$\boxed{V_{TH} = \frac{\ln 2}{2} \approx 0.3466}$$

Q2.14.2 MCQ 2014 2M Ans: C ● ● ○

Theory: For coherent orthogonal FSK in an AWGN channel, the bit error probability is

$$P_e = Q\left(\sqrt{\frac{A_c^2 T_b}{2N_0}}\right),$$

where A_c is the signal amplitude, T_b is the bit duration, and N_0 is the noise PSD parameter.

Given,

$$A_c = 4\,\text{mV} = 4 \times 10^{-3}\,\text{V},$$

$$R_b = 500\,\text{kbps} = 500 \times 10^3\,\text{bits/s},$$

$$T_b = \frac{1}{R_b} = \frac{1}{500 \times 10^3}\,\text{s},$$

$$\frac{N_0}{2} = 0.5 \times 10^{-12}\,\text{W/Hz}.$$

Therefore,

$$N_0 = 10^{-12}\,\text{W/Hz}.$$

Substituting into the FSK error probability expression,

$$P_e = Q\left(\sqrt{\frac{(4 \times 10^{-3})^2\left(\frac{1}{500 \times 10^3}\right)}{2 \times 10^{-12}}}\right).$$

Since

$$(4 \times 10^{-3})^2 = 16 \times 10^{-6},$$

$$\frac{16 \times 10^{-6}}{(500 \times 10^3)(2 \times 10^{-12})} = 16,$$

we get

$$P_e = Q\left(\sqrt{16}\right) = Q(4).$$

Hence,

$$\boxed{(C)\,P_e = Q(4)}$$

Q2.14.3 NAT 2014 2M Ans: 199 to 201 ● ○ ○

Theory: In PCM, the minimum data rate is obtained using the minimum number of quantization bits that can accommodate the signal range.

Given,

$$m(t) = \sin(100\pi t) + \cos(100\pi t).$$

Using the identity,

$$\sin\theta + \cos\theta = \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right),$$

the maximum amplitude of $m(t)$ is

$$m_{\max} = \sqrt{2}.$$

Hence,

$$V_{max} = \sqrt{2}, \qquad V_{min} = -\sqrt{2}.$$

The quantizer step size is

$$\Delta = 0.75 \text{ V}.$$

For a uniform quantizer,

$$\Delta = \frac{V_{max} - V_{min}}{2^n}.$$

Substituting the values,

$$0.75 = \frac{\sqrt{2} - (-\sqrt{2})}{2^n} = \frac{2\sqrt{2}}{2^n}.$$

Therefore,

$$2^n = \frac{2\sqrt{2}}{0.75} = 3.77.$$

Hence,

$$n = 2 \text{ bits/sample}.$$

The highest frequency component is

$$f_{max} = \frac{100\pi}{2\pi} = 50 \text{ Hz}.$$

Since sampling is at the Nyquist rate,

$$f_s = 2f_{max} = 100 \text{ samples/s}.$$

Therefore, the PCM data rate is

$$R_b = nf_s = 2 \times 100 = 200 \text{ bits/s}.$$

$$\boxed{R_b = 200 \text{ bits/s}}$$

Q2.14.4 MCQ 2014 2M Ans: C ●○○

Theory: For binary detection in AWGN with unequal prior probabilities, the optimum threshold is given by the MAP criterion. The threshold shifts toward the less probable symbol to minimize the overall probability of error.

Given,

$$x_1 = +1, \qquad x_2 = -1$$

and

$$P(x_1) = 0.75, \qquad P(x_2) = 0.25.$$

The optimum threshold is

$$\tau_{opt} = \frac{\sigma^2}{x_1 - x_2}\left[\ln\left(\frac{P(x_2)}{P(x_1)}\right) + \frac{x_1^2 - x_2^2}{2\sigma^2}\right].$$

Substituting

$$x_1 = 1, \qquad x_2 = -1,$$

we get

$$x_1 - x_2 = 1 - (-1) = 2,$$

and

$$x_1^2 - x_2^2 = 1 - 1 = 0.$$

Therefore,

$$\tau_{\text{opt}} = \frac{\sigma^2}{2} \ln\left(\frac{0.25}{0.75}\right).$$

Since

$$\ln\left(\frac{0.25}{0.75}\right) = \ln\left(\frac{1}{3}\right) < 0,$$

it follows that

$$\tau_{\text{opt}} < 0.$$

Hence, the optimum threshold is always strictly negative.

$(C)\ \tau_{\text{opt}}$ is strictly negative

Q2.14.5 NAT | 2014 | 2M | Ans: 15.9 to 16.1 | ● ○ ○

Theory: For an M-PSK system with a raised-cosine spectrum, the required bandwidth depends on the symbol rate and roll-off factor. The minimum value of M is obtained by satisfying the bandwidth constraint.

Given,

$$R_b = 200 \text{ kbps},$$

$$B = 100 \text{ kHz},$$

and 100% excess bandwidth,

$$\alpha = 1.$$

For M-PSK,

$$R_s = \frac{R_b}{\log_2 M}.$$

The raised-cosine bandwidth is

$$B = R_s(1 + \alpha).$$

Substituting R_s,

$$B = \frac{R_b}{\log_2 M}(1 + \alpha).$$

Substituting the given values,

$$100 = \frac{200}{\log_2 M}(1+1).$$

$$100 = \frac{400}{\log_2 M}.$$

Hence,

$$\log_2 M = 4.$$

Therefore,

$$M = 2^4 = 16.$$

$$\boxed{M = 16}$$

Q2.14.6 MCQ 2014 2M Ans: C

Theory: For antipodal signaling in Gaussian noise, the BER is determined by the effective distance of the received mean from the decision threshold. A signal-dependent noise mean shifts this distance and hence changes the BER.

Given,

$$Y = X + Z,$$

where

$$X \in \{-a, +a\},$$

and Z is Gaussian with

$$\mu_Z = \beta X, \qquad \sigma_Z^2 = 1.$$

For $\beta = 0$,

$$Y = X + Z,$$

and hence

$$\text{BER} = Q(a) = 10^{-8}.$$

Using the given approximation

$$Q(v) \approx e^{-v^2/2}, \qquad v > 1,$$

we obtain

$$10^{-8} = e^{-a^2/2}.$$

Taking natural logarithm,

$$-\frac{a^2}{2} = \ln(10^{-8}) = -8 \ln 10.$$

Thus,

$$a^2 = 16 \ln 10 \approx 36.84,$$

$$a \approx 6.$$

For

$$\beta = -0.3,$$

the mean of the noise becomes

$$\mu_Z = \beta a = (-0.3)(6) = -1.8.$$

Therefore, the effective received mean is

$$\mu_Y = a + \mu_Z = 6 - 1.8 = 4.2.$$

Hence,

$$\text{BER} = Q(4.2).$$

Using

$$Q(v) \approx e^{-v^2/2},$$

$$Q(4.2) \approx e^{-4.2^2/2} = e^{-8.82}.$$

$$Q(4.2) \approx 1.48 \times 10^{-4}.$$

$$\boxed{(C)\text{BER} \approx 10^{-4}}$$

Q2.14.7 NAT 2014 1M Ans: 8

Theory: In a synchronous CDMA system, the maximum number of mutually orthogonal signature sequences equals the length of the spreading code. Given,

$$N = 8 \text{ chips}.$$

For a spreading sequence of length N,

$$\boxed{\text{Maximum number of mutually orthogonal signature sequences} = N.}$$

Therefore, the required answer is

$$\boxed{8}$$

Key Points

- Orthogonal signature sequences are obtained using Walsh-Hadamard codes.
- An $N \times N$ Walsh-Hadamard matrix provides N mutually orthogonal sequences.
- The maximum number of synchronous CDMA users equals the spreading code length.

Mistakes to be avoided

- Do not confuse the number of chips with the processing gain.
- This result is valid for mutually orthogonal synchronous CDMA systems.
- Do not assume the number of users is 2^N; it is N.

Q2.14.8 NAT 2014 1M Ans: 0.24 to 0.26 ●●○

Theory: For a uniform quantizer, the maximum quantization error is equal to half the quantization step size.

Given,

$$V_{\max} = 8\ \text{V}, \qquad V_{\min} = 0\ \text{V}.$$

Hence, the dynamic range is

$$V_{\max} - V_{\min} = 8 - 0 = 8\ \text{V}.$$

The range is divided into

$$L = 16$$

equal quantization intervals.
Therefore, the quantization step size is

$$\Delta = \frac{\text{Dynamic Range}}{L} = \frac{8}{16} = 0.5\ \text{V}.$$

The maximum quantization error is

$$Q_e^{(\max)} = \frac{\Delta}{2}.$$

Substituting $\Delta = 0.5$,

$$Q_e^{(\max)} = \frac{0.5}{2} = 0.25\ \text{V}.$$

$$\boxed{Q_e^{(\max)} = 0.25\ \text{V}}$$

Q2.14.9 NAT 2014 1M Ans: 10 ●○○

Theory: Although $y(t)$ is a bandpass signal, the question asks for recovering the baseband signal $m(t)$; hence only the Nyquist rate of the message signal is required.

Given,

$$y(t) = m(t)\cos(40000\pi t).$$

The carrier frequency is

$$2\pi f_c = 40000\pi$$

or,

$$f_c = 20000\ \text{Hz} = 20\ \text{kHz}.$$

The message signal has frequency components less than

$$5\ \text{kHz}.$$

Hence,

$$f_{m,\max} = 5 \text{ kHz}.$$

To recover the message signal without aliasing, the sampling frequency must satisfy the Nyquist criterion,

$$f_s \geq 2f_{m,\max}.$$

Therefore,

$$f_s = 2 \times 5 = 10 \text{ kHz}.$$

Hence,

$$\boxed{f_s = 10 \text{ kHz}}$$

Mistakes to be avoided

- Do not use the carrier frequency (20 kHz) to calculate the sampling rate.
- Read the question carefully: it asks for the sampling rate required to recover $m(t)$, not to reconstruct the bandpass signal $y(t)$.
- Do not apply the bandpass sampling theorem unnecessarily.
- Use the highest message frequency (5 kHz) and apply the Nyquist criterion directly.

Q2.13.1 MCQ 2013 1M Ans: D ●●○

Theory: A matched filter has frequency response proportional to the complex conjugate of the signal spectrum. Hence, in the frequency domain, the output spectrum is the product of the signal spectrum and the matched filter response.

Given,

$$g(t) = e^{-\pi t^2}.$$

Using the standard Fourier transform pair,

$$e^{-\pi t^2} \overset{\mathcal{F}}{\longleftrightarrow} e^{-\pi f^2}.$$

Therefore,

$$G(f) = e^{-\pi f^2}.$$

Since $h(t)$ is matched to $g(t)$,

$$H(f) = G^*(f).$$

As $G(f)$ is real,

$$H(f) = e^{-\pi f^2}.$$

The output spectrum is

$$Y(f) = G(f)\,H(f).$$

Substituting,

$$Y(f) = e^{-\pi f^2}\, e^{-\pi f^2}.$$

Hence, the Fourier transform of the output is

$$\boxed{(D)\, Y(f) = e^{-2\pi f^2}}.$$

Q2.13.2 MCQ 2013 1M Ans: D ● ● ○

Theory: For ISI-free transmission using Nyquist pulse shaping, the minimum baseband bandwidth is $R_s/2$, where R_s is the symbol rate. In M-QAM, each symbol carries $\log_2 M$ bits.

Given,

$$R_b = R \text{ kbits/s},$$

and

$$M = 32.$$

Since

$$\log_2(32) = 5,$$

the symbol rate is

$$R_s = \frac{R_b}{\log_2 M} = \frac{R}{5}.$$

For zero ISI, the minimum baseband bandwidth is

$$(BW)_{\text{baseband}} = \frac{R_s}{2} = \frac{R}{10}.$$

For QAM, the passband bandwidth is twice the baseband bandwidth:

$$(BW)_{\text{QAM}} = 2(BW)_{\text{baseband}}.$$

Substituting,

$$(BW)_{32\text{-QAM}} = 2\left(\frac{R}{10}\right) = \frac{R}{5}.$$

Therefore,

$$\boxed{(D)\ (BW)_{32\text{-QAM}} = \frac{R}{5} \text{ kHz}}$$

Q2.12.1 MCQ 2012 1M Ans: C ● ● ○

Theory: For a raised-cosine pulse with roll-off factor α, the bandwidth and symbol rate are related by

$$BW = \frac{R_s}{2}(1 + \alpha).$$

For baseband signaling, the symbol rate itself is the signaling rate.

Given,

$$BW = 3500 \text{ Hz},$$

and

$$\alpha = 0.75.$$

Using

$$BW = \frac{R_s}{2}(1+\alpha),$$

we get

$$3500 = \frac{R_s}{2}(1+0.75).$$

$$3500 = \frac{R_s}{2}(1.75).$$

$$R_s = \frac{2\times 3500}{1.75}.$$

$$R_s = 4000 \text{ symbols/s}.$$

Hence, the maximum possible signaling rate is

$$\boxed{(C)4000 \text{ symbols/s}}.$$

Q2.12.2 MCQ 2012 2M Ans: B ●●○

Theory: In coherent BPSK, a phase mismatch θ reduces the effective signal energy by a factor of $\cos^2\theta$, which directly affects the BER.

The probability of error for coherent BPSK is

$$P_e = Q\left(\sqrt{\frac{2E}{N_0}}\right).$$

Given the phase difference is

$$\theta = 45^\circ,$$

the effective received energy becomes

$$E_{\text{eff}} = E\cos^2(45^\circ).$$

Since

$$\cos^2(45^\circ) = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2},$$

we have

$$E_{\text{eff}} = \frac{E}{2}.$$

Substituting this reduced energy into the BPSK BER expression,

$$P_e = Q\left(\sqrt{\frac{2E_{\text{eff}}}{N_0}}\right) = Q\left(\sqrt{\frac{2(E/2)}{N_0}}\right).$$

$$P_e = Q\left(\sqrt{\frac{E}{N_0}}\right).$$

Hence,

$$\boxed{(B)P_e = Q\left(\sqrt{\frac{E}{N_0}}\right)}$$

Q2.12.3 MCQ 2012 2M Ans: D ●○○

Theory: For an optimum (MAP) receiver, the decision is made in favor of the symbol having the larger posterior probability. For the given BSC,

$$P(X=0) = \frac{9}{10}, \qquad P(X=1) = \frac{1}{10}.$$

Since the cross-over probability is

$$p = \frac{1}{8},$$

the transition probabilities are

$$P(y_0|x_0) = P(y_1|x_1) = \frac{7}{8},$$

and

$$P(y_1|x_0) = P(y_0|x_1) = \frac{1}{8}.$$

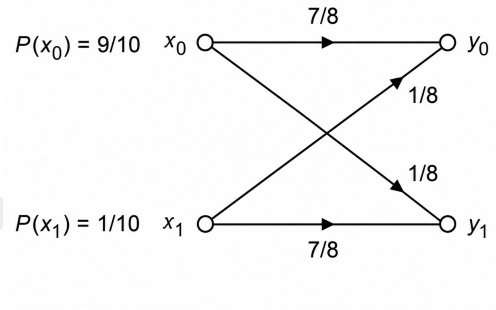


Fig. Solution Q2.12.3

Calculating the MAP metrics for received symbol y_0,

$$P(y_0|x_0)P(x_0) = \frac{7}{8} \times \frac{9}{10} = \frac{63}{80},$$

$$P(y_0|x_1)P(x_1) = \frac{1}{8} \times \frac{1}{10} = \frac{1}{80}.$$

Since

$$\frac{63}{80} > \frac{1}{80},$$

the optimum receiver decides in favor of x_0 whenever y_0 is received.
Similarly, for received symbol y_1,

$$P(y_1|x_0)P(x_0) = \frac{1}{8} \times \frac{9}{10} = \frac{9}{80},$$

$$P(y_1|x_1)P(x_1) = \frac{7}{8} \times \frac{1}{10} = \frac{7}{80}.$$

Since

$$\frac{9}{80} > \frac{7}{80},$$

the optimum receiver again decides in favor of x_0.
Thus, irrespective of the received symbol, the optimum receiver always decides

$$\hat{X} = x_0.$$

Therefore, an error occurs only when x_1 was actually transmitted.
Hence,

$$P_e = P(x_1) = \frac{1}{10}.$$

$$\boxed{(D)P_e = \frac{1}{10}}$$

Mistakes to be avoided

- For an optimum receiver, do not use only the channel transition probabilities. Use the MAP rule:

  $$\text{Decide } x_i \quad \text{if} \quad P(y|x_i)P(x_i) \text{ is maximum.}$$

- A symbol with a very large prior probability can dominate the MAP decision even when the received symbol suggests otherwise.
- Do not assume that the probability of error equals the cross-over probability.

  $$P_e \neq p$$

  in general.
- For equiprobable symbols,

  $$P(x_0) = P(x_1),$$

  MAP and ML decisions become identical. Otherwise they are different.

Q2.11.1 MCQ | 2011 | 2M | Ans: D | ● ● ○

Theory: For an M-PSK constellation, the minimum distance between adjacent symbols is

$$d = 2r \sin\left(\frac{\pi}{M}\right),$$

where $r = \sqrt{E_s}$ is the radius of the constellation.
For 4-PSK,

$$d = 2r_1 \sin\left(\frac{\pi}{4}\right)$$

$$r_1 = \frac{d}{2\sin(\pi/4)}$$

$$r_1 = \frac{d}{\sqrt{2}}$$

$$\boxed{r_1 = 0.707\,d}$$

For 8-PSK,

$$d = 2r_2 \sin\left(\frac{\pi}{8}\right)$$

$$r_2 = \frac{d}{2\sin(\pi/8)}$$

$$\boxed{r_2 = 1.307\,d}$$

Hence,

$$\boxed{(D) r_1 = 0.707\,d, \qquad r_2 = 1.307\,d}$$

Q2.11.2 MCQ 2011 2M Ans: D ● ● ○

Theory: For high SNR, equal error probability implies equal minimum distance between neighboring constellation points. Therefore, the required symbol energy is proportional to the square of the constellation radius.
Using the result from the previous question,

$$r_1 = 0.707d, \qquad r_2 = 1.307d$$

Since symbol energy is

$$E_s = r^2,$$

$$\frac{(E_s)_{8\text{PSK}}}{(E_s)_{4\text{PSK}}} = \left(\frac{r_2}{r_1}\right)^2$$

$$= \left(\frac{1.307}{0.707}\right)^2$$

$$= 3.42$$

Converting this energy ratio into dB,

$$(E_s)_{8\text{PSK(dB)}} - (E_s)_{4\text{PSK(dB)}} = 10\log_{10}(3.42)$$

$$= 5.33\,\text{dB}$$

Therefore,

$$\boxed{(E_s)_{8\text{PSK(dB)}} = (E_s)_{4\text{PSK(dB)}} + 5.33\,\text{dB}}$$

or equivalently,

$$\boxed{(D)\text{Additional energy required} = 5.33\,\text{dB}}$$

Q2.11.3 MCQ 2011 1M Ans: D ●●○

Theory: When all quantization levels are equally probable, the information contained in each sample is equal to the entropy of the source.

Since there are 4 quantization levels and all are equiprobable,

$$H = \log_2 4 = 2 \text{ bits/sample.}$$

Given that the transmission rate is

$$r = 2 \text{ samples/sec,}$$

the information rate is

$$R = rH.$$

Substituting the values,

$$R = 2 \times 2 = 4 \text{ bits/sec.}$$

Therefore,

$$\boxed{(D)\, R = 4 \text{ bits/sec}}$$

Q2.10.1 MCQ 2010 2M Ans: B ●●○

Theory: The output noise PSD is obtained by multiplying the input noise PSD with the squared magnitude response of the filter.

Given,

$$S_N(f) = 10^{-20} \text{ W/Hz}$$

and

$$S_{NO}(f) = S_N(f)\, |H(f)|^2.$$

Since the ideal LPF has unity gain,

$$|H(f)| = 1,$$

hence

$$S_{NO}(f) = 10^{-20} \text{ W/Hz.}$$

The filter bandwidth extends from

$$-1 \text{ MHz} \quad \text{to} \quad +1 \text{ MHz,}$$

therefore

$$B = 2 \times 10^6 \text{ Hz.}$$

The output noise power is

$$P_N = \int_{-\infty}^{\infty} S_{NO}(f)\, df$$

$$= 10^{-20} \times 2 \times 10^{6}$$

$$= 2 \times 10^{-14} \text{ W}.$$

Since mean-square value equals power,

$$\frac{2}{\alpha^2} = 2 \times 10^{-14}.$$

Therefore,

$$\alpha^2 = 10^{14}$$

and

$$\boxed{(B)\alpha = 10^{7}}.$$

Q2.10.2 MCQ 2010 2M Ans: D

Theory: The probability of error is obtained by integrating the noise PDF over the region where the received sample crosses the decision threshold.

Given,

$$Y_k = \begin{cases} N_k, & b_k = 0 \\ a + N_k, & b_k = 1 \end{cases}$$

The threshold is

$$Z = \frac{a}{2} = 10^{-6}$$

Hence,

$$a = 2 \times 10^{-6}.$$

From the previous result,

$$\frac{2}{\alpha^2} = 2 \times 10^{-14}$$

which gives

$$\alpha = 10^{7}.$$

When a '1' is transmitted:

$$Y_k = a + N_k$$

For an error,

$$Y_k < 10^{-6}$$

$$2 \times 10^{-6} + N_k < 10^{-6}$$

$$N_k < -10^{-6}.$$

Therefore,

$$P(0|1) = \int_{-\infty}^{-10^{-6}} P_{N_k}(n)\, dn$$

$$= \int_{-\infty}^{-10^{-6}} 0.5\,\alpha e^{\alpha n}\, dn$$

with

$$\alpha = 10^7.$$

Hence,

$$P(0|1) = 0.5\, e^{-10}.$$

When a '0' is transmitted:

$$Y_k = N_k$$

For an error,

$$Y_k > 10^{-6}$$

Thus,

$$P(1|0) = \int_{10^{-6}}^{\infty} P_{N_k}(n)\, dn = 0.5\, e^{-10}.$$

Since the bits are equiprobable,

$$P(0) = P(1) = \frac{1}{2}.$$

Therefore, the probability of bit error is

$$P_e = P(1)P(0|1) + P(0)P(1|0)$$

$$= \frac{1}{2}\Big[P(0|1) + P(1|0)\Big]$$

$$= \frac{1}{2}\Big[0.5e^{-10} + 0.5e^{-10}\Big]$$

$$= 0.5e^{-10}.$$

$$\boxed{(D)\, P_e = 0.5\, e^{-10}}$$

Q2.10.3 MCQ 2010 1M Ans: C ●●○

Theory: For a matched filter, the impulse response is the time-reversed and delayed version of the signal, i.e.,

$$h(t) = s(T - t).$$

A delay by T shifts the signal to the left, and time reversal produces its mirror image about the y-axis.

Given signal $s(t)$:

$$s(t) = \begin{cases} \frac{t}{T}, & 0 \le t \le T \\ 0, & \text{otherwise} \end{cases}$$

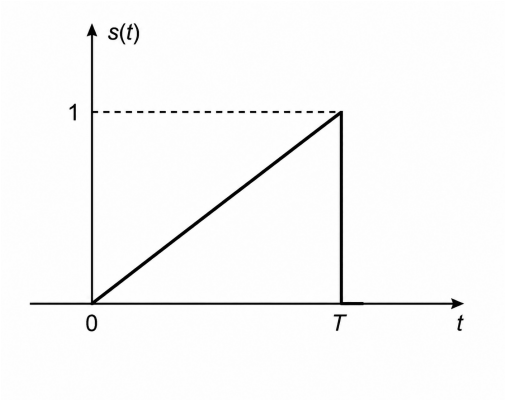


Fig. Solution Q2.10.3

The matched filter impulse response is

$$h(t) = s(T - t).$$

First, consider a left shift of $s(t)$ by T:

$$s(T + t) = \begin{cases} 1 + \frac{t}{T}, & -T \le t \le 0 \\ 0, & \text{otherwise} \end{cases}$$

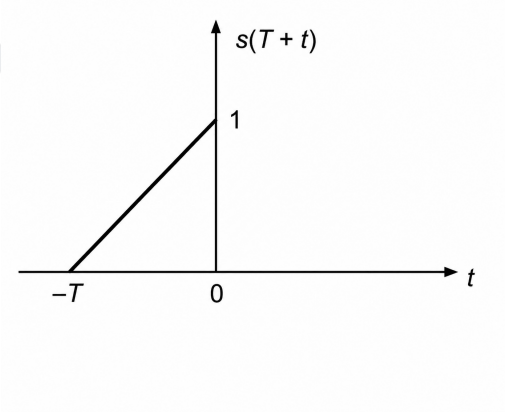


Fig. Solution Q2.10.3

Now, taking the mirror image about the y-axis,

$$h(t) = s(T - t) = \begin{cases} 1 - \frac{t}{T}, & 0 \le t \le T \\ 0, & \text{otherwise} \end{cases}$$

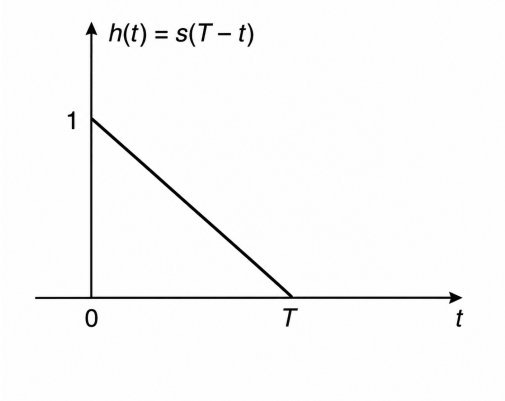


Fig. Solution Q2.10.3

Thus, $h(t)$ is a linearly decreasing ramp starting from 1 at $t = 0$ and reaching 0 at $t = T$.

$$(C)\, h(t) = \begin{cases} 1 - \dfrac{t}{T}, & 0 \le t \le T \\ 0, & \text{otherwise} \end{cases}$$

Q2.09.1 MCQ 2009 2M Ans: C ● ● ○

Theory: For a uniform quantizer, the quantization noise power is

$$N_0 = \frac{\Delta^2}{12}.$$

The SQNR is defined as the ratio of signal power to quantization noise power.

The random signal is uniformly distributed over the interval

$$-5 \le x \le 5.$$

Hence, its probability density function is

$$f_X(x) = \frac{1}{10}, \qquad -5 \le x \le 5.$$

The signal power is

$$S = E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx$$

$$= \int_{-5}^{5} \frac{1}{10} x^2\, dx$$

$$= \frac{1}{10}\left[\frac{x^3}{3}\right]_{-5}^{5}$$

$$= \frac{1}{10}\left(\frac{125}{3} + \frac{125}{3}\right)$$

$$= \frac{25}{3}.$$

Given,

$$\text{SQNR} = 43.5 \text{ dB}.$$

Converting into linear scale,

$$\text{SQNR} = 10^{43.5/10} = 10^{4.35}.$$

Since

$$\text{SQNR} = \frac{S}{N_0},$$

and

$$N_0 = \frac{\Delta^2}{12},$$

we get

$$\frac{\Delta^2}{12} = \frac{S}{\text{SQNR}}.$$

Therefore,

$$\Delta = \sqrt{\frac{12S}{\text{SQNR}}}.$$

Substituting the values,

$$\Delta = \sqrt{\frac{12\left(\frac{25}{3}\right)}{10^{4.35}}}.$$

$$= \sqrt{\frac{100}{10^{4.35}}}.$$

$$\Delta \approx 0.0668 \text{ V}.$$

Hence,

$$\boxed{(C)\Delta \approx 0.0667 \text{ V}}$$

Q2.09.2 MCQ 2009 2M Ans: C ●●○

Theory: Quantization noise power for a uniform quantizer is

$$\sigma_q^2 = \frac{\Delta^2}{12}.$$

Since the positive and negative halves use different step sizes, the overall quantization noise power is the probability-weighted average of the two noise powers.

Given,

$$\Delta_1 = 0.05 \text{ V} \qquad \text{(for positive values)}$$

$$\Delta_2 = 0.1 \text{ V} \qquad \text{(for negative values)}$$

Since the signal amplitude is uniformly distributed between −5 V and 5 V,

$$P(\Delta_1) = P(\Delta_2) = \frac{1}{2}.$$

Hence, the quantization noise power is

$$N_0 = \frac{\Delta_1^2}{12}P(\Delta_1) + \frac{\Delta_2^2}{12}P(\Delta_2)$$

$$= \frac{1}{24}\left(\Delta_1^2 + \Delta_2^2\right)$$

$$= \frac{1}{24}\left(0.05^2 + 0.1^2\right)$$

$$= \frac{1}{24}(0.0025 + 0.01)$$

$$= \frac{0.0125}{24} = \frac{125}{24 \times 10000} = \frac{1}{1920}.$$

The signal power of a uniformly distributed random variable over $[-5, 5]$ is

$$S = E[X^2] = \frac{A^2}{3} = \frac{5^2}{3} = \frac{25}{3}.$$

Therefore,

$$\text{SQNR} = \frac{S}{N_0} = \frac{\frac{25}{3}}{\frac{1}{1920}} = \frac{1920 \times 25}{3} = 16000.$$

In decibels,

$$\text{SQNR}_{dB} = 10\log_{10}(16000)$$

$$= 42.04 \text{ dB}.$$

$$\boxed{(C)\text{SQNR} \approx 42 \text{ dB}}$$

Q2.08.1 MCQ | 2008 | 2M | Ans: D | ● ● ○

Theory: In TDM, each message must be sampled at least at its Nyquist rate. The total pulse rate is the sum of all individual sampling rates.

For the four messages,

$$f_{s1} = 2W$$

$$f_{s2} = 2W$$

$$f_{s3} = 2(2W) = 4W$$

$$f_{s4} = 2(3W) = 6W$$

Hence, the total sampling rate is

$$f_s = f_{s1} + f_{s2} + f_{s3} + f_{s4}$$

$$f_s = 2W + 2W + 4W + 6W$$

$$f_s = 14W.$$

For minimum bandwidth transmission, the minimum Nyquist bandwidth is

$$B_{\min} = \frac{R_b}{2}.$$

Taking one bit per sample,

$$R_b = f_s = 14W.$$

Therefore,

$$B_{\min} = \frac{14W}{2} = 7W.$$

Hence,

$$\boxed{(D)\, B_{\min} = 7W}$$

Q2.08.2 MCQ 2008 2M Ans: B ●●○

At the Nyquist sampling rate,

$$f_s = 2 \times 4\,\text{kHz} = 8\,\text{kHz}.$$

Each sample uses eight bits, so

$$R_b = 8 f_s = 64\,\text{kbit/s}.$$

For ideal distortion-free binary transmission, the minimum Nyquist bandwidth is half the symbol rate. There is one bit per binary symbol, hence

$$B_{\min} = \frac{R_b}{2} = \boxed{32\,\text{kHz}},$$

which is option (B). The first-null bandwidth of an unshaped rectangular pulse is a different bandwidth measure, not the minimum Nyquist bandwidth requested here.

Q2.08.3 MCQ 2008 2M Ans: C ●●○

Theory: For a uniformly distributed signal and a uniform quantizer, the quantization SNR increases by approximately 6 dB for every additional bit.

The signal-to-quantization-noise ratio is given by

$$\left(\frac{S}{N_q}\right)_{\text{dB}} = 6n\ \text{dB},$$

where

$$n = \text{number of bits per sample.}$$

Given,

$$n = 8.$$

Therefore,

$$\left(\frac{S}{N_q}\right)_{\text{dB}} = 6 \times 8 = 48 \text{ dB}.$$

Hence,

$$\boxed{(C)\text{SNR} = 48 \text{ dB}}$$

Q2.08.4 MCQ 2008 2M Ans: B ●●○

Theory: Quantization noise power is proportional to the square of the quantization step size:

$$N_q = \frac{\Delta^2}{12}.$$

Therefore, reducing the quantization noise by a factor of 4 requires reducing the step size by a factor of 2.

Given,

$$N_q = \frac{\Delta^2}{12}.$$

Let the new quantization noise be

$$N_q' = \frac{N_q}{4}.$$

Then,

$$\frac{\Delta'^2}{12} = \frac{1}{4} \cdot \frac{\Delta^2}{12}.$$

Hence,

$$\Delta'^2 = \frac{\Delta^2}{4}$$

$$\Rightarrow \Delta' = \frac{\Delta}{2}.$$

The quantization step size is

$$\Delta = \frac{V_{pp}}{L},$$

where

$$V_{pp} = 10\,\text{V}$$

and L is the number of quantization levels.
Initially, each sample is represented by 8 bits, hence

$$L = 2^8 = 256.$$

Since the step size must be reduced by a factor of 2,

$$\Delta' = \frac{V_{pp}}{L'} = \frac{\Delta}{2} = \frac{V_{pp}}{2L}.$$

Therefore,

$$L' = 2L$$

$$L' = 2 \times 256$$

$$L' = 512.$$

Hence, the required number of quantization levels is

$$\boxed{(B)512}$$

Q2.07.1 MCQ 2007 2M Ans: D ● ● ○

Theory: Slope overload distortion occurs when the input signal changes faster than the delta modulator can track. Increasing the step size allows the staircase approximation to follow steeper signal variations.

The condition to avoid slope overload distortion in Delta Modulation is

$$\frac{\Delta}{T_s} \geq \left|\frac{dm(t)}{dt}\right|_{\max}$$

where

$$\Delta = \text{step size}, \qquad T_s = \text{sampling period}.$$

The maximum tracking slope of the DM system is

$$\frac{\Delta}{T_s}.$$

To reduce slope overload distortion, this tracking slope must be increased.
Since T_s is fixed,

$$\frac{\Delta}{T_s} \uparrow \implies \Delta \uparrow$$

$$\boxed{(D)\text{Increase the step size}}$$

Q2.07.2 MCQ 2007 2M Ans: C ● ● ○

Theory: For a raised cosine pulse, values at singular points are evaluated using L'Hospital's rule whenever direct substitution leads to the indeterminate form $0/0$.

Given,

$$p(t) = \frac{\sin(4\pi Wt)}{4\pi Wt\left(1 - 16W^2t^2\right)}$$

At

$$t = \frac{1}{4W},$$

direct substitution gives

$$p\left(\frac{1}{4W}\right) = \frac{\sin(\pi)}{\pi(1-1)} = \frac{0}{0}.$$

Hence, apply L'Hospital's rule:

$$\lim_{t\to\frac{1}{4W}} p(t) = \lim_{t\to\frac{1}{4W}} \frac{4\pi W\cos(4\pi Wt)}{4\pi W - 64\pi W^3(3t^2)}.$$

Substituting $t = \frac{1}{4W}$,

$$= \frac{4\pi W\cos(\pi)}{4\pi W - 64\pi W^3\left(3\cdot\frac{1}{16W^2}\right)}$$

$$= \frac{4\pi W(-1)}{4\pi W - 12\pi W}$$

$$= \frac{-4\pi W}{-8\pi W}$$

$$= \frac{1}{2}.$$

Therefore,

$$\boxed{(C)\, p\left(\frac{1}{4W}\right) = 0.5}$$

Q2.07.3 MCQ 2007 2M Ans: B ● ● ○

Theory: With frequency reuse factor $\frac{1}{5}$, only one-fifth of the total allocated spectrum is available to a single cell.

Given,

$$\text{Total allocated bandwidth} = 5\ \text{MHz}$$

$$\text{Frequency reuse factor} = \frac{1}{5}$$

Hence, bandwidth available per cell is

$$B_{\text{cell}} = 5\times\frac{1}{5} = 1\ \text{MHz}$$

Also,

$$8\ \text{channels} \longrightarrow 200\ \text{kHz}$$

Therefore, the number of channels in 1 MHz is

$$N = \frac{1\ \text{MHz}}{200\ \text{kHz}}\times 8$$

$$= \frac{1000}{200}\times 8$$

$$= 5\times 8$$

Hence, the maximum number of simultaneous channels in one cell is

$$\boxed{(B)40}$$

Q2.07.4 MCQ 2007 2M Ans: A ● ○ ○

Theory: In a Direct Sequence CDMA system, the processing gain is the ratio of chip rate to data rate. To achieve a minimum required processing gain, the data rate must not exceed the corresponding limit.

Given,

$$R_c = 1.2288 \times 10^6 \text{ chips/s}$$

$$\text{Processing Gain} = \frac{R_c}{R_b}$$

Since the required processing gain is at least 100,

$$\frac{R_c}{R_b} \geq 100$$

$$R_b \leq \frac{R_c}{100}$$

Substituting $R_c = 1.2288 \times 10^6$,

$$R_b \leq \frac{1.2288 \times 10^6}{100}$$

$$R_b \leq 1.2288 \times 10^4$$

$$R_b \leq 12.288 \times 10^3 \text{ bits/s}$$

Therefore, the data rate must be less than or equal to

$$\boxed{(A)12.288 \times 10^3 \text{ bits/s}}$$

Q2.07.5 MCQ 2007 2M Ans: C ● ● ○

Theory: Since bit errors occur independently with probability p, the number of errors in a block of n bits follows a binomial distribution.

Probability of **no error** in n bits:

$$P(0 \text{ errors}) = (1-p)^n$$

Probability of **exactly one error** in n bits:

$$P(1 \text{ error}) = \binom{n}{1} p(1-p)^{n-1} = np(1-p)^{n-1}$$

Therefore, the probability of **at most one bit in error** is

$$P(\leq 1 \text{ error}) = P(0 \text{ errors}) + P(1 \text{ error})$$

$$= (1-p)^n + np(1-p)^{n-1}$$

Hence,

$$\boxed{(C)P(\leq 1 \text{ error}) = np(1-p)^{n-1} + (1-p)^n}$$

Q2.07.6 MCQ 2007 2M Ans: C ● ● ○

Theory: Two signals are orthogonal if their inner product over the observation interval is zero.

$$\int_{-\infty}^{\infty} f(t)g(t)\,dt = 0$$

Since all signals are zero outside $[0, T]$,

$$\int_{-\infty}^{\infty} f_2(t)f_3(t)\,dt = \int_0^T f_2(t)f_3(t)\,dt$$

Evaluating over the three intervals:

$$0 \leq t < \frac{T}{3}$$

$$f_2(t) = 1, \qquad f_3(t) = 1$$

$$\int_0^{T/3} f_2(t)f_3(t)\,dt = \int_0^{T/3} (1)(1)\,dt = \frac{T}{3}$$

$$\frac{T}{3} \leq t < \frac{2T}{3}$$

$$f_2(t) = -1, \qquad f_3(t) = -3$$

$$\int_{T/3}^{2T/3} f_2(t)f_3(t)\,dt = \int_{T/3}^{2T/3} 3\,dt = T$$

$$\frac{2T}{3} \leq t \leq T$$

$$f_2(t) = 2, \qquad f_3(t) = 1$$

$$\int_{2T/3}^{T} f_2(t)f_3(t)\,dt = \int_{2T/3}^{T} 2\,dt = \frac{2T}{3}$$

Using the sign convention from the waveform segments,

$$\int_0^T f_2(t)f_3(t)\,dt = \frac{T}{3} - T + \frac{2T}{3} = 0$$

Hence,

$$\boxed{\int_0^T f_2(t)f_3(t)\,dt = 0}$$

Therefore,

$(C)\, f_2(t)$ and $f_3(t)$ are orthogonal

Q2.07.7 MCQ 2007 2M Ans: A ●●○

Theory: For maximum entropy, all quantizer output levels must be equiprobable. Since the quantizer has 6 levels and the pdf is symmetric, each pair of symmetric regions must have probability $1/3$.

For the outer region,

$$\int_1^5 p_X(x)\,dx = \frac{1}{3}$$

$$\int_1^5 b\,dx = \frac{1}{3}$$

$$b[x]_1^5 = \frac{1}{3}$$

$$b(5-1) = \frac{1}{3}$$

$$4b = \frac{1}{3}$$

$$b = \frac{1}{12}$$

Similarly, for the central region,

$$\int_{-1}^1 p_X(x)\,dx = \frac{1}{3}$$

$$\int_{-1}^1 a\,dx = \frac{1}{3}$$

$$a[x]_{-1}^1 = \frac{1}{3}$$

$$a\big(1-(-1)\big) = \frac{1}{3}$$

$$2a = \frac{1}{3}$$

$$a = \frac{1}{6}$$

Hence,

$$(A)\, a = \frac{1}{6}, \qquad b = \frac{1}{12}$$

Q2.07.8 MCQ | 2007 | 2M | Ans: D | ● ● ○

Theory: For a maximum-entropy quantizer, all output levels are equiprobable. Hence, each of the 6 quantization regions has probability $1/6$.

Since the pdf is symmetric,

$$2a(1) + 2b(4) = 1$$

$$a + 4b = \frac{1}{2}$$

Also, the probability of the central region $(-1, 1)$ is

$$P(-1 < x < 1) = 2a = \frac{2}{6} = \frac{1}{3}$$

$$a = \frac{1}{6}$$

Substituting into $a + 4b = \frac{1}{2}$,

$$\frac{1}{6} + 4b = \frac{1}{2}$$

$$4b = \frac{1}{3}$$

$$b = \frac{1}{12}$$

Signal Power:

$$S = E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)\, dx$$

Using symmetry,

$$S = 2\left[\int_0^1 ax^2\, dx + \int_1^5 bx^2\, dx\right]$$

$$= 2\left[a\frac{x^3}{3}\Big|_0^1 + b\frac{x^3}{3}\Big|_1^5\right]$$

$$= 2\left[\frac{a}{3} + \frac{124b}{3}\right]$$

$$= 2\left[\frac{1}{18} + \frac{124}{36}\right]$$

$$= 7$$

Quantization Noise Power:

$$N_Q = \sum_{i=1}^{2} \frac{\Delta_i^2}{12} P(\Delta_i)$$

For the central region,

$$\Delta_1 = 1, \qquad P(\Delta_1) = P(-1 < x < 1) = \frac{1}{3}$$

For the outer regions,

$$\Delta_2 = 2, \qquad P(\Delta_2) = 1 - \frac{1}{3} = \frac{2}{3}$$

Hence,

$$N_Q = \frac{1}{3}\left(\frac{1^2}{12}\right) + \frac{2}{3}\left(\frac{2^2}{12}\right)$$

$$= \frac{1}{36} + \frac{2}{9}$$

$$= \frac{1}{4}$$

Therefore,

$$SQNR = \frac{S}{N_Q} = \frac{7}{1/4} = 28$$

$$\boxed{(D) SQNR = 28}$$

Q2.07.9 MCQ 2007 2M Ans: B ●●○

Theory: The average symbol energy is obtained by averaging the energies of all equiprobable constellation points.

Average energy of Constellation 1:

$$E_1 = \frac{0 + 4a^2 + 4a^2 + 8a^2}{4} = 4a^2$$

Average energy of Constellation 2:

$$E_2 = \frac{a^2 + a^2 + a^2 + a^2}{4} = a^2$$

Therefore,

$$\frac{E_1}{E_2} = \frac{4a^2}{a^2} = 4$$

$$\boxed{(B) \frac{E_1}{E_2} = 4}$$

Q2.07.10 MCQ 2007 2M Ans: A ●●○

Theory: For an AWGN channel, the probability of symbol error decreases as the average symbol energy increases (for comparable minimum-distance structures).

For Constellation 1, the four signal points are

$$(0, 0), \quad (-2\sqrt{2}a, 0), \quad (-\sqrt{2}a, \sqrt{2}a), \quad (-\sqrt{2}a, -\sqrt{2}a)$$

Their energies are

$$0, \quad 8a^2, \quad 4a^2, \quad 4a^2$$

Hence, the average symbol energy is

$$E_{s1} = \frac{1}{4}\left(0 + 8a^2 + 4a^2 + 4a^2\right) = 4a^2$$

For Constellation 2, the four signal points are

$$(a, 0), \quad (-a, 0), \quad (0, a), \quad (0, -a)$$

Each point has energy

$$a^2$$

Therefore,

$$E_{s2} = \frac{1}{4}\left(a^2 + a^2 + a^2 + a^2\right) = a^2$$

Thus,

$$E_{s1} > E_{s2}$$

Since symbol error probability decreases with increasing average symbol energy,

$$P_{e,1} < P_{e,2}$$

Hence,

(A)Probability of symbol error for Constellation 1 is lower

Q2.06.1 MCQ 2006 2M Ans: B ●●○

Theory: To avoid slope-overload distortion in Delta Modulation, the staircase approximation must be able to track the maximum slope of the input signal:

$$\frac{\Delta}{T_s} \geq \max\left|\frac{dx(t)}{dt}\right|$$

Given

$$x(t) = 125\,[u(t) - u(t-1)] + (250 - 125t)\,[u(t-1) - u(t-2)]$$

For $1 < t < 2$,

$$x(t) = 250 - 125t$$

Hence,

$$\frac{dx(t)}{dt} = -125$$

Therefore,

$$\max\left|\frac{dx(t)}{dt}\right| = 125$$

Sampling frequency:

$$f_s = 32 \times 10^3 \text{ samples/s}$$

Thus,

$$T_s = \frac{1}{f_s} = \frac{1}{32 \times 10^3}$$

Applying the slope-overload condition,

$$\frac{\Delta}{T_s} \geq 125$$

$$\Delta \geq \frac{125}{32 \times 10^3}$$

Since

$$32 \times 10^3 = 2^5 \times (125 \times 2^3) = 125 \times 2^8$$

we get

$$\Delta \geq \frac{125}{125 \times 2^8} = \frac{1}{2^8}$$

$$\boxed{(B)\Delta_{\min} = 2^{-8}}$$

Q2.06.2 MCQ 2006 2M Ans: B ●●○

Theory: For the most efficient source encoder, the average bit rate approaches the source entropy rate.

Entropy of the source:

$$H = \sum_i p_i \log_2\left(\frac{1}{p_i}\right)$$

$$H = 0.25 \log_2\left(\frac{1}{0.25}\right) + 0.25 \log_2\left(\frac{1}{0.25}\right) + 0.5 \log_2\left(\frac{1}{0.5}\right)$$

$$= 0.25(2) + 0.25(2) + 0.5(1)$$

$$= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} = 1.5 \text{ bits/symbol}$$

Given symbol rate,

$$R_s = 3000 \text{ symbols/s}$$

Average bit rate:

$$R_b = H R_s$$

$$R_b = 1.5 \times 3000 = 4500 \text{ bits/s}$$

$$\boxed{(B)R_b = 4500 \text{ bits/s}}$$

Q2.06.3 MCQ | 2006 | 2M | Ans: C | ●●○

Theory: The impulse response of a matched filter is the time-reversed version of the signal,

$$h(t) = s(T - t)$$

where T is the signal duration.
Given

$$p(t) = u(t) - u(t - 1)$$

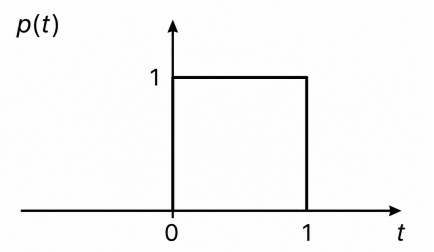


Fig. Solution Q2.06.3

and

$$g(t) = p(t) * p(t)$$

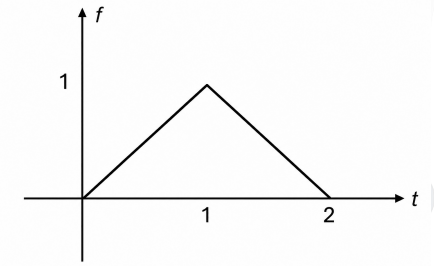


Fig. Solution Q2.06.3

Also,

$$s(t) = g(t) - \delta(t - 2) * g(t)$$

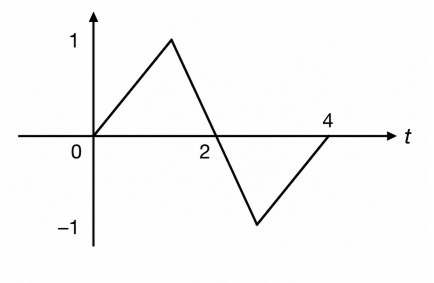


Fig. Solution Q2.06.3

Using the shifting property of convolution,

$$\delta(t-2) * g(t) = g(t-2)$$

Hence,

$$s(t) = g(t) - g(t-2)$$

The signal $s(t)$ is nonzero over $0 \le t \le 4$, therefore

$$T = 4$$

Thus, the matched filter impulse response is

$$h(t) = s(4-t)$$

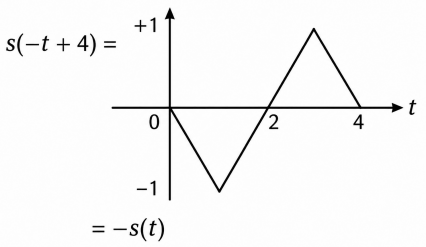


Fig. Solution Q2.06.3

From the symmetry of $s(t)$,

$$s(4-t) = -s(t)$$

$$\boxed{(C)\, h(t) = -s(t)}$$

Q2.06.4 MCQ 2006 2M Ans: B

Theory: For a uniform quantizer, the output level changes only when the input crosses a decision threshold. Hence, the minimum offset required to force a change in the quantizer output is half the quantization step size.

Let the quantizer step size be

$$\Delta$$

The maximum distance of a quantized value from the nearest decision boundary is

$$\frac{\Delta}{2}$$

Therefore, to ensure that the second quantizer produces a different output level,

$$C \ge \frac{\Delta}{2}$$

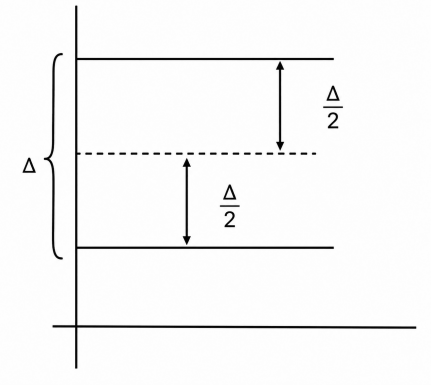


Fig. Solution Q2.06.4

Hence, the minimum value of the constant is

$$\boxed{(B)C_{\min} = \frac{\Delta}{2}}$$

Q2.06.5 MCQ 2006 1M Ans: C ●○○

Theory: Multiplication in time domain corresponds to convolution in frequency domain. If a signal of bandwidth B is raised to the n^{th} power, its bandwidth becomes nB.

Given

$$x(t) = 5\left(\frac{\sin(2\pi 1000t)}{\pi t}\right)^3 + 7\left(\frac{\sin(2\pi 1000t)}{\pi t}\right)^2$$

The signal

$$\frac{\sin(2\pi 1000t)}{\pi t}$$

has bandwidth

$$B = 1000 \text{ Hz}$$

Hence,

$$\left(\frac{\sin(2\pi 1000t)}{\pi t}\right)^2$$

has bandwidth

$$2B = 2000 \text{ Hz}$$

and

$$\left(\frac{\sin(2\pi 1000t)}{\pi t}\right)^3$$

has bandwidth

$$3B = 3000 \text{ Hz}$$

Therefore, the highest frequency present in $x(t)$ is

$$f_m = 3000 \text{ Hz}$$

By the Nyquist sampling theorem,

$$f_s \geq 2f_m$$

$$f_s \geq 2 \times 3000$$

$$f_s \geq 6000 \text{ samples/sec}$$

$$\boxed{(C)\, f_s = 6 \times 10^3 \text{ samples/sec}}$$

Q2.05.1 MCQ | 2005 | 2M | Ans: A | ● ○ ○

Theory: The quantization noise power in a quantization interval of width Δ is

$$N_q = \frac{\Delta^2}{12}$$

and the total noise power is obtained by weighting each interval with its probability of occurrence.

Given

$$a = \frac{2}{3}$$

Hence the central quantization region is

$$-\frac{2}{3} \leq x \leq \frac{2}{3}$$

with reconstruction level at

$$x_q = 0$$

Therefore,

$$N_q = E[X^2] = \int_{-a}^{a} x^2 f_X(x)\, dx$$

Using

$$f_X(x) = \frac{1}{4}, \qquad a = \frac{2}{3},$$

$$N_q = 2\int_0^{2/3} x^2 \left(\frac{1}{4}\right) dx$$

$$= \frac{1}{2}\int_0^{2/3} x^2\, dx$$

$$= \frac{1}{2}\left[\frac{x^3}{3}\right]_0^{2/3}$$

$$= \frac{1}{2} \cdot \frac{1}{3}\left(\frac{2}{3}\right)^3$$

$$= \frac{1}{6} \cdot \frac{8}{27}$$

$$= \frac{4}{81}$$

$$\boxed{(A) N_q = \frac{4}{81}}$$

Q2.05.2 MCQ 2005 2M Ans: B ● ○ ○

Theory: For maximum entropy, all quantization regions must be equiprobable.

Since there are three quantization levels,

$$P(\text{Region 1}) = P(\text{Region 2}) = P(\text{Region 3}) = \frac{1}{3}$$

For Region 1,

$$\int_{-a}^{a} f_X(x)\, dx = \frac{1}{3}$$

Given

$$f_X(x) = \frac{1}{4}, \qquad -a \le x \le a$$

Therefore,

$$\frac{1}{4} \int_{-a}^{a} dx = \frac{1}{3}$$

$$\frac{1}{4}(2a) = \frac{1}{3}$$

$$\frac{a}{2} = \frac{1}{3}$$

$$a = \frac{2}{3}$$

$$\boxed{(B) a = \frac{2}{3}}$$

Q2.05.3 MCQ 2005 2M Ans: C ● ○ ○

Theory: The impulse response of a matched filter is the time-reversed version of the signal shifted by the signal duration:

$$h(t) = s(T - t)$$

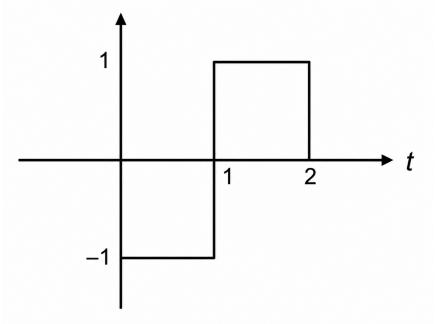


Fig. Solution Q2.05.3

where $T = 2$.

The output of the matched filter is

$$y(t) = s(t) * h(t) = \int_{-\infty}^{\infty} s(\tau)\, h(t-\tau)\, d\tau$$

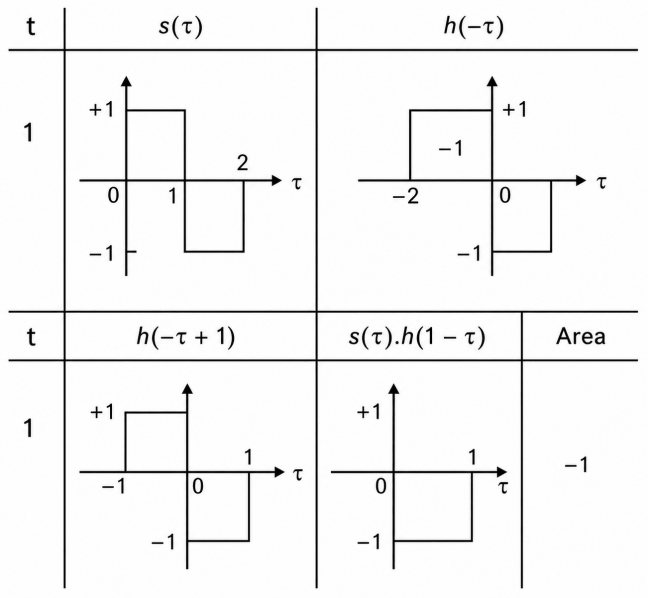


Fig. Solution Q2.05.3

Using graphical convolution, at $t = 1$,

$$y(1) = \int_{-\infty}^{\infty} s(\tau)\, h(1-\tau)\, d\tau$$

The overlap region contributes an area

$$y(1) = -1$$

Since the matched-filter output must satisfy

$$y(1) = -1,$$

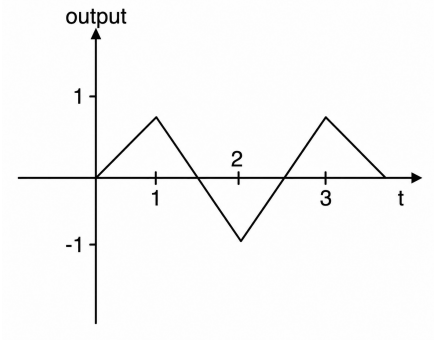


Fig. Solution Q2.05.3

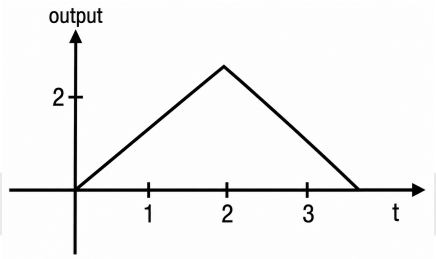


Fig. Solution Q2.05.3

option (A) and (D) is not correct.
Also, the support of the convolution must extend from

$$t = 0 \quad \text{to} \quad t = 4,$$

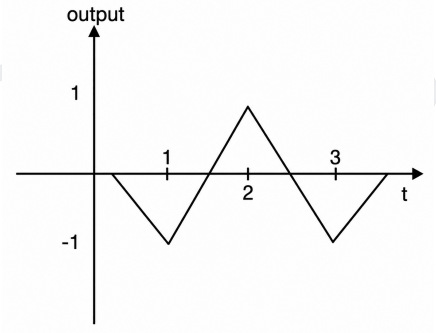


Fig. Solution Q2.05.3

hence option (B) is not correct.

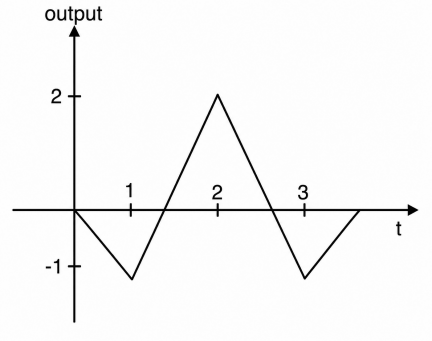


Fig. Solution Q2.05.3

Among the given waveforms, only option (C) satisfies

$$y(1) = -1$$

and has the correct support length.
Therefore,

(C) is correct and it's peak value is y(2)=+2

Q2.04.1 MCQ 2004 2M Ans: C ●○○

Theory: For PSK systems, the required transmission bandwidth is proportional to the symbol rate. Since QPSK carries 2 bits per symbol, its symbol rate is half that of BPSK for the same bit rate.
Given,

$$R_b = 10 \text{ kbps}$$

For BPSK,

$$B_1 = 2R_b$$

$$B_1 = 2(10 \text{ kHz})$$

$$B_1 = 20 \text{ kHz}$$

For QPSK,

$$R_s = \frac{R_b}{2}$$

and

$$B_2 = 2R_s$$

$$B_2 = 2\left(\frac{R_b}{2}\right)$$

$$B_2 = R_b$$

$$B_2 = 10 \text{ kHz}$$

Hence,

$$\boxed{(C)B_1 = 20 \text{ kHz}, \qquad B_2 = 10 \text{ kHz}}$$

Q2.04.2 MCQ 2004 2M Ans: C ● ○ ○

Theory: In PCM-TDM systems, each signal is sampled at its Nyquist rate, and the multiplexed bit rate equals the total sampling rate multiplied by the number of bits per sample.
The Nyquist sampling rates are

$$f_{s1} = 2(1200) = 2400 \text{ Hz}$$

$$f_{s2} = 2(600) = 1200 \text{ Hz}$$

$$f_{s3} = 2(600) = 1200 \text{ Hz}$$

Hence, the total sampling rate after TDM is

$$f_s = f_{s1} + f_{s2} + f_{s3}$$

$$f_s = 2400 + 1200 + 1200$$

$$f_s = 4800 \text{ Hz} = 4.8 \text{ kHz}$$

Since each sample is encoded using $n = 12$ bits,

$$R_b = nf_s$$

$$R_b = 12 \times 4800$$

$$R_b = 57600 \text{ bps}$$

$$R_b = 57.6 \text{ kbps}$$

Therefore,

$$\boxed{(C)R_b = 57.6 \text{ kbps}}$$

Q2.04.3 MCQ 2004 2M Ans: A ● ○ ○

Theory: The average bit error probability is the weighted sum of the conditional error probabilities. Since 0's and 1's are equally likely, each occurs with probability 1/2.

$$P(0) = \frac{1}{2}, \qquad P(1) = \frac{1}{2}$$

Let the detector threshold be

$$V_T = 0.2 \text{ V}$$

For transmitted binary 0,

$$z \sim U(-0.25,\ 0.25)$$

An error occurs when

$$z > 0.2$$

Hence,

$$P(1|0) = \frac{0.25 - 0.2}{0.25 - (-0.25)} = \frac{0.05}{0.5} = 0.1$$

For transmitted binary 1,

$$z \sim U(0,\ 1)$$

An error occurs when

$$z < 0.2$$

Thus,

$$P(0|1) = \frac{0.2 - 0}{1 - 0} = 0.2$$

Therefore, the average bit error probability is

$$P_e = P(0)P(1|0) + P(1)P(0|1)$$

$$P_e = \frac{1}{2}(0.1) + \frac{1}{2}(0.2)$$

$$P_e = \frac{0.3}{2} = 0.15$$

Hence,

$$\boxed{(A)P_e = 0.15}$$

Q2.04.4 MCQ 2004 2M Ans: A ● ○ ○

Theory: The received SNR is obtained by dividing the received signal power by the total noise power over the signal bandwidth. Cable loss reduces the SNR by the same amount in dB.
Given,

$$P_s = 1 \text{ mW} = 10^{-3} \text{ W}$$

$$B = 100 \text{ MHz} = 100 \times 10^6 \text{ Hz}$$

$$N_0 = 10^{-20} \text{ W/Hz}$$

The noise power is

$$N = N_0 B$$

Hence,

$$\text{SNR} = \frac{P_s}{N_0 B} = \frac{10^{-3}}{10^{-20} \times 100 \times 10^6}$$

$$\text{SNR} = 10^9$$

Converting to decibels,

$$(\text{SNR})_{\text{dB}} = 10 \log_{10}(10^9) = 90 \text{ dB}$$

Given cable loss,

$$L = 40 \text{ dB}$$

Therefore, the SNR at the receiver is

$$(\text{SNR})_{R_x} = 90 - 40 = 50 \text{ dB}$$

Hence,

$$\boxed{(B)(\text{SNR})_{R_x} = 50 \text{ dB}}$$

Q2.04.5 MCQ 2004 2M Ans: B ● ○ ○

Theory: The impulse response of a matched filter is the time-reversed and shifted version of the signal:

$$h(t) = x(T - t)$$

where $T = 4$ s.
Thus,

$$h(t) = x(4 - t)$$

Time reversal changes the sign of the slope. From the given waveform, the segment of $x(t)$ over

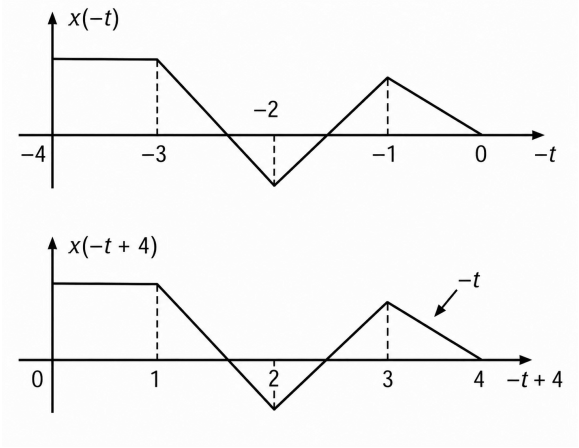


Fig. Solution Q2.04.5

$$0 < t < 1$$

has slope

$$\frac{1-0}{1-0} = 1\ \text{s}^{-1}.$$

After time reversal and shifting, this segment appears in

$$3 < t < 4$$

of $h(t)$, with slope

$$-1\ \text{s}^{-1}.$$

Therefore,

$$\boxed{\frac{dh(t)}{dt} = -1\ \text{s}^{-1}, \qquad 3 < t < 4}$$

Q2.04.6 MCQ 2004 2M Ans: A ● ○ ○

Theory: In Delta Modulation (DM), alternating positive and negative output pulses indicate that the input signal variation between successive samples is smaller than the step size. Hence, the input is approximately constant over that interval.
If the change in the input signal satisfies

$$|\Delta x| < \Delta,$$

where Δ is the DM step size, the modulator output alternates between positive and negative pulses.
Therefore, during the interval

$$t_1 \le t \le t_2,$$

the input signal is essentially constant.
Hence,

$$\boxed{(A)\text{The input to the modulator is essentially constant}}$$

Q2.04.7 MCQ 2004 2M Ans: C ● ○ ○

Theory: In a PCM system, the signal-to-quantization-noise ratio (SQNR) is proportional to the square of the number of quantization levels:

$$\text{SQNR} \propto L^2$$

where

$$L = 2^n$$

and n is the number of bits per sample.
Initially,

$$n_1 = 6$$

and finally,

$$n_2 = 8.$$

The improvement factor in SQNR is

$$\frac{\text{SQNR}_2}{\text{SQNR}_1} = \left(\frac{L_2}{L_1}\right)^2 = \left(\frac{2^8}{2^6}\right)^2$$

$$= \left(2^2\right)^2 = 2^4 = 16.$$

Equivalently, each additional bit improves SQNR by a factor of 4; therefore, an increase of 2 bits improves SQNR by

$$4^2 = 16.$$

Hence,

$$\boxed{(C)16}$$

Q2.03.1 MCQ 2003 2M Ans: B ● ○ ○

Theory: For a PCM system with an n-bit uniform quantizer and a full-scale sinusoidal input,

$$\text{SNR}_q(\text{dB}) \approx 1.76 + 6.02n$$

and the PCM bit rate is

$$R = nf_s.$$

Given

$$f_s = 8 \text{ kHz}, \qquad n = 8.$$

Therefore, the bit rate is

$$R = nf_s = 8 \times 8 \times 10^3 = 64 \times 10^3 \text{ bps} = 64 \text{ kbps}.$$

The quantization SNR is

$$\text{SNR}_q = 1.76 + 6.02(8)$$

$$= 1.76 + 48.16$$

$$= 49.92 \text{ dB} \approx 49.8 \text{ dB}.$$

Hence,

$$\boxed{(B)R = 64 \text{ kbps}, \qquad \text{SNR}_q \approx 49.8 \text{ dB}}$$

Q2.03.2 MCQ 2003 2M Ans: B ● ○ ○

Theory: Coherent BPSK requires carrier synchronization at the receiver, and with Nyquist pulse shaping its bandwidth efficiency is

$$\rho = \frac{R_b}{BW}.$$

For BPSK,

$$BW = R_b = \frac{1}{T_b}.$$

Hence,

$$\rho = \frac{R_b}{BW} = \frac{R_b}{R_b} = 1.$$

Also, since it is *coherent* BPSK, carrier synchronization is essential.

$$\therefore \quad S \text{ is required.}$$

Thus,

$$\boxed{(B)\rho = 1.0, \qquad S \text{ is required}}$$

Q2.03.3 MCQ 2003 2M Ans: B ● ○ ○

Theory: For a sinusoidal input to a Delta Modulator,

$$m(t) = E_m \sin(\omega_m t),$$

the maximum slope is

$$\left|\frac{dm(t)}{dt}\right|_{\max} = E_m \omega_m.$$

Slope overload occurs when the input slope exceeds the maximum tracking slope of the DM:

$$E_m \omega_m > \frac{\Delta}{T_s} = \Delta f_s.$$

Given

$$\Delta = 0.628, \qquad f_s = 40 \times 10^3 \text{ Hz},$$

$$\frac{\Delta}{T_s} = 0.628 \times 40 \times 10^3 = 25120.$$

Now evaluate each option:

(A) $E_m \omega_m = 2\pi(0.3)(8 \times 10^3) = 15079 < 25120$

No slope overload.

(B) $E_m \omega_m = 2\pi(1.5)(4 \times 10^3) = 37699 > 25120$

Slope overload occurs.

(C) $E_m \omega_m = 2\pi(1.5)(2 \times 10^3) = 18850 < 25120$

No slope overload.

(D) $E_m \omega_m = 2\pi(0.3)(1 \times 10^3) = 1885 < 25120$

No slope overload.
Therefore, only option (B) satisfies the slope-overload condition.

$$\boxed{(B)E_m = 1.5 \text{ V}, \qquad f_m = 4 \text{ kHz}}$$

Q2.03.4 MCQ 2003 2M Ans: D ● ○ ○

Theory: For a binary digital communication system using a matched filter in AWGN, the peak output SNR is

$$\text{SNR}_{\text{out}} = \frac{2E_b}{N_0}.$$

Given

$$E_b = 10^{-5} \text{ W-s}, \qquad N_0 = 10^{-6} \text{ W/Hz}.$$

Therefore,

$$\text{SNR}_{\text{out}} = \frac{2 \times 10^{-5}}{10^{-6}} = 20.$$

Converting to decibels,

$$(\text{SNR}_{\text{out}})_{\text{dB}} = 10 \log_{10}(20)$$

Hence,

$$\boxed{(D) \text{SNR}_{\text{out}} \approx 13 \text{ dB}}$$

Q2.03.5 MCQ 2003 2M Ans: C ● ○ ○

Theory: For a uniform quantizer, the quantization-noise power is

$$P_N = \frac{\Delta^2}{12},$$

where Δ is the quantization step size.
Given,

$$V_{pp} = 1.536 \text{ V}, \qquad L = 128.$$

For a mid-rise uniform quantizer,

$$\Delta = \frac{V_{pp}}{L} = \frac{1.536}{128} = 0.012 \text{ V}.$$

Hence,

$$P_N = \frac{\Delta^2}{12} = \frac{(0.012)^2}{12} = \frac{144 \times 10^{-6}}{12} = 12 \times 10^{-6} \text{ V}^2.$$

Therefore,

$$\boxed{(C) P_N = 12 \times 10^{-6} \text{ V}^2}$$

Q2.03.6 MCQ 2003 2M Ans: B ● ○ ○

Theory: For coherent binary modulation schemes in AWGN, the probability of error is

$$P_e(\text{BPSK}) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

and

$$P_e(\text{BFSK}) = Q\left(\sqrt{\frac{E_b}{N_0}}\right).$$

To achieve the same probability of error, the arguments of the Q-functions must be equal:

$$\sqrt{\frac{2E_{b,\text{BPSK}}}{N_0}} = \sqrt{\frac{E_{b,\text{BFSK}}}{N_0}}.$$

Hence,

$$E_{b,\text{BFSK}} = 2E_{b,\text{BPSK}}.$$

Therefore,

$$\frac{E_{b,\text{BFSK}}}{E_{b,\text{BPSK}}} = 2.$$

Expressing this ratio in decibels,

$$10\log_{10}(2) = 3.01 \text{ dB} \approx 3 \text{ dB}.$$

Thus, coherent BFSK requires 3 dB more energy per bit than coherent BPSK for the same error probability.

(B)Coherent BFSK is inferior to coherent BPSK by 3 dB

Q2.02.1 MCQ 2002 2M Ans: D ● ○ ○

Theory: Ideal sampling replicates the spectrum at integer multiples of the sampling frequency. Any spectral component lying within the LPF passband appears at the output.
Given

$$x(t) = 100\cos\left(24\pi \times 10^3\, t\right)$$

Comparing with

$$A\cos(2\pi f_m t),$$

we get

$$2\pi f_m = 24\pi \times 10^3$$

$$f_m = 12 \text{ kHz}.$$

The sampling period is

$$T_s = 50\ \mu s.$$

Hence,

$$f_s = \frac{1}{T_s} = \frac{1}{50 \times 10^{-6}} = 20 \text{ kHz}.$$

After sampling, spectral replicas occur at

$$f = nf_s \pm f_m.$$

For $n = 1$,

$$f = 20 \pm 12$$

$$f = 8 \text{ kHz},\ 32 \text{ kHz}.$$

The ideal LPF has cutoff frequency

$$f_c = 15 \text{ kHz}.$$

Therefore, the frequency components that lie within the passband are

$$8 \text{ kHz}$$

and

$$12 \text{ kHz}.$$

All higher-frequency replicas are rejected.

$$\boxed{(D)\text{Frequencies present at the filter output: } 12 \text{ kHz and } 8 \text{ kHz}}$$

Q2.02.2 MCQ 2002 2M Ans: A

Theory: For coherent BFSK, the two carrier frequencies must satisfy the orthogonality condition

$$f_H - f_L = \frac{n}{2T_b},$$

where n is a positive integer.
Given the bit rate

$$R_b = 8 \text{ kbps},$$

the bit duration is

$$T_b = \frac{1}{R_b} = \frac{1}{8000}.$$

Hence,

$$f_H - f_L = \frac{n}{2T_b} = \frac{nR_b}{2} = n \times 4 \text{ kHz}.$$

For the best frequency utilization, the separation should be the minimum possible, i.e.,

$$n = 1.$$

Therefore,

$$f_H - f_L = 4 \text{ kHz}.$$

Checking the options:

$$20 \text{ kHz} - 16 \text{ kHz} = 4 \text{ kHz}$$

which satisfies the coherent BFSK orthogonality condition with minimum separation.

$$\boxed{(A) f_L = 16 \text{ kHz}, \qquad f_H = 20 \text{ kHz}}$$

Q2.02.3 MCQ | 2002 | 2M | Ans: C | ● ○ ○

Theory: Sampling by an impulse train replicates the spectrum at multiples of the sampling frequency. An ideal LPF passes only the baseband component and scales it by the sampling constant.
Given

$$y(t) = 5 \times 10^{-6} x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

with

$$x(t) = 10\cos(8\pi \times 10^3 t)$$

and

$$T_s = 100\ \mu s.$$

The sampling frequency is

$$f_s = \frac{1}{T_s} = \frac{1}{100 \times 10^{-6}} = 10\ \text{kHz}.$$

The signal frequency is

$$f_m = \frac{8\pi \times 10^3}{2\pi} = 4\ \text{kHz}.$$

Since

$$f_m = 4\ \text{kHz} < f_c = 5\ \text{kHz},$$

the ideal LPF passes the baseband component.
The spectrum of the sampled signal is

$$Y(f) = \frac{5 \times 10^{-6}}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s).$$

Hence, the LPF output is

$$z(t) = \frac{5 \times 10^{-6}}{T_s} x(t).$$

Substituting $T_s = 100 \times 10^{-6}$,

$$z(t) = \frac{5 \times 10^{-6}}{100 \times 10^{-6}} x(t) = \frac{1}{20} x(t).$$

Therefore,

$$z(t) = \frac{1}{20}\left(10\cos(8\pi \times 10^3 t)\right) = \frac{10}{20}\cos(8\pi \times 10^3 t).$$

$$z(t) = 5 \times 10^{-1}\cos(8\pi \times 10^3 t).$$

Hence,

$$\boxed{(C) z(t) = 5 \times 10^{-1}\cos(8\pi \times 10^3 t)}$$

Q2.01.1 MCQ 2001 2M Ans: C ● ○ ○

Theory: The function

$$\text{sinc}(at) = \frac{\sin(\pi at)}{\pi at}$$

has a bandwidth of

$$\frac{a}{2} \text{ Hz.}$$

Given

$$x(t) = \text{sinc}(700t) + \text{sinc}(500t),$$

the bandwidths of the two components are

$$B_1 = \frac{700}{2} = 350 \text{ Hz},$$

$$B_2 = \frac{500}{2} = 250 \text{ Hz}.$$

Hence, the highest frequency present in the signal is

$$f_{\max} = 350 \text{ Hz}.$$

Therefore, the Nyquist sampling rate is

$$f_s = 2f_{\max} = 2(350) = 700 \text{ Hz}.$$

The Nyquist sampling interval is

$$T_s = \frac{1}{f_s} = \frac{1}{700} \text{ sec}.$$

Thus,

$$\boxed{(C)\, T_s = \frac{1}{700} \text{ sec}}$$

Q2.01.2 MCQ 2001 2M Ans: C ● ○ ○

Theory: The data rate is obtained by multiplying the number of frames per second, the number of pixels per frame, and the number of bits required per pixel.
Given

$$\text{Frames per second} = 625$$

$$\text{Pixels per frame} = 400 \times 400 = 160000$$

Since there are 64 intensity levels,

$$\text{Bits per pixel} = \log_2(64) = 6.$$

Hence, the data rate is

$$R = 625 \times 400 \times 400 \times 6$$

$$= 625 \times 160000 \times 6$$

$$= 600 \times 10^6 \text{ bits/s}$$

Therefore,

$$\boxed{(C) R = 600 \text{ Mbps}}$$

Q2.00.1 MCQ 2000 2M Ans: B ● ○ ○

Theory: Two FSK waveforms are orthogonal when their inner product over one bit interval is zero. For coherent BFSK, an equivalent shortcut is that the bit duration is the reciprocal of the greatest common divisor (GCD) of the two frequencies.
Given

$$f_1 = 25 \text{ kHz}, \qquad f_2 = 10 \text{ kHz}$$

Orthogonality condition:

$$\int_0^{T_b} \cos(2\pi f_1 t)\, \cos(2\pi f_2 t)\, dt = 0$$

Using the shortcut,

$$f_0 = \gcd(f_1, f_2) = \gcd(25, 10)\,\text{kHz} = 5 \text{ kHz}$$

Hence,

$$T_b = \frac{1}{f_0} = \frac{1}{5 \times 10^3}$$

$$T_b = 2 \times 10^{-4} \text{ s}$$

$$\boxed{(B) T_b = 200\ \mu\text{s}}$$

Q2.99.1 MCQ 1999 2M Ans: 43.9 ● ○ ○

Theory: SQNR of a PCM system is given by the ratio of signal power to quantization noise power. If the input amplitude is less than full-scale, the SQNR decreases accordingly.
Given,

$$n = 8, \qquad V_{pp} = 2 \text{ V}, \qquad x(t) = 0.5 \cos(\omega_m t).$$

The quantizer full-scale peak value is

$$V_{\max} = \frac{V_{pp}}{2} = 1 \text{ V}.$$

Hence, the quantization step size is

$$\Delta = \frac{V_{pp}}{2^n} = \frac{2}{256} = \frac{1}{128} \text{ V}.$$

The signal power is

$$P_s = \frac{A^2}{2} = \frac{(0.5)^2}{2} = 0.125.$$

The quantization noise power is

$$P_q = \frac{\Delta^2}{12} = \frac{1}{12(128)^2} = 5.086 \times 10^{-6}.$$

Therefore,

$$\text{SQNR} = \frac{P_s}{P_q} = \frac{0.125}{5.086 \times 10^{-6}} = 24576.$$

Hence,

$$\text{SQNR}_{\text{dB}} = 10 \log_{10}(24576) = 43.9 \text{ dB}.$$

Therefore,

$$\boxed{\text{SQNR} \approx 43.9 \text{ dB}.}$$

Note: The calculated SQNR is 43.9 dB, which does not match any of the given options. This indicates that the listed options are incorrect. Based on the given data the correct value is 43.9 dB.

Key Points

- The standard expression $1.76 + 6.02n$ dB is valid only for a full-scale sinusoidal input.
- Here, the input amplitude is 0.5 V while the full-scale peak is 1 V, resulting in a 6 dB reduction in SQNR.
- Always compute the actual signal power when the input is not full-scale.

Mistakes to be avoided

- Do not directly use $1.76 + 6.02n$ unless the sinusoid is full-scale.
- Do not confuse the peak-to-peak range with the peak amplitude.
- Use the quantization noise power $P_q = \Delta^2/12$ for a uniform quantizer.

Q2.99.2 MCQ 1999 2M Ans: D ● ○ ○

Theory: For a matched filter, the maximum output occurs at the sampling instant and is equal to the *energy of the signal*:

$$y_{\max} = E_s = \int_0^{T_b} s^2(t)\, dt$$

Given

$$s(t) = 10 \sin(2\pi \times 10^6 t), \qquad 0 < t < 10^{-4} \text{ s}$$

For a sinusoidal signal of amplitude A,

$$E_s = \frac{A^2 T_b}{2}$$

Substituting

$$A = 10, \qquad T_b = 10^{-4} \text{ s}$$

$$E_s = \frac{10^2}{2} \times 10^{-4}$$

$$= 50 \times 10^{-4}$$

$$= 5 \times 10^{-3} \text{ V}$$

$$= 5 \text{ mV}$$

Hence, the peak value of the matched filter output is

$$\boxed{5 \times 10^{-3} \text{ V}}$$

$$\boxed{(D) 5 \text{ mV}}$$

Q2.99.3 MCQ 1999 2M Ans: A ●○○

Theory: For Time Division Multiplexing (TDM), each message signal is sampled at its Nyquist rate, and the total transmitted sampling rate is the sum of the individual sampling rates.
Given bandwidths:

$$B_1 = 100 \text{ Hz}, \quad B_2 = 100 \text{ Hz}, \quad B_3 = 200 \text{ Hz}, \quad B_4 = 400 \text{ Hz}$$

Nyquist sampling rates:

$$f_{s1} = 2B_1 = 200 \text{ Hz}$$

$$f_{s2} = 2B_2 = 200 \text{ Hz}$$

$$f_{s3} = 2B_3 = 400 \text{ Hz}$$

$$f_{s4} = 2B_4 = 800 \text{ Hz}$$

Hence, the minimum transmitted sampling rate is

$$(f_s)_{\min} = f_{s1} + f_{s2} + f_{s3} + f_{s4}$$

$$= 200 + 200 + 400 + 800$$

$$= 1600 \text{ Hz}$$

Therefore,

$$\boxed{(A)(f_s)_{\min} = 1600 \text{ Hz}}$$

Q2.98.1 MCQ 1998 1M Ans: C ● ○ ○

Theory: In a uniform PCM system, quantization noise power is proportional to the square of the quantization step size. Increasing the number of bits by 1 doubles the number of levels, thereby reducing the quantization noise power by a factor of 4.
For a uniform quantizer,

$$P_q = \frac{\Delta^2}{12}$$

When the number of bits increases from n to $n + 1$,

$$L = 2^n \rightarrow 2^{n+1}$$

Hence,

$$\Delta \rightarrow \frac{\Delta}{2}$$

Therefore,

$$P'_q = \frac{\left(\frac{\Delta}{2}\right)^2}{12} = \frac{\Delta^2}{48} = \frac{P_q}{4}$$

Thus, the quantization noise power is reduced by a factor of

$$\boxed{(C)4}$$

Q2.98.2 MCQ 1998 1M Ans: D ● ○ ○

Theory: Quadrature multiplexing transmits two signals simultaneously over the same carrier frequency using two orthogonal carriers ($\cos \omega_c t$ and $\sin \omega_c t$). Hence it is fundamentally different from both FDM and TDM.
FDM separates signals in the frequency domain.
TDM separates signals in the time domain.
In quadrature multiplexing,

$$s(t) = m_1(t) \cos \omega_c t + m_2(t) \sin \omega_c t$$

where

$$\int_0^T \cos \omega_c t \ \sin \omega_c t \, dt = 0$$

showing that the two carriers are orthogonal.
Since both signals occupy the same frequency band and the same time interval while remaining orthogonal,

$$\boxed{(D)\text{Quadrature multiplexing is quite different from FDM and TDM}}$$

Q2.98.3 MCQ 1998 1M Ans: A ● ○ ○

Theory: Companding techniques such as μ-law and A-law compress large-amplitude signals before quantization so that low-amplitude signals receive finer quantization.
In PCM systems,

$$\text{Compression} \rightarrow \mu\text{-law or A-law}$$

The compression characteristic reduces the dynamic range by compressing larger amplitudes more than smaller amplitudes.

Hence, compression in PCM refers to the relative compression of

$$\boxed{(A)\text{Higher signal amplitudes}}$$

Q2.98.4 MCQ 1998 1M Ans: C ● ○ ○

Theory: For a given bit rate, both BPSK and OOK (a special case of ASK) require the same transmission bandwidth.
For BPSK,

$$B_p = \frac{2}{T_b}$$

Since

$$r_b = \frac{1}{T_b}$$

we obtain

$$B_p = 2r_b$$

For OOK (ON-OFF Keying),

$$B_0 = \frac{2}{T_b}$$

or

$$B_0 = 2r_b$$

Therefore,

$$B_p = B_0$$

Hence,

$$\boxed{(C)B_p = B_0}$$

Q2.97.1 MCQ 1997 1M Ans: C ● ○ ○

Theory: A line code has zero DC component if its average value is zero. In AMI (Alternate Mark Inversion), successive binary 1's are transmitted with alternating positive and negative polarities, making the average value zero.
For AMI signaling,

$$1 \rightarrow +A, -A, +A, -A, \ldots$$

$$0 \rightarrow 0$$

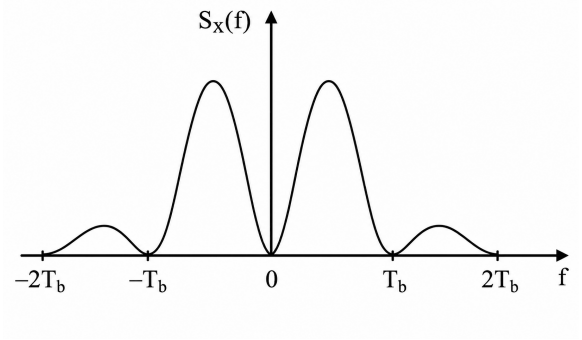


Fig. Solution Q2.97.1

Hence, over a long random data sequence,

$$\text{Average value} = 0$$

Therefore,

$$S_x(f)\big|_{f=0} = 0$$

which implies that there is no DC component.
Thus, the line code having zero DC component is

$$\boxed{(C)\text{Alternate Mark Inversion (AMI)}}$$

Q2.97.2 MCQ 1997 1M Ans: B ●○○

Theory: For a practical signal, the minimum sampling frequency is often chosen based on the frequency range containing most of the signal energy. From the given spectrum, approximately 90% of the signal energy lies within the main lobe extending from −1 kHz to +1 kHz.
Hence, the effective highest frequency is

$$f_m = 1 \text{ kHz}$$

By the Nyquist sampling criterion,

$$(f_s)_{\min} = 2f_m$$

Substituting $f_m = 1$ kHz,

$$(f_s)_{\min} = 2 \times 1 \text{ kHz}$$

$$(f_s)_{\min} = 2 \text{ kHz}$$

Therefore, the minimum sampling frequency required is

$$\boxed{(B)2 \text{ kHz}}$$

Q2.96.1 MCQ | 1996 | 2M | Ans: D | ● ○ ○

Theory: For a uniform PCM system, the quantization noise power is

$$N_q = \frac{\Delta^2}{12}$$

Since the quantization step size is

$$\Delta = \frac{V_{\max} - V_{\min}}{2^n}$$

substituting into the expression for N_q,

$$N_q = \frac{(V_{\max} - V_{\min})^2}{12 \cdot 2^{2n}}$$

Hence, the signal-to-quantization-noise ratio is

$$\frac{S}{N_q} = \frac{\text{Signal Power}}{N_q}$$

$$\frac{S}{N_q} = \frac{\text{Signal Power} \times 12 \times 2^{2n}}{(V_{\max} - V_{\min})^2}$$

Therefore,

$$\frac{S}{N_q} \propto 2^{2n}$$

Now, when the number of bits increases from n to $n + 1$,

$$\left(\frac{S}{N_q}\right)_{n+1} \propto 2^{2(n+1)}$$

Thus, the improvement factor is

$$\frac{\left(\frac{S}{N_q}\right)_{n+1}}{\left(\frac{S}{N_q}\right)_n} = \frac{2^{2(n+1)}}{2^{2n}}$$

$$= 2^2 = 4$$

Hence, for every one-bit increase in PCM word length, the SQNR improves by a factor of 4, independent of n.

$$\boxed{\frac{\left(\frac{S}{N_q}\right)_{n+1}}{\left(\frac{S}{N_q}\right)_n} = 4}$$

(D): Independent of n

Q2.95.1 MCQ | 1995 | 1M | Ans: B | ● ○ ○

Theory: For a uniform PCM system, each additional bit halves the quantization step size, reducing quantization noise power by a factor of 4, which corresponds to an improvement of 6 dB in SQNR.
For an n-bit PCM system,

$$\left(\frac{S}{N_q}\right)_{\text{dB}} = 1.8 + 6n$$

For $(n + 1)$ bits,

$$\left(\frac{S}{N_q}\right)_{\text{dB}} = 1.8 + 6(n + 1)$$

Hence, the improvement is

$$\Delta\left(\frac{S}{N_q}\right)_{\text{dB}} = \left[1.8 + 6(n + 1)\right] - \left[1.8 + 6n\right]$$

Therefore,

$$\boxed{(B)6 \text{ dB}}$$

Q2.95.2 MCQ 1995 1M Ans: C ● ○ ○

Theory: Increasing the number of quantization bits reduces quantization noise power and hence improves the signal-to-quantization noise ratio (SQNR).
For an n-bit PCM system,

$$\left(\frac{S}{N_q}\right)_{\text{dB}} = 1.8 + 6n$$

where

$$1.8 \text{ dB}$$

is the constant term for a full-scale sinusoidal input.
Since

$$\left(\frac{S}{N_q}\right)_{\text{dB}} \propto n$$

the SQNR increases as the number of bits increases.
Therefore,

$$\boxed{(C)\text{SQNR increases with increasing value of } n}$$

Q2.94.1 NAT 1994 1M Ans: 3 ● ○ ○

Theory: In PCM, the transmission bandwidth is proportional to the number of bits per sample. Since the number of bits is $n = \log_2 L$, increasing the quantization levels increases the required bandwidth.
Given,

$$L_1 = 4 = 2^2$$

$$\Rightarrow n_1 = 2$$

and

$$L_2 = 64 = 2^6$$

$$\Rightarrow n_2 = 6$$

The PCM bandwidth is

$$(BW)_{\text{PCM}} = \frac{nf_s}{2}$$

Therefore,

$$\frac{(BW)_2}{(BW)_1} = \frac{n_2}{n_1} = \frac{6}{2} = 3$$

Hence, the bandwidth increases by a factor of

$$\boxed{3}$$

Q2.92.1 MCQ | 1992 | 2M | Ans: C | ● ○ ○

Theory: In DPSK using the delay-and-XOR method, the encoded bit is obtained by XORing the current input bit with the previous encoded bit (reference bit).
Given:

$$b(t) = 01001$$

Reference bit:

$$b'_0 = 1$$

Differential encoding rule:

$$b'_k = b_k \oplus b'_{k-1}$$

$b(t)$	0	1	0	0	1
$b'(t)$	1	0	0	0	1

Ignoring the reference bit, the encoded sequence is

$$10001$$

Given phase assignment:

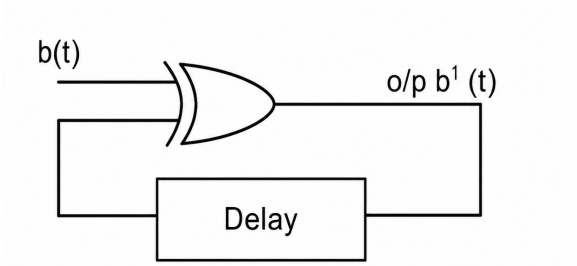
$$1 \rightarrow 0$$

$$0 \rightarrow \pi$$

Therefore,

$$10001 \Rightarrow 0, \pi, \pi, \pi, 0$$

Hence, the transmitted phase sequence is



b(t)	0	1	0	0	1
b_1(t)(Ref.bit)	1	1	0	0	0
Phase	0	π	π	π	0

Fig. Solution Q2.92.1

$(C) 0\ \pi\ \pi\ \pi\ 0$

Q2.92.2 MCQ 1992 2M Ans: A ● ○ ○

Theory: Coherent FSK detection requires a locally generated carrier having the same frequency and phase as the transmitted signal. Such detection is implemented using a correlation receiver (or equivalently a matched-filter receiver).
Hence,

(A) Correlation Receiver

Q2.92.3 MCQ 1992 2M Ans: A ● ○ ○

Theory: Source encoding removes redundancy from the source output and represents the information using the minimum number of bits, thereby improving transmission efficiency.

The source signal is converted into a digital sequence of 1's and 0's by a source encoder.

Efficient representation of the source output results in

No Redundancy

and hence improves the utilization of the communication channel.
Therefore, source encoding is used to

(A) Enhance the information transmission

Q2.91.1 NAT 1991 2M Ans: 2.88 , 3.88 ● ○ ○

Theory: For a discrete memoryless source, the average codeword length is bounded by the source entropy. For block (extended) coding of length n,

$$H(X) \le \overline{L} \le H(X) + 1$$

where $H(X)$ is the entropy per source symbol.

Given probabilities,

$$P(X_1) = 0.8, \qquad P(X_2) = 0.2$$

The source entropy is

$$H(X) = -\sum p_i \log_2 p_i$$

$$H(X) = -0.8\log_2(0.8) - 0.2\log_2(0.2)$$

$$H(X) = 0.7219 \approx 0.72 \text{ bits/symbol}$$

For an extension coding scheme using blocks of 4 symbols,

$$H(X^4) = 4H(X)$$

$$= 4(0.72) = 2.88 \text{ bits/4-symbol block}$$

Using the source coding bound,

$$H(X^4) \le \overline{L} \le H(X^4) + 1$$

$$2.88 \le \overline{L} \le 3.88$$

Hence,

Lower bound = 2.88 bits

Upper bound = 3.88 bits

Q2.91.2 NAT 1991 2M Ans: QPSK ● ○ ○

Theory: A constellation having four equally spaced points in a two-dimensional signal space represents Quadrature Phase Shift Keying (QPSK).

The constellation diagram contains four signal points

$$S_1,\ S_2,\ S_3,\ S_4$$

distributed in a two-dimensional signal space formed by the orthogonal basis functions

$$\cos\left(\frac{2\pi n}{T}t\right) \quad \text{and} \quad \sin\left(\frac{2\pi n}{T}t\right)$$

where T is the symbol duration.

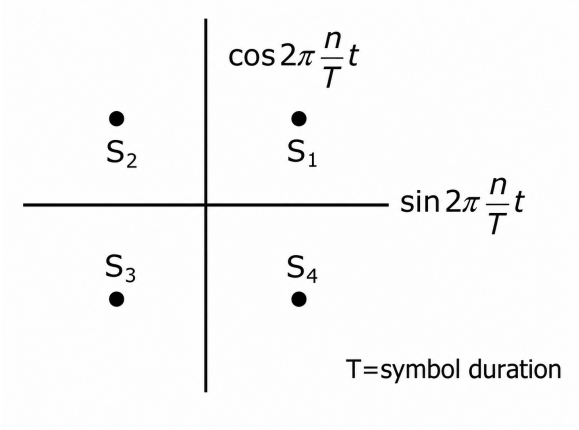


Fig. Solution Q2.91.2

Since there are four possible signal states, the carrier assumes four distinct phases:

$$\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Thus the modulation scheme represented by the constellation is

Quadrature Phase Shift Keying (QPSK)

Q2.90.1 MCQ 1990 2M Ans: B ●○○

Theory: A phase mismatch between the received BPSK signal and the local oscillator reduces the detected signal amplitude by a factor of $\cos\phi$.

Let the phase error between the received carrier and local oscillator be ϕ.
Then the detected signal amplitude becomes

$$A_{out} = A\cos\phi$$

Since power is proportional to the square of amplitude,

$$P_{out} \propto (A\cos\phi)^2$$

$$P_{out} \propto A^2\cos^2\phi$$

Therefore, the output SNR is reduced by the factor

$$\cos^2\phi$$

Given

$$\phi = 20^\circ$$

Hence,

(B)SNR reduction factor $= \cos^2 20^\circ$

Q2.89.1 MCQ 1989 2M Ans: A ● ○ ○

Theory: DPSK is preferred over PSK because it eliminates the need for carrier phase synchronization at the receiver.

Non-coherent detection of PSK is generally not possible since accurate carrier phase information is required at the receiver.

To overcome this difficulty, Differential PSK (DPSK) is used, where information is conveyed through the phase difference between successive symbols rather than the absolute phase.

(A) A coherent carrier is not required to be generated at the receiver

Q2.88.1 MCQ 1988 2M Ans: A ● ○ ○

Theory: A Zero-Order Hold (ZOH) holds each sampled value constant over one sampling interval T, producing a rectangular pulse response.

For an impulse input,

$$x(t) = \delta(t)$$

$$X(s) = 1$$

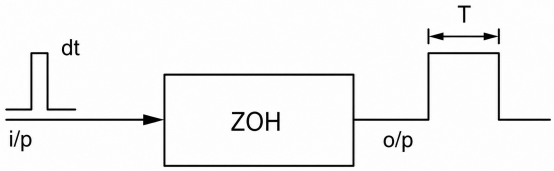


Fig. Solution Q2.88.1

The output of a Zero-Order Hold is a rectangular pulse of duration T,

$$y(t) = u(t) - u(t - T)$$

Taking Laplace transform,

$$Y(s) = \frac{1 - e^{-Ts}}{s}$$

Hence, the transfer function is

$$H(s) = \frac{Y(s)}{X(s)}$$

$$H(s) = \frac{\frac{1 - e^{-Ts}}{s}}{1}$$

$$H(s) = \frac{1 - e^{-Ts}}{s}$$

$$\boxed{(A) H(s) = \frac{1 - e^{-Ts}}{s}}$$

Q2.88.2 MCQ | 1988 | 2M | Ans: C | ● ○ ○

Theory: In DPSK, information is represented by phase changes relative to the previous symbol rather than by absolute phase.

Given message sequence,

$$b_k = 1,\ 1,\ 0,\ 0,\ 1,\ 1$$

The carrier phase during reception of the first two bits is

$$\pi,\ \pi$$

Since

$$d_k = 0 \iff \text{phase } \pi$$

the first two encoded bits are

$$d_1 = 0, \qquad d_2 = 0$$

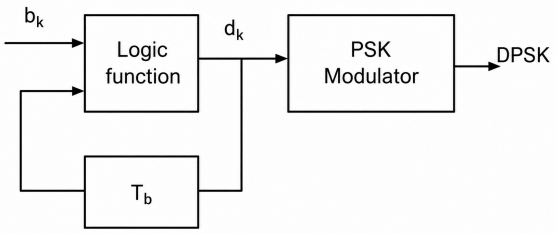


Fig. Solution Q2.88.2

For DPSK encoding,

$$d_k = b_k \odot d_{k-1}$$

where $\odot$ denotes EX-NOR operation.
Computing the remaining encoded bits:

$$d_3 = 0 \odot 0 = 1$$

$$d_4 = 0 \odot 1 = 0$$

$$d_5 = 1 \odot 0 = 0$$

$$d_6 = 1 \odot 0 = 0$$

Hence,

$$d_k = 0,\ 0,\ 1,\ 0,\ 0,\ 0$$

Using the mapping

$$d_k = 1 \rightarrow 0$$

$$d_k = 0 \rightarrow \pi$$

the corresponding carrier phases become

$$\pi,\ \pi,\ 0,\ \pi,\ \pi,\ \pi$$

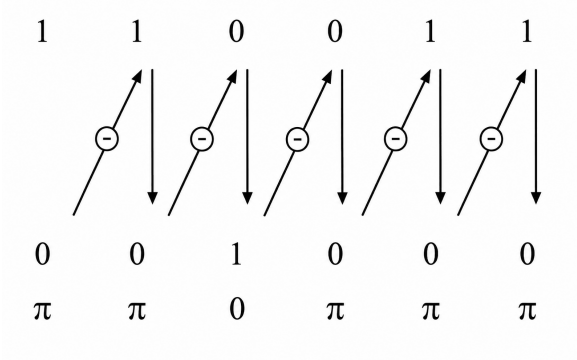


Fig. Solution Q2.88.2

The first two phases are already given as

$$\pi,\ \pi$$

Therefore, the carrier phases for the remaining four message bits are

$$\boxed{(C)0,\ \pi,\ \pi,\ \pi}$$

Q2.88.3 MCQ 1988 2M Ans: C ● ○ ○

Theory: For a uniformly distributed signal, the signal-to-quantization noise ratio depends on the quantization step size, which in turn depends on the number of quantization levels.

Given that the signal amplitude is uniformly distributed over

$$(-V, +V)$$

The probability density function is

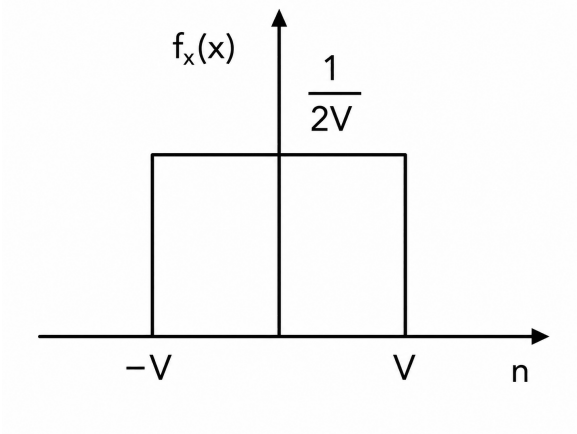


Fig. Solution Q2.88.3

$$f_X(x) = \frac{1}{2V}, \qquad -V \le x \le V$$

Signal power:

$$E[X^2] = \int_{-V}^{V} x^2 f_X(x)\, dx$$

$$= \frac{1}{2V} \int_{-V}^{V} x^2\, dx$$

$$= \frac{1}{2V} \left[\frac{x^3}{3}\right]_{-V}^{V}$$

$$= \frac{V^2}{3}$$

For uniform quantization,

$$N_q = \frac{\Delta^2}{12}$$

where Δ is the quantization step size.
Since

$$\Delta = \frac{\text{Peak-to-Peak Value}}{\text{Number of Levels}} = \frac{2V}{L}$$

where L is the number of quantization levels,

$$N_q = \frac{1}{12}\left(\frac{2V}{L}\right)^2 = \frac{V^2}{3L^2}$$

Therefore,

$$\frac{S}{N_q} = \frac{\frac{V^2}{3}}{\frac{V^2}{3L^2}} = L^2$$

Hence, the signal-to-quantization noise ratio depends only on the number of quantization levels.

$$\boxed{\frac{S}{N_q} = L^2}$$

(C) Number of quantization levels

Q2.88.4 MCQ 1988 2M Ans: B ●○○

Theory: Since each bit transmission is independent with error probability p, the number of errors in 8 transmitted bits follows a binomial distribution.
Let

$$X = \text{number of bit errors in 8 bits}$$

Then

$$X \sim \text{Binomial}(8, p)$$

We require the probability of at most one error:

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

Probability of zero errors:

$$P(X = 0) = \binom{8}{0} p^0 (1-p)^8 = (1-p)^8$$

Probability of exactly one error:

$$P(X = 1) = \binom{8}{1} p^1 (1-p)^7 = 8p(1-p)^7$$

Therefore,

$$P(X \leq 1) = (1-p)^8 + 8p(1-p)^7$$

$$\boxed{(B) P(X \leq 1) = (1-p)^8 + 8p(1-p)^7}$$

Q2.87.1 MCQ 1987 2M Ans: C ● ○ ○

Theory: Companding allocates finer quantization levels to small-amplitude signals, thereby improving the sigsonal-to-quantization noise ratio for weak signals.

In uniform quantization, the quantization step size remains constant for all amplitudes.

Hence, low-amplitude signals experience relatively larger quantization error compared to high-amplitude signals.

Companding compresses the dynamic range before quantization and expands it after reception, effectively reducing the quantization step size for small-amplitude signals.

Therefore, the improvement in signal-to-quantization noise ratio is obtained mainly for lower-amplitude signals.

$$\boxed{\text{(C) Lower amplitudes only}}$$

SOLUTIONS

III

Information Theory

Topics

1. Entropy and Mutual Information
2. Channel Capacity Theorem

Unit Snapshot*

Stream: EC	**Questions:** 24	**MCQ:** 10	**MSQ:** 2	**NAT:** 12
1M: 8	**2M:** 16	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–5M(2025)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q3.26.1 MCQ | 2026 | 2M | Ans: C | ● ● ○

Theory: Shannon's capacity for an AWGN channel is given by

$$C = B\log_2(1 + \text{SNR})$$

where B is the channel bandwidth and SNR is the signal-to-noise ratio.

Given,

$$B = 1\,\text{MHz} = 10^6\,\text{Hz}$$

$$kT = -174\,\text{dBm/Hz}$$

$$S = -80\,\text{dBm}$$

Calculating the noise power

$$N = kT + 10\log_{10}(B)$$

$$N = -174 + 10\log_{10}(10^6)$$

$$N = -174 + 60 = -114\,\text{dBm}$$

Calculating the signal-to-noise ratio

$$\text{SNR}_{\text{dB}} = S - N$$

$$= -80 - (-114) = 34\,\text{dB}$$

Converting into linear scale,

$$\text{SNR} = 10^{34/10} = 10^{3.4} \approx 2512$$

Calculating the Shannon capacity

$$C = B\log_2(1 + \text{SNR})$$

$$= 10^6\log_2(2513)$$

$$= 10^6\left(\frac{\ln(2513)}{\ln(2)}\right)$$

$$\approx 10^6(11.3)$$

$$C \approx 11.3 \times 10^6\,\text{bps}$$

$$C \approx 11.3\,\text{Mbps} = 11.3B$$

Hence,

$$\boxed{\text{(C) } C > 3B}$$

Mistakes to be avoided

- Do not use -174 dBm/Hz directly as noise power. It is a noise power spectral density and must be multiplied by bandwidth.
- Remember that
$$1\,\text{MHz} = 10^6\,\text{Hz}.$$
- Convert SNR from dB to linear scale before applying Shannon's formula:
$$\text{SNR} = 10^{\text{SNR}_{\text{dB}}/10}.$$
- Use
$$C = B\log_2(1 + \text{SNR}),$$
not $\log_{10}$.
- Capacity is measured in bits/s, whereas bandwidth is measured in Hz.

Q3.25.1 NAT | 2025 | 2M | Ans: 0 | ● ● ○

Theory: Mutual information measures the amount of information shared between two random variables and is given by

$$I(X;Y) = \sum_x \sum_y P(x,y)\log_2\left(\frac{P(x,y)}{P(x)P(y)}\right).$$

If

$$P(x,y) = P(x)P(y),$$

then X and Y are independent and

$$I(X;Y) = 0.$$

Calculating the marginal probabilities,

$$P(X=0) = 0.06 + 0.14 = 0.20$$

$$P(X=1) = 0.24 + 0.56 = 0.80$$

$$P(Y=0) = 0.06 + 0.24 = 0.30$$

$$P(Y=1) = 0.14 + 0.56 = 0.70$$

Checking each joint probability,

$$P(X=0,Y=0) = 0.06 = (0.20)(0.30) = P(X=0)P(Y=0)$$

$$P(X=0,Y=1) = 0.14 = (0.20)(0.70) = P(X=0)P(Y=1)$$

$$P(X=1,Y=0) = 0.24 = (0.80)(0.30) = P(X=1)P(Y=0)$$

$$P(X=1,Y=1)=0.56=(0.80)(0.70)=P(X=1)P(Y=1)$$

Hence,

$$P(x,y)=P(x)P(y)$$

for all possible values of x and y.
Therefore, X and Y are independent random variables.
For independent random variables,

$$I(X;Y)=0.$$

$$\boxed{I(X;Y)=0.00 \text{ bits}}$$

Mistakes to be avoided

- Do not directly apply the mutual information formula before checking whether the variables are independent.
- Remember that if
$$P(x,y)=P(x)P(y)$$
for all (x,y), then
$$I(X;Y)=0.$$
- Mutual information is always non-negative:
$$I(X;Y)\geq 0.$$

Q3.25.2 MSQ 2025 2M Ans: B,C ●●○

Theory: For a discrete random variable, entropy is maximum when all outcomes are equally likely.

The entropy of X is

$$g(\alpha)=-\alpha\log_2\alpha-(1-2\alpha)\log_2(1-2\alpha)-\alpha\log_2\alpha$$

$$g(\alpha)=-2\alpha\log_2\alpha-(1-2\alpha)\log_2(1-2\alpha).$$

Since X has three possible outcomes, entropy is maximized when

$$P(X=-1)=P(X=0)=P(X=1).$$

Using

$$P(X=-1)=P(X=1)=\alpha,$$

and

$$P(X=0)=1-2\alpha,$$

we get

$$\alpha=1-2\alpha.$$

Hence,

$$3\alpha = 1$$

$$\alpha = \frac{1}{3}.$$

Therefore, $g(\alpha)$ attains its maximum value at

$$\alpha = \frac{1}{3}.$$

Now,

$$0.3 < \frac{1}{3} < 0.4,$$

and 0.3 is closer to $\frac{1}{3}$ than 0.4.
Hence,

$$g(0.3) > g(0.4).$$

Also,

$$0.25 < 0.3 < \frac{1}{3},$$

and entropy increases as α moves toward $\frac{1}{3}$.
Therefore,

$$g(0.3) > g(0.25).$$

Thus

(B) g(0.3)>g(0.4) and (C) g(0.3)>g(0.25)are true

Q3.25.3 MCQ 2025 1M Ans: A ●○○

Theory: For an AWGN channel, Shannon's capacity depends on both the bandwidth and the signal-to-noise ratio (SNR). Increasing bandwidth increases capacity, but at the same time reduces SNR per unit bandwidth when the transmit power is fixed.

The Shannon capacity of an AWGN channel is

$$C = W \log_2\left(1 + \frac{P_{av}}{WN_0}\right) \quad \text{bits/s.}$$

As $W \to 0$,

$$C \to 0,$$

since the available transmission bandwidth tends to zero.
For very large bandwidth,

$$\frac{P_{av}}{WN_0} \to 0.$$

Using the approximation

$$\log_2(1 + x) \approx \frac{x}{\ln 2}, \qquad x \ll 1,$$

we obtain

$$C \approx W\left(\frac{P_{av}}{WN_0 \ln 2}\right) = \frac{P_{av}}{N_0 \ln 2}.$$

Thus,

$$\lim_{W\to\infty} C = \frac{P_{av}}{N_0 \ln 2},$$

which is a finite constant.
Therefore, capacity increases with bandwidth initially but gradually approaches a constant value as bandwidth becomes very large.
Hence, the C-versus-W curve is an increasing concave curve that saturates to a finite limit.

$$\boxed{\text{(A)}}$$

Mistakes to be avoided

- Do not assume that capacity increases linearly with bandwidth.
- Remember that increasing W decreases the SNR term
$$\frac{P_{av}}{WN_0}.$$
- Do not conclude that capacity grows without bound when P_{av} is fixed.
- For large W, use
$$\log_2(1+x) \approx \frac{x}{\ln 2}$$
only when
$$x \ll 1.$$
- Always check the limiting behavior at both
$$W \to 0$$
and
$$W \to \infty.$$

Q3.24.1 NAT 2024 1M Ans: 4 ● ○ ○

Theory: The entropy of a source is maximum when all symbols are equally likely.

For an alphabet of size M, the maximum entropy is

$$H_{\max} = \log_2 M.$$

Substituting $M = 16$,

$$H_{\max} = \log_2 16 = \log_2(2^4) = 4 \text{ bits}.$$

$$\boxed{H_{\max} = 4 \text{ bits}}$$

Mistakes to be avoided

- Maximum entropy occurs only when all symbols are equiprobable.
- Use base-2 logarithm when entropy is required in bits.

- Do not confuse alphabet size M with the number of transmitted symbols.
- Remember:

$$\log_2(2^n) = n.$$

Q3.23.1 NAT 2023 2M Ans: 1.98 ● ○ ○

Theory: The minimum average number of yes/no questions required to identify a symbol is equal to the entropy of the source.

The entropy is

$$H(X) = \sum_i P(x_i) \log_2\left(\frac{1}{P(x_i)}\right).$$

Substituting the given probabilities,

$$H(X) = \frac{1}{2}\log_2 2 + \frac{1}{4}\log_2 4 + \frac{1}{8}\log_2 8 + \frac{1}{16}\log_2 16$$

$$+\frac{1}{32}\log_2 32 + \frac{1}{64}\log_2 64 + \frac{1}{128}\log_2 128 + \frac{1}{128}\log_2 128.$$

Using

$$\log_2 2 = 1, \quad \log_2 4 = 2, \quad \log_2 8 = 3, \quad \ldots, \quad \log_2 128 = 7,$$

we get

$$H(X) = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + \frac{1}{32}(5) + \frac{1}{64}(6) + 2\left(\frac{1}{128}(7)\right).$$

$$H(X) = 0.5 + 0.5 + 0.375 + 0.25 + 0.15625 + 0.09375 + 0.109375$$

$$H(X) = 1.984375.$$

Therefore, the minimum average number of yes/no questions is

$$\boxed{1.98}$$

Mistakes to be avoided

- Round only at the final step.

Q3.22.1 NAT 2022 2M Ans: 1 ● ● ●

Theory: Channel capacity is the maximum mutual information between the input and output. For a strongly symmetric channel, capacity is obtained using a uniform input distribution.

The channel transition matrix is

$$P(Y|X) = \begin{bmatrix} 1-\alpha & \alpha & 0 \\ 0 & 1-\alpha & \alpha \\ \alpha & 0 & 1-\alpha \end{bmatrix}.$$

Since each row is a permutation of the others and all column sums are equal, the channel is strongly symmetric.

Therefore,

$$C = \log_2 3 - H\left(\frac{Y}{X}\right).$$

The conditional entropy of any row is

$$H\left(\frac{Y}{X}\right) = -(1-\alpha)\log_2(1-\alpha) - \alpha\log_2\alpha.$$

Hence,

$$C = \log_2 3 + (1-\alpha)\log_2(1-\alpha) + \alpha\log_2\alpha.$$

$$H\left(\frac{Y}{X}\right) = -\alpha\log_2\alpha - (1-\alpha)\log_2(1-\alpha)$$

$$C = \log_2 3 + \alpha\log_2\alpha + (1-\alpha)\log_2(1-\alpha)$$

$$C = f(\alpha)$$

For maximum/minimum points,

$$\frac{dC}{d\alpha} = 0$$

$$\frac{d}{d\alpha}\left[\alpha\log_2\alpha + (1-\alpha)\log_2(1-\alpha)\right] = 0$$

Using

$$\frac{d}{d\alpha}(\alpha\ln\alpha) = \ln\alpha + 1,$$

and

$$\frac{d}{d\alpha}((1-\alpha)\ln(1-\alpha)) = -\ln(1-\alpha) - 1,$$

we get

$$\ln\alpha + 1 - \ln(1-\alpha) - 1 = 0$$

$$\ln\alpha - \ln(1-\alpha) = 0$$

$$\ln\left(\frac{\alpha}{1-\alpha}\right) = 0$$

$$\frac{\alpha}{1-\alpha} = 1$$

$$\alpha = 1 - \alpha$$

$$2\alpha = 1$$

$$\boxed{\alpha = \frac{1}{2}}$$

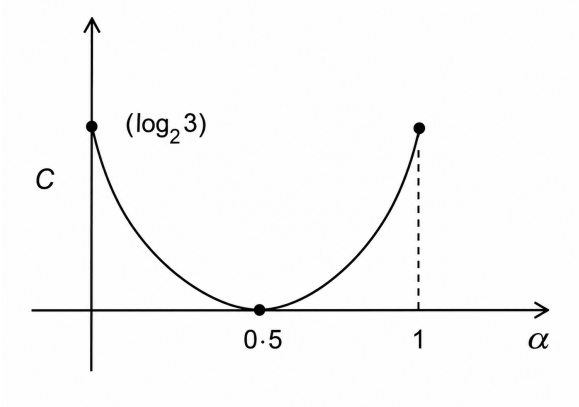


Fig. Solution Q3.22.1

Thus, $\alpha = \frac{1}{2}$ is the stationary point and corresponds to the minimum value of C.
Since

$$\alpha \in [0.25, 1],$$

the maximum capacity occurs at the boundary point

$$\boxed{\alpha = 1}.$$

Q3.22.2 MSQ 2022 1M Ans: A,B,D ● ● ○

Theory: If $Y = g(X)$ is a one-to-one transformation of a discrete random variable X, then entropy is preserved:

$$H(Y) = H(X).$$

Also, for a discrete random variable having K possible values,

$$H(X) \leq \log_2 K,$$

with equality when all outcomes are equiprobable.

Option (A): Since X can take at most K distinct values,

$$H(X) \leq \log_2 K.$$

Hence, option (A) is true.

Option (B): The transformation

$$Y = 2X$$

is one-to-one.
Therefore,

$$H(2X) = H(X).$$

Hence,

$$H(X) \leq H(2X)$$

is true.

Option (C): The transformation

$$Y = X^2$$

need not be one-to-one.
Consider

$$P(X = -1) = \frac{1}{4}, \qquad P(X = 0) = \frac{1}{2}, \qquad P(X = 1) = \frac{1}{4}.$$

Then

$$X^2 = \begin{cases} 0, & X = 0, \\ 1, & X = \pm 1. \end{cases}$$

Thus,

$$P(Y = 0) = \frac{1}{2}, \qquad P(Y = 1) = \frac{1}{2}.$$

Hence,

$$H(X^2) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1.$$

But

$$H(X) = -\frac{1}{4}\log_2\frac{1}{4} - \frac{1}{2}\log_2\frac{1}{2} - \frac{1}{4}\log_2\frac{1}{4} = 1.5.$$

Therefore,

$$H(X) > H(X^2),$$

so option (C) is not necessarily true.

Option (D): The transformation

$$Y = 2^X$$

is one-to-one.
Therefore,

$$H(2^X) = H(X).$$

Hence,

$$H(X) \le H(2^X)$$

is true.

(A), (B) and (D) are correct

Mistakes to be avoided

- Do not assume that every function of X preserves entropy.
- One-to-one transformations preserve entropy:

$$H(g(X)) = H(X).$$

- Many-to-one transformations can reduce entropy.

- The mapping
$$X \mapsto X^2$$
is not one-to-one because
$$(-x)^2 = x^2.$$
- Maximum entropy for K outcomes is
$$\log_2 K,$$

Q3.17.1 MCQ | 2017 | 2M | Ans: C | ● ○ ○

Theory: A channel is called *useless* (or a zero-capacity channel) if the output is statistically independent of the input, i.e.,

$$I(X;Y) = 0$$

for every possible input distribution.

From the transition diagram,

$$P(Y|X) = \begin{bmatrix} 0.25 & 0.75 \\ 0.25 & 0.75 \end{bmatrix}.$$

Let

$$P(X) = \begin{bmatrix} \alpha & 1-\alpha \end{bmatrix}, \qquad 0 \le \alpha \le 1.$$

The output distribution is

$$P(Y) = P(X)P(Y|X)$$

$$= \begin{bmatrix} \alpha & 1-\alpha \end{bmatrix} \begin{bmatrix} 0.25 & 0.75 \\ 0.25 & 0.75 \end{bmatrix} = \begin{bmatrix} 0.25 & 0.75 \end{bmatrix}.$$

Observe that $P(Y)$ is independent of α.
Also, both rows of $P(Y|X)$ are identical:

$$P(Y|X=0) = P(Y|X=1).$$

Hence, knowing the output provides no information about the input.
Therefore,

$$P(X|Y) = P(X),$$

which implies

$$H(X|Y) = H(X).$$

Thus,

$$I(X;Y) = H(X) - H(X|Y) = H(X) - H(X) = 0.$$

Since the mutual information is zero for every input distribution,

$$C = \max I(X;Y) = 0.$$

(C) Useless

Key Points

- **Noiseless Channel:**

  A channel is noiseless if every output uniquely identifies the transmitted input.

  $$H(X|Y) = 0$$

  and hence

  $$I(X;Y) = H(X).$$

  The receiver can recover the transmitted symbol without any ambiguity.

- **Lossless Channel:**

  A channel is lossless if every input can be uniquely determined from the received output.

  $$H(X|Y) = 0$$

  Equivalently,

  $$I(X;Y) = H(X).$$

  Note: Every noiseless channel is lossless, but the terms are often used differently in channel classification problems.

- **Deterministic Channel:**

  A channel is deterministic if each input produces exactly one output with probability 1.

  $$H(Y|X) = 0$$

  The output is completely determined once the input is known.

  $$P(y_j|x_i) \in \{0, 1\}$$

  for all i, j.

- **Useless Channel (Zero-Capacity Channel):**

  A channel is useless if the output provides no information about the input.

  $$I(X;Y) = 0$$

  for every input distribution.

  Equivalently,

  $$P(y|x) = P(y)$$

  for all inputs x.

  Thus,

$$C = \max I(X;Y) = 0.$$

The output and input are statistically independent.

- **Do Not Confuse Deterministic and Noiseless Channels:**
  A deterministic channel satisfies

$$H(Y|X) = 0,$$

  whereas a noiseless channel satisfies

$$H(X|Y) = 0.$$

  These are different conditions.

- **Quick Summary:**

Noiseless Channel: $H(X|Y) = 0$

Lossless Channel: $H(X|Y) = 0$

Deterministic Channel: $H(Y|X) = 0$

Useless Channel: $I(X;Y) = 0$

PrepFusion

Q3.17.2 MCQ 2017 1M Ans: C ●○○

Theory: The capacity of a Binary Symmetric Channel (BSC) is equal to one minus its binary entropy. As the crossover probability increases toward 0.5, uncertainty increases and capacity decreases.

For a Binary Symmetric Channel with crossover probability p,

$$C = 1 - H_b(p),$$

where

$$H_b(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

is the binary entropy function.
Therefore,

$$C = 1 + p \log_2 p + (1-p) \log_2 (1-p).$$

Evaluating at important points,

$$C(0) = 1,$$

since

$$H_b(0) = 0.$$

Also,

$$C\left(\frac{1}{2}\right) = 1 - H_b\left(\frac{1}{2}\right) = 1 - 1 = 0.$$

Further,

$$C(1) = 1,$$

since

$$H_b(1) = 0.$$

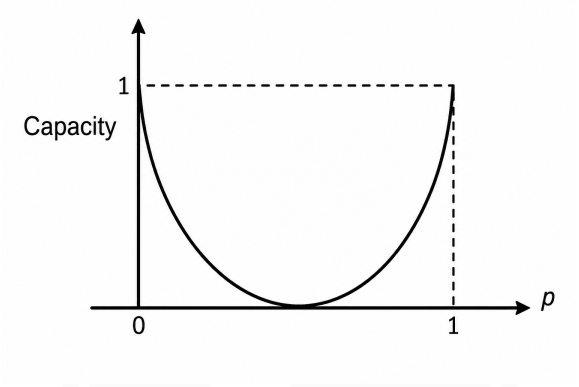


Fig. Solution Q3.17.2

Hence, the graph is a symmetric U-shaped curve with minimum at

$$p = \frac{1}{2}.$$

(C)

Mistakes to be avoided

- Do not use
$$C = H_b(p).$$
The correct expression is
$$C = 1 - H_b(p).$$
- Remember that a BSC with
$$p = \frac{1}{2}$$
has zero capacity because the output is independent of the input.
- Do not assume
$$p = 1$$
gives zero capacity. When $p = 1$, every bit is flipped, and the receiver can perfectly recover the input by inverting the output.
- The capacity curve is symmetric about
$$p = \frac{1}{2}.$$

Q3.17.3 NAT 2017 1M Ans: 0 ● ○ ○

Theory: Mutual information measures the amount of information one random variable provides about another.
For independent random variables,

$$P_{X_1,X_2}(x_1,x_2) = P_{X_1}(x_1)\,P_{X_2}(x_2).$$

Hence, the mutual information is zero:

$$I(X_1;X_2) = 0.$$

The given means and variances do not affect the answer because independence alone is sufficient.

$$\boxed{I(X_1;X_2) = 0 \text{ bits}}$$

Q3.16.1 NAT 2016 1M Ans: 360 to 363 ● ○ ○

Theory: Information rate is given by

$$R = H \times r,$$

where H is the entropy per sample and r is the sampling rate.
Since the signal is bandlimited to 100 Hz, the Nyquist sampling rate is

$$r = 2f_m = 2(100) = 200 \text{ samples/sec.}$$

Given

$$p_1 = p_4 = \frac{1}{8}, \qquad p_2 = p_3.$$

Using

$$p_1 + p_2 + p_3 + p_4 = 1,$$

we get

$$\frac{1}{8} + p_2 + p_3 + \frac{1}{8} = 1$$

$$p_2 + p_3 = \frac{3}{4}.$$

Since $p_2 = p_3$,

$$p_2 = p_3 = \frac{3}{8}.$$

Calculating the entropy per sample,

$$H = -\sum_{i=1}^{4} p_i \log_2 p_i$$

$$= -\left[\frac{1}{8}\log_2\frac{1}{8} + \frac{3}{8}\log_2\frac{3}{8} + \frac{3}{8}\log_2\frac{3}{8} + \frac{1}{8}\log_2\frac{1}{8}\right]$$

$$= \frac{1}{8}\log_2 8 + \frac{1}{8}\log_2 8 + \frac{3}{8}\log_2\frac{8}{3} + \frac{3}{8}\log_2\frac{8}{3}$$

$$H \approx 1.811 \text{ bits/sample.}$$

Therefore,

$$R = H \times r$$

$$R = 1.811 \times 200$$

$$R \approx 362.2 \text{ bits/sec.}$$

$$\boxed{R \approx 362.2 \text{ bits/sec}}$$

Q3.16.2 NAT 2016 2M Ans: 1.8 to 2.2 ● ○ ○

Theory: For a discrete memoryless source, entropy is

$$H = \sum_{i=0}^{\infty} P_i \log_2\left(\frac{1}{P_i}\right).$$

Given

$$P_i = \left(\frac{1}{2}\right)^{i+1}, \qquad i = 0, 1, 2, \ldots$$

Therefore,

$$H = \sum_{i=0}^{\infty} \left(\frac{1}{2}\right)^{i+1} \log_2\left(2^{i+1}\right).$$

Since

$$\log_2\left(2^{i+1}\right) = i + 1,$$

we get

$$H = \sum_{i=0}^{\infty} (i+1)\left(\frac{1}{2}\right)^{i+1}.$$

Let

$$H = \frac{1}{2} + 2\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right)^3 + \cdots \tag{1}$$

Multiplying by $\frac{1}{2}$,

$$\frac{H}{2} = \left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right)^3 + 3\left(\frac{1}{2}\right)^4 + \cdots \tag{2}$$

Subtracting (2) from (1),

$$\frac{H}{2} = \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots$$

The right-hand side is a geometric series with first term

$$a = \frac{1}{2}$$

and common ratio

$$r = \frac{1}{2}.$$

Hence,

$$\frac{H}{2} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1.$$

Therefore,

$$H = 2.$$

$$\boxed{H = 2 \text{ bits/symbol}}$$

Q3.16.3 NAT 2016 2M Ans: 0.8 to 0.82 ●○○

Theory: If the transmitted symbol is known, the uncertainty is only due to the possible received symbols. Hence, the required information is the conditional entropy

$$H(Y|x_0) = -\sum_j P(y_j|x_0) \log_2 P(y_j|x_0).$$

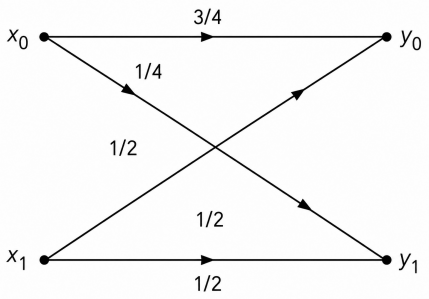


Fig. Solution Q3.16.3

Given

$$P(y_0|x_0) = \frac{3}{4}.$$

Therefore,

$$P(y_1|x_0) = 1 - \frac{3}{4} = \frac{1}{4}.$$

Calculating the conditional entropy,

$$H(Y|x_0) = -\left[\frac{3}{4}\log_2\left(\frac{3}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right].$$

Using

$$\log_2\left(\frac{3}{4}\right) = -\log_2\left(\frac{4}{3}\right), \qquad \log_2\left(\frac{1}{4}\right) = -2,$$

we obtain

$$H(Y|x_0) = \frac{3}{4}\log_2\left(\frac{4}{3}\right) + \frac{1}{4}\log_2(4).$$

$$H(Y|x_0) = \frac{3}{4}\log_2\left(\frac{4}{3}\right) + \frac{1}{2}.$$

$$H(Y|x_0) \approx 0.811 \text{ bits.}$$

$$\boxed{0.811 \text{ bits}}$$

Q3.14.1 MCQ | 2014 | 2M | Ans: A | ● ● ○

Theory: For an AWGN channel,

$$C = W\log_2\left(1 + \frac{P}{\sigma^2 W}\right).$$

When bandwidth tends to infinity, the SNR per unit bandwidth becomes very small. Hence, we use the standard limit

$$\lim_{x\to\infty} x\log_2\left(1 + \frac{1}{x}\right) = \log_2 e.$$

Calculating the infinite-bandwidth capacity,

$$C_\infty = \lim_{W\to\infty} W\log_2\left(1 + \frac{P}{\sigma^2 W}\right).$$

Rearranging,

$$C_\infty = \frac{P}{\sigma^2}\lim_{W\to\infty}\frac{\sigma^2 W}{P}\log_2\left(1 + \frac{P}{\sigma^2 W}\right).$$

Let

$$x = \frac{\sigma^2 W}{P}.$$

As $W \to \infty$,

$$x \to \infty.$$

Therefore,

$$C_\infty = \frac{P}{\sigma^2}\lim_{x\to\infty} x\log_2\left(1 + \frac{1}{x}\right).$$

Using the limit,

$$C_\infty = \frac{P}{\sigma^2}\log_2 e.$$

Given

$$\frac{P}{\sigma^2} = 1000,$$

$$C_\infty = 1000 \log_2 e.$$

Since

$$\log_2 e \approx 1.4427,$$

$$C_\infty \approx 1000(1.4427) = 1442.7 \text{ bps}.$$

$$\boxed{(A) C_\infty \approx 1.44 \text{ kbps}}$$

Mistakes to be avoided

- Do not substitute $W = \infty$ directly into

$$C = W \log_2\left(1 + \frac{P}{\sigma^2 W}\right).$$

  It results in the indeterminate form $\infty \times 0$.
- Infinite bandwidth does *not* imply infinite capacity for fixed power.
- For an AWGN channel with $W \to \infty$,

$$C_\infty = \frac{P}{\sigma^2} \log_2 e \approx 1.44 \frac{P}{\sigma^2}.$$

Q3.14.2 NAT 2014 2M Ans: 1.99 to 2.01 ● ○ ○

Theory: The number of tosses required to obtain the first Head follows a geometric distribution.
For a fair coin,

$$P(L = n) = \left(\frac{1}{2}\right)^n, \qquad n = 1, 2, 3, \ldots$$

The entropy is

$$H(L) = \sum_{n=1}^{\infty} P(L = n) \log_2\left(\frac{1}{P(L = n)}\right).$$

Substituting $P(L = n) = 2^{-n}$,

$$H(L) = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n \log_2(2^n).$$

Since

$$\log_2(2^n) = n,$$

$$H(L) = \sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Expanding,

$$H(L) = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \frac{4}{2^4} + \cdots \quad (1)$$

Multiplying by $\frac{1}{2}$,

$$\frac{H(L)}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + \frac{4}{2^5} + \cdots \quad (2)$$

Subtracting (2) from (1),

$$H(L) - \frac{H(L)}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots$$

Hence,

$$\frac{H(L)}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \cdots$$

The right-hand side is a geometric series with

$$a = \frac{1}{2}, \qquad r = \frac{1}{2}.$$

Therefore,

$$\frac{H(L)}{2} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1.$$

Thus,

$$\boxed{H(L) = 2 \text{ bits}}$$

Q3.14.3 NAT 2014 2M Ans: 0.80 ● ○ ○

Theory: For a received symbol, the probability that a particular symbol was transmitted is obtained using Bayes' theorem by combining prior probabilities and channel transition probabilities.

Let

$$P(X = 0) = P(X = 1) = \frac{1}{2}.$$

From the Z-channel,

$$P(Y = 0|X = 0) = 1,$$

$$P(Y = 0|X = 1) = 0.25.$$

Therefore,

$$P(X = 0, Y = 0) = P(Y = 0|X = 0)P(X = 0) = 1 \times \frac{1}{2} = 0.5.$$

Similarly,

$$P(X = 1, Y = 0) = P(Y = 0|X = 1)P(X = 1) = 0.25 \times \frac{1}{2} = 0.125.$$

Hence,

$$P(Y=0) = 0.5 + 0.125 = 0.625.$$

Using Bayes' theorem,

$$P(X=0|Y=0) = \frac{P(X=0, Y=0)}{P(Y=0)} = \frac{0.5}{0.625} = 0.8.$$

$$\boxed{P(X=0|Y=0) = 0.8}$$

Mistakes to be avoided

- Do not confuse $P(Y=0|X=0)$ with $P(X=0|Y=0)$.
- Always compute $P(Y=0)$ using total probability before applying Bayes' theorem.
- Remember that the inputs are equiprobable: $P(X=0) = P(X=1) = \frac{1}{2}$.
- Do not directly use the transition probability 1 as the final answer.

Q3.14.4 NAT | 2014 | 2M | Ans: 2 | ● ○ ○

Theory: A BSC with crossover probability 1/2 completely randomizes its output, making the output symbols equiprobable irrespective of the input distribution.

Given,

$$P(X=1) = \frac{1}{3}, \qquad P(X=0) = \frac{2}{3}.$$

Since the first BSC has crossover probability

$$p = \frac{1}{2},$$

we have

$$P(Y_1=1) = P(X=1)\cdot\frac{1}{2} + P(X=0)\cdot\frac{1}{2}.$$

Substituting,

$$P(Y_1=1) = \frac{1}{3}\cdot\frac{1}{2} + \frac{2}{3}\cdot\frac{1}{2} = \frac{1}{2}.$$

Hence,

$$P(Y_1=0) = \frac{1}{2}.$$

Therefore,

$$H(Y_1) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1 \text{ bit}.$$

Since the second BSC is also identical with crossover probability

$$p = \frac{1}{2},$$

its output is also equiprobable.
Thus,

$$P(Y_2=1) = P(Y_2=0) = \frac{1}{2}.$$

Hence,

$$H(Y_2) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1 \text{ bit}.$$

Therefore,

$$H(Y_1) + H(Y_2) = 1 + 1 = 2 \text{ bits}.$$

$$\boxed{H(Y_1) + H(Y_2) = 2 \text{ bits}}$$

Q3.14.5 NAT 2014 1M Ans: 0 ● ○ ○

Theory: The capacity of a Binary Symmetric Channel (BSC) is

$$C = 1 - H_b(p),$$

where p is the cross-over probability and

$$H_b(p) = -p \log_2 p - (1-p) \log_2 (1-p)$$

is the binary entropy function.
For

$$p = 0.5,$$

the binary entropy becomes

$$H_b(0.5) = -0.5 \log_2(0.5) - 0.5 \log_2(0.5).$$

Since

$$\log_2(0.5) = -1,$$

$$H_b(0.5) = -0.5(-1) - 0.5(-1) = 0.5 + 0.5 = 1.$$

Therefore,

$$C = 1 - H_b(0.5) = 1 - 1 = 0.$$

$$\boxed{C = 0 \text{ bits/channel use}}$$

Q3.12.1 MCQ 2012 1M Ans: D ● ○ ○

Theory: Entropy measures uncertainty. For a fixed total probability, entropy is maximum when the probabilities are as equal (uniform) as possible.
Suppose initially the first two symbols have equal probability:

$$p_1 = p_2 = p.$$

The encoder changes them to

$$p_1 = p + \varepsilon, \qquad p_2 = p - \varepsilon,$$

where $\varepsilon > 0$.
Notice that the total probability remains unchanged:

$$(p + \varepsilon) + (p - \varepsilon) = 2p.$$

However, the two probabilities are no longer equal.
To understand what happens to entropy, consider a simple example.
Let

$$p_1 = p_2 = 0.5.$$

Then the entropy contribution of these two symbols is

$$H_{\text{before}} = -0.5\log_2(0.5) - 0.5\log_2(0.5) = 1 \text{ bit.}$$

Now let

$$p_1 = 0.6, \qquad p_2 = 0.4.$$

Then

$$H_{\text{after}} = -0.6\log_2(0.6) - 0.4\log_2(0.4)$$

$$= 0.971 \text{ bits.}$$

Therefore,

$$H_{\text{after}} < H_{\text{before}}.$$

The uncertainty has reduced because one symbol has become more likely and the other less likely.
The same argument holds for any value of N. Making two equal probabilities unequal always makes the overall distribution less uniform, thereby reducing entropy.

(D) Entropy decreases

Q3.09.1 MCQ | 2009 | 2M | Ans: B | ● ○ ○

Theory: For an AWGN channel, Shannon capacity is

$$C = B\log_2(1 + \text{SNR}).$$

Since

$$\text{SNR} \gg 1,$$

we can use the approximation

$$1 + \text{SNR} \approx \text{SNR}.$$

Therefore,

$$C_1 = B\log_2(\text{SNR}).$$

When the SNR is doubled,

$$\text{SNR}' = 2\,\text{SNR}.$$

Hence,

$$C_2 = B\log_2(2\,\text{SNR}).$$

Using the logarithmic property

$$\log_2(ab) = \log_2 a + \log_2 b,$$

we obtain

$$C_2 = B\left[\log_2 2 + \log_2(\text{SNR})\right].$$

Since

$$\log_2 2 = 1,$$

$$C_2 = B + B\log_2(\text{SNR}).$$

But

$$C_1 = B\log_2(\text{SNR}),$$

therefore

$$\boxed{(B)C_2 \approx C_1 + B}$$

Mistakes to be avoided

- Capacity is proportional to the logarithm of SNR, not directly proportional to SNR:

  $$C = B\log_2(1 + \text{SNR}).$$

  Doubling SNR does **not** double capacity.
- For high SNR,

  $$1 + \text{SNR} \approx \text{SNR},$$

  which greatly simplifies the calculation.
- The result

  $$C_2 = C_1 + B$$

  is valid only under the approximation

  $$\text{SNR} \gg 1.$$

Q3.08.1 MCQ | 2008 | 2M | Ans: A | ● ○ ○

Theory: Since all n symbols are equally likely,

$$p_1 = p_2 = \cdots = p_n = \frac{1}{n}.$$

The entropy of a discrete memoryless source is

$$H = -\sum_{i=1}^{n} p_i \log_2 p_i.$$

Substituting

$$p_i = \frac{1}{n},$$

gives

$$H = -\sum_{i=1}^{n} \frac{1}{n} \log_2\left(\frac{1}{n}\right).$$

Since all n terms are identical,

$$H = -n\left(\frac{1}{n}\right) \log_2\left(\frac{1}{n}\right).$$

Therefore,

$$H = -\log_2\left(\frac{1}{n}\right).$$

Using

$$\log_2\left(\frac{1}{n}\right) = -\log_2 n,$$

we obtain

$$H = \log_2 n.$$

Hence, the entropy increases logarithmically with the number of symbols.

$$\boxed{H = \log_2 n}$$

Therefore,

$$\boxed{\text{(A) Entropy increases as } \log n}$$

Q3.90.1 MCQ | 1990 | 2M | Ans: B | ● ○ ○

Theory: Maximum entropy occurs when all possible intensity levels are equally likely.
Since each pixel can take one of

$$8$$

intensity levels,

$$H_{\text{pixel}} = \log_2 8 = 3 \text{ bits/pixel}.$$

Calculating the total number of pixels,

$$N = 512 \times 512 = 262144.$$

Since each pixel contributes a maximum of

$$3$$

bits,

$$H_{\max} = N \times H_{\text{pixel}}.$$

Substituting the values,

$$H_{\max} = 262144 \times 3.$$

Hence,

$$\boxed{(B) H_{\max} = 786432 \text{ bits}}.$$

Q3.89.1 MCQ | 1989 | 2M | Ans: B | ● ○ ○

Theory: The efficiency of a source code is defined as

$$\eta = \frac{H}{L},$$

where

$$H = \text{source entropy}, \qquad L = \text{average codeword length}.$$

Calculating the source entropy,

$$H = -\sum_i p_i \log_2 p_i.$$

Substituting the given probabilities,

$$H = -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right].$$

Using

$$\log_2\frac{1}{2} = -1, \qquad \log_2\frac{1}{4} = -2, \qquad \log_2\frac{1}{8} = -3,$$

we obtain

$$H = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3).$$

$$H = \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{3}{8}.$$

$$H = \frac{7}{4} = 1.75 \text{ bits/symbol}.$$

Given

$$L = 2 \text{ bits/symbol},$$

therefore

$$\eta = \frac{H}{L} = \frac{1.75}{2} = 0.875.$$

Hence,

$$\boxed{(B) \eta = \frac{7}{8}}.$$

SOLUTIONS

IV

Analog Communications

Topics

1. Amplitude Modulation and Demodulation
2. Spectra of AM Signals
3. Angle Modulation - FM and PM
4. Demodulation of FM and PM
5. Spectra of FM Signals
6. Superheterodyne Receivers

Unit Snapshot*

Stream: EC	**Questions:** 84	**MCQ:** 63	**MSQ:** 3	**NAT:** 18
1M: 36	**2M:** 48	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–5M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MSQ | 2026 | 2M | Ans: A,B,C | ● ● ○

Theory: The general narrowband representation of a bandpass signal is

$$x(t) = A(t)\cos(2\pi f_c t + \theta(t)),$$

where $A(t)$ is the envelope and $\theta(t)$ is the phase.
Since

$$f_c \gg f_m,$$

the signal is a narrowband bandpass signal.

To verify option (A), choose

$$A(t) = A_c$$

(constant amplitude) and let

$$\theta(t) \in \{0, \pi\}$$

for BPSK.
Then

$$x(t) = A_c \cos(2\pi f_c t + \theta(t)),$$

which is a PSK signal.
Hence, option **A** is correct.

To verify option (B), choose

$$A(t) = A_c[1 + k_a m(t)]$$

and

$$\theta(t) = 0.$$

Then

$$x(t) = A_c[1 + k_a m(t)]\cos(2\pi f_c t),$$

which is the standard AM signal.
Hence, option **B** is correct.

To verify option (C), use the identity

$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$

Thus,

$$x(t) = A(t)\cos\theta(t)\cos(2\pi f_c t) - A(t)\sin\theta(t)\sin(2\pi f_c t).$$

Define

$$x_I(t) = A(t)\cos\theta(t),$$

$$x_Q(t) = A(t)\sin\theta(t).$$

Then

$$x(t) = x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t).$$

If $x_I(t)$ and $x_Q(t)$ are low-pass Gaussian random processes, then $x(t)$ becomes a band-limited Gaussian noise process.
Hence, option **C** is correct.

Now consider option (D).
For narrowband FM,

$$x(t) = A_c \cos\left[2\pi f_c t + k_f \int m(\tau)\, d\tau\right].$$

Comparing with

$$x(t) = A(t)\cos(2\pi f_c t + \theta(t)),$$

choose

$$A(t) = A_c,$$

and

$$\theta(t) = k_f \int m(\tau)\, d\tau.$$

Then the given signal becomes a narrowband FM signal.
Therefore the statement

"$x(t)$ never represents a narrowband FM signal"

is false.
Hence, option **D** is incorrect.

(A) $x(t)$ represents a PSK modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
(B) $x(t)$ represents an amplitude modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
(C) $x(t)$ represents a band-limited Gaussian noise process.

Mistakes to be avoided

- Do not associate

  $$x(t) = A(t)\cos(2\pi f_c t + \theta(t))$$

  with only one modulation scheme. It is a very general narrowband representation.
- PSK is obtained when the amplitude is constant and the phase carries information.

  $$A(t) = A_c.$$

- AM is obtained when the phase is fixed and the envelope carries information.

  $$\theta(t) = 0.$$

- FM and PM are both angle-modulation schemes and can be represented through a suitable choice of

  $$\theta(t).$$

- A bandpass Gaussian noise process can be expressed using Gaussian in-phase and quadrature components:

$$x(t) = x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t).$$

Q4.25.1 MSQ 2025 1M Ans: A,D ● ○ ○

Theory: An envelope detector works correctly only when the envelope of the transmitted signal uniquely represents the message signal and does not suffer from phase reversals. For the DSB-SC signal,

$$f_{\text{DSB}}(t) = A_c m(t)\cos(2\pi f_c t),$$

the envelope is

$$E_{\text{DSB}}(t) = A_c|m(t)|.$$

Hence, if $m(t) > 0$ for all t, then $|m(t)| = m(t)$ and an envelope detector can recover $m(t)$. Therefore, statement (i) is TRUE. For the AM signal,

$$f_{\text{AM}}(t) = A_c(1 + \mu m(t))\cos(2\pi f_c t),$$

an envelope detector works when there is no overmodulation, i.e.,

$$1 + \mu m(t) > 0 \quad \text{for all } t.$$

It is not necessary that $m(t) > 0$ for all t. Thus, statement (ii) is FALSE.

(i) is True
(ii) is FALSE

Mistakes to be avoided

- Do not assume that a positive message signal automatically allows envelope detection in DSB-SC.
- For standard AM, the correct condition is

$$1 + \mu m(t) > 0,$$

not

$$m(t) > 0.$$

- Confusing "positive message" with "positive envelope" is a common mistake.
- Overmodulation occurs when

$$1 + \mu m(t) < 0,$$

causing envelope distortion and failure of envelope detection.

Q4.25.2 MCQ | 2025 | 1M | Ans: A | ● ○ ○

Theory: For an FM signal of the form

$$s(t) = A_c \cos\big(2\pi f_c t + \phi(t)\big),$$

the instantaneous frequency deviation is

$$\Delta f(t) = \frac{1}{2\pi}\frac{d\phi(t)}{dt}.$$

The peak frequency deviation is the maximum absolute value of $\Delta f(t)$.

Comparing with the given signal,

$$\phi(t) = 3\sin(2\pi f_1 t) + 4\sin(6\pi f_1 t).$$

Differentiating,

$$\frac{d\phi(t)}{dt} = 3(2\pi f_1)\cos(2\pi f_1 t) + 4(6\pi f_1)\cos(6\pi f_1 t),$$

$$\frac{d\phi(t)}{dt} = 6\pi f_1 \cos(2\pi f_1 t) + 24\pi f_1 \cos(6\pi f_1 t).$$

Therefore,

$$\Delta f(t) = \frac{1}{2\pi}\left[6\pi f_1 \cos(2\pi f_1 t) + 24\pi f_1 \cos(6\pi f_1 t)\right].$$

Hence,

$$\Delta f(t) = 3f_1 \cos(2\pi f_1 t) + 12 f_1 \cos(6\pi f_1 t).$$

Since

$$|\cos(\cdot)| \le 1,$$

the maximum possible value of $|\Delta f(t)|$ is

$$\Delta f_{\text{peak}} = 3f_1 + 12f_1.$$

Thus,

$$\boxed{(A)\Delta f_{\text{peak}} = 15 f_1}$$

Mistakes to be avoided

- For

  $$\sin(6\pi f_1 t),$$

  the angular frequency is

  $$6\pi f_1 = 2\pi(3f_1),$$

  so its frequency is $3f_1$, not $6f_1$.

Q4.24.1 NAT | 2024 | 1M | Ans: 2.5 | ● ● ○

Theory: The given signal is an AM wave consisting of one carrier and two sidebands.
For a sinusoid

$$V\cos(\omega t),$$

across a 1 Ω resistance, the power is

$$P = \frac{V^2}{2}.$$

Calculating the carrier amplitude:

$$P_c = \frac{A^2}{2} = 50.$$

Hence,

$$A^2 = 100,$$

$$A = 10.$$

Calculating the sideband power:
Each sideband has amplitude B, therefore

$$P_{\text{USB}} = \frac{B^2}{2},$$

$$P_{\text{LSB}} = \frac{B^2}{2}.$$

Thus,

$$P_{\text{SB}} = P_{\text{USB}} + P_{\text{LSB}} = B^2.$$

The total transmitted power is

$$P_t = P_c + P_{\text{SB}} = 50 + B^2.$$

Given,

$$\frac{P_{\text{SB}}}{P_t} = \frac{1}{9}.$$

Substituting,

$$\frac{B^2}{50 + B^2} = \frac{1}{9}.$$

Cross-multiplying,

$$9B^2 = 50 + B^2.$$

$$8B^2 = 50.$$

$$B^2 = \frac{25}{4}.$$

$$B = \frac{5}{2}.$$

Therefore,

$$\boxed{B = 2.50 \text{ V}}$$

Q4.23.1 MCQ 2023 2M Ans: B ●●○

Theory: When an FM signal passes through a squaring device,

$$x^2(t) = A^2 \cos^2 \theta(t) = \frac{A^2}{2} + \frac{A^2}{2} \cos\big(2\theta(t)\big),$$

where

$$\theta(t) = \omega_c t + k_f \int_{-\infty}^{t} m(\lambda)\, d\lambda.$$

Thus, squaring produces

- a DC component,
- an FM signal centered at $2\omega_c$.

The frequency multiplier effect doubles the carrier frequency and frequency deviation.

Let the peak frequency deviation of $x(t)$ be Δf.
Using Carson's rule,

$$B_T = 2(\Delta f + W).$$

Hence,

$$\Delta f = \frac{B_T}{2} - W.$$

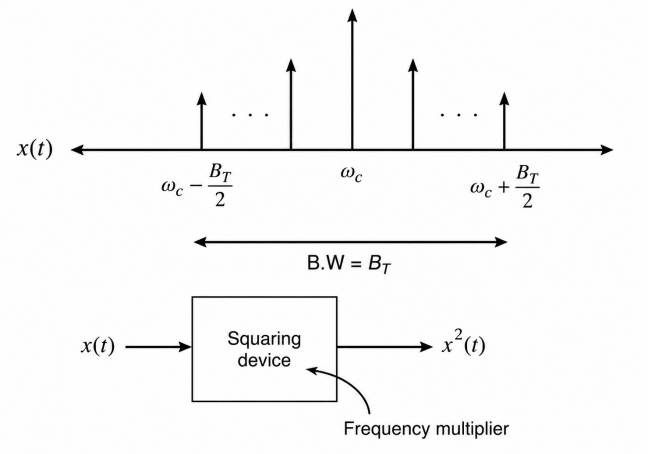


Fig. Solution Q4.23.1

The spectrum of $x(t)$ occupies

$$\omega_c - \frac{B_T}{2} \quad \text{to} \quad \omega_c + \frac{B_T}{2}.$$

After squaring,

$$\omega_c' = 2\omega_c,$$

$$\Delta f' = 2\Delta f.$$

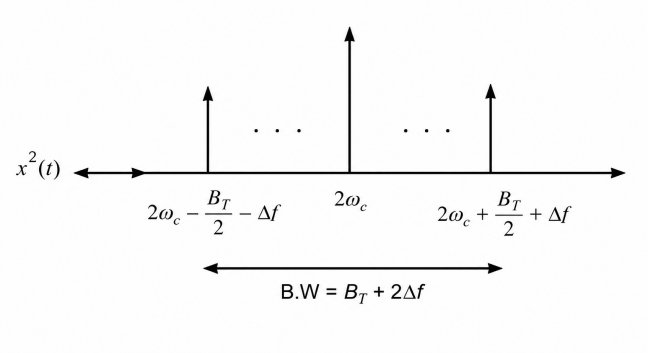


Fig. Solution Q4.23.1

Therefore,

$$BW[x^2(t)] = 2(\Delta f' + W) = 2(2\Delta f + W) = B_T + 2\Delta f.$$

Substituting

$$\Delta f = \frac{B_T}{2} - W,$$

gives

$$BW[x^2(t)] = B_T + 2\left(\frac{B_T}{2} - W\right) = 2B_T - 2W.$$

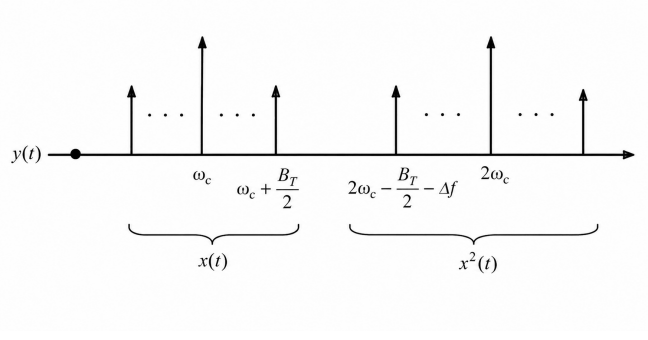


Fig. Solution Q4.23.1

To recover $x(t)$ using a bandpass filter, the spectrum around ω_c and the spectrum around $2\omega_c$ must not overlap. Hence,

$$2\omega_c - \left(\frac{B_T}{2} + \Delta f\right) > \omega_c + \frac{B_T}{2}.$$

Substituting

$$\Delta f = \frac{B_T}{2} - W,$$

$$2\omega_c - B_T + W > \omega_c + \frac{B_T}{2}.$$

Therefore,

$$\omega_c > \frac{3}{2}B_T - W.$$

Since

$$W \ll B_T$$

for a practical wideband FM signal,

$$(\omega_c)_{\min} \approx \frac{3}{2}B_T.$$

Thus,

$$\boxed{(B)(\omega_c)_{\min} = \frac{3}{2}B_T}$$

Q4.22.1 MSQ | 2022 | 2M | Ans: A | ● ○ ○

Theory: For perfect compensation,

$$H_{de}(\omega) = \frac{1}{H_{pe}(\omega)}.$$

The given pre-emphasis filter is

$$H_{pe}(\omega) = 1 + j\frac{\omega}{\omega_0}.$$

Hence,

$$\left|H_{pe}(\omega)\right|^2 = 1 + \left(\frac{\omega}{\omega_0}\right)^2.$$

For the RC low-pass network,

$$H_{de}(\omega) = \frac{1}{1 + j\omega RC}.$$

Therefore,

$$|H_{de}(\omega)|^2 = \frac{1}{1 + (\omega RC)^2}.$$

Since the de-emphasis filter must be the inverse of the pre-emphasis filter,

$$\left|H_{pe}(\omega)\right|^2 = \frac{1}{|H_{de}(\omega)|^2}.$$

Substituting the expressions,

$$1 + \left(\frac{\omega}{\omega_0}\right)^2 = 1 + (\omega RC)^2.$$

Thus,

$$\frac{1}{\omega_0} = RC.$$

Given

$$\omega_0 = 10^4 \text{ rad/s},$$

$$RC = \frac{1}{10^4} = 10^{-4} \text{ s}.$$

Checking the options:

$$(1\,\text{k}\Omega)(0.1\,\mu\text{F}) = 10^3 \times 10^{-7} = 10^{-4} \text{ s}$$

which satisfies the condition.

$$\boxed{(A) R = 1\,\text{k}\Omega, \quad C = 0.1\,\mu\text{F}}$$

Q4.21.1 NAT | 2021 | 2M | Ans: 1400 | ● ○ ○

Theory: In a superheterodyne receiver,

$$f_{IF} = |f_{LO} - f_s|$$

where f_s is the desired signal frequency. Given,

$$f_s = 600\,\text{kHz}$$

$$f_{LO} = 1000\,\text{kHz}$$

Calculating the intermediate frequency,

$$f_{IF} = 1000 - 600$$

$$f_{IF} = 400\,\text{kHz}$$

The image frequency is located on the opposite side of the local oscillator at the same separation f_{IF}.
Hence,

$$f_{si} = f_{LO} + f_{IF}$$

$$f_{si} = 1000 + 400$$

$$f_{si} = 1400\,\text{kHz}$$

Equivalently,

$$f_{si} = f_s + 2f_{IF}$$

$$f_{si} = 600 + 2(400)$$

$$f_{si} = 1400\,\text{kHz}$$

Therefore,

$$\boxed{f_{si} = 1400\,\text{kHz}}$$

Q4.21.2 NAT | 2021 | 2M | Ans: 1011313.7 | ● ● ○

Given,

$$V_{\text{rms}} = 4\ \text{V}$$

$$f_m = 1\ \text{kHz}$$

$$K_p = 2\ \text{rad/V}$$

$$f_c = 10^6\ \text{Hz}$$

The peak value of the message signal is

$$V_m = \sqrt{2}\,V_{\text{rms}}$$

$$V_m = 4\sqrt{2}\ \text{V}$$

Hence,

$$m(t) = 4\sqrt{2}\sin(2\pi \times 10^3 t).$$

For phase modulation,

$$s(t) = A_c \cos\left(2\pi f_c t + K_p m(t)\right).$$

The instantaneous frequency is

$$f_i(t) = f_c + \frac{K_p}{2\pi}\frac{dm(t)}{dt}.$$

Differentiating $m(t)$,

$$\frac{dm(t)}{dt} = 4\sqrt{2}\,(2\pi \times 10^3)\cos(2\pi \times 10^3 t).$$

The maximum value occurs when

$$\cos(2\pi \times 10^3 t) = 1.$$

Therefore,

$$\left(\frac{dm(t)}{dt}\right)_{\text{max}} = 4\sqrt{2}\,(2\pi \times 10^3).$$

Substituting into the instantaneous frequency expression,

$$(f_i)_{\text{max}} = 10^6 + \frac{2}{2\pi}\left[4\sqrt{2}\,(2\pi \times 10^3)\right].$$

$$(f_i)_{\max} = 10^6 + 8000\sqrt{2}.$$

$$(f_i)_{\max} = 10^6 + 11313.7085.$$

$$(f_i)_{\max} = 1011313.7085 \text{ Hz}.$$

Hence,

$$\boxed{(f_i)_{\max} = 1011313.7 \text{ Hz}}$$

Mistakes to be avoided

- For PM,
$$f_i(t) = f_c + \frac{K_p}{2\pi}\frac{dm(t)}{dt},$$
so frequency deviation depends on the derivative of the message signal.
- Do not use
$$\Delta f = K_p m(t),$$
which corresponds to phase deviation, not frequency deviation.
- Convert RMS value to peak value correctly:
$$V_m = \sqrt{2}\, V_{\text{rms}}.$$
- The carrier amplitude $A_c = 2$ has no effect on the instantaneous frequency.

Q4.21.3 NAT | 2021 | 1M | Ans: 94.4 | ●○○

Given,

$$\mu = 0.5$$

For a conventional AM signal,

$$P_t = P_c\left(1 + \frac{\mu^2}{2}\right)$$

Substituting $\mu = 0.5$,

$$P_t = P_c\left(1 + \frac{(0.5)^2}{2}\right) = P_c(1.125)$$

The power in each sideband is

$$P_{SB} = \frac{P_c\mu^2}{4}$$

When the carrier and one sideband are suppressed, the power saved is

$$P_{\text{saved}} = P_c + \frac{P_c\mu^2}{4}.$$

Therefore,

$$\frac{P_{\text{saved}}}{P_t} = \frac{P_c + \frac{P_c\mu^2}{4}}{P_c\left(1 + \frac{\mu^2}{2}\right)} = \frac{1 + \frac{\mu^2}{4}}{1 + \frac{\mu^2}{2}}.$$

Substituting $\mu = 0.5$,

$$\frac{P_{\text{saved}}}{P_t} = \frac{1 + \frac{0.25}{4}}{1 + \frac{0.25}{2}} = \frac{1.0625}{1.125} = 0.9444.$$

Hence,

$$\%\ \text{Power Saved} = 0.9444 \times 100 = 94.44\%.$$

$$\boxed{\%\ \text{Power Saved} = 94.4\%}$$

Mistakes to be avoided

- For AM,
$$P_t = P_c\left(1 + \frac{\mu^2}{2}\right),$$
not $P_t = P_c(1 + \mu^2)$.
- Each sideband contains
$$\frac{P_c\mu^2}{4}$$
power.
- If the carrier and one sideband are removed, the remaining signal is SSB-SC.

Q4.20.1 NAT | 2020 | 2M | Ans: 2 | ●●○

For a signal

$$s(t) = A_c \cos\theta(t),$$

the instantaneous frequency is

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt}.$$

For the PM signal,

$$S_{PM}(t) = \cos\left(1000\pi t + K_p m(t)\right),$$

the instantaneous frequency is

$$f_{i,PM} = \frac{1}{2\pi}\frac{d}{dt}\left(1000\pi t + K_p m(t)\right) = 500 + \frac{K_p}{2\pi}\frac{dm(t)}{dt}.$$

Hence,

$$(f_{i,PM})_{\max} = 500 + \frac{K_p}{2\pi}\left(\frac{dm(t)}{dt}\right)_{\max}.$$

From the given waveform,

$$m(t) \text{ rises from 0 V to 10 V in } (3 - 1) = 2 \text{ s}.$$

Therefore,

$$\left(\frac{dm(t)}{dt}\right)_{\max} = \frac{10 - 0}{3 - 1} = 5.$$

Thus,

$$(f_{i,PM})_{\max} = 500 + \frac{5K_p}{2\pi}.$$

For the FM signal,

$$S_{FM}(t) = \cos\left(1000\pi t + K_f \int_{-\infty}^{t} m(\tau)\, d\tau\right),$$

the instantaneous frequency is

$$f_{i,FM} = \frac{1}{2\pi}\frac{d}{dt}\left(1000\pi t + K_f \int_{-\infty}^{t} m(\tau)\, d\tau\right).$$

Using

$$\frac{d}{dt}\left(\int_{-\infty}^{t} m(\tau)\, d\tau\right) = m(t),$$

we obtain

$$f_{i,FM} = 500 + \frac{K_f}{2\pi}m(t).$$

The maximum value of $m(t)$ from the graph is

$$m_{\max} = 10.$$

Hence,

$$(f_{i,FM})_{\max} = 500 + \frac{10K_f}{2\pi}.$$

Given that the highest instantaneous frequencies are equal,

$$(f_{i,PM})_{\max} = (f_{i,FM})_{\max}.$$

Therefore,

$$500 + \frac{5K_p}{2\pi} = 500 + \frac{10K_f}{2\pi}.$$

Cancelling common terms,

$$5K_p = 10K_f.$$

Hence,

$$\boxed{\frac{K_p}{K_f} = 2 \text{ s}}$$

Mistakes to be avoided

- For PM,
$$f_i(t) = f_c + \frac{K_p}{2\pi}\frac{dm(t)}{dt},$$
so the instantaneous frequency depends on the slope of $m(t)$.
- For FM,
$$f_i(t) = f_c + \frac{K_f}{2\pi}m(t),$$

so the instantaneous frequency depends on the amplitude of $m(t)$.

- Do not use $m_{\max}$ for PM; PM requires
$$\left(\frac{dm(t)}{dt}\right)_{\max}.$$
- From a piecewise-linear waveform, compute slope as
$$\frac{\Delta m}{\Delta t},$$
not from the peak value directly.

Q4.20.2 MCQ | 2020 | 2M | Ans: B

Given
$$x(t) = m(t)\cos(2\pi f_c t),$$
with
$$m(t) = 4\cos(1000\pi t) = 4\cos(2\pi \cdot 500\, t).$$
The mixer output is
$$v(t) = x(t)\cos\big(2\pi(f_c + 40)t\big).$$
Substituting $x(t)$,
$$v(t) = m(t)\cos(2\pi f_c t)\cos\big(2\pi(f_c + 40)t\big).$$
Using
$$\cos A \cos B = \frac{1}{2}\left[\cos(A+B) + \cos(A-B)\right],$$
we obtain
$$v(t) = \frac{m(t)}{2}\left[\cos\big(2\pi(2f_c + 40)t\big) + \cos(2\pi \cdot 40\, t)\right].$$
Substituting $m(t) = 4\cos(2\pi \cdot 500\, t)$,
$$v(t) = 2\cos(2\pi \cdot 500\, t)\left[\cos\big(2\pi(2f_c + 40)t\big) + \cos(2\pi \cdot 40\, t)\right].$$
Expanding the low-frequency term,
$$2\cos(2\pi \cdot 500\, t)\cos(2\pi \cdot 40\, t) = \cos(2\pi \cdot 540\, t) + \cos(2\pi \cdot 460\, t).$$
Hence,
$$v(t) = \underbrace{\cos(2\pi \cdot 540\, t) + \cos(2\pi \cdot 460\, t)}_{\text{baseband terms}} + \underbrace{2\cos(2\pi \cdot 500\, t)\cos\big(2\pi(2f_c + 40)t\big)}_{\text{high-frequency terms}}.$$
The LPF passes frequencies up to 510 Hz.
Since
$$540 \text{ Hz} > 510 \text{ Hz},$$
the component $\cos(2\pi \cdot 540\, t)$ is rejected.

Also, all components around $2f_c$ are removed.
Only

$$\cos(2\pi \cdot 460\, t)$$

remains.
Therefore,

$$\boxed{(B)\, y(t) = \cos(920\pi t)}.$$

Q4.19.1 NAT 2019 1M Ans: 0.75 ● ○ ○

For a PM signal,

$$s(t) = A_c \cos\big(2\pi f_c t + k\, m(t)\big),$$

the instantaneous phase is

$$\theta(t) = 2\pi f_c t + k\, m(t).$$

Hence, the instantaneous frequency is

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt} = f_c + \frac{k}{2\pi}\frac{dm(t)}{dt}.$$

Given

$$k = 10\pi,$$

therefore,

$$f_i(t) = f_c + 5\frac{dm(t)}{dt}.$$

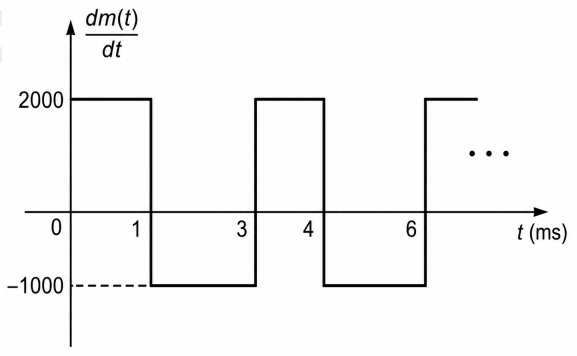


Fig. Solution Q4.19.1

From the waveform:

$$\frac{dm(t)}{dt} = \frac{1-(-1)}{1\text{ ms}} = \frac{2}{10^{-3}} = 2000.$$

during the rising sections, and

$$\frac{dm(t)}{dt} = \frac{-1-1}{2\text{ ms}} = \frac{-2}{2\times 10^{-3}} = -1000.$$

during the falling sections.

Therefore,

$$f_{i,\max} = 50\,000 + 5(2000) = 60\,000 \text{ Hz} = 60 \text{ kHz},$$

and

$$f_{i,\min} = 50\,000 + 5(-1000) = 45\,000 \text{ Hz} = 45 \text{ kHz}.$$

Hence,

$$\frac{f_{i,\min}}{f_{i,\max}} = \frac{45}{60} = 0.75.$$

$$\boxed{\frac{f_{i,\min}}{f_{i,\max}} = 0.75}$$

Q4.18.1 MCQ 2018 2M Ans: D ● ● ○

Theory: A square-law device generates the product term

$$m(t)\cos(2\pi f_c t),$$

which is the basic mechanism used to generate AM signals.

The input to the non-linear device is

$$v_i(t) = A_c \cos(2\pi f_c t) + \cos(2\pi f_m t).$$

The output is

$$v_o(t) = a\, v_i(t) + b\, v_i^2(t).$$

Substituting $v_i(t)$,

$$v_o(t) = a\Big[A_c \cos(2\pi f_c t) + \cos(2\pi f_m t)\Big] + b\Big[A_c \cos(2\pi f_c t) + \cos(2\pi f_m t)\Big]^2.$$

Expanding,

$$\begin{aligned} v_o(t) = {} & aA_c \cos(2\pi f_c t) + a\cos(2\pi f_m t) \\ & + bA_c^2 \cos^2(2\pi f_c t) + b\cos^2(2\pi f_m t) \\ & + 2bA_c \cos(2\pi f_c t)\cos(2\pi f_m t). \end{aligned}$$

The BPF centered at f_c removes all terms except those around f_c.
Therefore,

$$y(t) = aA_c \cos(2\pi f_c t) + 2bA_c \cos(2\pi f_c t)\cos(2\pi f_m t).$$

Factoring out $aA_c \cos(2\pi f_c t)$,

$$y(t) = aA_c \left[1 + \frac{2b}{a}\cos(2\pi f_m t)\right]\cos(2\pi f_c t).$$

Comparing with the standard AM form

$$s_{AM}(t) = A_c\,[1 + \mu\cos(2\pi f_m t)]\cos(2\pi f_c t),$$

the modulation index is

$$\mu = \frac{2b}{a}.$$

For a standard AM wave,

$$P_{SB} = \frac{\mu^2}{2} P_C.$$

Given

$$P_{SB} = \frac{1}{2} P_C,$$

hence

$$\frac{\mu^2}{2} = \frac{1}{2}.$$

Therefore,

$$\mu^2 = 1$$

and since $\mu > 0$,

$$\mu = 1.$$

Using

$$\mu = \frac{2b}{a},$$

we get

$$\frac{2b}{a} = 1.$$

Thus,

$$a = 2b.$$

Hence,

$$\boxed{(D)\ \frac{a}{b} = 2}.$$

Q4.18.2 NAT | 2018 | 1M | Ans: 0.12 to 0.13 | ● ○ ○

The carrier is the middle-frequency term, $4\cos(2400\pi t)$, and each sideband has amplitude 1. Using the same reference resistance for every component,

$$P_c = \frac{4^2}{2} = 8, \qquad P_{\text{SB}} = \frac{1^2}{2} + \frac{1^2}{2} = 1.$$

The message-carrying power in the modulated wave is the total sideband power. Therefore,

$$\frac{P_{\text{SB}}}{P_c} = \frac{1}{8} = \boxed{0.125}.$$

Equivalently, the modulation index is $\mu = 2(1)/4 = 0.5$, so $P_{\text{SB}}/P_c = \mu^2/2 = 0.125$.

Q4.17.1 NAT | 2017 | 2M | Ans: 70 | ● ● ○

For a phase modulator, the instantaneous frequency is

$$f_i(t) = f_c + \frac{k_p}{2\pi} \frac{dx(t)}{dt}.$$

Given

$$x(t) = 5\sin\left(4\pi \times 10^3 t - 10\pi\cos\left(2\pi \times 10^3 t\right)\right)\text{ V}, \qquad k_p = 5\,\text{rad/V},$$

and

$$f_c = 20\,\text{kHz}.$$

Let

$$\phi(t) = 4\pi \times 10^3 t - 10\pi\cos\left(2\pi \times 10^3 t\right).$$

Then

$$\frac{dx(t)}{dt} = 5\cos\phi(t)\left[4\pi \times 10^3 + 20\pi^2 \times 10^3 \sin\left(2\pi \times 10^3 t\right)\right].$$

At $t = 0.5\,\text{ms}$,

$$\sin\left(2\pi \times 10^3 t\right) = \sin\pi = 0,$$

and

$$\phi(0.5\,\text{ms}) = 2\pi + 10\pi = 12\pi.$$

Hence,

$$\frac{dx}{dt} = 5(1)(4\pi \times 10^3) = 20\pi \times 10^3\,\text{V/s}.$$

Therefore,

$$f_i = 20 \times 10^3 + \frac{5}{2\pi}(20\pi \times 10^3) = 70 \times 10^3\,\text{Hz}.$$

$$\boxed{70\,\text{kHz}}$$

Key Points

- In phase modulation, frequency deviation is proportional to the time derivative of the modulating signal: $\Delta f(t) = \frac{k_p}{2\pi}\frac{dx(t)}{dt}$.
- The carrier frequency f_c must be added to the instantaneous frequency.

Q4.17.2 NAT 2017 2M Ans: 5.208 ●●○

Theory: For a single-tone AM signal,

$$P_t = P_c\left(1 + \frac{\mu^2}{2}\right),$$

where P_t is the total transmitted power and P_c is the carrier power.

Initially,

$$P_c = 5\,\text{kW}, \qquad \mu = 0.5.$$

Hence, the maximum transmitter power rating is

$$P_{t,\max} = 5\left(1 + \frac{(0.5)^2}{2}\right).$$

$$= 5\left(1 + \frac{0.25}{2}\right) = 5(1.125) = 5.625\,\text{kW}.$$

This is the maximum power the transmitter can handle without overloading.

Now the modulation index is reduced to

$$\mu = 0.4.$$

Let the new allowable carrier power be $P_{c,\max}$.Since the transmitter rating remains unchanged,

$$P_{c,\max}\left(1 + \frac{(0.4)^2}{2}\right) = 5.625.$$

Therefore,

$$P_{c,\max} = \frac{5.625}{1 + \frac{0.16}{2}}.$$

$$= \frac{5.625}{1.08} = 5.2083 \text{ kW}.$$

Hence,

$$\boxed{P_{c,\max} = 5.208 \text{ kW}}$$

Q4.16.1 NAT 2016 1M Ans: 5 ● ○ ○

Theory: For high-side injection,

$$f_{LO} = f_s + f_{IF}$$

and the image frequency is

$$f_{si} = f_s + 2f_{IF}.$$

The image frequency must lie outside the receiver tuning range.

Given tuning range:

$$58 \text{ MHz} \le f_s \le 68 \text{ MHz}.$$

The image frequency is

$$f_{si} = f_s + 2f_{IF}.$$

To ensure that *all* image frequencies lie outside the band, even the smallest image frequency must be greater than 68 MHz. The smallest image frequency occurs when

$$f_s = 58 \text{ MHz}.$$

Therefore,

$$58 + 2f_{IF} > 68.$$

$$2f_{IF} > 10.$$

$$f_{IF} > 5 \text{ MHz}.$$

Hence, the minimum required intermediate frequency is

$$\boxed{f_{IF,\min} = 5 \text{ MHz}}.$$

Mistakes to be avoided

- Do not use
$$f_{si} = f_s + f_{IF}.$$
The image frequency is separated from the desired signal by $2f_{IF}$, so
$$f_{si} = f_s + 2f_{IF}.$$
- The condition $f_{IF} \leq f_{LO}$ implies high-side injection:
$$f_{LO} = f_s + f_{IF}.$$
- Always check the *worst-case* image frequency. For high-side injection, the smallest image occurs at the lowest tuned frequency (58 MHz).
- Do not use 68 MHz while computing the minimum image frequency. The minimum image corresponds to
$$f_s = 58 \text{ MHz}.$$
- The image frequency must be completely outside the tuning band, so use
$$58 + 2f_{IF} > 68,$$
not $58 + 2f_{IF} = 68$.

Q4.16.2 NAT 2016 1M Ans: 3485

Theory: In a superheterodyne receiver,
$$f_{IF} = |f_s - f_{LO}|.$$
The image frequency is the other frequency that also produces the same intermediate frequency.

Given,
$$f_{IF} = 15 \text{ MHz}, \qquad f_{LO} = 3.5 \text{ GHz} = 3500 \text{ MHz}.$$
Since the received signal frequency is greater than the local oscillator frequency,
$$f_s = f_{LO} + f_{IF}.$$
Thus,
$$f_s = 3500 + 15 = 3515 \text{ MHz}.$$
The image frequency lies on the opposite side of the local oscillator by the same IF spacing:
$$f_{si} = f_{LO} - f_{IF}.$$
Hence,
$$f_{si} = 3500 - 15.$$
$$\boxed{f_{si} = 3485 \text{ MHz}}$$

Mistakes to be avoided

- Do not use
$$f_{si} = f_s + 2f_{IF}$$
blindly. First determine whether the signal is above or below the local oscillator.
- Convert GHz to MHz before substitution:
$$3.5\ \text{GHz} = 3500\ \text{MHz}.$$
- Remember that the image frequency is symmetric about the local oscillator frequency.
- Since
$$f_s > f_{LO},$$
the image must be below the local oscillator.
- A quick shortcut for this case is:
$$f_{si} = f_{LO} - f_{IF}.$$

Q4.16.3 MCQ 2016 1M Ans: 0.5 ● ○ ○

Theory: For a single-tone AM signal,

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c\mu}{2}\cos\big(2\pi(f_c - f_m)t\big) + \frac{A_c\mu}{2}\cos\big(2\pi(f_c + f_m)t\big).$$

Thus,

$$\text{Carrier amplitude} = A_c, \qquad \text{Sideband amplitude} = \frac{A_c\mu}{2}.$$

Given,

$$s(t) = 5\cos(1600\pi t) + 20\cos(1800\pi t) + 5\cos(2000\pi t).$$

The middle frequency component has the largest amplitude and hence is the carrier:

$$A_c = 20.$$

The lower and upper sidebands have amplitudes

$$\frac{A_c\mu}{2} = 5.$$

Substituting $A_c = 20$,

$$\frac{20\mu}{2} = 5.$$

$$10\mu = 5.$$

$$\mu = \frac{5}{10} = 0.5.$$

Therefore,

$$\boxed{\mu = 0.5}$$

Q4.16.4 MCQ 2016 1M Ans: A ● ● ○

Theory: In a PLL frequency synthesizer with a divide-by-N counter in the feedback path, the phase detector compares

$$f_{in} \quad \text{and} \quad \frac{f_{VCO}}{N}.$$

When the PLL is locked, the two frequencies must be equal:

$$f_{in} = \frac{f_{VCO}}{N}.$$

Therefore,

$$f_{VCO} = N f_{in}.$$

Thus, the PLL acts as a frequency multiplier by a factor N.

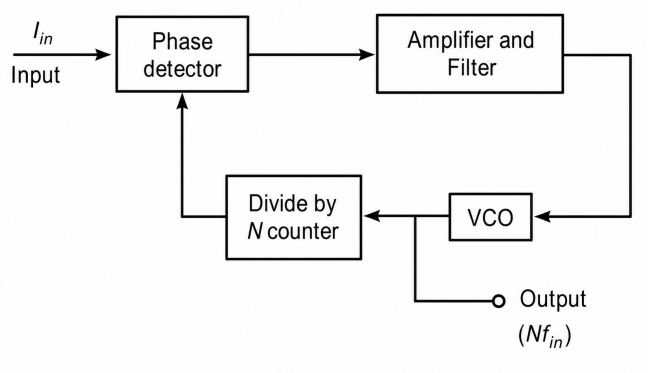


Fig. Solution Q4.16.4

Given,

$$f_{in} = 5 \text{ kHz}$$

and the divider can be set to

$$N = 2,\ 4,\ 8,\ 16.$$

The switch moves in the order

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4.$$

Hence:

$$N = 2$$

$$f_{VCO} = 2(5) = 10 \text{ kHz}$$

$$N = 4$$

$$f_{VCO} = 4(5) = 20 \text{ kHz}$$

$$N = 8$$

$$f_{VCO} = 8(5) = 40 \text{ kHz}$$

$$N = 16$$

$$f_{VCO} = 16(5) = 80 \text{ kHz}$$

f_{in}	VCO output (Nf_{in})	Divide by N counter
5 kHz	10 kHz	2
5 kHz	20 kHz	4
5 kHz	40 kHz	8
5 kHz	80 kHz	16

Fig. Solution Q4.16.4

Therefore, the synthesized frequencies are

$$\boxed{(A)10\text{ kHz},\ 20\text{ kHz},\ 40\text{ kHz},\ 80\text{ kHz}}$$

Q4.15.1 MCQ 2015 2M Ans: C ●●○

Theory: A bandpass signal can be represented as

$$x(t) = \Re\{\tilde{x}(t)e^{j2\pi f_c t}\},$$

where $\tilde{x}(t)$ is the complex envelope.

Given,

$$x(t) = -\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)\sin\left(\pi t - \frac{\pi}{4}\right)$$

Since

$$f_c = \frac{1}{2}\text{ Hz},$$

we have

$$2\pi f_c t = \pi t.$$

Therefore,

$$x(t) = -\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)\sin\left(2\pi f_c t - \frac{\pi}{4}\right).$$

Using

$$-\sin(\theta) = \cos\left(\theta + \frac{\pi}{2}\right),$$

we obtain

$$x(t) = \sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)\cos\left(2\pi f_c t + \frac{\pi}{4}\right).$$

Comparing with

$$x(t) = \Re\{\tilde{x}(t)e^{j2\pi f_c t}\},$$

the amplitude term is

$$\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)$$

and the phase shift is

$$+\frac{\pi}{4}.$$

Hence,

$$\tilde{x}(t) = \sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{j\frac{\pi}{4}}.$$

Therefore,

$$(C)\tilde{x}(t) = \sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{j\frac{\pi}{4}}$$

Mistakes to be avoided

- Do not directly take the coefficient multiplying the sine term as the complex envelope.
- First convert the signal into the standard cosine form

$$x(t) = A(t)\cos(2\pi f_c t + \phi(t)).$$

- Remember

$$-\sin\theta = \cos\left(\theta + \frac{\pi}{2}\right),$$

  which changes the phase by $+\pi/2$.
- Do not forget the $\sqrt{2}$ factor while writing the complex envelope.
- The complex envelope contains both amplitude and phase:

$$\tilde{x}(t) = A(t)e^{j\phi(t)}.$$

  Many students correctly identify the phase but miss the amplitude scaling.

Q4.15.2 NAT 2015 2M Ans: 350 ● ● ●

Theory: To recover only $10x(t)$ at the output, the spectrum generated by the nonlinear term $x^2(t)$ must lie completely outside the passband of the filter.

Given,

$$x(t) = m(t)\cos(2400\pi t).$$

Since

$$2400\pi = 2\pi f_c,$$

we obtain

$$f_c = 1200 \text{ Hz}.$$

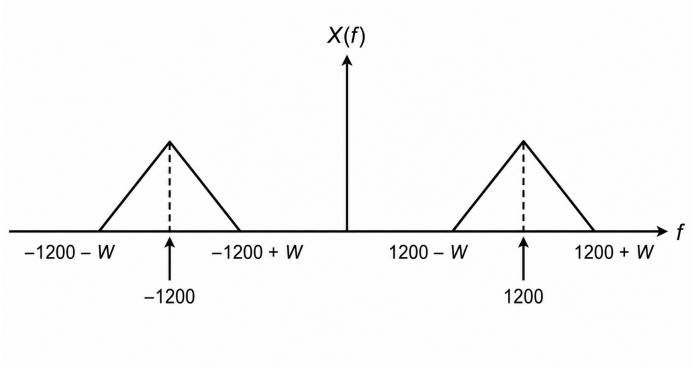


Fig. Solution Q4.15.2

The amplifier output is

$$y(t) = 10x(t) + x^2(t).$$

Substituting $x(t)$,

$$y(t) = 10m(t)\cos(2400\pi t) + m^2(t)\cos^2(2400\pi t).$$

Using

$$\cos^2\theta = \frac{1+\cos 2\theta}{2},$$

$$y(t) = 10m(t)\cos(2400\pi t) + \frac{m^2(t)}{2} + \frac{m^2(t)}{2}\cos(4800\pi t).$$

Thus $y(t)$ contains three spectral components:

1. Baseband term

   $$\frac{m^2(t)}{2}$$

   with bandwidth $2W$.

2. Desired bandpass term

   $$10m(t)\cos(2400\pi t)$$

   centered at 1200 Hz with bandwidth W.

3. Undesired term

   $$\frac{m^2(t)}{2}\cos(4800\pi t)$$

   centered at 2400 Hz with bandwidth $2W$.

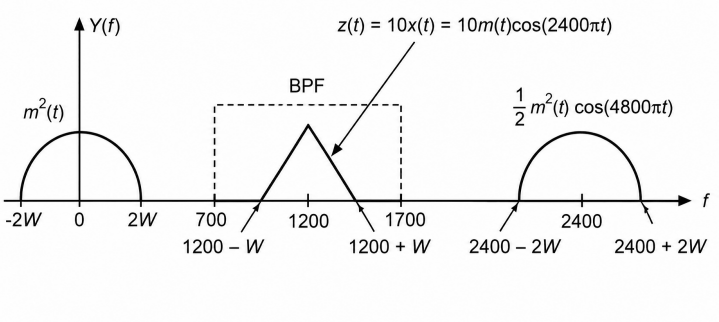


Fig. Solution Q4.15.2

The desired spectrum occupies

$$1200 - W \le f \le 1200 + W.$$

Since the filter passband is

$$700 \le |f| \le 1700,$$

the entire desired spectrum must remain inside the passband.
Hence,

$$1200 - W > 700$$

$$W < 500.$$

Also,

$$1200 + W < 1700$$

$$W < 500.$$

Now consider the baseband component arising from

$$\frac{m^2(t)}{2}.$$

Its bandwidth is

$$2W.$$

To prevent it from entering the filter passband,

$$2W < 700.$$

Therefore,

$$W < 350.$$

Next consider the high-frequency component

$$\frac{m^2(t)}{2}\cos(4800\pi t).$$

Its spectrum extends from

$$2400 - 2W$$

to

$$2400 + 2W.$$

To keep it completely outside the passband,

$$2400 - 2W > 1700.$$

Thus,

$$700 > 2W$$

$$W < 350.$$

Combining all constraints,

$$W < 500, \qquad W < 350, \qquad W < 350.$$

Hence,

$$\boxed{W < 350 \text{ Hz}}.$$

Mistakes to be avoided

- Do not forget that squaring a signal doubles its bandwidth.
  If $m(t)$ has bandwidth W,

  $$m^2(t)$$

  has bandwidth

  $$2W.$$

- Always use

  $$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$

  before drawing the spectrum.

Q4.15.3 MCQ | 2015 | 1M | Ans: C | ● ○ ○

Theory: For a PM signal, Carson's rule gives

$$B \approx 2(\Delta f + f_m),$$

where Δf is the peak frequency deviation.

Given

$$y(t) = A_c \cos(2\pi f_c t + A_m \sin(2\pi f_m t)).$$

Comparing with the standard PM expression,

$$s_{PM}(t) = A_c \cos\left(2\pi f_c t + k_p m(t)\right),$$

we observe that the peak phase deviation is

$$\Delta\phi = A_m.$$

For PM,

$$\Delta f = \Delta\phi\, f_m.$$

Therefore,

$$\Delta f = A_m f_m.$$

Using Carson's rule,

$$B = 2(\Delta f + f_m) = 2(A_m f_m + f_m).$$

Taking f_m common,

$$B = 2(A_m + 1) f_m.$$

From the above expression,

$$B \propto (A_m + 1) f_m.$$

Hence, the bandwidth depends on both A_m and f_m.
Therefore,

$$\boxed{\text{(C) depends on both } A_m \text{ and } f_m}$$

is correct.

Mistakes to be avoided

- Do not confuse PM with FM.

  For FM,

  $$\Delta f = k_f A_m,$$

  whereas for PM,

  $$\Delta f = \Delta\phi\, f_m.$$

- In PM, increasing the modulating frequency increases the frequency deviation.

  Students often incorrectly assume Δf is independent of f_m.

- Do not conclude bandwidth depends only on A_m because A_m appears inside the phase term.

  The frequency deviation is

  $$\Delta f = A_m f_m,$$

  which contains both parameters.

- Always apply Carson's rule

  $$B \approx 2(\Delta f + f_m)$$

  before deciding bandwidth dependence.

- Remember that for a single-tone PM signal,

  $$\beta_{PM} = A_m,$$

  but bandwidth still depends on both A_m and f_m.

Q4.15.4 MCQ 2015 1M Ans: C ● ○ ○

Theory: Any signal of the form

$$s(t) = m(t)\cos(2\pi f_c t) \pm \hat{m}(t)\sin(2\pi f_c t)$$

represents an SSB (Single Sideband) signal generated using the phase-shift method.

Given

$$s(t) = m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t).$$

Using the standard SSB expressions,

$$s_{LSB}(t) = m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t),$$

$$s_{USB}(t) = m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t).$$

Hence,

$$s(t) = s_{LSB}(t),$$

which is a Lower Sideband (LSB) SSB signal.

Since SSB modulation shifts the spectrum of the baseband signal $m(t)$ to a frequency region centered around f_c,

$$S(f)$$

occupies frequencies near

$$\pm f_c.$$

Therefore, the signal energy is concentrated in a finite band around the carrier frequency rather than around DC. Hence, $s(t)$ is a *band-pass signal*.

(C) Band-pass signal

Mistakes to be avoided

- Remember the standard identities:

$$m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t) \rightarrow \text{LSB}$$

$$m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t) \rightarrow \text{USB}$$

- Whenever you see a Hilbert transform together with cos and sin carrier terms, immediately think of the phase-shift method of SSB generation.

Q4.14.1 NAT 2014 2M Ans: 60 ● ● ○

Theory: Multiplication by a cosine shifts the spectrum, while filtering selects desired spectral components; hence the final bandwidth is obtained by tracking the spectrum through each block.

Given

$$A = 100 \text{ Hz}, \qquad B = 40 \text{ Hz}.$$

The message spectrum $M(f)$ occupies

$$B \le |f| \le A.$$

Thus,

$$40 \le |f| \le 100.$$

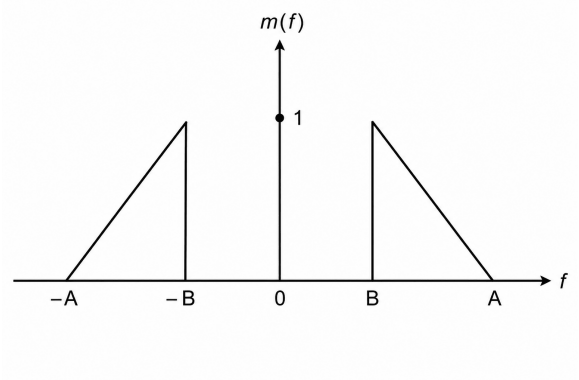


Fig. Solution Q4.14.1

The signal is multiplied by

$$v(t) = \cos(2\pi f_c t).$$

Hence,

$$x(t) = m(t)\cos(2\pi f_c t).$$

In frequency domain,

$$X(f) = \frac{1}{2}\Big[M(f - f_c) + M(f + f_c)\Big].$$

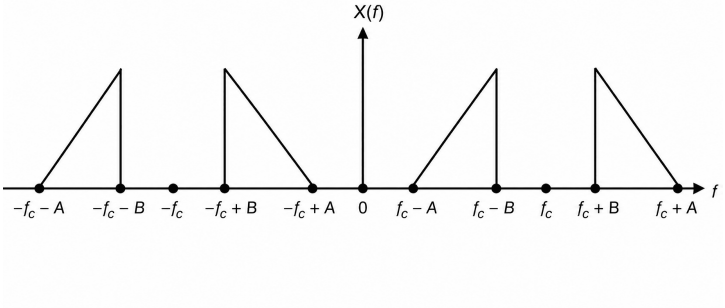


Fig. Solution Q4.14.1

Therefore, the spectrum is shifted to

$$f_c - B \text{ to } f_c - A,$$

and

$$f_c + B \text{ to } f_c + A,$$

along with corresponding negative-frequency components.

The high-pass filter has cutoff frequency f_c.Hence, only the spectral component lying above f_c survives. Therefore,

$$Y(f) \text{ occupies } f_c + B \le f \le f_c + A.$$

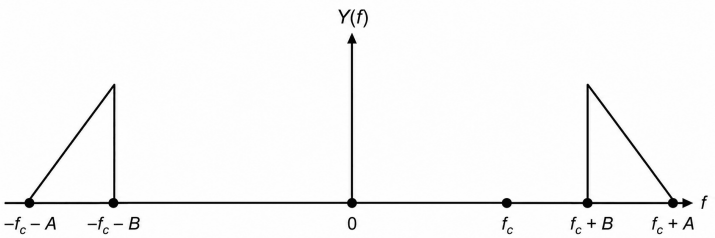


Fig. Solution Q4.14.1

Now the signal is multiplied by

$$w(t) = \cos\left(2\pi(f_c + A)t\right).$$

Therefore,

$$z(t) = y(t)\cos\left(2\pi(f_c + A)t\right).$$

In frequency domain,

$$Z(f) = \frac{1}{2}\left[Y(f - (f_c + A)) + Y(f + (f_c + A))\right].$$

The positive-frequency band

$$f_c + B \le f \le f_c + A$$

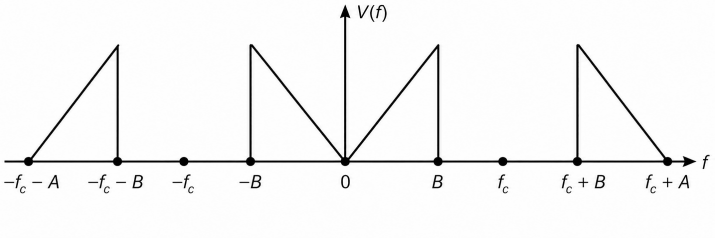


Fig. Solution Q4.14.1

gets shifted to baseband as

$$(f_c + B) - (f_c + A) \le f \le (f_c + A) - (f_c + A).$$

Thus,

$$B - A \le f \le 0.$$

Similarly,

$$0 \le f \le A - B.$$

Hence the low-frequency spectrum extends from

$$-(A - B) \quad \text{to} \quad (A - B).$$

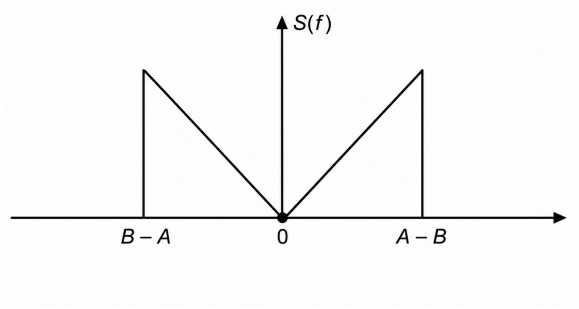


Fig. Solution Q4.14.1

The low-pass filter passes only the baseband component.
Hence the output spectrum occupies

$$-(A - B) \le f \le (A - B).$$

Therefore, the output bandwidth is

$$A - B.$$

Substituting the values,

$$\text{Bandwidth} = 100 - 40 = 60 \text{ Hz}.$$

$$\boxed{60 \text{ Hz}}$$

$$\boxed{\text{Answer} = 60}$$

Q4.14.2 NAT 2014 1M Ans: 0.5 ● ○ ○

Theory: In an AM signal, the modulation index can be directly obtained from the maximum and minimum values of the envelope because the envelope amplitude varies as $A_c(1 \pm \mu)$.

For an AM wave,

$$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}.$$

Given,

$$A_{\max} = 3 \text{ V}, \qquad A_{\min} = 1 \text{ V}.$$

Substituting,

$$\mu = \frac{3 - 1}{3 + 1} = \frac{2}{4} = 0.5.$$

Hence,

$$\boxed{\mu = 0.5}$$

Mistakes to be avoided

- Do not use RMS values; the formula requires envelope maximum and minimum amplitudes.
- The relation $\mu = \dfrac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$ is valid only when there is no overmodulation.

Q4.14.3 MCQ 2014 1M Ans: A ● ○ ○

Theory: For an FM signal, the instantaneous frequency is obtained by differentiating the total phase with respect to time.
Given,

$$f(t) = \cos[2\pi f_c t + \beta_1 \sin 2\pi f_1 t + \beta_2 \sin 2\pi f_2 t] .$$

The instantaneous phase is

$$\theta(t) = 2\pi f_c t + \beta_1 \sin 2\pi f_1 t + \beta_2 \sin 2\pi f_2 t.$$

Hence,

$$f_i(t) = \frac{1}{2\pi}\frac{d\theta(t)}{dt}.$$

Therefore,

$$f_i(t) = f_c + \beta_1 f_1 \cos 2\pi f_1 t + \beta_2 f_2 \cos 2\pi f_2 t.$$

Thus,

$$\Delta f = \beta_1 f_1 \cos 2\pi f_1 t + \beta_2 f_2 \cos 2\pi f_2 t.$$

The maximum deviation occurs when both cosine terms are equal to 1.
Hence,

$$\boxed{(A)(\Delta f)_{\max} = \beta_1 f_1 + \beta_2 f_2.}$$

Q4.14.4 NAT 2014 1M Ans: 4 ● ○ ○

Theory: In conventional AM, sideband power is proportional to the square of the modulation index μ.
For DSB-FC AM,

$$P_{SB} = \frac{\mu^2}{2} P_c.$$

Therefore,

$$\frac{P_{SB}}{P_c} = \frac{\mu^2}{2}.$$

Suppose initially the modulation index is μ.
Then,

$$\left(\frac{P_{SB}}{P_c}\right)_1 = \frac{\mu^2}{2}.$$

If the modulation index is doubled,

$$\mu' = 2\mu.$$

Hence,

$$\left(\frac{P_{SB}}{P_c}\right)_2 = \frac{(2\mu)^2}{2} = \frac{4\mu^2}{2}.$$

Therefore,

$$\frac{\left(\frac{P_{SB}}{P_c}\right)_2}{\left(\frac{P_{SB}}{P_c}\right)_1} = \frac{\frac{4\mu^2}{2}}{\frac{\mu^2}{2}} = 4.$$

Thus, the ratio increases by a factor of

$$\boxed{4}$$

Q4.12.1 MCQ | 2012 | 2M | Ans: B | ● ● ○

Theory: For the same message signal, PM phase deviation depends on the maximum amplitude of $m(t)$, whereas FM phase deviation depends on the maximum accumulated area under $m(t)$.

For the phase modulator,

$$\phi_{PM}(t) = 2\pi f_c t + k_p m(t).$$

Hence, the maximum phase deviation is

$$(\phi_D)_{\max} = k_p \max |m(t)|.$$

Since

$$\max |m(t)| = 2,$$

we get

$$(\phi_D)_{\max} = 2k_p.$$

For the frequency modulator,

$$\phi_{FM}(t) = 2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau.$$

The maximum accumulated area under $m(t)$ occurs over the interval

$$0 \le t \le 2,$$

where

$$m(t) = 2.$$

Therefore,

$$\left[\int_0^t m(\tau)\, d\tau\right]_{\max} = \int_0^2 2\, dt = 4.$$

Hence, the maximum phase deviation for FM is

$$(\phi'_D)_{\max} = 2\pi k_f (4) = 8\pi k_f.$$

Since the maximum phase deviations are the same,

$$2k_p = 8\pi k_f.$$

Therefore,

$$\boxed{(B)\ \frac{k_p}{k_f} = 4\pi \text{ rad/Hz}.}$$

Mistakes to be avoided

- In PM, phase deviation depends on the maximum value of $m(t)$, not on its integral.
- In FM, phase deviation depends on the accumulated area under $m(t)$, not merely on its peak amplitude.
- Remember the FM phase expression:

$$\phi_{FM}(t) = 2\pi f_c t + 2\pi k_f \int_0^t m(\tau)\, d\tau.$$

Q4.11.1 MCQ 2011 2M Ans: B ● ○ ○

Theory: For proper envelope detection, the detector time constant must be much larger than the carrier period and much smaller than the message period.

The design condition for an envelope detector is

$$\frac{1}{f_c} \ll RC \ll \frac{1}{f_{m,\max}}.$$

Here,

$$f_c = 1 \text{ MHz},$$

and the highest message frequency is

$$f_{m,\max} = 2 \text{ kHz},$$

since

$$m(t) = \cos 2000\pi t + 4\cos 4000\pi t.$$

Therefore,

$$\frac{1}{f_c} = \frac{1}{10^6} = 1\ \mu\text{s},$$

and

$$\frac{1}{f_{m,\max}} = \frac{1}{2000} = 0.5 \text{ ms}.$$

Hence,

$$\boxed{(B) 1\ \mu\text{s} \ll RC < 0.5 \text{ ms}.}$$

Mistakes to be avoided

- Always use the highest message frequency $f_{m,\max}$, not the lowest frequency component.
- Do not confuse the carrier period with the carrier frequency.
- Remember the envelope detector condition:

$$\frac{1}{f_c} \ll RC \ll \frac{1}{f_{m,\max}}.$$

- The factor 4 multiplying the second cosine affects only its amplitude, not its frequency.

Q4.11.2 MCQ 2011 1M Ans: B ● ○ ○

Theory: Different modulation schemes are designed to optimize different performance metrics such as power efficiency, bandwidth efficiency, and receiver complexity.

	Column-I		Column-II
P	Power efficient transmission of signals	1	Conventional AM
Q	Most bandwidth efficient transmission of voice signals	2	FM
R	Simplest receiver structure	3	VSB
S	Bandwidth efficient transmission of signals with Significant dc component	4	SSB-SC

Fig. Solution Q4.11.2

Matching each attribute:

$$\boxed{P \rightarrow 2}$$

FM has constant envelope transmission, allowing highly efficient nonlinear power amplifiers and hence better power efficiency.

$$\boxed{Q \rightarrow 4}$$

SSB-SC occupies only one sideband, making it the most bandwidth-efficient modulation scheme for voice signals.

$$\boxed{R \rightarrow 1}$$

Conventional AM has the simplest receiver since it can be demodulated using a simple envelope detector.

$$\boxed{S \rightarrow 3}$$

VSB retains a vestige of one sideband, making it suitable for transmitting signals containing significant low-frequency (DC) components while maintaining bandwidth efficiency.
Hence,

$$\boxed{(B) P-2,\ Q-4,\ R-1,\ S-3.}$$

Mistakes to be avoided

- Do not confuse **power efficiency** with **bandwidth efficiency**; FM is power efficient, whereas SSB-SC is bandwidth efficient.
- Conventional AM is preferred for its simple receiver, not for bandwidth or power efficiency.
- VSB is specifically used for signals having significant low-frequency (DC) components, such as television video signals.
- Remember that SSB-SC is the most bandwidth-efficient analog modulation scheme for voice communication.

Q4.10.1 MCQ 2010 1M Ans: B ●○○

Theory: In angle modulation (FM/PM), only the angle of the carrier changes while the amplitude remains constant; therefore, the average transmitted power depends only on the carrier amplitude.

The given signal is

$$x(t) = 6\cos\left[2\pi \times 10^6 t + 2\sin(8000\pi t) + 4\cos(8000\pi t)\right].$$

The carrier amplitude is

$$A_c = 6.$$

The average power of an angle-modulated signal is

$$P = \frac{A_c^2}{2}.$$

Hence,

$$P = \frac{6^2}{2} = \frac{36}{2} = 18\ \text{W}.$$

Therefore,

$$\boxed{(B) P = 18\ \text{W}.}$$

Mistakes to be avoided

- In FM and PM, the modulation indices or phase terms do **not** affect the average transmitted power.
- Always identify the carrier amplitude A_c directly from the signal before computing power.
- Do not confuse angle modulation with AM. In AM, transmitted power depends on the modulation index, whereas in

FM/PM it depends only on A_c.

- Remember the formula

$$P = \frac{A_c^2}{2},$$

assuming a 1 Ω load.

Q4.10.2 MCQ 2010 1M Ans: C ●●○

Theory: A conventional AM signal avoids over-modulation only if its modulation index satisfies $\mu \leq 1$.

The general expression for a conventional AM signal is

$$x(t) = A_c\,[1 + k_a m(t)]\cos(2\pi f_c t),$$

where the modulation index is

$$\mu = k_a A_m,$$

and A_m is the peak amplitude of the message signal.
Since

$$m(t) = 2\cos(2\pi f_m t),$$

the message amplitude is
Options (A) and (D) are not conventional AM signals.

$$A_m = 2.$$

For option (B),

$$x(t) = A_c\,[1 + m(t)]\cos(2\pi f_c t).$$

Here,

$$k_a = 1,$$

therefore,

$$\mu = k_a A_m = 1 \times 2 = 2 > 1.$$

Hence, option (B) represents an over-modulated AM signal.
For option (C),

$$x(t) = A_c\cos(2\pi f_c t) + \frac{A_c}{4} m(t)\cos(2\pi f_c t).$$

Rearranging,

$$x(t) = A_c\left[1 + \frac{m(t)}{4}\right]\cos(2\pi f_c t).$$

Thus,

$$k_a = \frac{1}{4}.$$

Hence,

$$\mu = k_a A_m = \frac{1}{4} \times 2 = \frac{1}{2} < 1.$$

Therefore,

$$\boxed{(C) x(t) = A_c \cos(2\pi f_c t) + \frac{A_c}{4} m(t) \cos(2\pi f_c t)}$$

is a conventional AM signal without over-modulation.

Mistakes to be avoided

- Always identify the peak amplitude A_m of the message signal before calculating the modulation index.
- Compare the given signal with the standard AM form

$$x(t) = A_c \left[1 + k_a m(t)\right] \cos(2\pi f_c t).$$

- Do not assume $k_a = 1$; determine it from the coefficient multiplying $m(t)$.
- Remember that over-modulation occurs whenever

$$\mu > 1.$$

Q4.09.1 MCQ 2009 2M Ans: C ●○○

Theory: For a multi-tone AM signal, the overall modulation index is obtained by the root-sum-square of the individual modulation indices, and the transmission efficiency is

$$\eta = \frac{m_a^2}{2 + m_a^2} \times 100\%.$$

Given,

$$m(t) = \left(\frac{1}{2}\right) \cos \omega_1 t - \left(\frac{1}{2}\right) \sin \omega_2 t.$$

Hence,

$$m_{a1} = \frac{1}{2}, \qquad m_{a2} = \frac{1}{2}.$$

Therefore, the equivalent modulation index is

$$m_a = \sqrt{m_{a1}^2 + m_{a2}^2} = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{2}}.$$

Thus,

$$m_a^2 = \frac{1}{2}.$$

The power efficiency is

$$\eta = \frac{m_a^2}{2 + m_a^2} \times 100\% = \frac{\frac{1}{2}}{2 + \frac{1}{2}} \times 100\% = \frac{0.5}{2.5} \times 100\% = 20\%.$$

$$\boxed{(C) \eta = 20\%}$$

Mistakes to be avoided

- Do not add the modulation indices directly; use the root-sum-square for multiple independent tones.
- Do not forget that the efficiency formula uses m_a^2, not m_a.

- Compute the equivalent modulation index first, then substitute it into the efficiency formula.
- Do not confuse transmission efficiency with modulation index.

Q4.09.2 MCQ 2009 1M Ans: C ● ○ ○

Theory: An SSB signal contains only one sideband (either USB or LSB) while suppressing both the carrier and the other sideband.
Given,

$$m(t) = \cos(2\pi f_m t).$$

The four options represent:

$$\cos(2\pi f_m t)\cos(2\pi f_c t) = \frac{1}{2}\cos[2\pi(f_c + f_m)t] + \frac{1}{2}\cos[2\pi(f_c - f_m)t]\,.$$

Hence, Option (A) is a DSB-SC signal since it contains both upper and lower sidebands.
Option (B),

$$\cos(2\pi f_c t),$$

is only the carrier.
Option (C),

$$\cos[2\pi(f_c + f_m)t]\,,$$

contains only the upper sideband (USB), and therefore represents an SSB signal.
Option (D),

$$[1 + \cos(2\pi f_m t)]\cos(2\pi f_c t),$$

is a conventional AM signal containing the carrier along with both sidebands.
Hence,

$$\boxed{(C)\ \cos[2\pi(f_c + f_m)t]}$$

Mistakes to be avoided

- Do not confuse DSB-SC with SSB; DSB-SC contains both sidebands whereas SSB contains only one.
- Remember that a pure carrier is not an SSB signal.
- Conventional AM always contains the carrier and both sidebands.
- Use the identity $\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$ to identify the spectral components.

Q4.08.1 MCQ 2008 2M Ans: B ● ○ ○

Theory: Under the narrowband approximation, the same signal can be represented as both a narrowband FM signal and a narrowband AM signal with an appropriate phase shift.
Given,

$$s(t) = \cos\omega_c t + 0.5\cos\omega_m t\ \sin\omega_c t.$$

For a narrowband FM signal,

$$s_{NBFM}(t) = \cos\omega_c t - \beta\sin\omega_c t\sin\omega_m t.$$

Using the identity

$$\sin \omega_m t = \cos\left(\omega_m t - \frac{\pi}{2}\right),$$

the given signal can be written as

$$s(t) = \cos \omega_c t + 0.5 \sin \omega_c t \sin\left(\omega_m t + \frac{\pi}{2}\right),$$

which is the standard narrowband FM form with an appropriate phase-shifted message signal.
Also,

$$\cos A + \mu \cos B \ \sin A = \cos A + \mu \sin A \cos B,$$

is the quadrature representation of an AM signal with a suitable phase-shifted carrier/message.
Hence, the signal can be interpreted as both a narrowband FM signal and an AM signal.

(B) Both AM and FM

Mistakes to be avoided

- Do not assume every signal containing $\sin \omega_c t$ is purely FM.
- Remember that narrowband FM and AM can have identical mathematical forms after an appropriate phase shift.
- This equivalence is valid only under the narrowband approximation.

Q4.08.2 MCQ 2008 2M Ans: B ●●○

Theory: For a multi-tone FM signal, the modulation index is computed using the maximum frequency deviation divided by the highest modulating frequency.
Given,

$$s(t) = 10 \cos\left[2\pi \times 10^5 t + 5 \sin(2\pi \times 1500t) + 7.5 \sin(2\pi \times 1000t)\right].$$

The instantaneous phase is

$$\theta(t) = 2\pi \times 10^5 t + 5 \sin(2\pi \times 1500t) + 7.5 \sin(2\pi \times 1000t).$$

The instantaneous angular frequency is

$$\omega_i(t) = \frac{d\theta(t)}{dt} = 2\pi \times 10^5 + 5(2\pi \times 1500) \cos(2\pi \times 1500t) + 7.5(2\pi \times 1000) \cos(2\pi \times 1000t).$$

The maximum angular frequency deviation is

$$\Delta\omega_{\max} = 2\pi\,(5 \times 1500 + 7.5 \times 1000) = 2\pi(15000).$$

Hence,

$$\Delta f = \frac{\Delta\omega_{\max}}{2\pi} = 15000 \text{ Hz}.$$

The highest modulating frequency is

$$f_{m,\max} = 1500 \text{ Hz}.$$

Therefore,

$$m_f = \frac{\Delta f}{f_{m,\max}} = \frac{15000}{1500} = 10.$$

$$\boxed{(B) m_f = 10}$$

Q4.08.3 MCQ 2008 1M Ans: A ●○○

Theory: For proper envelope detection, the modulation index of a conventional AM signal must satisfy $\mu \leq 1$ so that over-modulation does not occur.
Given,

$$s(t) = A_c \cos \omega_c t + 2 \cos \omega_m t \cos \omega_c t.$$

Factor out the carrier term,

$$s(t) = A_c \left(1 + \frac{2}{A_c} \cos \omega_m t\right) \cos \omega_c t.$$

Comparing with the standard AM signal,

$$s(t) = A_c \left(1 + \mu \cos \omega_m t\right) \cos \omega_c t,$$

the modulation index is

$$\mu = \frac{2}{A_c}.$$

For envelope detection,

$$\mu \leq 1.$$

Hence,

$$\frac{2}{A_c} \leq 1$$

or,

$$A_c \geq 2.$$

Therefore, the minimum value of A_c is

$$\boxed{(A) 2}$$

Mistakes to be avoided

- Always rewrite the given signal in the standard AM form before identifying the modulation index.
- Do not assume the coefficient of $\cos \omega_m t$ is the modulation index directly; compare with the standard AM expression.
- Remember that an envelope detector works correctly only when $\mu \leq 1$.
- The question asks for the *minimum* value of A_c; apply the equality condition $\mu = 1$.

Q4.07.1 MCQ 2007 2M Ans: A ●●●

Theory: The Hilbert transform suppresses one sideband by introducing a 90° phase shift, thereby generating an SSB signal whose spectrum is obtained by selecting only one shifted copy of the message spectrum.
From the given block diagram,

$$y(t) = m(t) \cos(2\pi B t) + \hat{m}(t) \sin(2\pi B t),$$

where $\hat{m}(t)$ is the Hilbert transform of $m(t)$.
Using Fourier transform properties,

$$m(t) \xleftrightarrow{\mathcal{F}} M(f),$$

$$\hat{m}(t) \xleftrightarrow{\mathcal{F}} \hat{M}(f) = -j \ \text{sgn}(f)\, M(f).$$

Hence,

$$Y_1(f) = \frac{1}{2}M(f-B) + \frac{1}{2}M(f+B),$$

and

$$Y_2(f) = \frac{1}{2j}\left[\hat{M}(f-B) - \hat{M}(f+B)\right].$$

Substituting the Hilbert transform property,

$$Y_2(f) = \frac{1}{2}\left[-\operatorname{sgn}(f-B)M(f-B) + \operatorname{sgn}(f+B)M(f+B)\right].$$

Therefore,

$$\boxed{Y(f) = \frac{1}{2}\left[1 - \operatorname{sgn}(f-B)\right]M(f-B) + \frac{1}{2}\left[1 + \operatorname{sgn}(f+B)\right]M(f+B)}$$

Now evaluate the sign functions.
For the shifted spectrum around $+B$,

$$1 - \operatorname{sgn}(f-B) = \begin{cases} 2, & f < B, \\ 0, & f > B. \end{cases}$$

Hence only the portion of $M(f-B)$ lying in

$$0 \le f \le B$$

is retained.
Similarly,

$$1 + \operatorname{sgn}(f+B) = \begin{cases} 0, & f < -B, \\ 2, & f > -B. \end{cases}$$

Hence only the portion of $M(f+B)$ lying in

$$-B \le f \le 0$$

is retained.
Thus, the output spectrum consists of

$$[-B, 0] \quad \text{and} \quad [0, B],$$

forming two triangular lobes meeting at the origin.
This spectrum exactly matches Option (A).

Mistakes to be avoided

- Do not treat the given structure as ordinary DSB-SC modulation; it is the canonical Hilbert transform method for SSB generation.
- Remember that $\hat{M}(f) = -j\ \operatorname{sgn}(f)M(f)$.
- Carefully evaluate the signum functions after frequency shifting by $\pm B$.
- Do not forget that only one sideband from each shifted spectrum survives after combining the two branches.
- Draw the shifted spectra $M(f-B)$ and $M(f+B)$ before applying the signum multipliers; this avoids selecting the wrong sideband.

Q4.06.1 MCQ 2006 2M Ans: B ●●○

Theory: In AM, the average sideband power depends on the modulation index, while the noise power is obtained by integrating the given PSD over the receiver bandwidth.

Given,

$$A_c = 10, \qquad \mu = 0.5.$$

Average sideband power is

$$P_s = P_c \frac{\mu^2}{2} = \frac{A_c^2}{2} \cdot \frac{\mu^2}{2}.$$

Substituting the values,

$$P_s = \frac{10^2}{2} \cdot \frac{(0.5)^2}{2} = 50 \cdot \frac{0.25}{2} = \frac{25}{4} \text{ W}.$$

From the given PSD,

$$S_n(f) = \frac{N_0}{2}$$

over a bandwidth $2B$.

Hence,

$$P_N = \int_{f_c - B}^{f_c + B} \frac{N_0}{2}\, df = \frac{N_0}{2}(2B) = N_0 B.$$

Therefore,

$$\frac{P_s}{P_N} = \frac{\frac{25}{4}}{N_0 B}$$

$$\boxed{(B)\ \frac{25}{4N_0 B}}.$$

Mistakes to be avoided

- Do not use the total transmitted power instead of the sideband power.
- Remember that $P_s = P_c \frac{\mu^2}{2}$, not $P_c \mu^2$.
- Compute the noise power by integrating the PSD over the receiver bandwidth.
- Do not forget that the PSD level is $\frac{N_0}{2}$.

Q4.06.2 MCQ 2006 2M Ans: C ●●○

Theory: The total sideband power of a conventional AM signal is determined only by the carrier power and modulation index.

Given,

$$A_c = 10, \qquad \mu = 0.5.$$

Carrier power is

$$P_c = \frac{A_c^2}{2} = \frac{10^2}{2} = 50 \text{ W}.$$

The total average sideband power is

$$P_s = P_c \frac{\mu^2}{2}.$$

Hence,

$$P_s = 50 \cdot \frac{(0.5)^2}{2} = 50 \cdot \frac{0.25}{2} = 50 \cdot 0.125 = 6.25 \text{ W}.$$

Therefore,

$$\boxed{(C) P_s = 6.25 \text{ W}.}$$

Mistakes to be avoided

- Do not confuse sideband power with total transmitted power.
- Always compute the carrier power first using $P_c = \frac{A_c^2}{2}$.
- Remember that the factor $\frac{1}{2}$ appears in $P_s = P_c \frac{\mu^2}{2}$.
- The result is the *total* sideband power (USB + LSB), not the power of a single sideband.

Q4.06.3 MCQ 2006 2M Ans: A ●●○

Theory: A conventional AM signal is identified by comparing it with the standard form $A_c[1 + \mu m(t)] \cos \omega_c t$, where the coefficient of the message determines the modulation index.
The given signal is

$$X_{AM}(t) = 100\,(p(t) + 0.5g(t)) \cos \omega_c t.$$

For $0 \le t \le 1$,

$$p(t) = 1.$$

Also,

$$g(t) = p(t) * p(t) = t, \qquad 0 \le t \le 1.$$

Hence,

$$X_{AM}(t) = 100\,(1 + 0.5t) \cos \omega_c t.$$

Comparing with the standard AM signal,

$$s(t) = A_c\,[1 + \mu m(t)] \cos \omega_c t,$$

gives

$$\mu m(t) = 0.5t.$$

Hence, the correct answer is

$$\boxed{(A) m(t) = t, \qquad \mu = 0.5.}$$

Mistakes to be avoided

- Use the interval $0 \le t \le 1$; within this interval $p(t) = 1$.
- Remember that the convolution $p(t) * p(t)$ is triangular, so $g(t) = t$ only for $0 \le t \le 1$.
- Compare the given signal with the standard AM form $A_c[1 + \mu m(t)] \cos \omega_c t$.
- Do not confuse the modulation index μ with the message signal $m(t)$.

Q4.06.4 MCQ 2006 2M Ans: A ● ○ ○

Theory: The envelope detector time constant should be large enough to remove carrier ripple but small enough to follow the variations of the message envelope, thereby preventing diagonal clipping.
For proper operation of an envelope detector,

$$\frac{1}{\omega_c} < RC < \frac{1}{W},$$

where

$$W = \text{message bandwidth}, \qquad \omega_c = \text{carrier angular frequency}.$$

To avoid diagonal clipping, the capacitor must discharge fast enough to track the highest variation of the message envelope. Hence,

$$\boxed{(A) RC < \frac{1}{W}.}$$

Mistakes to be avoided

- Do not confuse the message bandwidth W with the carrier angular frequency ω_c.
- Remember that both conditions must hold simultaneously: $\frac{1}{\omega_c} < RC < \frac{1}{W}$.
- A very large RC causes diagonal clipping because the capacitor cannot discharge fast enough.
- A very small RC causes excessive carrier ripple at the detector output.

Q4.06.5 MCQ 2006 2M Ans: C ● ● ○

Theory: In narrowband FM, the bandwidth is approximately twice the highest frequency present in the modulating signal, irrespective of the FM carrier frequency.
The message bandwidth is

$$B = 10 \text{ kHz}.$$

After LSB-SSB modulation with carrier frequency

$$f_{c1} = 10^6 \text{ Hz} = 1000 \text{ kHz},$$

the modulating signal applied to the NBFM has frequency components from

$$1000 - 10 = 990 \text{ kHz}$$

to

$$1000 \text{ kHz}.$$

Hence, the highest modulating frequency is

$$f_m = 1000 + 10 = 1010 \text{ kHz} = 1.01 \times 10^6 \text{ Hz}.$$

For a Narrow-Band FM signal,

$$\text{BW} \approx 2 f_m.$$

Therefore,

$$\text{BW} = 2 \times 1010 \text{ kHz} = 2020 \text{ kHz} = 2.02 \times 10^6 \text{ Hz} \approx 2 \times 10^6 \text{ Hz}.$$

Hence, the correct answer is

$$\boxed{(C) 2 \times 10^6 \text{ Hz}.}$$

Mistakes to be avoided

- Do not use the original message bandwidth (10 kHz) directly for the NBFM calculation.
- Remember that the SSB output becomes the input (modulating signal) to the FM modulator.
- The carrier frequency $f_{c2} = 10^9$ Hz does not affect the bandwidth of an NBFM signal.
- For NBFM, use BW $\approx 2f_m$, where f_m is the highest frequency present in the modulating signal.

Q4.05.1 MCQ 2005 2M Ans: D ●●○

Theory: In phase modulation, if the phase deviation remains constant, the modulation index remains constant and the frequency deviation is directly proportional to the modulating frequency.
For PM,

$$m_f = \Delta\phi = \frac{\Delta f}{f_m}.$$

Initially,

$$\Delta f = 10 \text{ kHz}, \qquad f_m = 1 \text{ kHz}.$$

Hence,

$$m_f = \frac{10}{1} = 10.$$

Since the phase deviation remains unchanged,

$$m_f = 10.$$

When

$$f_m = 2 \text{ kHz},$$

the new frequency deviation is

$$\Delta f = m_f f_m = 10 \times 2 = 20 \text{ kHz}.$$

Using Carson's rule,

$$BW = 2(\Delta f + f_m) = 2(20 + 2) = 44 \text{ kHz}.$$

Hence, the correct answer is

$$\boxed{44 \text{ kHz}.}$$

Mistakes to be avoided

- In PM, constant phase deviation implies constant modulation index, not constant frequency deviation.
- Recalculate the frequency deviation when the modulating frequency changes.
- Use Carson's rule only after finding the new value of Δf.
- Do not confuse PM with FM, where the frequency deviation remains constant for a fixed message amplitude.

Q4.05.2 MCQ 2005 2M Ans: A ● ○ ○

Theory: Squaring an FM signal doubles its instantaneous phase, thereby doubling the frequency deviation while leaving the modulating frequency unchanged.
The input FM signal has

$$\Delta f = 90 \text{ kHz}, \qquad f_m = 5 \text{ kHz}.$$

Since

$$y(t) = x^2(t),$$

the instantaneous phase doubles. Hence,

$$\Delta f_{\text{out}} = 2\Delta f = 2 \times 90 = 180 \text{ kHz}.$$

The modulating signal bandwidth remains

$$f_m = 5 \text{ kHz}.$$

Using Carson's rule,

$$BW = 2(\Delta f_{\text{out}} + f_m) = 2(180 + 5) = 370 \text{ kHz}.$$

Hence, the correct answer is

$$\boxed{(A) 370 \text{ kHz}.}$$

Mistakes to be avoided

- Remember that squaring an FM signal doubles the instantaneous phase and hence doubles the frequency deviation.
- Do not double the modulating frequency or message bandwidth; only the frequency deviation changes.
- Apply Carson's rule using the updated value of Δf.
- Do not confuse frequency doubling of the carrier with doubling of the message bandwidth.

Q4.05.3 MCQ 2005 1M Ans: C ● ○ ○

Theory: SSB transmits only one sideband while suppressing the carrier, thereby minimizing both transmitted power and required bandwidth.
Among the given modulation schemes,

AM : Carrier + two sidebands,
DSB-SC : Two sidebands only,
VSB : One complete sideband and a vestige of the other,
SSB : Only one sideband.

Since SSB transmits only a single sideband,

$$\boxed{\text{It requires the minimum transmitted power and the minimum channel bandwidth.}}$$

Hence, the correct answer is

$$\boxed{(C) \text{SSB}.}$$

Mistakes to be avoided

- Do not confuse SSB with DSB-SC; DSB-SC still transmits both sidebands.
- VSB requires more bandwidth than SSB because it includes a vestige of the second sideband.
- Conventional AM has the highest transmitted power due to the carrier component.

- Remember that minimum bandwidth and minimum transmitted power are achieved simultaneously by SSB.

Q4.05.4 MCQ 2005 1M Ans: D ●●○

Theory: Each analog modulation scheme has a unique mathematical representation that directly identifies whether the message varies amplitude, phase, or frequency.
The standard forms are

$$A_c\,[1 + k_a m(t)]\sin(\omega_c t) \quad \Rightarrow \quad \text{AM},$$

$$A_c\,m(t)\sin(\omega_c t) \quad \Rightarrow \quad \text{DSB-SC},$$

$$A_c \sin\big(\omega_c t + k_p m(t)\big) \quad \Rightarrow \quad \text{PM},$$

$$A_c \sin\left(\omega_c t + k_f \int m(t)\,dt\right) \quad \Rightarrow \quad \text{FM}.$$

Hence,

$$\boxed{(D)P - Y,\ Q - Z,\ R - W,\ S - X.}$$

Mistakes to be avoided

- Do not confuse AM with DSB-SC; AM contains the carrier term $(1 + k_a m(t))$, whereas DSB-SC does not.
- Remember that PM contains $m(t)$ directly inside the phase.
- Remember that FM contains the integral of $m(t)$.
- Match the mathematical expression with the standard modulation form before selecting the answer.

Q4.04.1 MCQ 2004 2M Ans: B ●●○

Theory: Balanced modulation shifts the spectrum by the carrier frequency, while the HPF removes frequencies below its cutoff before the second frequency translation.
From the given spectrum, the positive spectral peak of $X(f)$ is at

$$f = 1 \text{ kHz}.$$

After modulation with the 10 kHz carrier,

$$f = 10 \pm 1 = 9 \text{ kHz},\ 11 \text{ kHz}.$$

The ideal HPF has cutoff frequency 10 kHz, hence only

$$11 \text{ kHz}$$

is passed.
The second balanced modulator shifts this component by 13 kHz:

$$f = 13 \pm 11 = 2 \text{ kHz},\ 24 \text{ kHz}.$$

Hence, the positive spectral peaks of $Y(f)$ are

$$\boxed{(B)2 \text{ kHz and } 24 \text{ kHz}.}$$

Q4.04.2 MCQ | 2004 | 2M | Ans: D | ● ● ○

Theory: An ideal frequency detector produces an output proportional to the instantaneous frequency, i.e., the time derivative of the instantaneous phase.
The input signal is

$$x(t) = \cos(2\pi f_c t) + \cos\big(2\pi(f_c + f_m)t\big),$$

where

$$f_c = 10\ \text{kHz}, \qquad f_c + f_m = 10.1\ \text{kHz}.$$

Using

$$\cos(A + B) = \cos A \cos B - \sin A \sin B,$$

$$\begin{aligned} x(t) &= \cos(2\pi f_c t) + \cos(2\pi f_c t)\cos(2\pi f_m t) \\ &\quad - \sin(2\pi f_c t)\sin(2\pi f_m t) \\ &= [1 + \cos(2\pi f_m t)]\cos(2\pi f_c t) - \sin(2\pi f_m t)\sin(2\pi f_c t). \end{aligned}$$

Hence, the instantaneous phase is

$$\phi(t) = \tan^{-1}\left(\frac{\sin(2\pi f_m t)}{1 + \cos(2\pi f_m t)}\right).$$

Using

$$\sin 2\theta = 2\sin\theta\cos\theta, \qquad 1 + \cos 2\theta = 2\cos^2\theta,$$

$$\phi(t) = \tan^{-1}\left(\frac{2\sin(\pi f_m t)\cos(\pi f_m t)}{2\cos^2(\pi f_m t)}\right) = \tan^{-1}\left(\tan(\pi f_m t)\right) = \pi f_m t.$$

The output of an ideal frequency detector is

$$y(t) = k\frac{d\phi(t)}{dt},$$

where k is a constant.
Therefore,

$$y(t) = k\pi f_m,$$

which is a constant.
Hence, the correct answer is

(D) a constant.

Mistakes to be avoided

- Do not assume the detector output is the beat frequency directly.
- Remember that an ideal frequency detector differentiates the instantaneous phase.

Q4.04.3 MCQ | 2004 | 2M | Ans: C ●●○

Theory: A balanced modulator generates DSB-SC, the HPF retains only the upper sideband, and adding a quadrature carrier changes the resultant envelope.

The balanced modulator output is

$$\cos(2\pi 10^6 t)\cos(2\pi 100\times 10^6 t) = \frac{1}{2}\cos(2\pi 101\times 10^6 t) + \frac{1}{2}\cos(2\pi 99\times 10^6 t).$$

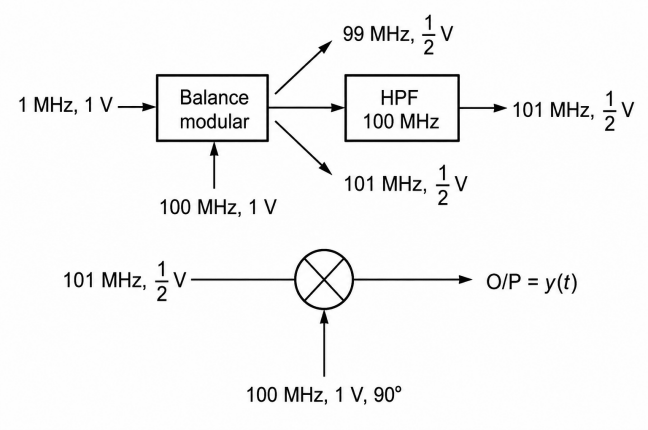


Fig. Solution Q4.04.3

The ideal HPF with cut-off frequency 100 MHz removes the 99 MHz component. Hence,

$$y_1(t) = \frac{1}{2}\cos(2\pi 101\times 10^6 t).$$

The 100 MHz carrier with 90° phase shift is

$$y_2(t) = \cos\left(2\pi 100\times 10^6 t + \frac{\pi}{2}\right) = -\sin(2\pi 100\times 10^6 t).$$

Therefore,

$$y(t) = \frac{1}{2}\cos(2\pi 101\times 10^6 t) - \sin(2\pi 100\times 10^6 t).$$

Using

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$

$$y(t) = \frac{1}{2}\cos(2\pi 100\times 10^6 t)\cos(2\pi 10^6 t) - \frac{1}{2}\sin(2\pi 100\times 10^6 t)\sin(2\pi 10^6 t) - \sin(2\pi 100\times 10^6 t).$$

Grouping the carrier components,

$$y(t) = A\cos(2\pi 100\times 10^6 t) + B\sin(2\pi 100\times 10^6 t),$$

where

$$A = \frac{1}{2}\cos(2\pi 10^6 t),$$

$$B = -\left(1 + \frac{1}{2}\sin(2\pi 10^6 t)\right).$$

Hence, the envelope is

$$\sqrt{A^2 + B^2} = \sqrt{\frac{1}{4}\cos^2(2\pi 10^6 t) + \left(1 + \frac{1}{2}\sin(2\pi 10^6 t)\right)^2}.$$

Using

$$\cos^2\theta + \sin^2\theta = 1,$$

$$\boxed{(C)\text{Envelope} = \sqrt{\frac{5}{4} - \sin(2\pi \times 10^6 t)}.}$$

Mistakes to be avoided

- Do not forget that the balanced modulator generates both upper and lower sidebands with amplitude halved.
- Apply the HPF before adding the quadrature carrier.
- While computing the envelope, first express the signal as $A\cos\omega_c t + B\sin\omega_c t$, then use $\sqrt{A^2 + B^2}$.

Q4.04.4 MCQ 2004 1M Ans: B

Theory: Adding an AM signal and an NBFM signal with the same carrier and message causes one sideband to cancel, leaving a single sideband along with the carrier.
For a narrow-band FM signal,

$$s_{\text{NBFM}}(t) = A_c\cos\omega_c t - \frac{A_c\beta}{2}\cos(\omega_c + \omega_m)t + \frac{A_c\beta}{2}\cos(\omega_c - \omega_m)t.$$

For an AM signal,

$$s_{\text{AM}}(t) = A_c\cos\omega_c t + \frac{A_c m_a}{2}\cos(\omega_c + \omega_m)t + \frac{A_c m_a}{2}\cos(\omega_c - \omega_m)t.$$

Since

$$m_a = \beta = 0.1,$$

adding the two signals gives

$$\begin{aligned} s(t) &= s_{\text{AM}}(t) + s_{\text{NBFM}}(t) \\ &= 2A_c\cos\omega_c t + A_c\beta\cos(\omega_c - \omega_m)t. \end{aligned}$$

The upper sideband cancels while the lower sideband remains along with the carrier.
Hence, the resultant signal is

$$\boxed{\text{(B) SSB with carrier.}}$$

Q4.04.5 MCQ 2004 1M Ans: B ● ○ ○

Theory: The envelope detector time constant must be large enough to suppress carrier ripple and small enough to follow the message envelope.

For proper envelope detection,

$$\frac{1}{f_c} < RC < \frac{1}{f_m}.$$

Given,

$$f_c = 1 \text{ MHz}, \qquad f_m = 2 \text{ kHz}.$$

Hence,

$$1\ \mu\text{s} < RC < 500\ \mu\text{s}.$$

Among the given options,

$$20\ \mu\text{s}$$

satisfies the required condition.
Therefore,

$$\boxed{(B) RC = 20\ \mu\text{s}.}$$

Mistakes to be avoided

- Do not reverse the inequalities for the envelope detector time constant.
- Remember that $RC > \frac{1}{f_c}$ suppresses carrier ripple.
- Remember that $RC < \frac{1}{f_m}$ avoids diagonal clipping.
- Check that the selected option satisfies both inequalities simultaneously.

Q4.03.1 MCQ 2003 2M Ans: A ● ● ○

Theory: In a superheterodyne receiver, the local oscillator tracks the tuned frequency while maintaining a constant intermediate frequency, and the image frequency lies $2f_{IF}$ away from the desired signal.
For a tuned LC oscillator,

$$f_{LO} = \frac{1}{2\pi\sqrt{LC}}.$$

Since L is constant,

$$f_{LO} \propto \frac{1}{\sqrt{C}}$$

which gives

$$R = \frac{C_{\max}}{C_{\min}} = \left(\frac{f_{LO,\max}}{f_{LO,\min}}\right)^2.$$

Using high-side injection,

$$f_{LO,\max} = 1650 + 450 = 2100 \text{ kHz},$$

$$f_{LO,\min} = 550 + 450 = 1000 \text{ kHz}.$$

Hence,

$$R = \left(\frac{2100}{1000}\right)^2 = 4.41.$$

For a receiver tuned to

$$f_s = 700 \text{ kHz},$$

the image frequency is

$$I = f_s + 2f_{IF} = 700 + 2(450) = 1600 \text{ kHz}.$$

Therefore,

$$\boxed{(A) R = 4.41, \quad I = 1600 \text{ kHz}.}$$

Mistakes to be avoided

- Do not use the RF tuning range directly; first determine the local oscillator frequency range.
- Remember that $C \propto 1/f^2$, so the capacitance ratio is the square of the frequency ratio.
- For high-side injection, use $f_{LO} = f_s + f_{IF}$.
- The image frequency is $f_s + 2f_{IF}$ for high-side injection, not $f_s + f_{IF}$.

Q4.03.2 MCQ 2003 2M Ans: B ●●○

Theory: Each communication block performs a specific function such as modulation, demodulation, frequency conversion, or clock recovery.
The correct matching is

P :	Ring modulator	$\rightarrow$	Generation of DSB-SC	(6),
Q :	VCO	$\rightarrow$	Generation of FM	(5),
R :	Foster-Seeley discriminator	$\rightarrow$	Demodulation of FM	(2),
S :	Mixer	$\rightarrow$	Frequency conversion	(3).

Hence,

$$\boxed{(B) P - 6,\ Q - 5,\ R - 2,\ S - 3}$$

Key Points

- Ring modulator is used for generating DSB-SC signals.
- VCO is the basic block for FM generation.
- Foster-Seeley discriminator is an FM demodulator.
- A mixer performs frequency translation (frequency conversion).

Mistakes to be avoided

- Do not confuse a mixer with an adder; a mixer performs multiplication and frequency conversion.
- VCO generates FM, not DSB-SC.
- Foster-Seeley discriminator is used for FM demodulation, not FM generation.

Q4.03.3 MCQ 2003 2M Ans: D ●●○

Theory: In an FM signal, the amplitude of the n^{th} sideband is $A_c J_n(\beta)$, where $\beta = \Delta\omega/\omega_m$.
For an AM signal,

$$BW_{AM} = 2f_m.$$

Given,

$$\Delta f = 3BW_{AM} = 3(2f_m) = 6f_m.$$

Hence,

$$\beta = \frac{\Delta f}{f_m} = 6.$$

The FM signal is

$$s_{FM}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos[(\omega_c + n\omega_m)t]\,.$$

Here,

$$\omega_c = 2\pi \times 10^6, \qquad \omega_m = 4\pi \times 10^3 = 2\pi \times 2 \times 10^3.$$

The required frequency is

$$1008 \times 10^3 \text{ Hz}.$$

Therefore,

$$10^6 + n(2 \times 10^3) = 1008 \times 10^3,$$

which gives

$$n = 4.$$

Since $A_c = 5$ and $\beta = 6$,

$$\boxed{\text{(D) Coefficient} = 5J_4(6).}$$

Mistakes to be avoided

- Do not confuse AM transmission bandwidth with FM bandwidth; here $BW_{AM} = 2f_m$.
- Use $\beta = \Delta f / f_m = \Delta\omega/\omega_m$.
- Find the sideband order n from the given frequency before selecting the Bessel coefficient.
- Multiply the Bessel coefficient by the carrier amplitude A_c.

Q4.03.4 MCQ 2003 2M Ans: D ● ● ○

Theory: In conventional AM, the total sideband power depends only on the modulation index and equals $\frac{m^2}{2}$ times the carrier power.
For an AM signal,

$$P_c = \frac{A_c^2}{2}.$$

The total sideband power is

$$P_{SB} = \frac{A_c^2}{2} \cdot \frac{m^2}{2}.$$

Hence,

$$\frac{P_{SB}}{P_c} = \frac{\left(\frac{A_c^2}{2}\right)\left(\frac{m^2}{2}\right)}{\frac{A_c^2}{2}} = \frac{m^2}{2}.$$

Given,

$$m = 0.5,$$

therefore,

$$\frac{P_{SB}}{P_c} = \frac{(0.5)^2}{2} = \frac{0.25}{2}$$

$$\boxed{(D)\ \frac{1}{8}.}$$

Mistakes to be avoided

- Do not use the total transmitted power formula when only sideband power is asked.
- Remember that the ratio is $\frac{m^2}{2}$, not $\frac{m^2}{4}$.
- Use the given modulation index directly; carrier amplitude cancels in the ratio.
- Distinguish between total sideband power and power in a single sideband.

Q4.03.5 MCQ 2003 2M Ans: C ● ● ○

Theory: A non-linear device generates cross-product terms, and the required DSB-SC component is obtained by selecting the appropriate frequency component using a band-pass filter. Given,

$$V_0 = a_0 v_i + a_1 v_i^3,$$

with

$$v_i = A_c' \cos(2\pi f_c' t) + m(t).$$

Expanding,

$$\begin{aligned} V_0 = {} & a_0 A_c' \cos(2\pi f_c' t) + a_0 m(t) \\ & + a_1 A_c'^3 \cos^3(2\pi f_c' t) + a_1 m^3(t) \\ & + 3a_1 A_c'^2 \cos^2(2\pi f_c' t) m(t) + 3a_1 A_c' \cos(2\pi f_c' t) m^2(t). \end{aligned}$$

The required DSB-SC term comes from

$$3a_1A_c'^2\cos^2(2\pi f_c't)m(t).$$

Using

$$\cos^2\theta = \frac{1+\cos 2\theta}{2},$$

we obtain

$$3a_1A_c'^2\frac{1+\cos(4\pi f_c't)}{2}m(t).$$

Hence, the modulated component is centered at

$$2f_c'.$$

For the required DSB-SC carrier frequency,

$$2f_c' = 1 \text{ MHz}.$$

Therefore,

$$\boxed{(C)\, f_c' = 0.5 \text{ MHz}.}$$

Mistakes to be avoided

- Do not assume the carrier frequency appears directly; identify which nonlinear term generates the DSB-SC component.
- Always expand $\cos^2\theta$ using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$.
- Remember that the DSB-SC spectrum is centered at $2f_c'$, not at f_c'.
- Ignore terms such as $m(t)$, $m^2(t)$, and $m^3(t)$ since they do not produce the required DSB-SC carrier component.

Q4.03.6 MCQ 2003 1M Ans: A ● ○ ○

Theory: A coherent detector multiplies the received signal with a synchronized carrier, so only the in-phase noise component contributes to the output while the quadrature component is rejected. In coherent detection, the received signal is multiplied by a locally generated carrier that is phase synchronized with the transmitted carrier.
The received noise can be resolved into two orthogonal components:

$$n(t) = n_I(t)\cos\omega_c t - n_Q(t)\sin\omega_c t,$$

where

$$n_I(t) = \text{In-phase component}, \qquad n_Q(t) = \text{Quadrature component}.$$

During coherent demodulation,

$$n_I(t)$$

appears at the detector output after low-pass filtering, whereas

$$n_Q(t)$$

is rejected because it is orthogonal to the reference carrier.
Hence, the detector output contains only the in-phase component of noise.

$$\boxed{\text{(A) Noise at the detector output is the in-phase component.}}$$

Mistakes to be avoided

- Do not confuse coherent detection with envelope detection.
- Remember that only the in-phase noise component passes through the coherent detector.
- The quadrature component is rejected due to orthogonality with the reference carrier.
- The detector output is not noise-free; the in-phase noise component still remains.

Q4.02.1 MCQ 2002 2M Ans: C ●●○

Theory: Coherent detection shifts the desired DSB-SC signal back to baseband, while the added bandpass noise is translated and only the components within the LPF bandwidth appear at the output. The output of the first multiplier is

$$m(t)s(t) = \frac{2\sin 2\pi t}{t}\cos 200\pi t = \frac{1}{t}(\sin 202\pi t - \sin 198\pi t).$$

After addition of noise,

$$x(t) = m(t)s(t) + n(t).$$

The second multiplier gives

$$x(t)\cos 200\pi t = \left[\frac{1}{t}(\sin 202\pi t - \sin 198\pi t) + \frac{\sin 199\pi t}{t}\right]\cos 200\pi t.$$

Using

$$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)],$$

we obtain

$$x(t)\cos 200\pi t = \frac{1}{2t}(\sin 402\pi t + \sin 2\pi t - \sin 398\pi t + \sin 2\pi t) + \frac{1}{2t}(\sin 399\pi t - \sin \pi t).$$

The low-pass filter has cutoff frequency 1 Hz, so it removes the high-frequency terms

$$\sin 402\pi t,\ \sin 399\pi t,\ \sin 398\pi t,$$

and passes only

$$\frac{1}{2t}(\sin 2\pi t + \sin 2\pi t - \sin \pi t).$$

Thus,

$$y(t) = \frac{\sin 2\pi t}{t} - \frac{\sin \pi t}{2t}.$$

Using

$$2\sin A\cos B = \sin(A+B) + \sin(A-B),$$

with

$$A = \frac{\pi t}{2}, \qquad B = \frac{3\pi t}{2},$$

gives

$$\sin \pi t = 2\sin\frac{\pi t}{2}\cos\frac{3\pi t}{2}.$$

Hence,

$$\boxed{(C)\ y(t) = \frac{\sin 2\pi t}{t} + \frac{\sin 0.5\pi t}{t}\cos 1.5\pi t.}$$

Mistakes to be avoided

- Do not forget that coherent detection multiplies the received signal by the same carrier before low-pass filtering.
- Apply the identity $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$ correctly.
- Remove all frequency components outside the 1 Hz LPF passband.
- Use the identity $2\sin A \cos B = \sin(A+B) + \sin(A-B)$ to obtain the final expression in the required form.

Q4.02.2 MCQ 2002 2M Ans: D ●●○

Theory: The maximum phase deviation equals the peak phase term, while the maximum frequency deviation is obtained from the maximum value of the derivative of the instantaneous phase.

The instantaneous phase is

$$\phi(t) = 2\pi\left(2\times10^{6}t + 30\sin 150t + 40\cos 150t\right).$$

Hence,

$$\frac{d\phi(t)}{dt} = 2\pi(2\times10^{6}) + 2\pi(4500\cos 150t) - 2\pi(6000\sin 150t).$$

The maximum angular frequency deviation is

$$\Delta\omega = 2\pi\sqrt{4500^2 + 6000^2} = 2\pi(7500)\ \text{rad/s}.$$

Therefore,

$$\Delta f = \frac{\Delta\omega}{2\pi} = 7500\ \text{Hz} = 7.5\ \text{kHz}.$$

The maximum phase deviation is

$$\Delta\phi = 2\pi\sqrt{30^2 + 40^2} = 2\pi(50) = 100\pi\ \text{rad}.$$

Hence,

$$\boxed{(D)\ \Delta f = 7.5\ \text{kHz}, \qquad \Delta\phi = 100\pi\ \text{rad}.}$$

Mistakes to be avoided

- Do not treat the coefficients of the sine and cosine terms separately; combine them using the root-sum-square value.
- Differentiate the instantaneous phase to obtain frequency deviation.
- Remember that $\Delta f = \frac{\Delta\omega}{2\pi}$.
- Phase deviation is obtained directly from the peak phase term, not by differentiation.

Q4.02.3 MCQ 2002 1M Ans: C ●○○

Theory: A symmetrical square wave contains only odd harmonics, so AM sidebands appear only at carrier frequencies shifted by odd multiples of the message frequency.

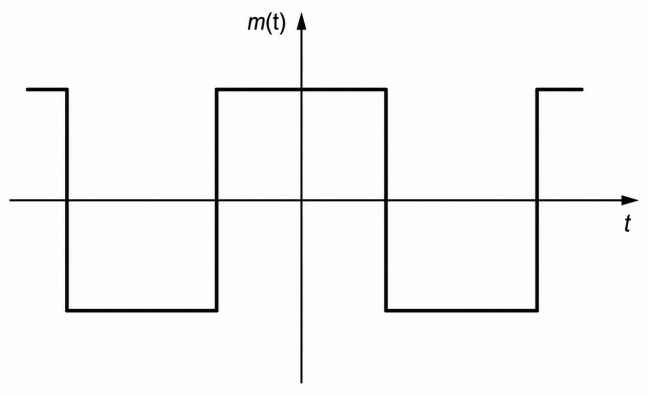


Fig. Solution Q4.02.3

The period of the square wave is

$$T_0 = 100\ \mu\text{sec}.$$

Hence, the fundamental message frequency is

$$f_m = \frac{1}{T_0} = \frac{1}{100 \times 10^{-6}} = 10\ \text{kHz}.$$

Since the message is a symmetrical square wave, its Fourier series contains only odd harmonics:

$$10\ \text{kHz},\ 30\ \text{kHz},\ 50\ \text{kHz}, \ldots$$

The carrier frequency is

$$f_c = 1\ \text{MHz} = 1000\ \text{kHz}.$$

Therefore, the AM sidebands occur at

$$f_c \pm (2n-1) f_m, \qquad n = 1, 2, 3, \ldots$$

Thus, the frequencies present are

$$1000 \pm 10 = 990,\ 1010\ \text{kHz},$$
$$1000 \pm 30 = 970,\ 1030\ \text{kHz},$$
$$1000 \pm 50 = 950,\ 1050\ \text{kHz},$$

and so on.
Since 20 kHz is an even harmonic, the frequency

$$1000 + 20 = 1020\ \text{kHz}$$

will not be present.

$$\boxed{(C)\, 1020\ \text{kHz}}$$

Mistakes to be avoided

- Remember that a symmetrical square wave contains only odd harmonics.
- Do not assume sidebands occur at every multiple of the message frequency.
- Compute the message frequency correctly using $f_m = 1/T_0$.
- Sidebands appear at $f_c \pm n f_m$, where only odd values of n exist for a symmetrical square wave.

Q4.00.1 MCQ 2000 2M Ans: A ● ● ○

Theory: Raising an FM signal to the n^{th} power multiplies the carrier frequency and frequency deviation by n, while the modulating frequency remains unchanged.
Adjacent spectral components are separated by

$$f_m = 5 \text{ kHz}.$$

Using Carson's rule,

$$BW = 2(\Delta f + f_m) = 1 \text{ MHz}.$$

Hence,

$$\Delta f + f_m = 500 \text{ kHz}$$

$$\Delta f = 500 - 5 = 495 \text{ kHz}.$$

Since

$$y(t) = (\text{FM signal})^3,$$

the non-linearity triples the carrier frequency and frequency deviation:

$$(\Delta f)_{\text{new}} = 3 \times 495 = 1485 \text{ kHz},$$

$$(f_c)_{\text{new}} = 300 \text{ MHz}.$$

The modulating frequency remains

$$f_m = 5 \text{ kHz}.$$

Therefore,

$$BW_{\text{new}} = 2\,(1485 + 5) \times 10^3 = 2.98 \text{ MHz} \approx 3 \text{ MHz}.$$

The spacing between adjacent spectral components remains

$$5 \text{ kHz}.$$

Hence,

$$\boxed{(A) 3 \text{ MHz},\ 5 \text{ kHz}}$$

Mistakes to be avoided

- Do not multiply the modulating frequency by the non-linearity order.
- Only the carrier frequency and frequency deviation scale by the factor n.
- Apply Carson's rule after computing the new frequency deviation.
- Spectral spacing in FM is always equal to the modulating frequency.

Q4.00.2 MCQ 2000 2M Ans: D ● ● ○

Theory: Multiplication with a coherent carrier followed by low-pass filtering retains only the baseband component scaled by the phase mismatch factor.
The multiplier output is

$$m(t) \cos \omega_0 t \cdot \cos(\omega_0 t + \theta).$$

Using

$$\cos A \cos B = \frac{1}{2} \left[\cos(A - B) + \cos(A + B)\right],$$

we get

$$= \frac{m(t)}{2} \left[\cos(2\omega_0 t + \theta) + \cos\theta\right].$$

The ideal LPF removes the high-frequency term, therefore

$$v_o(t) = \frac{m(t)}{2}\cos\theta.$$

Hence,

$$P_{\text{out}} = \lim_{T\to\infty}\frac{1}{T}\int v_o^2(t)\,dt$$

$$= \lim_{T\to\infty}\frac{1}{T}\int\left(\frac{m(t)}{2}\cos\theta\right)^2 dt$$

$$= \frac{\cos^2\theta}{4}\left[\lim_{T\to\infty}\frac{1}{T}\int m^2(t)\,dt\right].$$

Since

$$P_m = \lim_{T\to\infty}\frac{1}{T}\int m^2(t)\,dt,$$

$$\boxed{(D)\,P_{\text{out}} = \frac{P_m\cos^2\theta}{4}}$$

Mistakes to be avoided

- Do not forget the factor of $\frac{1}{2}$ obtained from the product-to-sum identity.
- The LPF removes the $2\omega_0$ component completely.
- Power is proportional to the square of amplitude, hence $\cos^2\theta$ appears instead of $\cos\theta$.
- Use the given message power P_m directly after evaluating the average of $m^2(t)$.

Q4.00.3 MCQ 2000 1M Ans: C ● ○ ○

Theory: When an AM signal is overmodulated, the envelope detector follows the magnitude of the envelope instead of its signed value.

The AM signal is

$$s(t) = A_c[1 + K_a m(t)]\cos\omega_c t.$$

Since

$$|K_a m(t)| > 1,$$

overmodulation occurs.

Hence, the envelope detector produces

$$\boxed{(C)\,|A_c\,[1 + K_a m(t)]|}.$$

Mistakes to be avoided

- Envelope detectors cannot preserve the sign after envelope inversion.
- Overmodulation occurs when $|K_a m(t)| > 1$.
- Envelope is always the magnitude of the signal envelope.

Q4.99.1 MCQ 1999 1M Ans: D ● ○ ○

Theory: The transmission bandwidth is determined by the highest baseband frequency present in the quadrature modulated signal.
Highest transmitted frequency

$$= f_c + 15 \text{ kHz}.$$

Lowest transmitted frequency

$$= f_c - 15 \text{ kHz}.$$

Hence,

$$BW = (f_c + 15) - (f_c - 15)$$

$$= 30 \text{ kHz}.$$

Therefore,

$$\boxed{(D) 30 \text{ kHz}}$$

Mistakes to be avoided

- Do not add the bandwidths of $m_1(t)$ and $m_2(t)$.
- Overall bandwidth is decided by the larger message bandwidth.
- Remember that DSB transmission bandwidth is twice the maximum baseband bandwidth.
- Ignore the carrier frequency while calculating bandwidth.

Q4.98.1 MCQ 1998 1M Ans: B ● ○ ○

Theory: The envelope of a DSB-SC signal is obtained from the magnitude of its in-phase and quadrature components.
The signal is

$$s(t) = x(t)\cos(\omega_c t + \theta)$$

Expanding,

$$s(t) = x(t)\cos\theta\cos\omega_c t - x(t)\sin\theta\sin\omega_c t.$$

Hence,

$$\text{Envelope} = \sqrt{[x(t)\cos\theta]^2 + [x(t)\sin\theta]^2}$$

$$= \sqrt{x^2(t)(\cos^2\theta + \sin^2\theta)}$$

$$= |x(t)|.$$

Therefore,

$$\boxed{(B) |x(t)|}$$

Mistakes to be avoided

- Include both in-phase and quadrature components while evaluating the envelope.

- Use $\cos^2\theta + \sin^2\theta = 1$.
- Envelope is always non-negative.

Q4.98.2 MCQ 1998 1M Ans: D ●○○

Theory: Image frequency must be rejected before the mixer, since after frequency conversion it becomes indistinguishable from the desired signal.
Image rejection is performed before the mixer by the RF front-end.
Hence, image channel selectivity depends upon

(D) Pre selector and RF amplifiers only

Mistakes to be avoided

- IF amplifiers cannot reject image frequency because image and desired signal produce the same IF.
- Image rejection always occurs before the mixer.

Q4.96.1 MCQ 1996 1M Ans: B ●○○

Theory: The image frequency must be rejected before mixing; therefore image channel rejection is achieved only by the RF section.

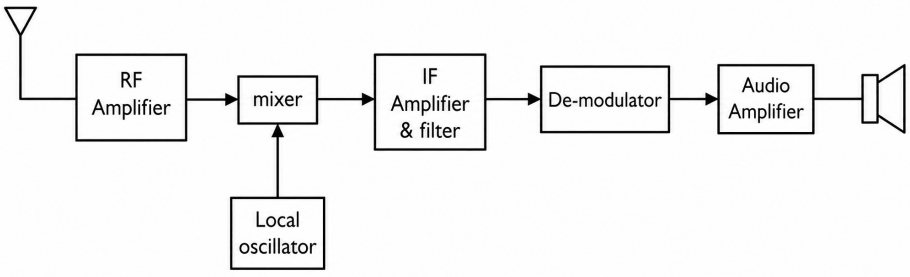


Fig. Solution Q4.96.1

The image frequency component must be removed before the mixer. Once mixing takes place, both the desired signal and the image signal are translated to the same intermediate frequency. Therefore, neither the IF amplifier nor the detector can distinguish between them. Hence, image channel rejection is provided only by the RF stages.

(B) RF stages only

Key Points

- Image rejection occurs before the mixer.
- RF stages provide image channel rejection.
- IF and detector stages cannot reject image signals after frequency conversion.

Mistakes to be avoided

- Do not include IF stages in image rejection.
- Do not confuse image rejection with adjacent channel selectivity.
- Remember that after mixing, the desired and image signals have the same IF.

Q4.96.2 MCQ 1996 2M Ans: D ● ○ ○

Theory: A frequency multiplier multiplies both the carrier frequency and the frequency deviation by the same factor, while the modulating frequency remains unchanged.
Given,

$$\beta = 9$$

For a frequency tripler,

$$n = 3$$

Frequency multiplication changes the carrier frequency and frequency deviation by the factor n.
Hence,

$$\Delta f' = 3\Delta f$$

Since the modulating frequency remains unchanged,

$$\beta' = \frac{\Delta f'}{f_m} = \frac{3\Delta f}{f_m} = 3\beta$$

Therefore,

$$\beta' = 3 \times 9 = 27.$$

(D) 27

Key Points

- Frequency multipliers increase both carrier frequency and frequency deviation by the multiplication factor.
- The modulating frequency remains unchanged.
- Modulation index increases by the same multiplication factor.

Q4.95.1 MCQ 1995 1M Ans: B ● ○ ○

Theory: PAM is recovered by extracting the average value of the sampled pulses using a low-pass filter or an integrator.
A PAM signal consists of pulses whose amplitudes follow the message signal.

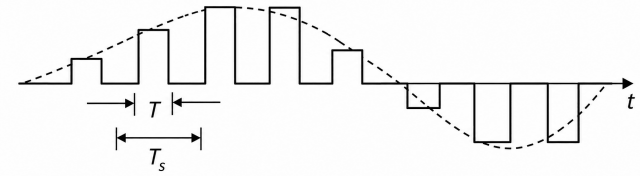


Fig. Solution Q4.95.1

An integrator performs averaging over each sampling interval and reconstructs the envelope of the sampled signal.

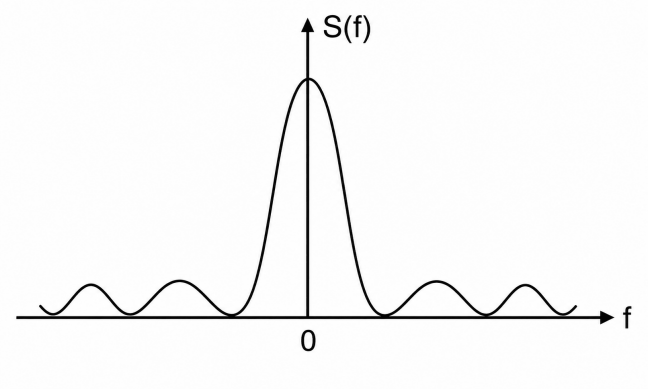


Fig. Solution Q4.95.1

Hence, an integrator (or equivalently a low-pass filter) is used to detect PAM signals.

(B) an integrator

Key Points

- PAM information is carried in pulse amplitudes.
- An integrator smooths the sampled waveform.
- A low-pass filter performs the same reconstruction function.

Mistakes to be avoided

- Do not use a band-pass or high-pass filter for PAM detection.
- ADC is not the analog detector of PAM.
- Remember that reconstruction requires averaging of pulse amplitudes.

Q4.95.2 MCQ 1995 1M Ans: C

Theory: A PLL converts frequency variations of an FM signal into a proportional output voltage, thereby recovering the message signal.

A Phase Locked Loop (PLL) consists of a phase detector, low-pass filter and voltage controlled oscillator (VCO).

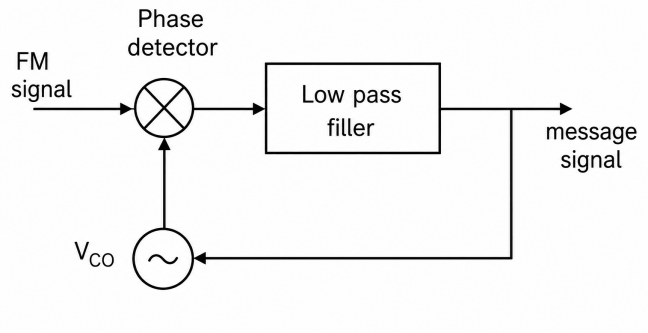


Fig. Solution Q4.95.2

The phase detector generates an error voltage proportional to the instantaneous frequency difference between the incoming FM signal and the VCO output.

After low-pass filtering, this error voltage is the recovered message signal.

(C) FM signals

Key Points

- PLL is an FM demodulator.
- Phase detector generates the error signal.
- Low-pass filter extracts the message signal.

Mistakes to be avoided

- Do not confuse PLL demodulation with coherent AM or DSB-SC detection.
- PCM and PAM are not demodulated using a PLL.
- Remember that the PLL tracks frequency variations, not amplitude variations.

Q4.95.3 MCQ 1995 1M Ans: B

Theory: Image frequency must be rejected before the mixer; therefore image channel selectivity depends only on the RF front-end.

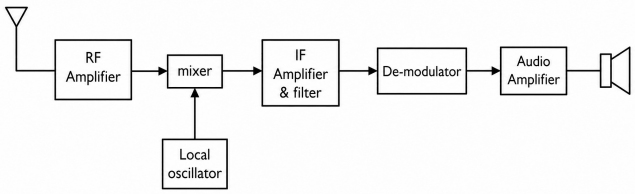


Fig. Solution Q4.95.3

The image frequency component must be filtered before reaching the mixer.
After the mixer, both the desired signal and image signal are converted to the same intermediate frequency (IF). Hence, the IF amplifier cannot distinguish between them.
Therefore, image channel selectivity is determined only by the preselector and RF amplifier present before the mixer.

(B) the pre selector and RF amplifiers

Key Points

- Image rejection must occur before frequency conversion.
- The preselector and RF amplifier determine image channel selectivity.
- IF stages cannot reject the image frequency after mixing.

Q4.94.1 NAT 1994 2M Ans: A-3,B-1,C-2 ● ○ ○

Theory: Each modulation scheme requires a detector that matches its signal structure for proper demodulation.

List I	List II
a) SSB	1) Envelope detector
b) AM	2) Integrate and dump
c) BPSK	3) Hilbert transform
	4) Ratio detector
	5) PLL

Fig. Solution Q4.94.1

For SSB signals, the Hilbert transform method is used for generation and detection.
For AM signals, an envelope detector recovers the message from the signal envelope.
For BPSK signals, an integrate-and-dump receiver performs optimum coherent detection.
Hence, the correct matching is

$$\boxed{a-3, \qquad b-1, \qquad c-2.}$$

Key Points

- Hilbert transform is associated with SSB generation/detection.
- Envelope detector is used for AM demodulation.
- Integrate-and-dump receiver is used for coherent BPSK detection.
- PLL is primarily used for FM demodulation.
- Ratio detector is also used for FM demodulation.

Mistakes to be avoided

- Do not confuse the Hilbert transform with envelope detection.
- Do not use an envelope detector for BPSK or SSB signals.
- Remember that PLL and ratio detector are FM demodulators, not AM or BPSK demodulators.

Q4.94.2 MCQ 1994 1M Ans: D ● ○ ○

Theory: A narrowband FM signal can be identified by comparing the given expression with the standard NBFM approximation.
The given signal is

$$v(t) = 5\left[\cos(10^6\pi t) - \sin(10^3\pi t)\sin(10^6\pi t)\right].$$

The standard expression for a narrowband FM signal is

$$s(t) = A_c\cos(2\pi f_c t) - A_c\beta\sin(2\pi f_c t)\sin(2\pi f_m t).$$

The given signal matches the standard form of NBFM.
Hence,

$$\boxed{\text{(D)Narrow band FM signal}}$$

Key Points

- NBFM consists of a carrier and a small quadrature term.
- Compare the given expression with the standard NBFM approximation.
- NBFM is obtained for modulation index $\beta \ll 1$.

Q4.94.3 NAT 1994 1M Ans: D ●○○

Theory: Carson's rule estimates the transmission bandwidth of an FM signal using the maximum frequency deviation and the highest modulating frequency.

Given,

$$f_m = 500 \text{ Hz}, \qquad \Delta f = 50 \text{ kHz}.$$

The modulation index is

$$\beta = \frac{\Delta f}{f_m} = \frac{50 \times 10^3}{500} = 100.$$

Using Carson's rule,

$$BW = 2(\beta + 1) f_m$$

$$= 2(100 + 1) \times 500$$

$$= 101000 \text{ Hz}$$

$$\boxed{101 \text{ kHz}}$$

Key Points

- Carson's rule is $BW = 2(\Delta f + f_m)$.
- Alternatively, $BW = 2(\beta + 1) f_m$.
- Use the highest modulating frequency in Carson's rule.

Mistakes to be avoided

- Do not use the carrier frequency while calculating FM bandwidth.
- Do not forget to convert units before substitution.
- Do not omit the +1 term while using $BW = 2(\beta + 1) f_m$.

Q4.93.1 MCQ 1993 2M Ans: C ●○○

Theory: Over-modulated AM signals cannot be recovered using envelope or square-law detectors, whereas coherent (synchronous) detection always recovers the message.

The given signal is

$$x(t) = 5(1 + 2\cos 200\pi t)\cos 20000\pi t.$$

Comparing with the standard AM signal,

$$x(t) = A_c(1 + \mu \cos 2\pi f_m t)\cos 2\pi f_c t,$$

we obtain

$$\mu = 2.$$

Since

$$\mu > 1,$$

the signal is over-modulated. Hence, envelope and square-law detectors fail due to envelope distortion.
A synchronous demodulator correctly recovers the message even under over-modulation.
Therefore,

$$\boxed{\text{(C) Synchronous demodulator}}$$

Key Points

- Envelope detector requires $\mu \leq 1$.
- Over-modulated AM requires coherent detection.
- Synchronous detection works irrespective of modulation index.

Mistakes to be avoided

- Do not use an envelope detector for over-modulated AM.
- Do not assume a square-law detector works for all AM signals.
- Always calculate the modulation index before selecting the detector.

Q4.93.2 NAT 1993 2M Ans: 2110 ●○○

Theory: In a super-heterodyne receiver, the image frequency is separated from the desired signal by twice the intermediate frequency.
Given,

$$f_s = 1200 \text{ kHz},$$

$$IF = 455 \text{ kHz}.$$

The image frequency is

$$f_{si} = f_s + 2IF$$

$$= 1200 + 2(455)$$

$$= 1200 + 910$$

$$= 2110 \text{ kHz}.$$

Hence,

$$\boxed{f_{si} = 2110 \text{ kHz}}$$

Key Points

- Image frequency is separated from the desired signal by $2IF$.
- For high-side injection, $f_{si} = f_s + 2IF$.
- RF preselector rejects the image frequency.

Q4.92.1 MCQ | 1992 | 2M | Ans: B | ● ○ ○

Theory: The power efficiency of a conventional AM modulator is maximized at 100% modulation, giving a maximum efficiency of 50%.

For an AM signal,

$$\eta = \frac{K_a^2 \left\langle m^2(t) \right\rangle}{1 + K_a^2 \left\langle m^2(t) \right\rangle},$$

where $\left\langle m^2(t) \right\rangle$ is the mean square value of the message signal.

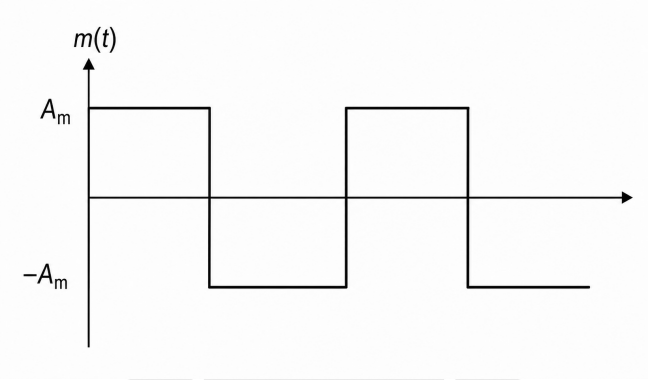


Fig. Solution Q4.92.1

For a square wave,

$$\left\langle m^2(t) \right\rangle = A_m^2.$$

Hence,

$$\eta = \frac{K_a^2 A_m^2}{1 + K_a^2 A_m^2}.$$

Using

$$\mu = K_a A_m,$$

we obtain

$$\eta = \frac{\mu^2}{1 + \mu^2}.$$

The maximum permissible modulation index for AM is

$$\mu_{\max} = 1.$$

Therefore,

$$\eta_{\max} = \frac{1}{1+1} = \frac{1}{2} = 50\%.$$

Hence,

$$\boxed{(B)\eta_{\max} = 50\%}$$

Key Points

- AM efficiency depends on the modulation index.
- Maximum efficiency occurs at $\mu = 1$ (100% modulation).
- The maximum efficiency of conventional AM is 50%.

Mistakes to be avoided

- Do not use $\mu > 1$, as it leads to overmodulation.
- Do not confuse transmission efficiency with power efficiency.
- Remember that conventional AM cannot exceed 50% efficiency.

Q4.90.1 MCQ 1990 2M Ans: C ●○○

Theory: Analog TV transmission uses VSB for video to reduce bandwidth while preserving low-frequency components, whereas FM provides better noise immunity for audio. The picture (video) signal is transmitted using **Vestigial Sideband (VSB)** modulation because the video signal occupies a wide bandwidth and VSB provides a good compromise between bandwidth saving and ease of demodulation. The speech (audio) signal is transmitted using **Frequency Modulation (FM)** due to its superior noise immunity and better audio quality.

Hence,

$$\boxed{\text{(C) VSB and FM}}$$

Key Points

- Picture (video) signal uses VSB modulation.
- Speech (audio) signal uses FM modulation.
- VSB reduces bandwidth while retaining low-frequency information.

Q4.89.1 MCQ 1989 2M Ans: A ●○○

Theory: The FM modulation index equals the ratio of frequency deviation to the modulating frequency.

Given,

$$k_f = 10 \text{ kHz/volt},$$

$$A_m = 2 \text{ V}.$$

From

$$x(t) = 2\cos(\pi \times 10^4 t),$$

$$\omega_m = \pi \times 10^4,$$

therefore,

$$f_m = \frac{\omega_m}{2\pi} = 5 \text{ kHz}.$$

The frequency deviation is

$$\Delta f = k_f A_m = 10 \times 2 = 20 \text{ kHz}.$$

Hence,

$$\beta = \frac{\Delta f}{f_m} = \frac{20}{5} = 4.$$

Therefore,

$$\boxed{(A)\beta = 4}$$

Key Points

- Frequency deviation is $\Delta f = k_f A_m$.
- FM modulation index is $\beta = \dfrac{\Delta f}{f_m}$.

- Convert angular frequency to frequency before applying the formula.

Mistakes to be avoided

- Do not use ω_m directly in place of f_m.
- Do not forget to multiply sensitivity by message amplitude to obtain Δf.
- Remember that $f_m = \dfrac{\omega_m}{2\pi}$.

Q4.89.2 MCQ 1989 2M Ans: C ●○○

Theory: Threshold effect is a phenomenon specific to FM receivers in which performance degrades abruptly when the input SNR falls below a certain level. The threshold effect occurs when the receiver input SNR becomes very low, making the noise comparable to or larger than the desired signal. Among the given schemes, this phenomenon is associated with FM detection using a discriminator. Hence, the correct answer is

(C) FM detection using a discriminator

Key Points

- Threshold effect is characteristic of FM receivers.
- It occurs at low input SNR.
- AM and SSB detectors do not exhibit the FM threshold effect.

Mistakes to be avoided

- Do not associate threshold effect with AM envelope detection.
- Do not confuse threshold effect with the capture effect.
- Remember that discriminator-based FM receivers exhibit threshold behavior.

Q4.87.1 MCQ 1987 2M Ans: A ●●○

Theory: In FM, the instantaneous phase is obtained by integrating the message signal, resulting in a sine term for sinusoidal modulation. The FM signal is

$$V_{FM}(t) = A_c \cos\left[\omega_c t + k_f \int m(t)\, dt\right].$$

Given,

$$m(t) = E_m \cos \omega_m t.$$

Therefore,

$$\int m(t)\, dt = \int E_m \cos \omega_m t\, dt = \frac{E_m}{\omega_m} \sin \omega_m t.$$

Hence,

$$V_{FM}(t) = A_c \cos\left[\omega_c t + \frac{k_f E_m}{\omega_m} \sin \omega_m t\right].$$

Since

$$m_f = \frac{k_f E_m}{\omega_m},$$

the FM signal becomes

$$\boxed{V_{FM}(t) = A_c \cos\left[\omega_c t + m_f \sin \omega_m t\right]}$$

Hence, the correct answer is

$$\boxed{\text{(A) } A_c \cos\left[\omega_c t + m_f \sin \omega_m t\right]}$$

Key Points

- FM phase is proportional to the integral of the message signal.
- Integrating a cosine produces a sine.
- For a sinusoidal message, the standard FM expression contains $m_f \sin \omega_m t$.

Mistakes to be avoided

- Do not write $m_f \cos \omega_m t$; the integral introduces a sine term.
- Do not confuse FM with PM, where the message appears directly in the phase.
- Remember that $m_f = \frac{k_f E_m}{\omega_m}$.

Q4.87.2 MCQ 1987 2M Ans: A ●○○

Theory: Placing a Low Noise Amplifier (LNA) before a lossy component minimizes the overall receiver noise figure according to Friis' formula.

A preamplifier placed at the antenna side acts as a Low Noise Amplifier (LNA), providing high gain with a low noise figure.

According to Friis' formula for cascaded stages,

$$F = F_1 + \frac{F_2 - 1}{G_{a1}} + \frac{F_3 - 1}{G_{a1}G_{a2}} + \cdots + \frac{F_n - 1}{G_{a1}G_{a2}\cdots G_{a(n-1)}}.$$

If the waveguide (which has insertion loss) is placed before the amplifier, its loss directly increases the receiver noise figure. Placing the LNA before the waveguide provides gain before the lossy section, thereby suppressing the contribution of the waveguide and subsequent stages to the overall noise figure. Hence, placing the preamplifier on the antenna side reduces the overall receiver noise figure. The correct answer is:

$$\boxed{\text{(A) A reduction in the overall noise figure}}$$

Key Points

- An LNA should be placed as close to the antenna as possible.
- Friis' formula shows that the first stage dominates the overall noise figure.
- Gain before a lossy component minimizes the effect of its noise contribution.

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